MATHEMATICS

DIFFERENTIAL EQUATION

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DIFFERENTIAL EQUATION

JEE EXERCISES

Exercise J-I

ORDER, DEGREE AND FORMATION OF A DIFFERENTIAL EQUATION

1. The order and degree of the differential equation $\left[4+\left(\frac{dy}{dx}\right)^2\right]^{2/3}=\frac{d^2y}{dx^2}$ are :

अवकल समीकरण $\left[4+\left(\frac{dy}{dx}\right)^2\right]^{2/3}=\frac{d^2y}{dx^2}$ की कोटि व घात क्रमशः है :

(A) 2, 2 (B) 3, 3 (C) 2, 3 (D) 3, 2

Ans. C

Sol. $\left[4+\left(\frac{dy}{dx}\right)^2\right]^{2/3}=\left(\frac{d^2y}{dx^2}\right)^3$

O = 2 D = 3

2. The differential equation of all parabolas having their axis of symmetry coinciding with x-axis :

(A) $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$ (B) $x\frac{d^2x}{dy^2}+\left(\frac{dy}{dx}\right)^2=0$ (C) $y\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$ (D) none of these

सभी परवलयों जिनके सममित अक्ष X अक्ष के साथ सम्पाती है कि अवकल समीकरण है :

(A) $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$ (B) $x\frac{d^2x}{dy^2}+\left(\frac{dy}{dx}\right)^2=0$ (C) $y\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$ (D) इनमें से कोई नही

Ans. A

Sol. $y^2 = 4ax$

$$2y\frac{dy}{dx}=4a \Rightarrow y\frac{dy}{dx}=2a$$

$$\left(\frac{dy}{dx}\right)^2+y\frac{d^2y}{dx^2}=0$$

3. The differential equation of system of concentric circles with centre (1, 2) is :

उन सकेन्द्रीय वृत्त निकाय, जिनका केन्द्र (1, 2) है, की अवकल समीकरण होगी :

(A) $(x-2)+(y-2)\dfrac{dy}{dx}=0$ (B) $(x-1)+(y-2)\dfrac{dy}{dx}=0$

(C) $(x+1)\dfrac{dy}{dx}+(y-2)=0$ (D) $(x+2)\dfrac{dy}{dx}+(y-1)=0$

Ans. B

Sol. $(x-1)^2+(y-2)^2=r^2$

$$2(x-1)+2(y-2)\frac{dy}{dx}=0$$

$$(x-1)+(y-2)\frac{dy}{dx}=0$$

4. The order of the differential equation whose general solution is given by $y=(c_1+c_2)\cos(x+c_3)-c_4e^{x+c_5}$, where c_1, c_2, c_3, c_4, c_5 are arbitrary constant is :

अवकल समीकरण जिसका व्यापक हल $y=(c_1+c_2)\cos(x+c_3)-c_4e^{x+c_5}$, जहाँ c_1, c_2, c_3, c_4, c_5 स्वैच्छ अचर है, की कोटि gksxh:

(A) 5 (B) 4 (C) 3 (D) 2

Ans. C

Sol. $y=(c_1+c_2)\cos(x+c_3)-c_4e^{x+c_5}$

$$y=(c_1+c_2)\cos(x+c_3)-c_4e^{c_5}e^x$$

$$y=d_1\cos(x+c_3)+d_2e^x \qquad ...(1)$$

$$\frac{dy}{dx}=-d_1\sin(x+c_3)+d_2e^x \qquad ...(2)$$

$$\frac{d^2y}{dx^2}=-d_1\cos(x+c_3)+d_2e^x \qquad ...(3)$$

$$\frac{d^3y}{dx^3}=d_1\cos(x+c_3)+d_2e^x \Rightarrow \frac{d^3y}{dx^3}=y$$

5. The value of the constant 'm' and 'c' for which $y = mx + c$ is a solution of the differential equation $D^2y-3Dy-4y=-4x$:

(A) is $m = -1; c = 3/4$ (B) is $m = 1; c = -3/4$

(C) no such real m, c (D) is $m = 1;\ c = 3/4$

यदि $y = mx + c$ अवकल समीकरण $D^2y - 3Dy - 4y = -4x$ का एक हल है तो 'm' तथा 'c' के मान होंगे :

(A) $m = -1; c = 3/4$ (B) $m = 1; c = -3/4$

(C) कोई वास्तविक m एवं c नही होगा (D) $m = 1;\ c = 3/4$

Ans. B

Sol. $y = mx + c; \quad \frac{dy}{dx} = m; \quad \frac{d^2y}{dx^2} = 0$

substituting in $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} - 4y = -4x$

$0 - 3m - 4(mx + c) = -4x$

$-3m - 4c - 4mx = -4x$

$-(3m + 4c) = 4x(m - 1) \quad ....(1)$

(1) is true for all real x if

$m = +1$ and $c = -3/4 \quad \Rightarrow$ (B)

6. The order & the degree of the differential equation whose general solution is, $y = c(x - c)^2$, are respectively :

उस अवकल समीकरण की कोटि तथा घात क्रमशः होगें जिसका व्यापक हल $y = c(x - c)^2$, है :

(A) 1, 1 (B) 1, 2 (C) 1, 3 (D) 2, 1

Ans. C

Sol. $y = c\,(x - c)^2 \quad ...(1)$

$y_1 = 2c\,(x - c) \quad ...(2)$

(1) ÷ (2)

$$\frac{y}{y_1} = \frac{c(x-c)^2}{2c(x-c)}$$

$$\frac{2y}{y_1} = x - c$$

$$c = x - \frac{2y}{y_1}$$

$$y_1 = 2\left(x - \frac{2y}{y_1}\right)\left(\frac{2y}{y_1}\right)$$

$$(y_1)^3 = 2(xy_1 - 2y)(2y)$$

7. The differential equation, which represents the family of plane curves $y = e^{cx}$, is :

वह अवकल समीकरण जो समतल वक्र निकाय $y = e^{cx}$ को प्रदर्शित करती है, होगी :

(A) $y' = cy$ (B) $xy' - \log y = 0$ (C) $x \log y = yy'$ (D) $y \log y = xy'$

Ans. D

Sol. $y = e^{cx}$

$\log y = cx$ ..... (i)

$$\frac{1}{y}y' = c \Rightarrow y' = cy$$

$c = \frac{y'}{y}$ put in equation (i) $\log y = \frac{y'}{y}.x$

or $y \log y = xy'$

8. The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is :

वक्र निकाय $x^2 + y^2 - 2ay = 0$, जहाँ a स्वैच्छ नियतांक है, का अवकल समीकरण होगा :

(A) $2(x^2 - y^2)\, y' = xy$ (B) $2(x^2 + y^2)\, y' = xy$ (C) $(x^2 - y^2)\, y' = 2xy$ (D) $(x^2 + y^2)\, y' = 2xy$

Ans. C

Sol. Given family of curves is

$x^2 + y^2 - 2ay = 0$ ..... (1)

$2x + 2yy' - 2ay' = 0$ ..... (2)

Now put the value of 2a from (1) to in (2)

$$2x + 2yy' - \frac{x^2 + y^2}{y}.y' = 0$$

$2xy + (y^2 - x^2)y' = 0$ or $(x^2 - y^2)y' = 2xy$

9. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$, is a parameter, is of order and degree as follows :

(A) order 1, degree 2 (B) order 1, degree 1 (C) order 1, degree 3 (D) order 2, degree 2

वक्र कुल $y^2 = 2c(x + \sqrt{c})$, जहाँ $c > 0$ एक प्राचल है, को प्रदर्शित करने वाले अवकल समीकरण की कोटि (order) तथा घात (degree) निम्न है :

(A) कोटि 1, घात 2 (B) कोटि 1, घात 1 (C) कोटि 1, घात 3 (D) कोटि 2, घात 2

Ans. C

Sol. $y^2 = 2c(x + \sqrt{c})$ ..... (1)

$y^2 = 2cx + 2c\sqrt{c}$

$2y\frac{dy}{dx} = 2c \Rightarrow yy_1 = C$ Put in equation(1)

$\Rightarrow y^2 = 2yy_1(x + \sqrt{yy_1})$

$y^2 = -2yy_1x = 2yy_1\sqrt{yy_1}$ or $(y^2 - 2yy_1x)^2 = 4y^3y_1^3$

Degree = 3 order = 1

10. The differential equation whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitrary constants is of :

(A) first order and second degree (B) first order and first degree

(C) second order and first degree (D) second order and second degree

वह अवकल समीकरण, जिसका हल $Ax^2 + By^2 = 1$ है, जहाँ A तथा B स्वेच्छ अचर है, होगा :

(A) प्रथम कोटि तथा द्वितीय घात वाला अवकल समीकरण (B) प्रथम कोटि तथा प्रथम घात वाला अवकल समीकरण

(C) द्वितीय कोटि तथा प्रथम घात वाला अवकल समीकरण (D) द्वितीय कोटि तथा द्वितीय घात वाला अवकल समीकरण

Ans. C

Sol. Given $Ax^2 + By^2 = 1$ Divide by B

$\frac{A}{B}x^2 + y^2 = \frac{1}{B}$ Differentiate w.r.t x

$2x\frac{A}{B} + 2y\frac{dy}{dx} = 0$ ..... (i)

Again Differentiate w.r.t. x

$2\frac{A}{B} + 2\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] = 0$ ..... (ii)

Put $\frac{A}{B} = -\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right]$ in equation (i)

$-2x\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] + 2y\frac{dy}{dx} = 0$

or $xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$

It have second order and first degree.

11. The differential equation of all circles passing through the origin and having their centres on the x-axis is :

मूल बिन्दु से गुजरने वाले तथा x-अक्ष पर केन्द्र वाले सभी वृत्तों का अवकल समीकरण होगा :

(A) $x^2 = y^2 + xy\frac{dy}{dx}$ (B) $x^2 = y^2 + 3xy\frac{dy}{dx}$ (C) $y^2 = x^2 + 2xy\frac{dy}{dx}$ (D) $y^2 = x^2 - 2xy\frac{dy}{dx}$

Ans. C

Sol. Let the centre of circle is (h, 0) and radius will be also h

$\therefore$ equation of circle $(x-h)^2 + (y-0)^2 = h^2$

$\Rightarrow x^2 - 2hx + h^2 + y^2 = h^2$

$\Rightarrow x^2 - 2hx + y^2 = 0$ ..... (i)

Equation (i) passes through origin differentiating it w.r.t. x

$2x - 2h + 2y\frac{dy}{dx} = 0 \Rightarrow h = x + y\frac{dy}{dx}$ put in equation (i)

$$x^2 - 2x\left(x + y\frac{dy}{dx}\right) + y^2 = 0$$

$$\Rightarrow y^2 = x^2 + 2xy\frac{dy}{dx}$$

12. The differential equation of the family of circles with fixed radius 5 units and centre on the line $y = 2$ is :

उस वृत्त कुल, जिसकी स्थिर त्रिज्या 5 है तथा जिसका केन्द्र रेखा $y = 2$ पर स्थित हैं, का अवकल समीकरण है :

(A) $(y-2)y'^2 = 25 - (y-2)^2$ (B) $(y-2)^2 y'^2 = 25 - (y-2)^2$

(C) $(x-2)^2 y'^2 = 25 - (y-2)^2$ (D) $(x-2)y'^2 = 25 - (y-2)^2$

Ans. B

Sol. Equation of circle $(x-h)^2 + (y-2)^2 = 25$ .....(i)

Differentiate w.r.t. x

$$2(x-h) + 2(y-2)\frac{dy}{dx} = 0$$

$$(x-h) = -(y-2)\frac{dy}{dx} \text{ put in (i)}$$

$$(y-2)^2\left(\frac{dy}{dx}\right)^2 + (y-2)^2 = 25$$

or $(y-2)^2 (y')^2 + (y-2)^2 = 25$

13. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$ where c_1 and c_2 are arbitrary constants, is : **[AIEEE-2009]**

वक्र-कुल $y = c_1 e^{c_2 x}$ जहाँ c_1 और c_2 स्वेच्छ अचर हैं, को निरुपित करने वाला अवकल समीकरण है : **[AIEEE-2009]**

(A) $y' = y^2$ (B) $y'' = y' y$ (C) $yy' = y'$ (D) $yy'' = (y')^2$

Ans. D

Sol. $y = c_1 e^{c_2 X}$ .....(1)

$y' = c_1 c_2 e^{c_2 x}$ .....(2)

$y'' = c_1 c_2^2 e^{c_2 x}$

$y'' = c_2 y'$ .....(3)

Now $\frac{(2)}{(1)}$

$\frac{y'}{y} = c_2$

$\Rightarrow$ Put in (3)

$y'' = \frac{y'}{y} . y' \Rightarrow y'' y = (y')^2$

VARIABLE SEPARABLE

14. The solution of $\frac{dy}{dx} = e^x (\sin x + \cos x)$ is :

अवकल समीकरण $\frac{dy}{dx} = e^x (\sin x + \cos x)$ का हल है :

(A) $y = e^x (\sin x - \cos x) + c$ (B) $y = e^x (\cos x - \sin x) + c$

(C) $y = e^x \sin x + c$ (D) $y = e^x \cos x + c$

Ans. C

Sol. $\frac{dy}{dx} = e^x (\sin x + \cos x)$

$\int dy = \int e^x (\sin x + \cos x) dx$

$y = e^x \sin x + c$

15. The solution of $\frac{dy}{dx} = x \log x$ is :

(A) $y = x^2 \log x - \frac{x^2}{2} + c$ (B) $y = \frac{x^2}{2} \log x - x^2 + c$

(C) $y = \frac{1}{2}x^2 + \frac{1}{2}x^2 \log x + c$ (D) None of these

अवकल समीकरण $\frac{dy}{dx} = x \log x$ का हल है :

(A) $y = x^2 \log x - \frac{x^2}{2} + c$ (B) $y = \frac{x^2}{2}\log x - x^2 + c$

(C) $y = \frac{1}{2}x^2 + \frac{1}{2}x^2 \log x + c$ (D) इनमें से कोई नही

Ans. D

Sol. $\frac{dy}{dx} = x \log x$

$$\int dy = \int \underset{I}{x} \underset{II}{\log x}\, dx$$

$$y = \frac{x^2}{x}\log x - \int \frac{x^2}{2} \times \frac{1}{x} dx + C$$

$$y = \frac{x^2}{2}\log x - \frac{x^2}{4} + C$$

16. The solution of the differential equation $\frac{dy}{dx} = (1+x)(1+y^2)$ is :

अवकल समीकरण $\frac{dy}{dx} = (1 + x)(1 + y^2)$ का हल है :

(A) $y = \tan(x^2 + x + c)$ (B) $y = \tan(2x^2 + x + c)$

(C) $y = \tan(x^2 - x + c)$ (D) $y = \tan\left(\frac{x^2}{2} + x + c\right)$

Ans. D

Sol. $\frac{dy}{dx} = (1+x)(1+y^2)$

$$\int \frac{dy}{1+y^2} = \int (1+x)\, dx$$

$$\tan^{-1} y = x + \frac{x^2}{2} + C$$

$$y = \tan\left(x + \frac{x^2}{2} + C \right)$$

17. The general solution of the equation $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$ is :

(A) $y = \log x + c_1 x + c_2$ (B) $y = -\log x + c_1 x + c_2$

(C) $y = -\frac{1}{x} + c_1 x + c_2$ (D) None of these

अवकल समीकरण $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$ का व्यापक हल है :

(A) $y = \log x + c_1 x + c_2$ (B) $y = -\log x + c_1 x + c_2$

(C) $y = -\frac{1}{x} + c_1 x + c_2$ (D) इनमें से कोई नही

Ans. A

Sol. $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$

$$\int d^2y = \int -\frac{1}{x^2} dx^2$$

$$\int dy = \int \frac{dx}{x} + C_1$$

$$y = \ln x + c_1 x + c_2$$

18. The solution of $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ is :

(A) $\log\left[1 + \tan\left(\frac{x+y}{2}\right)\right] + c = 0$ (B) $\log\left[1 + \tan\left(\frac{x+y}{2}\right)\right] = x + c$

(C) $\log\left[1 - \tan\left(\frac{x+y}{2}\right)\right] = x + c$ (D) None of these

अवकल समीकरण $\frac{dy}{dx} = \sin(x + y) + \cos(x+y)$ का हल है :

(A) $\log\left[1+\tan\left(\frac{x+y}{2}\right)\right]+c=0$ (B) $\log\left[1+\tan\left(\frac{x+y}{2}\right)\right]=x+c$

(C) $\log\left[1-\tan\left(\frac{x+y}{2}\right)\right]=x+c$ (D) इनमें से कोई नही

Ans. B

Sol. $\frac{dy}{dx}=\sin(x+y)+\cos(x+y)$

$x+y=v$

$1+\frac{dy}{dx}=\frac{dv}{dx}$

$\frac{dv}{dx}=\sin v+\cos v+1$

$\frac{dv}{\sin v+\cos v+1}=dx$

$$\Rightarrow \frac{dv}{\frac{2\tan\frac{v}{2}}{1+\tan^2\frac{v}{2}}+\frac{1-\tan^2\frac{v}{2}}{1+\tan^2\frac{v}{2}}+1}=dx$$

$$\frac{\left(1+\tan^2\frac{v}{2}\right)dv}{2\tan\frac{v}{2}+1-\tan^2\frac{v}{2}+1+\tan^2\frac{v}{2}}=dx$$

$$\int\frac{\sec^2\frac{v}{2}dv}{2\left(1+\tan\frac{v}{2}\right)}=\int dx$$

$1+\tan\frac{v}{2}=t$

$$\frac{1}{2}\sec^2\frac{v}{2}dv = dt$$

$$\int\frac{dt}{t} = \int dx$$

$$\ln(t) = x + c$$

$$\ln\left(1+\tan\frac{x+y}{2}\right) = x + c$$

19. The solution of $(dy/dx) = (4x + y + 1)^2$ is :

(A) $4x - y + 1 = 2\tan(2x - 2c)$ (B) $4x - y - 1 = 2\tan(2x - 2c)$

(C) $4x + y + 1 = 2\tan(2x + 2c)$ (D) None of these

अवकल समीकरण $(dy/dx) = (4x + y + 1)^2$ का हल है :

(A) $4x - y + 1 = 2\tan(2x - 2c)$ (B) $4x - y - 1 = 2\tan(2x - 2c)$

(C) $4x + y + 1 = 2\tan(2x + 2c)$ (D) इनमें से कोई नही

Ans. C

Sol. $$\frac{dy}{dx} = (4x + y + 1)^2$$

$4x + y + 1 = t$

$4 + y_1 = dt$

$$\frac{dt}{dx} - 4 = t^2$$

$$\frac{dt}{dx} = t^2 + 4$$

$$\int\frac{dt}{4+t^2} = \int dx$$

$$\frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right) = x + c$$

$$\frac{t}{2} = \tan(2x + 2c)$$

$4x + y + 1 = 2\tan(2x + 2c)$

20. The solution of the differential equation $(\sin x + \cos x)\, dy + (\cos x - \sin x)dx = 0$ is :

अवकल समीकरण $(\sin x + \cos x)\, dy + (\cos x - \sin x)\, dx = 0$ का हल है :

(A) $e^x (\sin x + \cos x) + c = 0$ (B) $e^y (\sin x + \cos x) = c$

(C) $e^y (\cos x - \sin x) = c$ (D) $e^y (\sin x - \cos x) = c$

Ans. B

Sol. $(\sin x + \cos x)\, dy + (\cos x - \sin x)\, dx$

$$\frac{dy}{dx} = \frac{s - c}{s + c}$$

$$dy = \frac{s - c}{s + c} dx$$

$s + c = t$

$(c - s)\, dx = dt$

$$\int dy = \int -\frac{dt}{t}$$

$y = -\ln t + c$

$\ln (s + c) = -y + c$

$(s + c) = e^{-y+c} = e^{-y} \cdot e^c$

$e^y (s + c) = c_1$

21. The solution of $\left(x\sqrt{1+y^2}\right)dx + \left(y\sqrt{1+x^2}\right)dy = 0$ is :

(A) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$ (B) $\sqrt{1+x^2} - \sqrt{1+y^2} = c$

(C) $(1 + x^2)^{3/2} + (1 + y^2)^{3/2} = c$ (D) None of these

अवकल समीकरण $\left(x\sqrt{1+y^2}\right)dx + \left(y\sqrt{1+x^2}\right)dy = 0$ का हल है :

(A) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$ (B) $\sqrt{1+x^2} - \sqrt{1+y^2} = c$

(C) $(1 + x^2)^{3/2} + (1 + y^2)^{3/2} = c$ (D) इनमें से कोई नही

Ans. A

Sol. $\left(x\sqrt{1+y^2}\right)dx + \left(y\sqrt{1+x^2}\right)dy = 0$

$$\left(x\sqrt{1+y^2}\right)dx = -\left(y\sqrt{1+x^2}\right)dy$$

$$\frac{2ydy}{\sqrt{1+y^2}} = \frac{-2xdx}{\sqrt{1+x^2}}$$

$1 + y^2 = v$ $\qquad$ $1 + x^2 = u$

$2ydy = dv$ $\qquad$ $2xdx = du$

$$\frac{dv}{\sqrt{v}} = -\frac{du}{\sqrt{u}} \Rightarrow 2\sqrt{v} = -2\sqrt{u} + c$$

$$\sqrt{1+y^2} = -\sqrt{1+v^2} + c_1$$

22. Solution of differential equation $x = 1 + xy\frac{dy}{dx} + \frac{(xy)^2}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{(xy)^3}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots\ldots$ is :

अवकल समीकरण $x = 1 + xy\frac{dy}{dx} + \frac{(xy)^2}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{(xy)^3}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots\ldots$ का हल है :

(A) $y = \log_e(x) + C$ $\qquad$ (B) $y = (\log_e x)^2 + C$

(C) $y = \pm\sqrt{(\log_e x)^2 + 2C}$ $\qquad$ (D) $xy = x^y + k$

Ans. C

Sol. $x = 1 + xy\frac{dy}{dx} + \frac{(xy)^2}{\lfloor 2}\left(\frac{dy}{dx}\right)^2 + \frac{(xy)^3}{\lfloor 3}\left(\frac{dy}{dx}\right)^3 + \ldots\ldots$

$$x = (e)^{xy\frac{dy}{dx}}$$

$$\log x = xy\frac{dy}{dx}$$

$$\int ydy = \int \frac{\ln x}{x}dx$$

$\ln x = t$

$$\frac{1}{x}dx = dt$$

$$\frac{y^2}{2} = \int t\,dt$$

$$\frac{y^2}{2} = \frac{t^2}{2} + c$$

$$\frac{y^2}{2} = \frac{(\ln x)^2}{2} + c$$

$$y^2 = (\ln x)^2 + 2c$$

$$y = \pm\sqrt{(\ln x)^2 + 2c}$$

23. The solution of the differential equation $xdy - ydx = \sqrt{x^2 + y^2}\,dx$ is :

अवकल समीकरण $xdy - ydx = \sqrt{x^2 + y^2}\,dx$ का हल है :

(A) $x + \sqrt{x^2 + y^2} = cx^2$ (B) $y - \sqrt{x^2 + y^2} = cx$

(C) $x - \sqrt{x^2 + y^2} = cx$ (D) $y + \sqrt{x^2 + y^2} = cx^2$

Ans. D

Sol. $xdy - ydx = \sqrt{x^2 + y^2}dx$

$$xdy = ydx + \sqrt{x^2 + y^2}dx$$

$$xdy = (y + \sqrt{x^2 + y^2})dx$$

$$\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$$

$y = vx$

$$\frac{dy}{dx} = v + \frac{xdv}{dx}$$

$$v + \frac{xdv}{dx} = \frac{vx + \sqrt{x^2(1+v)^2}}{x} = v + \sqrt{1+v^2}$$

$$x\frac{dv}{dx} = \sqrt{1+v^2}$$

$$\int \frac{dv}{\sqrt{1+v^2}} = \int \frac{dx}{x}$$

$$\ell n\left(v+\sqrt{1+v^2}\right) = \ell nx + C$$

$$\ell n\left(\frac{y}{x}+\frac{\sqrt{x^2+y^2}}{x}\right) - \ell nx = \ell nC$$

$$\frac{y+\sqrt{x^2+y^2}}{x^2} = C$$

$$y+\sqrt{x^2+y^2} = Cx^2$$

24. A curve C has the property that if the tangent drawn at any point P on C meets the coordinate axis at A and B, then P is the midpoint of AB. If the curve passes through the point (1,1) then the equation of the curve is :

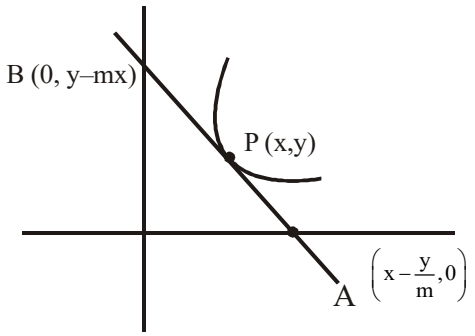
(A) $xy = 2$ (B) $xy = 3$ (C) $xy = 1$ (D) None of these

एक वक्र C इस प्रकार है कि यदि वक्र C के किसी बिन्दु P पर खींची गई स्पर्शरेखा निर्देशांक अक्षों को A तथा B पर मिलती है, तब P, AB का मध्य बिन्दु है। यदि वक्र बिन्दु (1,1) से गुजरता है तब वक्र का समीकरण होगा :

(A) $xy = 2$ (B) $xy = 3$ (C) $xy = 1$ (D) इनमें से कोई नही

Ans. C

Sol. $Y - y = m(X - x)$



$A \Rightarrow Y = 0$

$0 - y = m(X - x)$

$$-\frac{y}{m} = X - x$$

$$X = x - \frac{y}{m}$$

$B \Rightarrow X = 0$

$Y = y - mx$

P is midpoint

$$x = \frac{x - \frac{y}{m} \times 0}{2} \qquad \frac{y - mx + 0}{2} = y$$

$$2x = x - \frac{y}{m} \qquad mx = -y$$

$$m = -\frac{y}{x}$$

$$x = -\frac{y}{m} \qquad \frac{dy}{dx} = -\frac{y}{x}$$

$$m = -\frac{y}{m} \qquad \int \frac{dy}{y} = -\int \frac{dy}{x}$$

$$\ln y = -\ln x + c$$

$xy = e^c \Rightarrow xy = c\ (1, 1)$

$c = 1$

25. A curve is such that the area of the region bounded by the co-ordinate axes, the curve & the ordinate of any point on it is equal to the cube of that ordinate. The curve represents :

(A) a pair of straight lines (B) a circle

(C) a parabola (D) an ellipse

एक वक्र इस प्रकार है कि निर्देशांक अक्षों, वक्र एवं उस पर किसी बिन्दु की कोटि द्वारा परिबद्ध क्षेत्रफल कोटि के घन के बराबर है। वक्र i nf' kZ d j r kgS

(A) रेखा युग्म (B) वृत्त (C) परवलय (D) दीर्घवृत्त

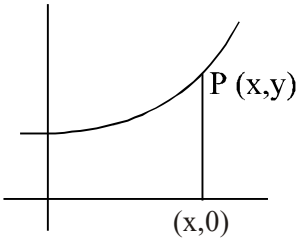
Ans. C

Sol. $\int_0^x f(x)\,dx = y^3$

Differentiating

$$f(x) = 3y^2 . \frac{dy}{dx}$$

$$y = 3y^2 \frac{dy}{dx} \quad \Rightarrow \quad y = 0 \text{ (rejected)}$$



or $3y\,dy = dx$

$$\frac{3y^2}{2} = x + c \Rightarrow \text{parabola} \Rightarrow C$$

26. In a chemical reaction a substance changes into another such that the rate of decomposition of a chemical substance x present at instant t is proportional to x itself i.e. amount of unchanged substance still present. If half of the substance present initially has been converted at the end of 1 minute then the time t in minutes at the end of which 99% of the substance will have changed lies in the interval :

(A) 5 and 6 (B) 6 and 7 (C) 7 and 8 (D) more than 10

एक रासायनिक पदार्थ एक दूसरे रासायनिक पदार्थ में इस प्रकार से बदल रहा है कि किसी भी समय t में पहले पदार्थ की शेष बची मात्रा x के अपघटन की दर x के ही समानुपाती है। यदि शुरू में मौजूद पदार्थ की आधी मात्र 1 मिनट में दूसरे पदार्थ में बदल जाती है, तो वह l e; t c i gy si nkHkZ h99 % मात्रा दूसरे पदार्थ में बदल जायेगी वह समय (मिनट में) किस अन्तराल में होगा :

(A) 5 से 6 (B) 6 से 7 (C) 7 से 8 (D) 10 से अधिक

Ans. B

Sol. $\frac{dx}{dt} = -kx \Rightarrow x = x_0 e^{-kt}$; where $k = l n 2$

27. The latus rectum of the conic passing through the origin and having the property that normal at each point (x, y) intersects the x–axis at ((x+1), 0) is :

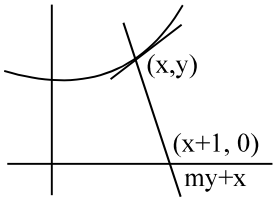
(A) 1 (B) 2 (C) 4 (D) None of these

उस शांकव के नाभिलम्ब की लम्बाई, जो मूल बिन्दु से गुजरता है तथा इसके प्रत्येक बिन्दु (x, y) से खीचें अभिलम्ब x-अक्ष को ((x+1), 0) पर मिलते है :

(A) 1 (B) 2 (C) 4 (D) कोई नहीं

Ans. B

Sol. Given $y\frac{dy}{dx} + x = x + 1$



$$\frac{y^2}{2} = x + c$$

$x = 0, y = 0 \Rightarrow c = 0$

$\therefore$ $y^2 = 2x \Rightarrow$ latus rectum$= 2 \Rightarrow B$

28. The equation of the curve through the point (1, 0), whose slope is $\frac{y-1}{x^2+x}$, is :

बिन्दु (1,0) से गुजरने वाले वक्र का समीकरण, जिसकी प्रवणता $\frac{y-1}{x^2+x}$ है, होगा :

(A) $(y-1)(x+1)+2x=0$ (B) $2x(y-1)+x+1=0$

(C) $x(y-1)(x+1)+2=0$ ` (D) $x(y+1)+y(x+1)=0$

Ans. A

Sol. Given $\frac{dy}{dx}=\frac{y-1}{x(x-1)}$ or $\int\frac{dy}{y-1}=\int\frac{dx}{x(x+1)}$

$$\log(y-1)=\log\left(\frac{x}{x+1}\right)+\log C$$

or $y-1=\frac{cx}{x+1}$ ..... (i)

Equation (i) passes through (1, 0)

$-1=\frac{C}{2}\Rightarrow C=-2$ Put in (i)

$(y-1)=\frac{-2x}{x+1}$ $(y-1)(x+1)+2x=0$

29. $\frac{dy}{dx}\left(\frac{1+\cos x}{y}\right)=-\sin x$ and $f\left(\frac{\pi}{2}\right)=-1$, then f(0) is :

$\frac{dy}{dx}\left(\frac{1+\cos x}{y}\right)=-\sin x$ तथा $f\left(\frac{\pi}{2}\right)=-1$ है, तो f(0) होगा :

(A) 2 (B) 1 (C) 3 (D) 4

Ans. A

Sol. $\frac{dy}{dx}\left(\frac{1+\cos x}{y}\right)=-\sin x$

$\int\frac{dy}{y}=\int\frac{-\sin x}{1+\cos x}dx$

$\ln(y)=\ln(1+\cos x)+c$

$\left(\frac{\pi}{2},-1\right)$

$c=0$

$y=1+\cos x$

$y=2$

30. The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is :

अवकल समीकरण $\frac{dy}{dx} = \frac{x+y}{x}$ जो प्रतिबन्ध $y(1) = 1$ को सन्तुष्ट करता है, का हल होगा :

(A) $y = x\,\ell n\,x + x^2$ (B) $y = x\,e^{(x-1)}$ (C) $y = x\,\ell n\,x + x$ (D) $y = \ell n\,x + x$

Ans. C

Sol. $\frac{dy}{dx} = 1 + \frac{y}{x}$ put $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$

$v + x\frac{dv}{dx} = 1 + \frac{vx}{x} \Rightarrow x\frac{dv}{dx} = 1$

$\int dv = \int \frac{dx}{x} \Rightarrow v = \log x + c$ or $\frac{y}{x} = \log x + c$ ... (i)

Given $y(1) = 1 \Rightarrow 1 = \log 1 + c \Rightarrow c = 1$ put (i)

$y = x\ell nx + x$

31. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with :

(A) variable radii and a fixed centre at (0, 1)

(B) variable radii and a fixed centre at (0, – 1)

(C) fixed radius 1 and variable centres along the x-axis

(D) fixed radius 1 and variable centres along the y-axis

अवकल समीकरण $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ एक वृत निकाय को प्रदर्शित करती है जिसकी :

(A) त्रिज्या चर एवं केन्द्र (0, 1) अचर है

(B) त्रिज्या चर एवं केन्द्र (0, – 1) अचर है

(C) अचर त्रिज्या इकाई है एवं केन्द्र x-अक्ष के अनुदिश चर है

(D) अचर त्रिज्या इकाई है एवं केन्द्र y-अक्ष के अनुदिश चर है

Ans. C

Sol. $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y} \Rightarrow \quad \int \frac{y}{\sqrt{1-y^2}}\,dy = \int dx$

$\Rightarrow \quad -\sqrt{1-y^2} = x + c$

$\Rightarrow \quad (x + c)^2 + y^2 = 1$

Centre (–c,0); radius = 1

32. If $\frac{dy}{dx} = y + 3 > 0$ and y(0) = 2, then y(In 2) is equal to : **[AIEEE-2011]**

यदि $\frac{dy}{dx} = y + 3 > 0$ तथा y(0) = 2, है, तो y(In 2) बराबर हैं : **[AIEEE-2011]**

(A) 13 (B) –2 (C) 7 (D) 5

Ans. C

Sol. $\frac{dy}{dx} = y + 3 > 0 \quad y(0) = 2, y(\log 2) = ?$

$$\int \frac{dy}{y+3} = \int dx$$

$\log |y + 3| = x + c$

$y(0) = 2$

$\log|2 + 3| = 0 + c \Rightarrow c = \log 5.$

$y.(\log 2) = ?$

$\log |y + 3| = \log 2 + \log 5$

$\log |y + 3| = \log 10$

$y + 3 = 10$

$y = 7$

33. Let I be the purchase value of an equipment and V(t) be the value after it has been used for t years. The value V(t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T - t)$, where k > 0 is a constant and T is the total life in years of the equipment. Then value V(T) of the equipment is : **[AIEEE-2011]**

यदि किसी उपकरण का क्रय मूल्य I है तथा t वर्ष के उपयोग के पश्चात् उसका मूल्य V(t) हैं, तो V(t) के अवमूल्यन की दर अवकल समीकरण $\frac{dV(t)}{dt} = -k(T - t)$ द्वारा प्रदत है जहाँ $k > 0$ एक अचर है तथा उपकरण का कुल जीवन काल T वर्ष है। तो V(T) के कबाड़ (scrap) का मूल्य है : **[AIEEE-2011]**

(A) $I - \frac{k(T-t)^2}{2}$ (B) e^{-kT} (C) $T^2 - \frac{I}{k}$ (D) $I - \frac{kT^2}{2}$

Ans. D

Sol. $\frac{dV}{dt} = -k(T-t)$

$\int dV = \int -K(T-t)dt$

$V = -K\left[Tt - \frac{t^2}{2}\right] + C$

At $t = 0$ $V = I \Rightarrow C = I$

$V = -Kt\left(T - \frac{t}{2}\right) + I$

$V(T) = -KT\left(T - \frac{T}{2}\right) + I = \frac{-KT^2}{2} + I$

34. The population p(t) at time t of a certain mouse species satisfies differential equation $\frac{dp(t)}{dt} = 0.5p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is : **[AIEEE-2012]**

चूहों की एक प्रजाति की किसी समय t पर जनसंख्या p(t) अवकल समीकरण $\frac{dp(t)}{dt} = 0.5p(t) - 450$ को संतुष्ट करती है। यदि $p(0) = 850$ है, तो वह समय जब यह शून्य हो गई है : **[AIEEE-2012]**

(A) In 18 (B) 2 In 18 (C) In 9 (D) $\frac{1}{2}$ In 18

Ans. B

Sol. $\frac{dP(t)}{dt} = \frac{1}{2}P(t) - 450$

integrate

$\int \frac{dP}{P-900} = \int \frac{1}{2}dt$

$\ell n|(P-900)| = \frac{1}{2}t + C$ ....(1)

given $t = 0 \rightarrow P = 850$

$\therefore$ $C = \ell n\, 50$

from (1)

$\ln|(P-900)| = \frac{1}{2}t + \ell n 50$

$\frac{1}{2}t = \ell n\left|\left(\frac{P-900}{50}\right)\right|$

$$t = 2\,\ell n\left|\left(\frac{P-900}{50}\right)\right|$$

at $P = 0$

$$t = 2\ell n\frac{900}{50}$$

$$t = 2\ell n18$$

35. At present a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workesrs x is given by $\frac{dp}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is : **[JEE(Main)2013]**

वर्तमान में एक फर्म 2000 नग बना रही है। यहद अनुमान लगाया गया है कि अतिरिक्त कामगारां की संख्या x के सापेक्ष उत्पादन P के i fj or Z d hnj $\frac{dp}{dx} = 100 - 12\sqrt{x}$ द्वारा प्रदत है। यदि फर्म 25 कामगार अधिक लगती है, तो नगों के उत्पादन का नया स्तर है : **[JEE(Main)2013]**

(A) 2500 (B) 3000 (C) 3500 (D) 4500

Ans. C

Sol. $P = 100x - 12\,x^{3/2}.\frac{2}{3} + C$

$x = 0, \quad P = 2000$

$C = 2000$

$p_{(x=25)} = 2500 - 1000 + 2000 = 3500$

36. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If $p(0) = 100$, then $p(t)$ equals : **[JEE(Main)2014]**

माना किसी समय t पर जीवित खरगोशों की जनसंख्या अवकल समीकरण $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$ द्वारा नियंत्रित है। यदि $p(0) = 100$है, तो $p(t)$ बराबर है : **[JEE(Main)2014]**

(A) $600 - 500\ e^{t/2}$ (B) $400 - 300\ e^{-t/2}$ (C) $400 - 300\ e^{t/2}$ (D) $300 - 200\ e^{-t/2}$

Ans. C

Sol. I.F. $= e^{-t/2}$

So solution is $P(t)e^{-t/2} = \int -200.e^{-t/2}dt = 400e^{-t/2} + k$

$\Rightarrow P(0) = 100 \Rightarrow k = -300$

$\Rightarrow P(t) = 400 - 300\ e^{t/2}$

37. If $(2+\sin x)\frac{dy}{dx}+(y+1)\cos x=0$ and $y(0)=1$, then $y\left(\frac{\pi}{2}\right)$ is equal to : **[JEE(Main)2017]**

यदि $(2+\sin x)\frac{dy}{dx}+(y+1)\cos x=0$ तथा $y(0)=1$ है, तो $y\left(\frac{\pi}{2}\right)$ बराबर है : **[JEE(Main)2017]**

(A) $-\frac{1}{3}$ (B) $\frac{4}{3}$ (C) $\frac{1}{3}$ (D) $-\frac{2}{3}$

Ans. C

Sol. $(2+\sin x)\frac{dy}{dx}=-(y+1)\cos x$

$$\int\frac{dy}{y+1}=-\int\frac{\cos x}{2+\sin x}dx$$

$$\ln(y+1)=\ln C-\ln(2+\sin x)$$

$$y+1=\frac{C}{2+\sin x}$$

$$y(0)=1 \Rightarrow C=4$$

$$y+1=\frac{4}{2+\sin x}$$

$$y\left(\frac{\pi}{2}\right)=\frac{1}{3}$$

HOMOGENEOUS DIFFERENTIAL EQUATION

38. The solution of the equation $x\frac{dy}{dx}=y-x\tan\left(\frac{y}{x}\right)$ is :

(A) $x\sin\left(\frac{y}{x}\right)+c=0$ (B) $x\sin y+c=0$ (C) $x\sin\left(\frac{y}{x}\right)=c$ (D) None of these

अवकल समीकरण $x\frac{dy}{dx}=y-x\tan\left(\frac{y}{x}\right)$ का हल है :

(A) $xy\sin\left(\frac{y}{x}\right)+c=0$ (B) $x\sin y+c=0$ (C) $x\sin\left(\frac{y}{x}\right)=c$ (D) इनमें से कोई नही

Ans. C

Sol. $x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$

$y = vx$

$\frac{dy}{dx} = x + \frac{xdv}{dx}$

$x\left(v + x\frac{dv}{dx}\right) = vx - x\tan v$

$vx + x^2\frac{dv}{dx} = vx - x\tan v$

$\int\frac{dv}{\tan v} = \int -\frac{dx}{x}$

$\ln(\sin v) = -\ln x + \ln c$

$x\sin v = c$

$x\sin\left(\frac{y}{x}\right) = c$

39. The equation of a curve passing through (1, 0) for which the product of the abscissa of a point P and the intercept made by a normal at P on the x-axis equals twice the square of the radius vector of the point P, is :

उस वक्र का समीकरण जो (1, 0) से गुजरता है तथा जिसके लिए किसी बिन्दु P के भुज तथा P पर अभिलम्ब द्वारा x-अक्ष पर निर्मित vUr%[k.Md kx qkuQy ml fcUnq ds f=kT; kl fn' kd sox Zd sa nqq sd scj kcj g\$gks k:

(A) $x^2 + y^2 = x^4$ (B) $x^2 + y^2 = 2x^4$ (C) $x^2 - y^2 = 4x^4$ (D) $x^2 - y^2 = x^4$

Ans. A

Sol. $Y - y = -\frac{1}{m}(X - x)$

when $m = \frac{dy}{dx}$

take, let $Y = 0$

$X = my + x$

hence $x(my + x) = 2(x^2 + y^2)$

$xy\frac{dy}{dx} = x^2 + 2y^2$ (Step-A)

Now put $y = vx$

$v + x\frac{dv}{dx} = \frac{x^2(1+2v^2)}{x^2v} = \frac{1+2v^2}{v}$

Alternatively :

Step – A

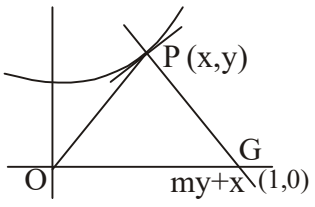
let $y^2 = t$

$2y\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{x}{2}\cdot\frac{dt}{dx} = x^2 + 2t$

$\frac{dt}{dx} - \frac{y}{x}t = 2x$

which is linear



$$x\frac{dv}{dx} = \frac{1+2v^2}{v} - v = \frac{1+v^2}{v}$$

$$\int \frac{v\,dv}{1+v^2} = \int \frac{dx}{x}$$

$$\frac{1}{2}\ ln\,(1 + v^2) = ln\,x + c$$

$$ln\left(\frac{1+v^2}{x^2}\right) = c$$

$x^2 + y^2 = cx^4 \quad \Rightarrow$ A

40. The solution of the differential equation $(x^2 - y^2)\,dx + 2xy\,dy = 0$ is :

अवकल समीकरण $(x^2 - y^2)\,dx + 2xy\,dy = 0$ का हल है :

(A) $x^2 + y^2 = cx$ (B) $x^2 - y^2 + cx = 0$ (C) $x^2 + 2xy = y^2 + cx$ (D) $x^2 + y^2 = 2xy + cx^2$

Ans. A

Sol. $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$ Put $y = vx \Rightarrow \frac{dy}{dx} = V + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{v^2x^2 - x^2}{2x.vx} = \frac{v^2 - 1}{2v}$$

$$\text{or} \quad \frac{xdv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$x\frac{dv}{dx} = \frac{-v^2 - 1}{2v} \quad \text{or} \quad -\int \frac{2v\,dv}{1+v^2} = \int \frac{dx}{x}$$

$-\log(1 + v^2) = \log x + c$

$\log x + \log(1 + v^2) = \log c$

$$\log x.\left(1 + \frac{y^2}{x^2}\right) = \log c \quad \text{or} \quad x\left(\frac{x^2 + y^2}{x^2}\right) = c$$

$$\frac{x^2 + y^2}{x} = c \quad \text{or} \quad x^2 + y^2 = cx$$

41. The x-intercept of the tangent to a curve is equal to the ordinate of the point of contact. The equation of the curve through the point (1, 1) is :

किसी वक्र की स्पर्श रेखा का x-अन्तःखण्ड स्पर्श बिन्दु के कोटि के बराबर है। वक्र का समीकरण जो (1, 1) से गुजरता है, होगा :

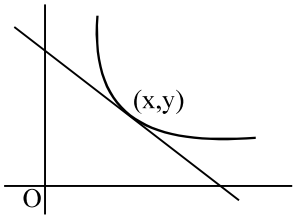
(A) $y e^{\frac{x}{y}} = e$ (B) $x e^{\frac{x}{y}} = e$ (C) $x e^{\frac{y}{x}} = e$ (D) $y e^{\frac{y}{x}} = e$

Ans. A

Sol. $Y - y = m(X - x)$

for X-intercept $Y = 0$

$$X = x - \frac{y}{m}$$



hence $x - \frac{y}{m} = y$

or $\frac{dy}{dx} = \frac{y}{x - y}$

put $y = Vx$

$$V + x\frac{dV}{dx} = \frac{V}{1 - V}$$

$$x\frac{dV}{dx} = \frac{V}{1 - V} - V = \frac{V - V + V^2}{1 - V}$$

$$\int \frac{1 - V}{V^2} dV = \int \frac{dx}{x}$$

$$-\frac{1}{V} - ln\, V = ln\, x + c$$

$$-\frac{x}{y} - ln\, \frac{y}{x} = ln\, x + c$$

$$-\frac{x}{y} = ln\, y + c$$

$$x = 1, y = 1 \Rightarrow c = -1$$

$$1 - \frac{x}{y} = ln\, y$$

$$y = e \cdot e^{-x/y}$$

$$e^{-x/y} = \frac{e}{y}$$

$$y e^{x/y} = e \Rightarrow (A)$$

Aliternative :

$$\frac{dx}{dy} = \frac{x - y}{y}$$

$$\frac{dx}{dy} - \frac{1}{y}x = -1$$

$$I.F. = e^{-\int \frac{1}{y} dy} = e^{ln\, y^{-1}} = \frac{1}{y}$$

$$x \cdot \frac{1}{y} = \int \frac{-1}{y} dy = -ln\, y + C$$

$$C = 1$$

$$\frac{x}{y} + ln\, y = 1$$

LINEAR DIFFERENTIAL EQUATION

42. Which of the following equation is linear :

निम्न में कौनसी अवकल समीकरण रैखिक है :

(A) $\left(\frac{d^2y}{dx^2}\right)^2 + x^2\left(\frac{dy}{dx}\right)^2 = 0$ (B) $y = \frac{dy}{dx} + \sqrt{1+\left(\frac{dy}{dx}\right)^2}$

(C) $\frac{dy}{dx} + \frac{y}{x} = \log x$ (D) $y\frac{dy}{dx} - 4 = x$

Ans. C

Sol. C is linear.

43. The solution of the differential equation $\frac{dy}{dx} + y\cot x = 2\cos x$ is :

अवकल समीकरण $\frac{dy}{dx} + y\cot x = 2\cos x$ का हल है :

(A) $y\sin x + \cos 2x = 2c$ (B) $2y\sin x + \cos x = c$

(C) $y\sin x + \cos x = c$ (D) $2y\sin x + \cos 2x = c$

Ans. D

Sol. $\frac{dy}{dx} + y\cot x = 2\cos x$

$I.F = e^{\int \cot x\, dx}$

$= e^{\ln \sin x} = \sin x$

$\int \sin x\frac{dy}{dx} + y\cos x = \int \sin 2x$

$y\sin x = -\frac{\cos 2x}{2} + C$

$2y\sin x + \cos 2x = C$

44. The solution of $(1 + y^2)\,dx = (\tan^{-1} y - x)\,dy$ is :

(A) $xe^{\tan^{-1} y} = e^{\tan^{-1} y}(\tan^{-1} y - 1) + c$ (B) $xe^{\tan^{-1} y} = (\tan^{-1} y + 1) - c$

(C) $xe^{\tan^{-1} y} = (\tan^{-1} y - 1) + c$ (D) None of these

अवकल समीकरण $(1 + y^2)\, dx = (\tan^{-1} y - x)dy$ का हल है :

(A) $xe^{\tan^{-1} y} = e^{\tan^{-1} y}(\tan^{-1} y - 1) + c$ (B) $x\, e^{\tan^{-1} y} = (\tan^{-1} y + 1) - c$

(C) $x\, e^{\tan^{-1} y} = (\tan^{-1} y - 1) + c$ (D) इनमें से कोई नही

Ans. A

Sol. $(1 + y^2)\, dx = (\tan^{-1} y - x)\, dy$

$$\frac{dy}{dx} = \frac{1+y^2}{\tan^{-1} y - x}$$

$$\frac{dx}{dy} = \frac{\tan^{-1} y - x}{1+y^2}$$

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1} y}{1+y^2}$$

I.F. $e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1} y}$

$$e^{\tan^{-1} y}\frac{dx}{dy} + \frac{e^{\tan^{-1} y}}{1+y^2} = \frac{\tan^{-1} y\, e^{\tan^{-} y}}{1+y^2}$$

$$x\, e^{\tan^{-1} y} = \int_{I\pi} te^t\, dx$$

$= e^t - e^t$

$$xe^{\tan^{-1} y} = \left(\tan^{-1} - 1\right)e^{\tan -1} + C$$

45. The solution of the equation $\frac{dy}{dx} + y \tan x = x^m \cos x$ is :

(A) $(m+1)\, y = x^{m+1} \cos x + c\,(m+1) \cos x$ (B) $my = (x^m + c) \cos x$

(C) $y = (x^{m+1} + c) \cos x$ (D) None of these

अवकल समीकरण $\frac{dy}{dx} + y \tan x = x^m \cos x$ का हल है :

(A) $(m+1)\, y = x^{m+1} \cos x + c\,(m+1) \cos x$ (B) $my = (x^m + c) \cos x$

(C) $y = (x^{m+1} + c) \cos x$ (D) इनमें से कोई नही

Ans. A

Sol. $\frac{dy}{dx} + y \tan x = x^m \cos x$

I.F. $= e^{\int \tan x dx} = e^{-\ln \cos x}$

$= \sec x$

$\int \sec x \frac{dy}{dx} + y \sec x \tan x = \int x^m$

$y \sec x = \frac{x^{m+1}}{m+1} + C$

46. The solution of the differential equation $\frac{dy}{dx} + \frac{3x^2}{1+x^3} y = \frac{\sin^2 x}{1+x^3}$ is :

अवकल समीकरण $\frac{dy}{dx} + \frac{3x^2}{1+x^3} y = \frac{\sin^2 x}{1+x^3}$ का हल है :

(A) $y(1 + x^3) = x + \frac{1}{2} \sin 2x + c$

(B) $y(1 + x^3) = cx + \frac{1}{2} \sin 2x$

(C) $y(1 + x^3) = cx - \frac{1}{2} \sin 2x$

(D) $y(1 + x^3) = \frac{x}{2} - \frac{1}{4} \sin 2x + c$

Ans. D

Sol. $\frac{dy}{dx} + \frac{3x^2}{1+x^3} y = \frac{\sin^2 x}{1+x^3}$

I.F. $= (e)^{\int \frac{3x^2}{1+x^3} dx} = e^{\ell n\left(1+x^3\right)} = 1 + x^3$

$\int \left(1+x^3\right) \frac{dy}{dx} + 3x^2 y = \int \sin^2 x$

$y\left(1+x^3\right) = \int \frac{1 - \cos 2x}{2}$

$y\left(1+x^3\right) = \frac{x}{2} - \frac{\sin 2x}{4} + C$

47. The solution of the differential equation $(1 + y^2) + \left(x - e^{\tan^{-1} y}\right) \frac{dy}{dx} = 0$, is :

अवकल समीकरण $(1 + y^2) + \left(x - e^{\tan^{-1} y}\right)\frac{dy}{dx} = 0$ का हल है :

(A) $xe^{2\tan^{-1} y} = e^{\tan^{-1} y} + K$ (B) $(x - 2) = K e^{\tan^{-1} y}$

(C) $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + K$ (D) $xe^{\tan^{-1} y} = \tan^{-1} y + K$

Ans. C

Sol. $(1 + y^2) = (e^{\tan^{-1} y} - x)\frac{dy}{dx}$ or $(1+ y^2)\frac{dx}{dy} + x + e^{\tan^{-1} y}$

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{e^{\tan^{-1} y}}{(1+y^2)}$$

Now I.F. $= e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1} y}$

$\therefore$ solution $x(e^{\tan^{-1} y}) = \int \frac{e^{\tan^{-1} y}}{1+y^2} e^{\tan^{-1} y}\, dy + C$

$$xe^{\tan^{-1} y} = \frac{e^{2\tan^{-1} y}}{2} + C$$

or $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + K$

48. A function $y = f(x)$ satisfies the condition $f'(x) \sin x + f(x) \cos x = 1$, $f(x)$ being bounded when $x \to 0$. If $I = \int_0^{\pi/2} f(x)\, dx$ then :

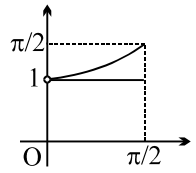
एक फलन $y = f(x)$ प्रतिबन्ध $f'(x) \sin x + f(x) \cos x = 1$, को संतुष्ट करता है जहाँ $f(x)$, $x \to 0$ पर परिबद्ध है। यदि $I = \int_0^{\pi/2} f(x)\, dx$ तो :

(A) $\frac{\pi}{2} < I < \frac{\pi^2}{4}$ (B) $\frac{\pi}{4} < I < \frac{\pi^2}{2}$ (C) $1 < I < \frac{\pi}{2}$ (D) $0 < I < 1$

Ans. A

Sol. $\sin x \frac{dy}{dx} + y \cos x = 1$

$$\frac{dy}{dx} + y\cot x = \operatorname{cosec} x$$



$$\text{I.F.} = e^{\int \cot x\,dx} = e^{\ln(\sin x)} = \sin x$$

$$y\sin x = \int \operatorname{cosec} x \cdot \sin x\,dx$$

$$y\sin x = x + C$$

if $x = 0$, y is finite

$$\therefore \quad C = 0$$

$$y = x(\operatorname{cosec} x) = \frac{x}{\sin x}$$

Now $I < \frac{\pi^2}{4}$ and $I > \frac{\pi}{2}$

Hence $\frac{\pi}{2} < I < \frac{\pi^2}{4} \Rightarrow$ (A)

49. The general solution of $\frac{dy}{dx} = 2y\tan x + \tan^2 x$ is :

$\frac{dy}{dx} = 2y\tan x + \tan^2 x$ का व्यापक हल होगा :

(A) $y\cos^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$

(B) $y\sec^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$

(C) $y\cos^2 x = \frac{x}{2} - \frac{\cos 2x}{4} + C$

(D) $y\cos^2 x = \frac{x}{2} - \frac{\sin 2x}{2} + C$

Ans. A

Sol. $\frac{dy}{dx} = 2y\tan x + \tan^2 x$

$$\frac{dy}{dx} - (2\tan x)y = \tan^2 x$$

$$\text{I.F} = e^{-\int 2\tan x dx} = e^{2\ln(\cos x)} = \cos^2 x$$

$$\cos^2 x\frac{dy}{dx} - (2\tan x\cos^2 x)y = \cos^2 x\tan x$$

$$\int \cos^2 x \frac{dy}{dx} - (2\sin x \cos x) y = \int \frac{1-\cos 2x}{2}$$

$$y\cos^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

50. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim\limits_{t\to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$.

Then $f(x)$ is :

माना $f(x)$ अन्तराल $(0, \infty)$ में अवकलनीय फलन इस प्रकार है कि $f(1) = 1$ और $\lim\limits_{t\to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1, \forall\ x > 0$. तब $f(x)$ है :

(A) $\frac{1}{3x} + \frac{2x^2}{3}$ (B) $\frac{-1}{3x} + \frac{4x^2}{3}$ (C) $\frac{-1}{x} + \frac{2}{x^2}$ (D) $\frac{1}{x}$

Ans. A

Sol. $\lim\limits_{t\to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$

Applying L-hospital, we get

$\Rightarrow \quad x^2 f'(x) - 2x f(x) + 1 = 0$

$\Rightarrow \quad \frac{x^2 f'(x) - 2x f(x)}{(x^2)^2} + \frac{1}{x^4} = 0$

$\Rightarrow \quad \frac{d}{dx}\left(\frac{f(x)}{x^2}\right) = -\frac{1}{x^4}$

$\Rightarrow \quad f(x) = cx^2 + \frac{1}{3x}$

Also $f(1) = 1 \Rightarrow c = \frac{2}{3}$

$\Rightarrow \quad f(x) = \frac{2x^2}{3} + \frac{1}{3x}$

51. For the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If $y(1) = 1$, then x is given by : **[AIEEE-2011]**

अवकल समीकरण $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$ पर विचार कीजिये। यदि $y(1) = 1$ है, तो x का मान है : **[AIEEE-2011]**

(A) $1-\frac{1}{y}+\frac{e^{\frac{1}{y}}}{e}$ (B) $4-\frac{2}{y}-\frac{e^{\frac{1}{y}}}{e}$ (C) $3-\frac{1}{y}+\frac{e^{\frac{1}{y}}}{e}$ (D) $1+\frac{1}{y}-\frac{e^{\frac{1}{y}}}{e}$

Ans. D

Sol. $y^2dx+\left(x-\frac{1}{y}\right)dy=0$

$\Rightarrow \quad y^2\frac{dx}{dy}+x=\frac{1}{y} \quad \Rightarrow \quad \frac{dx}{dy}+\frac{x}{y^2}+\frac{1}{y^3}$

$\therefore$ Integrating factor (I.F.) $= e^{\int\frac{1}{y^2}dy} = e^{-1/y}$

$\therefore$ General solution is –

$$x \cdot e^{-1/y}=\int\frac{1}{y^3}e^{-1/y}dy + c$$

Let $I_1=\int\frac{1}{y^3}e^{-1/y}dy$

put $\frac{-1}{y}=t$

$y^{-2}\,dy=dt$

$\therefore \quad I_1=-\int te^t dt$

$=-e^t(t-1)$

$=e^t(1-t)$

$\therefore$ General solution is

$$xe^{-1/y}=e^{-1/y}\left(1+\frac{1}{y}\right)+C$$

$\Rightarrow \quad x=1+\frac{1}{y}+Ce^{1/y}$

Put $x=1,\ y=1$

$\therefore \quad 1=1+\frac{1}{1}+Ce^{1/1}$

$\Rightarrow \quad C=-1/e$

$\therefore \quad x=1+\frac{1}{y}-\frac{e^{1/y}}{e}$

52. Let y(x) be the solution of the differential equation $(x\log x)\frac{dy}{dx}+y=2x\log x,(x\geq 1)$. Then y(e) is equal to :

[JEE(Main)2015]

माना अवकल समीकरण $(x\log x)\frac{dy}{dx}+y=2x\log x,(x\geq 1)$ का हल y(x) है, तो y(e) बराबर है : **[JEE(Main)2015]**

(A) e (B) 0 (C) 2 (D) 2e

Ans. C

Sol. $\ln x\frac{dy}{dx}+\frac{y}{x}=2\ln x$

$\int d(y\ln x)=2\int \ln x\,dx$

$y\ln x=2x(\ln x-1)+c$

$y(e)=2$ as $c=2$

53. Let y = y(x) be the solution o f the differential equation $\sin x\frac{dy}{dx}+y\cos x=4x,\ x\in(0,\pi)$. If $y\left(\frac{\pi}{2}\right)=0$, then

$y\left(\frac{\pi}{6}\right)$ is equal to : **[JEE(Main)2018]**

माना अवकल समीकरण $\sin x\frac{dy}{dx}+y\cos x=4x,\ x\in(0,\pi)$ का y = y(x) एक हल है। यदि $y\left(\frac{\pi}{2}\right)=0$ है, तो $y\left(\frac{\pi}{6}\right)$ बराबर है : **[JEE(Main)2018]**

(A) $-\frac{8}{9}\pi^2$ (B) $-\frac{4}{9}\pi^2$ (C) $\frac{4}{9\sqrt{3}}\pi^2$ (D) $\frac{-8}{9\sqrt{3}}\pi^2$

Ans. A

Sol. $\sin x\frac{dy}{dx}+y\cos x=4x$

$\int d(y\sin x)=\int 4x\,dx$

$y\sin x=2x^2+C$

$x=0,\ y=\frac{\pi}{2}\Rightarrow C=-\frac{\pi^2}{2}$

Put $x=\frac{\pi}{6}$

$$y \times \frac{1}{2} = 2\left(\frac{\pi^2}{36}\right) - \frac{\pi^2}{2}$$

$$y = \frac{-8\pi^2}{9}$$

BERNOULLI'S EQUATION

54. If the differentiable equation $\frac{dy}{dx} - y = y^2(\sin x + \cos x)$ with $y(0) = 1$ then $y(\pi)$ has the value equal to :

यदि अवकलनीय समीकरण $\frac{dy}{dx} - y = y^2(\sin x + \cos x)$ जबकि $y(0) = 1$ है तो $y(\pi)$ का मान होगा :

(A) ε^{π} (B) $\square\ \varepsilon^{\pi}$ (C) $\varepsilon^{-\pi}$ (D) $\square\ \varepsilon^{-\pi}$

Ans. A

Sol. $\frac{dy}{dx} - y = y^2(\sin x + \cos x)$

$$\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{y} = \sin x + \cos x \ ; \qquad \text{Let} \quad -\frac{1}{y} = t \quad \Rightarrow \quad \frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + t = \sin x + \cos x$$

I.F. e^x

$$\therefore \quad t \cdot e^x = \int e^x(\sin x + \cos x)\,dx \ ; \qquad -\frac{1}{y}e^x = e^x \sin x + C$$

if $x = 0,\ y = 1 \quad \Rightarrow \quad C = -1$

$$-\frac{e^x}{y} = e^x \sin x - 1$$

if $x = \pi$ then $-\frac{e^x}{y} = -1 \quad \Rightarrow \quad y = e^{\pi}$ Ans

55. Solution of the differential equation $\cos x\,dy = y(\sin x - y)dx,\ 0 < x < \frac{\pi}{2}$ is : **[AIEEE-2010]**

अवकल समीकरण $\cos x\,dy = y(\sin x - y)dx,\ 0 < x < \frac{\pi}{2}$ का हल है : **[AIEEE-2010]**

(A) $\sec x = (\tan x + c)\,y$ (B) $y \sec x = \tan x + c$

(C) $y \tan x = \sec x + c$ (D) $\tan x = (\sec x + c)y$

Ans. A

Sol. $\cos x\, dy = y(\sin x - y)dx$

$$\Rightarrow \frac{dy}{dx} = \frac{y\sin x - y^2}{\cos x}$$

$$\Rightarrow \frac{dy}{dx} = y\tan x - y^2 \sec x$$

$$\Rightarrow \frac{1}{y^2}\frac{dy}{dx} = \frac{1}{y}\tan x - \sec x$$

$$\Rightarrow \frac{1}{y^2}\frac{dy}{dx} - \frac{1}{y}\tan x = -\sec x$$

$$\Rightarrow -\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y}\tan x = \sec x \qquad \ldots (1)$$

Put $\frac{1}{y} = t$ in equation (1)

$$\Rightarrow -\frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx} \qquad \ldots (2)$$

From equation (1) & (2), we get,

$$\Rightarrow \frac{dt}{dx} + t\,.\tan x = \sec x$$

$\therefore$ I.F. $= e^{\int \tan x\, dx}$

$= e^{\log|\sec x|} = \sec x$

$\therefore$ solution of differential equation is :

$t.\sec x = \int \sec x\,.\sec x\,.\,dx + c$

$$\frac{1}{y}\sec x = \tan x + c$$

$\sec x = y(\tan x + c)$

56. If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy)dx = x\,dy$, then $f\left(-\frac{1}{2}\right)$ is equal to : **[JEE(Main)2016]**

यदि एक वक्र $y = f(x)$ बिन्दु $(1, -1)$ से होकर जाता है तथा अवकल समीकरण $y(1 + xy)dx = x\,dy$ को सन्तुष्ट करता है, तो $f\left(-\frac{1}{2}\right)$ बराबर है : **[JEE(Main)2016]**

(A) 2/5 (B) 4/5 (C) $-\frac{2}{5}$ (D) $-\frac{4}{5}$

Ans. B

Sol. $y\,dx + xy^2dx = xdy$

$ydx - xdy = -xy^2dx$

$$\int \frac{ydx - xdy}{y^2} = -\int xdx$$

$$\frac{x}{y} = \frac{-x^2}{2} + C$$

$$(1,-1) \Rightarrow \left(\frac{x^2+1}{2}\right)y = -1$$

$$y = -\frac{2x}{x^2+1}$$

CHANGE OF VARIABLE BY A SUITABLE SUBSTITUTION

57. The solution of the differential equation, $e^x(x + 1)dx + (ye^y - xe^x)dy = 0$ with initial condition $f(0) = 0$, is :

अवकल समीकरण $e^x(x + 1)dx + (ye^y - xe^x)dy = 0$ का हल, जबकि प्रारम्भिक अवस्था में $f(0) = 0$ है, होगा :

(A) $xe^x + 2y^2e^y = 0$ (B) $2xe^x + y^2e^y = 0$ (C) $xe^x - 2y^2e^y = 0$ (D) $2xe^x - y^2e^y = 0$

Ans. B

Sol. put $xe^x = t$

$$(e^x + xe^x)\frac{dx}{dy} = \frac{dt}{dy}$$

$$\therefore \quad \frac{dt}{dy} + (ye^y - t) = 0 \quad \Rightarrow \quad \frac{dt}{dy} - t + ye^y = 0$$

I.F. $e^{-\int dy} = e^{-y}$

$$t \cdot e^{-y} = -\int ye^y e^{-y}dy$$

$$x\,e^x\,e^{-y} = -\frac{y^2}{2} + C$$

$f(0) = 0 \Rightarrow C = 0;\ 2x\,e^x e^{-y} + y^2 = 0$ Ans

58. The solution of the differential equation, $x^2\frac{dy}{dx}.\cos\frac{1}{x} - y\sin\frac{1}{x} = -1$, where $y \to -1$ as $x \to \infty$ is :

अवकल समीकरण $x^2\frac{dy}{dx}.\cos\frac{1}{x} - y\sin\frac{1}{x} = -1$, यदि $y \to -1$ तथा $x \to \infty$, का हल है :

(A) $y = \sin\frac{1}{x} - \cos\frac{1}{x}$ (B) $y = \frac{x+1}{x\sin\frac{1}{x}}$ (C) $y = \cos\frac{1}{x} + \sin\frac{1}{x}$ (D) $y = \frac{x+1}{x\cos\frac{1}{x}}$

Ans. A

Sol. $\frac{dy}{dx} - \frac{y}{x^2}\tan\frac{1}{x} = -\sec\frac{1}{x}\cdot\frac{1}{x^2}$. IF $= e^{-\int\frac{-1}{x^2}\tan\frac{1}{x}\,dx} = \sec\frac{1}{x}$

$\Rightarrow y\cdot\sec\frac{1}{x} = -\int\sec^2\left(\frac{1}{x}\right)\frac{1}{x^2}\,dx = \tan\frac{1}{x} + c$

if $y \to -1$ then $x \to \infty \Rightarrow c = -1 \Rightarrow y = \sin\frac{1}{x} - \cos\frac{1}{x}$

59. If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is :

यदि $x\frac{dy}{dx} = y(\log y - \log x + 1)$, तो इस समीकरण का हल है :

(A) $y\log\left(\frac{x}{y}\right) = cx$ (B) $x\log\left(\frac{x}{y}\right) = cy$ (C) $\log\left(\frac{x}{y}\right) = cx$ (D) $\log\left(\frac{x}{y}\right) = cy$

Ans. C

Sol. $\frac{dy}{dx} = \frac{y}{x}\left(\log\frac{y}{x} + 1\right)$ which is homogeneous equ.

Put $y = vx$, $\frac{dy}{dx} = v + \frac{xdv}{dx}$

$v + x\frac{dv}{dx} = \frac{vx}{x}\left(\log\frac{vx}{x} + 1\right)$

$\frac{xdv}{dx} = v(\log v + 1) - v = v\log v + v - v$

$$\int \frac{dv}{v \log v} = \int \frac{dx}{x} \Rightarrow \log(\log v) = \log x + \log c$$

$$\Rightarrow \quad \log \frac{y}{x} = cx$$

EXACT DIFFERENTIALS BASED DIFFERENTIAL EQUATION

60. The solution of the differential equation $ydx + (x + x^2 y)\, dy = 0$ is :

अवकल समीकरण $ydx + (x + x^2 y)\, dy = 0$ का हल है :

(A) $-\frac{1}{xy} = C$ (B) $-\frac{1}{xy} + \log y = C$ (C) $\frac{1}{xy} + \log y = C$ (D) $\log y = Cx$

Ans. B

Sol. $ydx + (x + x^2y)dy = 0 \quad ydx + xdy = -x^2y\, dy$

$$\int \frac{d(xy)}{(xy)^2} = -\int \frac{dy}{y} \Rightarrow -\frac{1}{xy} = -\log y + c$$

$$\Rightarrow \quad \frac{-1}{xy} + \log y = c$$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

ORDER, DEGREE AND FORMATION OF A DIFFERENTAIL EQUATION

1. (a) Form the differential equation of the family of curves represented by, $c(y + c)^2 = x^3$; where c is any arbitrary constant.

(b) Form a differential equation for the family of curves represented by $ax^2 + by^2 = 1$, where a and b are arbitary constants.

(c) Obtain the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$; where g, f and c are arbitary constants.

(d) Obtain the differential equation associated with the primitive, $y = c_1e^{3x} + c_2e^{2x} + c_3e^x$, where c_1, c_2, c_3 are arbitrary constants.

(a) ब; ल़ बद्धे $= x^3$ से प्रदर्शित वक्र निकाय का अवकल समीकरण बनाओं। जहाँ पर c एक स्वैच्छिक नियतांक है।

(b) ˙ ग^2 + by^2 = 1 से प्रदर्शित वक्र निकाय का अवकलन समीकरण बताओं जहाँ पर a एवं b स्वैच्छिक नियतांक है।

(c) वृत निकाय $x^2 + y^2 + 2gx + 2fy + c = 0$; का अवकलन समीकरण बताओं जहाँ पर g, f व c स्वैच्छिक नियतांक है।

(d) पूर्वग $y = c_1e^{3x} + c_2e^{2x} + c_3e^x$ से सम्बन्धित अवकलन समीकरण बताओं जहाँ पर c_1, c_2, c_3 स्वैच्छिक नियतांक है।

Ans. (a) $12y\,(y')^2 = x[8(y')^3 - 27]$ (b) $xy\,\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$

(c) $[1 + (y')^2]\,.y''' - 3y'(y'')^2 = 0$ (d) $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$

Sol. (a) $c(y + c)^2 = x^3$ .....(1)

$2c\,(y + c)\,y' = 3x^2$ .....(2)

$2xy' = 3y + 3c$

$$C = \frac{2xy^1 - 3y}{3}$$

$$\left(\frac{2xy^1 - 3y}{3}\right)\left[y + \frac{2xy^1 - 3y}{3}\right]^2 = x^3$$

$$\left(\frac{2xy^1 - 3y}{3}\right)\left[\frac{2y + 2xy^1 - 3y}{3}\right]^2 = x^3$$

$$\frac{\left(2xy^1 - 3y\right)}{3} \times \frac{4x^2\left(y\right)^2}{9} = x \Rightarrow 4\left(2xy^1 - 3y\right)y^2 = 27x$$

$12y\,(y')^2 = x[8(y')^3 - 27]$

(b) $ax^2 + by^2 = 1$

$$\frac{a}{b}x^2 + y^2 = \frac{1}{b}$$

$$2\left(\frac{a}{b}\right)x + 2yy' = 0$$

$$\frac{a}{b} + (y')^2 + yy'' = 0$$

$$-\left[(y')^2 + yy''\right]x + yy' = 0$$

$yy' = x(y')^2 + xyy''$

(c) $x^2 + y^2 + 2gx + 2fy + c = 0$

$2x + 2yy_1 + 2g + 2fy_1 = 0$

$$\Rightarrow 1 + y_1^2 + yy_2 + fy_2 = 0$$

$$2y_1y_2 + y_1y_2 + yy_3 + fy_3 = 0$$

$$f = -\frac{3y_1y_2 + yy_3}{y_3}$$

$$1 + y_1^2 + yy_2 = y_2\left(\frac{3y_1y_2 + yy_3}{y_3}\right)$$

$$y_3\left[\left(1 + y_1^2 + yy_2\right) - yy_2\right] = 3y_1y_2^2$$

$$y_3\left(1 + y_1^2\right) = 3y_1y_2^2$$

(d) $y = c_1e^{3x} + c_2e^{2x} + c_3e^x$

$$\frac{dy}{dx} = 3c_1e^{3x} + 2c_2e^{2x} + c_3e^x$$

$$\frac{d^2y}{dx^2} = c_1c_1e^{3x} + 4c_2e^{2x} + c_3e^x$$

$$\frac{d^2y}{dx^3} = 27c_1e^{3x} + 8c_2e^{2x} + c_3e^x$$

$$\frac{d^3y}{dx^3} - \frac{6d^2y}{dx^3} + 11\frac{dy}{dx} - 6y = 0$$

VARIABLE SEPARABLE

Solve the following differential equation for Q.2 to Q.7 :

2. $\frac{\ell n\,(\sec x + \tan x)}{\cos x}\,dx = \frac{\ell n\,(\sec y + \tan y)}{\cos y}\,dy$

Ans. $ln^2\,(\sec x + \tan x) - ln^2\,(\sec y + \tan y) = c$

Sol. $\int \frac{\ell n\,(\sec x + \tan x)}{\cos x}\,dx = \int \frac{\ell n\,(\sec y + \tan y)}{\cos y}\,dy$

$\ln(\tan x + \sec x) = t \qquad \ln(\tan y + \sec y) = P$

$\int t\,dt = \int P\,dp$

$\frac{t^2}{2} = \frac{P^2}{2} + C$

$\ln^2(\sec x + \tan x) = \ln^3(\tan y + \sec y) + C$

3. $(1 - x^2)(1 - y)\,dx = xy(1 + y)\,dy$

Ans. $ln\,x\,(1-y)^2 = c - \frac{1}{2}\,y^2 - 2y + \frac{1}{2}\,x^2$

Sol. $(1 - x^2)(1 - y)\,dx = xy(1 + y)\,dy$

$\frac{y(1+y)}{1-y}dy = \frac{(1-x^2)}{x}dx$

$\left(\frac{y}{1-y} + \frac{y^2}{1-y}\right)dy = \left(\frac{1}{x} - x\right)dx$

$\left(\frac{y-1+1}{y-1} + \frac{y^2-1+1}{y-1}\right)dy = \left(x - \frac{1}{x}\right)dx$

$\int\left(1 + \frac{1}{y-1} + (y+1) + \frac{1}{y-1}\right)dy = \int\left(x - \frac{1}{x}\right)dx$

$y + 2\ln(y-1) + \frac{y^2}{2} + y = \frac{x^2}{2} - \ln x + C$

$$\ln\left[x(y-1)^2\right] = \frac{x^2}{2} - \frac{y^2}{2} - 2y + C$$

4. $\dfrac{dy}{dx} + \dfrac{\sqrt{(x^2-1)(y^2-1)}}{xy} = 0$

Ans. $\sqrt{x^2-1} - \sec^{-1}x + \sqrt{y^2-1} = c$

Sol. $\dfrac{dy}{dx} + \dfrac{\sqrt{(x^2-1)(y^2-1)}}{xy} = 0$

$$\frac{dy}{dx} = -\frac{\sqrt{x^2-1}\sqrt{y^2-1}}{xy}$$

$$\int \frac{1}{2}\frac{2y}{\sqrt{y^2-1}}dy = \int \frac{-\sqrt{x^2-1}}{x}dx$$

$$\sqrt{y^2-1} = -\int \frac{x^2-1}{x\sqrt{x^2-1}}dx = \int\left(\frac{1}{x\sqrt{x^2-1}} - \frac{x}{\sqrt{x^2-1}}\right)dx$$

$$\sqrt{y^2-1} = \sec^{-1}x - \sqrt{x^2-1} + c$$

$$\sqrt{x^2-1} - \sec^{-1}x + \sqrt{y^2-1} = c$$

5. $\dfrac{dy}{dx} = \sin(x+y) + \cos(x+y)$

Ans. $\ln\left[1 + \tan\dfrac{x+y}{2}\right] = x + c$

Sol. $\dfrac{dy}{dx} = \sin(x+y) + \cos(x+y)$

$x + y = t$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - 1 = \sin t + \cos t$$

$$\frac{dt}{1+\sin t+\cos t}=dx$$

$$\frac{dt}{1+\frac{2\tan\frac{t}{2}}{1+\tan^2\frac{t}{2}}+\frac{1-\tan^2\frac{t}{2}}{1+\tan^2\frac{t}{2}}}=dx$$

$$\frac{\sec^2\frac{t}{2}}{1+\tan^2\frac{t}{2}+2\tan\frac{t}{2}+1-\tan^2\frac{t}{2}}dt=dx$$

$$\int\frac{\frac{1}{2}\sec^2\left(\frac{t}{2}\right)}{1+\tan\left(\frac{t}{2}\right)}dt=\int dx$$

$$\ln\left[1+\tan\left(\frac{x+y}{2}\right)\right]=x+c$$

6. $\frac{dy}{dx}=\frac{x(2\ln x+1)}{\sin y+y\cos y}$

Ans. $y\sin y=x^2\,\ln x+c$

Sol. $\frac{dy}{dx}=\frac{x(2\ln x+1)}{\sin y+y\cos y}$

$$\int(\sin y+y\cos y)dy=\int(2x\ln x+x)dx$$

$$-\cos y+y\sin y-\int\sin y\,dy=\left[x^2\ln x-\int\frac{1}{x}x^2+\int x\right]dx$$

$$y\sin y=x^2\ln x+C$$

7. (a) $\frac{dy}{dx}+\sin\frac{x+y}{2}=\sin\frac{x-y}{2}$ (b) $\sin x\,.\,\frac{dy}{dx}=y\,.\,\ln y$ if $y=e$, when $x=\frac{\pi}{2}$

(a) $\frac{dy}{dx} + \sin\frac{x+y}{2} = \sin\frac{x-y}{2}$ (b) $\sin x \cdot \frac{dy}{dx} = y \cdot \ell n y$ यदि $y = e$, जब $x = \frac{\pi}{2}$

Ans. (a) $\ln\left|\tan\frac{y}{4}\right| = c - 2\sin\frac{x}{2}$ (b) $y = e^{\tan(x/2)}$

Sol. $\frac{dy}{dx} = \sin\frac{x-y}{2} - \sin\frac{x+y}{2}$

$\Rightarrow \quad \frac{dy}{dx} = -2\cos\frac{x}{2}\sin\frac{y}{2}$

$\Rightarrow \quad \text{cosec}\frac{y}{2}dy = -2\cos\frac{x}{2}dx$

$\Rightarrow \quad 2\left(\ell n\left|\tan\frac{y}{4}\right|\right) = 2\left(-2\sin\frac{x}{2}\right) + c$

$\Rightarrow \quad \ell n\left|\tan\frac{y}{4}\right| = c_1 - 2\sin\frac{x}{2}$

8. The population P of a town decreases at a rate proportional to the number by which the population exceeds 1000, proportionality constant being $k > 0$. Find

(a) Population at any time t, given initial population of the town being 2500.

(b) If 10 years later the population has fallen to 1900, find the time when the population will be 1500.

(c) Predict about the population of the town in the long run.

किसी शहर की जनसंख्या P, के घटने की दर उस संख्या के समानुपाती है, जितनी संख्या में जनसंख्या 1000 से अधिक हैं। अनुपातिक नियतांक $k > 0$ है।

(a) किसी समय t पर जनसंख्या ज्ञात कीजिए, जबकि प्रारम्भिक जनसंख्या 2500 है।

(b) यदि 10 वर्ष बाद जनसंख्या 1900 हो जाती है, तो कितने समय में 1500 होगी।

(c) लम्बे समय पश्चात् शहर की जनसंख्या अनुमानित करो।

Ans. (a) $P = 1000 + 1500e^{-kt}$ where $k = \frac{1}{10}\ell n\left(\frac{5}{3}\right)$ (b) $T = 10\log_{5/3}(3)$; (c) $P = 1000$ as $t \to \infty$

Sol. (a) $\frac{dp}{dt} = -k(P - 1000)$

$\int\frac{dp}{P-1000} = -k\int dt$

$\ln|P - 1000| = -kt + C$

$P = 2500$

$t = 0$

$\ln (1500) = 0 + C$

$\ln |P = 1000| = - kt + \ln 1500$

$$\ln\left(\frac{P-1000}{1500}\right) = -kt$$

$P - 1000 = 1500\ e^{-kt}$

$P = 1000 + 1500\ e^{-kt}$

(b) $t = 10$

$P = 1900$

$1900 = 1000 + 1500\ e^{-10k}$

$900 = 1500\ e^{-10k}$

$$e^{-10k} = \frac{3}{5}$$

$$-10k = \ln\left(\frac{3}{5}\right)$$

$$10k = -\ln\left(\frac{3}{5}\right)$$

$$10k = \ln\left(\frac{5}{3}\right) \qquad \text{.....(1)}$$

$1500 = 1000 + 1500\ e^{-kt}$

$500 = 1500\ e^{-kt}$

$$e^{-kt} = \frac{1}{3} \Rightarrow kt = \ln 3$$

$$t = \frac{\ln 3}{k} = \frac{(10 \ln 3)}{\ln \frac{5}{3}} = 10 \log_{\frac{5}{3}} 3$$

(C) $t \to \infty$

$P = 1000 + 1500\ e^{-kt}$

$t \to \infty, e^{-kt} \to 0$

$P = 1000 + 1500\ (0)$

$P = 1000$

9. It is known that the decay rate of radium is directly proportional to its quantity at each given instant. Find the law of variation of a mass of radium as a function of time if at $t = 0$, the mass of the radius was m_0 and during time t_0 α % of the original mass of radium decay.

रेडियम की क्षय दर किसी समय पर उसकी मात्रा के अनुपातिक है। रेडियम के द्रव्यमान के परिवर्तन की दर समय के फलन में ज्ञात कीजिए। यदि $t = 0$ पर रेडियम के द्रव्यमान m_0 था तथा t_0 समय में α % रेडियम क्षय होता है।

Ans. $m=m_0e^{-kt}$ where $k = -\frac{1}{t_0}\ln\left(1 - \frac{\alpha}{100}\right)$

Sol. $\frac{dm}{dt} = -\lambda m$

$\frac{1}{m}dm = -\lambda dt$

$\ell n\, m = -\lambda t + c$

$m = k e^{-\lambda t}$ (at $t = 0,\ m = m_0$)

$\Rightarrow \quad k = m_0$

$m = m_0e^{-\lambda t}$ (at $t = t_0,\ m = m_0 - \frac{\alpha m_0}{100}$)

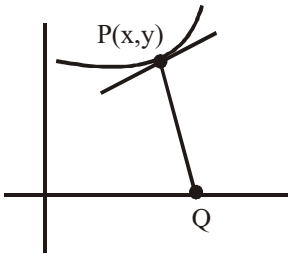
$\Rightarrow \quad \lambda = \frac{-1}{t_0}\ell n\left(1 - \frac{\alpha}{100}\right)$

10. A normal is drawn at a point P(x , y) of a curve. It meets the x– axis at Q. If PQ is of constant length k, then show that the differential equation describing such curves is, $y\frac{dy}{dx} = \pm\sqrt{k^2 - y^2}$. Find the equation of such a curve passing through (0, k).

किसी वक्र के बिन्दु P(x, y) पर अभिलम्ब खींचा जाता है। यह x– अक्ष को Q पर मिलता है। यदि PQ की लम्बाई नियत k तो दिखाओं की वक्र का अवकल समीकरण $y\frac{dy}{dx} = \pm\sqrt{k^2 - y^2}$ है। उस वक्र का समीकरण ज्ञात कीजिए जो (0, k) से गुजरता है।

Ans. $x^2 + y^2 = k^2$

Sol. $y\frac{dy}{dx} = \pm\sqrt{k^2 - y^2}$



$$Y - y = \frac{-1}{m}(X - x)$$

$0 + ym = X - x$

$X = x + my$

$Q\ (x + my,\ 0)$ $P(x, y)$

$PQ = k$

$$\sqrt{m^2y^2 + y^2} = k$$

$m^2y^2 + y^2 = k^2$

$m^2y^2 = k^2 - y^2$

$$y\frac{dy}{dx} = \pm\sqrt{(k^2 - y^2)}$$

$$(+)\ \int \frac{ydy}{\sqrt{k^2 - y^2}} = \int dx$$

$k^2 - y^2 = t$ $\qquad \frac{-1}{2}\int \frac{dt}{\sqrt{t}} = x + c$

$-2y\,dy = dt$ $\qquad -\sqrt{t} = x + c$

$$-\sqrt{k^2 - y^2} = x + c$$

$(0, k)$

$c = 0$

$$-\sqrt{k^2 - y^2} = x$$

$x^2 + y^2 = k^2$

11. $\frac{x\,dx - y\,dy}{x\,dy - y\,dx} = \sqrt{\frac{1 + x^2 - y^2}{x^2 - y^2}}$

Ans. $\sqrt{x^2 - y^2} + \sqrt{1 + x^2 - y^2} = \frac{c\,(x + y)}{\sqrt{x^2 - y^2}}$

Sol. $\frac{xdx - ydy}{xdy - ydx} = \sqrt{\frac{1+x^2-y^2}{x^2-y^2}}$

Put $x = r \sec\theta$, $y = r \tan\theta$

$\therefore \quad x^2 - y^2 = r^2$ ... (1)

and $\sin\theta = \frac{y}{x}$ ... (2)

On differentiating (1), $xdx - ydy = rdr$. .. (3)

On differentiating (2),

$\xi\delta\psi\square\ \psi\delta\xi = \xi^2 \cos\theta\, d\theta = r^2 \sec\theta\, d\theta$ ... (4)

Now $\frac{rdr}{r^2 \sec\theta d\theta} = \frac{\sqrt{1+r^2}}{r}$

$\Rightarrow \int \frac{dr}{\sqrt{1+r^2}} = \int \sec\theta\, d\theta$

$\therefore \quad \ell n(r + \sqrt{1+r^2}) = \ell n(\sec\theta + \tan\theta) + \ell nc$

$\Rightarrow r + \sqrt{1+r^2} = c\,(\sec\theta + \tan\theta)$

$$\sqrt{x^2 - y^2} + \sqrt{1+x^2-y^2} = c\left(\frac{x+y}{\sqrt{x^2-y^2}}\right)$$

12. A curve is such that the length of the polar radius of any point on the curve is equal to the length of the tangent drawn at this point. Form the differential equation and solve it to find the equation of the curve.

एक वक्र के किसी बिन्दु की ध्रुवीय त्रिज्या की लम्बाई, उस बिन्दु पर स्पर्श रेखा की लम्बाई के बराबर है। अवकल समीकरण ज्ञात कीजिए तथा हल करके वक्र का समीकरण ज्ञात कीजिए।

Ans. $y = kx$ or $xy = c$

Sol. $\sqrt{x^2 + y^2} = \left|\frac{y}{y'}\sqrt{1+y'^2}\right|$

$(y')^2 (x^2 + y^2) = y^2 (1 + y'^2)$

$x^2(y')^2 + y^2(y')^2 = y^2 + y^2 (y')^2$

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{y}{x}\right)^2$$

$$\frac{dy}{dx} = \pm\frac{y}{x}$$

(+)

$$\frac{dy}{dx} = \frac{y}{x}$$

$$\int\frac{dy}{y} = \int\frac{dx}{x}$$

$$\ln y = \ln x + c$$

$$y = cx$$

(–)

$$\frac{dy}{dx} = \frac{-y}{x}$$

$$\int\frac{dy}{y} = \int\frac{-dx}{x}$$

$$\ln y + \ln x = \ln c$$

$$xy = c$$

13. Tangent is drawn at the point (x_i, y_i) on the curve $y = f(x)$, which intersects the x-axis at $(x_{i+1}, 0)$. Now again a *tangent is drawn at* (x_{i+1}, y_{i+1}) on the curve which intersects the x-axis at $(x_{i+2}, 0)$ and the process is repeated *n* times i.e. i = 1, 2, 3, ..., n.

(a) If $x_1, x_2, x_3, ..., x_n$ form an arithmetic progression with common difference equal to $\log_2 e$ and curve passes through (0, 2), then find the equation of the curve.

(b) If $x_1, x_2, x_3, ..., x_n$ form a geometric progression with common ratio equal to 2 and the curve passes through (1, 2), then find the equation of the curve.

वक्र $y = f(x)$, के किसी बिन्दु (x_i, y_i) पर स्पर्श रेखा, x-अक्ष को बिन्दु $(x_{i+1}, 0)$ प्रतिच्छेदित करती है। पुनः (x_{i+1}, y_{i+1}) पर वक्र पर Li ' kZ[kh x-अक्ष को बिन्दु $(x_{i+2}, 0)$ पर प्रतिच्छेदित करती है। इस तरह से यह प्रक्रिया *n* बार दोहराई जाती है। i.e. i =1,2,3,.., n.

(a) यदि $x_1, x_2, x_3, ..., x_n$ एक समानान्तर श्रेणी को व्यक्त करती है जिसका सार्व अन्तर $\log_2 e$ हो तथा वक्र (0, 2) से गुजरता है तो वक्र का समीकरण बताओं।

(b) यदि $x_1, x_2, x_3, ..., x_n$ एक ज्यामितीय श्रेणी में हो जिसका सार्वअनुपात 2 हो तो (1, 2) से गुजरने वाले वक्र का समीकरण ज्ञात कीजिए।

Ans. (a) $y = 2^{1-x}$ (b) $xy = 2$

Sol. $y = mx + c \qquad (x_i, y_i)$

$$\frac{-c}{m} = x_{i+1}$$

$$\frac{mx_i - y_i}{m} = x_{i+1}$$

(a) $x - \frac{y}{m} = x + \log_2 e \Rightarrow -y = m \log_2 e$

$m = -y \ \ln 2$

$$\frac{dy}{dx} = -y \ln 2$$

$\ln y = (-\ln 2)\, x + C$

$(0, 2)$

$C = \ln 2$

$\ln y = -x \ln 2 + \ln 2$

$\ln y = \ln (2 \cdot 2^{-x})$

$y = 2^{1-x}$

(b) $x - \frac{y}{m} = 2x$

$$\frac{-y}{m} = x$$

$$\frac{dy}{dx} = \frac{-y}{x}$$

$$\int \frac{dy}{y} = \int -\frac{dx}{x}$$

$\ln y = -\ln x + \ln c$

$\ln (xy) = \ln c$

$xy = c$

$(1, 2) \qquad c = 2 \Rightarrow xy = 2$

14. Given $y(0) = 2000$ and $\frac{dy}{dx} = 32000 - 20y^2$, then find the value of $\lim\limits_{x \to \infty} y(x)$.

यदि $y(0) = 2000$ तथा $\frac{dy}{dx} = 32000 - 20y^2$ हो, तो $\lim\limits_{x \to \infty} y(x)$ ज्ञात कीजिए।

Ans. 40

Sol. $\frac{dy}{dx} = 32000 - 20y^2$

$= 20\,(1600 - y^2)$

$$\int \frac{dy}{(40)^2 - y^2} = \int 20dx$$

$$\frac{1}{2\times 40}\ln\left(\left|\frac{40+y}{40-y}\right|\right) = 20x + C$$

$(0, 2000)$

$$\frac{1}{80}\ln\left(\frac{2040}{1960}\right) = 0 + C$$

$$\frac{1}{80}\ln\left(\frac{204}{196}\right) = C$$

$$C = \frac{1}{80}\ln\left(\frac{51}{49}\right)$$

$$\frac{1}{80}\ln\left(\frac{y+40}{y-40}\right) = 20x + \frac{1}{80}\ln\left(\frac{51}{49}\right)$$

$$\ln\left(\frac{y+40}{y-40}\right) = 1600x + \ln\left(\frac{51}{49}\right)$$

$$\frac{y+40}{y-40} = (e)^{1600x + \ell n\left(\frac{51}{49}\right)}$$

C & D

$$\frac{2y}{80} = \frac{e^{1600x + \ln\frac{51}{49}} + 1}{e^{1600x + \ln\frac{51}{49}} - 1}$$

$$\lim_{x\to\infty} y = 40\left(\frac{e^{1600x+\ln\frac{51}{49}}+1}{e^{1600x+\ln\frac{51}{49}}-1}\right) \Rightarrow 40(1) = 40$$

15. Let $f(x)$ is a continuous function which takes positive values for $x \geq 0$ and satisfy $\int_0^x f(t)\,dt = x\sqrt{f(x)}$ with $f(1) = \frac{1}{2}$. Find the value of $f\left(\sqrt{2}+1\right)$.

यदि f(x) सतत् फलन हो जो $x \geq 0$ के लिये धनात्मक मान लेता हो तो तथा $\int_0^x f(t)\,dt = x\sqrt{f(x)}$ जहाँ पर $f(1) = \frac{1}{2}$ तब $f\left(\sqrt{2}+1\right)$ का मान ज्ञात कीजिए।

Ans. 1/4

Sol. $\int_0^x f(t)\,dt = x\sqrt{f(x)}$

$$f(x) = \sqrt{f(x)} + \frac{x}{2}\frac{f'(x)}{\sqrt{f(x)}}$$

$$2\left[f(x)\right]^{3/2} = 2f(x) + x\,f'(x)$$

$$xf'(x) = 2f(x)\left(\sqrt{f(x)}-1\right)$$

$$\frac{f'(x)}{2f(x)\left(\sqrt{f(x)}-1\right)} = \frac{1}{x}$$

$$\int \frac{dy}{2y\left(\sqrt{y}-1\right)} = \int \frac{dx}{x}$$

$$\sqrt{y} = t$$

$$\frac{1}{2\sqrt{y}}dy = dt$$

$$\int \frac{dt}{t(t-1)} = \ln x + C$$

$$\int \frac{t-(t-1)}{t(t-1)} dt = \ln x + C$$

$$\int \frac{dt}{t-1} - \int \frac{dt}{t} = \ln x + C$$

$$\ln\left|\frac{t-1}{t}\right| = \ln x + c \Rightarrow \ln\left|\frac{\sqrt{y}-1}{\sqrt{y}}\right| = \ln x + C$$

$$\left(1, \frac{1}{2}\right)$$

$$\ln\left|\frac{1-\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}\right| = \ln(1) + C$$

$$\ln\left(\sqrt{2}-1\right) = 0 + C$$

$$C = \ln\left(\sqrt{2}-1\right)$$

$$\ln\left(\left|\frac{1-\sqrt{y}}{\sqrt{y}}\right|\right) = \ln x + \ln\left(\sqrt{2}-1\right)$$

$$= \ln(2-1)$$

$$= \ln(1)$$

$$\ln\left(\frac{1-\sqrt{y}}{\sqrt{y}}\right) = 0$$

$$\frac{1-\sqrt{y}}{\sqrt{y}} = 1$$

$$1-\sqrt{y} = \sqrt{y}$$

$$2\sqrt{y} = 1$$

$$\sqrt{y} = \frac{1}{2}$$

$$y = \frac{1}{4}$$

HOMOGENEOUS DIFFERENTIAL EQUATION

16. $\frac{dy}{dx} = \frac{x^2 + xy}{x^2 + y^2}$

Ans. $c(x-y)^{2/3}(x^2 + xy + y^2)^{1/6} = \exp\left[\frac{1}{\sqrt{3}}\tan^{-1}\frac{x+2y}{x\sqrt{3}}\right]$ where $\exp x \equiv e^x$

Sol. $\frac{dy}{dx} = \frac{x^2 + xy}{x^2 + y^2}$

$y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + x\frac{dv}{dx} = \frac{x^2 + vx^2}{x^2 + v^2x^2} = \frac{1+v}{1+v^2}$$

$$x\frac{dv}{dx} = \frac{1+v}{1+v^2} - v \Rightarrow \frac{1+v-v-v^3}{1+v^2}$$

$$\frac{1+v^2}{1-v^3}dv = \frac{dx}{x}$$

$$\frac{1+v^2}{(1-v)(v^2+v+1)} = \frac{A}{1-v} + \frac{Bv+C}{v^2+v+1}$$

$(1 + v^2) = A(v^2 + v + 1) + (Bv + C)(1 - v)$

$A - B = 1$ $\quad A + B - C = 0$ $\quad A + C = 1$

$A - B = 1$ $\quad 2A - C = 1$

$B = \frac{-1}{3}$ $\quad 2A - C = 1$ $\quad 3A = 2$

$$A = \frac{2}{3} \qquad C = \frac{1}{3}$$

$$I = \frac{2}{3}\int \frac{dv}{1-v} - \frac{1}{3}\int \frac{v-1}{v^2+v+1}dv$$

$v - 1 = P(2v + 1) + Q$

$2P = 1$

$$P = \frac{1}{2}$$

$$Q = \frac{-3}{2}$$

$$I = \frac{-2}{3}\ln(v-1) - \frac{1}{3}\left[\frac{1}{2}\ln\left(v^2+v+1\right)\right] + \frac{1}{2}\int \frac{dv}{v^2+v+\frac{1}{4}+\frac{3}{4}}$$

$$\Rightarrow \frac{-2}{3}\ln\left(\frac{y}{x}-1\right) - \frac{1}{6}\ln\left(\frac{y^2+x^2+xy}{x^2}\right) + \frac{1}{2}\frac{1}{\frac{\sqrt{3}}{2}}\tan^{-1}\left(\frac{\frac{y}{x}+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) = \ln x + C$$

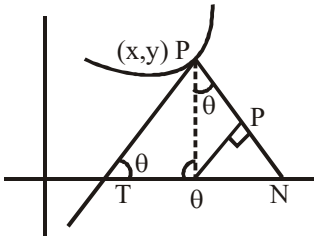
$$\Rightarrow \frac{-2}{3}\ln\left(\frac{y-x}{x}\right) - \frac{1}{6}\ln\left(\frac{x^2+y^2+xy}{x^2}\right) + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2y+x}{\sqrt{3}x}\right) = \ln x + C$$

17. Find the equation of a curve such that the projection of its ordinate upon the normal is equal to its abscissa.

वक्र का समीकरण ज्ञात कीजिए यदि इसके कोटि का अभिलम्ब पर प्रक्षेप इसके भुज के बराबर हो।

Ans. $\frac{y^2 \pm y\sqrt{y^2-x^2}}{x^2} = \ell n\left|\left(y \pm \sqrt{y^2-x^2}\right)\cdot\frac{c^2}{x^3}\right|$, where same sign has to be taken.

Sol.



$$\tan\theta = \frac{dy}{dx}$$

$$PR = PQ\cos\theta = x$$

$$\Rightarrow y\cos\theta = x$$

$$\sec\theta = \frac{y}{x}$$

$$\sqrt{1+\tan^2\theta} = \frac{y}{x}$$

$$1+\tan^2\theta = \frac{y^2}{x^2}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{y^2-x^2}{x^2}$$

$$\frac{dy}{dx} = \pm\sqrt{\left(\frac{y}{x}\right)^2 - 1}$$

$$\frac{y}{x} = t$$

$$y = tx$$

$$\frac{dy}{dx} = t + x\frac{dt}{dx}$$

$$t + x\frac{dt}{dx} = \pm\sqrt{t^2-1}$$

$$x\frac{dt}{dx} = -t \pm \sqrt{t^2-1}$$

$$= \int \frac{dt}{t \pm \sqrt{t^2-1}} = -\int \frac{dx}{x}$$

$$= \int \left(t \pm \sqrt{t^2-1}\right) dt = -\ln x + C$$

$$\frac{t^2}{2} \pm \left(\frac{t}{2}\sqrt{t^2-1} - \frac{1}{2}\ln\left(t+\sqrt{t^2-1}\right)\right) = -\ln x + C$$

$$\frac{y^2}{x^2} \pm \left(\frac{y}{x}\sqrt{\frac{y^2}{x^2}-1} - \ln\left(\frac{y}{x} + \sqrt{\frac{y^2}{x^2}-1} \right) \right) = -2\ln x + C$$

$$\frac{y^2 \pm y\sqrt{y^2-1}}{x^2} \qquad t \mp \left(\ln\left(\frac{y+\sqrt{y^2-x^2}}{x} \right) \right) + \ln x^2 = C$$

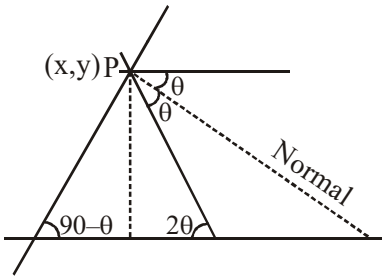
$$\frac{y^2 \pm y\sqrt{y^2-1}}{x^2} = \ln\left(\frac{y \pm \sqrt{y^2-x^2}}{x^3} C \right)$$

18. The light rays emanating from a point source situated at origin when reflected from the mirror of a search light are reflected as beam parallel to the x-axis. Show that the surface is parabolic, by first forming the differential equation and then solving it.

मूल बिन्दु पर स्थित स्रोत से निर्गत प्रकाश किरणें किसी सर्च लाइट के दर्पण से परावर्तित होकर x अक्ष के समानान्तर हो जाती है। पहले अवकल समीकरण बनाकर तथा इसको हल करके सिद्ध करो कि तल परवलयाकार है।

Ans. $y^2 = 2cx + c^2$

Sol. $\frac{dy}{dx} = \tan(90^\circ - \theta) = \cot\theta$



$$\tan 2\theta = \frac{y}{x}$$

$$\frac{dy}{dx} = \cot\theta = \frac{1+\cos 2\theta}{\sin 2\theta}$$

$$= \frac{1 - \frac{x}{\sqrt{x^2+y^2}}}{\frac{y}{\sqrt{x^2+y^2}}}$$

$$\frac{dy}{dx} = \frac{\sqrt{x^2+y^2} - x}{y}$$

$$y = vx$$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v+\frac{xdv}{dx}=\frac{\sqrt{x^2+v^2x^2}-x}{vx}=\frac{\sqrt{1+v^2}-1}{v}$$

$$x\frac{dv}{dx}=\frac{\sqrt{1+v^2}-1}{v}-v$$

$$x\frac{dv}{dx}=\frac{\sqrt{1+v^2}-\left(1+v^2\right)}{v}$$

$$\frac{vdv}{\left(1+v^2\right)-\sqrt{1+v^2}}=\frac{-dx}{x}$$

$1+v^2=P^2$

$2vdv=2P\,dP$

$$\frac{PdP}{P^2-P}=\frac{-dx}{x}$$

$$\int\frac{dP}{P-1}=\int-\frac{dx}{x}$$

$\ell n\,(P-1)=-\ell n\,x+\ell n\,c$

$\ell n(P-1)+\ell n\,x=\ell n\,C$

$(P-1)\,x=C$

$$\left(\sqrt{1+v^2}-1\right)x=C$$

$$\left(\sqrt{\frac{x^2+y^2}{x^2}}-1\right)x=C$$

$$\sqrt{x^2+y^2}-x=C$$

$$\sqrt{x^2+y^2}=x+C$$

$x^2+y^2=x^2+c^2+2cx$

$y^2=2cx+c^2$

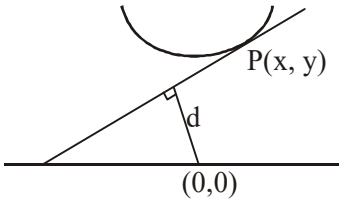
19. The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa of the point of contact. Find the equation of the curve satisfying the above condition and which passes through (1, 1).

मूल बिन्दु से लम्ब जो कि वक्र के किसी बिन्दु पर स्पर्श पर डाला जाता है, स्पर्श बिन्दु के भुज के बराबर है। वक्र का समीकरण बताओ जो (1, 1) से गुजरता है।

Ans. $x^2 + y^2 - 2x = 0$

Sol. $Y - y = m(X - x)$

$mX - Y - mx + y = 0$



$$d = \left|\frac{y - mx}{\sqrt{m^2 + 1}}\right| = 2$$

$d = x$

$$\frac{y - mx}{\sqrt{m^2 + 1}} = x$$

$$\frac{y^2 + m^2x^2 - 2mxy}{m^2 + 1} = x^2$$

$y^2 + m^2x^2 - 2mxy = m^2x^2 + x^2$

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

$y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + \frac{xdv}{dx} = \frac{v^2x^2 - v^2}{2x^2v} = \frac{v^2 - 1}{2v}$$

$$x\frac{dv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$= \frac{v^2 - 1 - 2v^2}{2v}$$

$$\frac{xdv}{dx} = -\frac{\left(1 + v^2\right)}{2v}$$

$$\int \frac{2v}{1+v^2}dv = \int -\frac{dx}{x}$$

$\ln|1 + v^2| = -\ln x + \ln C$

$$\ell n\left(1+\frac{y^2}{x^2}\right) + \ln x = \ln c$$

$$\ln\left(\frac{x^2+y^2}{x}\right) = C \Rightarrow \frac{x^2+y^2}{x} = C$$

$x^2 + y^2 - 2x = 0$

$(1, 1) \Rightarrow C = 2$

20. Show that the equation of the curve intersecting with the x- axis at the point x = 1 and for which the length of the subnormal at any point of the curve is equal to the arthemetic mean of the co-ordinates of this point is $(y - x)^2(x + 2y) = 1$.

यदि एक वक्र, x-अक्ष को x = 1 प्रतिच्छेदित करता है, तथा इसके किसी बिन्दु पर अधोलम्ब की लम्बाई इस बिन्दु के निर्देशांको का समानान्तर माध्य है, वक्र का समीकरण है $(y - x)^2(x + 2y) = 1$ दर्शाओं।

Ans.

Sol. $$yy' = \frac{x+y}{2}$$

$$y' = \frac{x+y}{2y}$$

$$\frac{dy}{dx} = \frac{x+y}{2y}$$

$$\frac{dx}{dy} = \frac{2y}{x+y}$$

$x = vy$

$$\frac{dx}{dy} = v + y\frac{dv}{dy}$$

$$v + y\frac{dv}{dy} = \frac{2y}{(v+1)y} \Rightarrow \frac{ydv}{dy} = \frac{2}{v+1} - v = \frac{2-v^2-v}{v+1}$$

$$\frac{v+1}{v^2+v-2}dv=\frac{-dy}{y}$$

$$\frac{v+1}{(v+2)(v-1)}dv=-\frac{dy}{y}$$

$$\int\left(\frac{1}{v+2}+2\cdot\frac{1}{(v+2)(v-1)}\right)dv=\int-\frac{dy}{y}$$

$$\ln|v+2|+\frac{2}{3}\ln\left(\frac{v-1}{v+2}\right)=-\ln y+\ln C$$

$$(v+2)\left(\frac{v-1}{v+2}\right)^{\frac{2}{3}}\cdot y=C$$

$$(v-1)^{\frac{2}{3}}(v+2)^{\frac{1}{3}}y=C$$

$(v-1)^2(v+2)y^3=C$

$$\frac{(x-y)^2}{y^2}\left(\frac{x+2y}{y}\right)y^3=C$$

$(x-y)^2(x+2y)=C$

$(1,0)$

$(1-0)^2(1+0)=C$

$c=1$

$(x-y)^2(x+2y)=1$

21. $$\left[x\cos\frac{y}{x}+y\sin\frac{y}{x}\right]y=\left[y\sin\frac{y}{x}-x\cos\frac{y}{x}\right]x\frac{dy}{dx}$$

Ans. $$xy\cos\frac{y}{x}=c$$

Sol. $$\left[x\cos\frac{y}{x}+y\sin\frac{y}{x}\right]y=x\left[y\sin\left(\frac{y}{x}\right)-x\cos\left(\frac{y}{x}\right)\right]\frac{dy}{dx}$$

$$x\left[\cos\frac{y}{x}+\frac{y}{x}\sin\frac{y}{x}\right]y=x^2\left[\frac{y}{x}\sin\frac{y}{x}-\cos\frac{y}{x}\right]\frac{dy}{dx}$$

$$\left[\cos\frac{y}{x}+\frac{y}{x}\sin\frac{y}{x}\right]\frac{y}{x}=\left[\frac{y}{x}\sin\frac{y}{x}-\cos\frac{y}{x}\right]\frac{dy}{dx}$$

$y = vx$

$$\frac{dy}{dx}=v+\frac{xdv}{dx}$$

$$(\cos v + v\sin v)v = (v\sin v - \cos v)\left(v+\frac{xdv}{dx}\right)$$

$$v+\frac{xdv}{dx}=\frac{(\cos v+v\sin v)v}{v\sin v-\cos v}$$

$$x\frac{dv}{dx}=\frac{v\cos v+v^2\sin v-v^2\sin v+v\cos v}{v\sin v-\cos v}$$

$$\frac{xdv}{dx}=\frac{2v\cos v}{v\sin v-\cos v}$$

$$\int\frac{v\sin v-\cos v}{v\cos v}dv=\int\frac{2dx}{x}$$

$$\int\tan v dv-\int\frac{1}{v}dv=2\ell nx+C$$

$\ln\sec v - \ln v = 2\ln x + C$

$$\ln(\sec v)=\ln vx^2+C\Rightarrow \ell n\left(\sec\frac{y}{x}\right)=\ell nxy+C$$

$$\ell n\left(xy\right)-\ell n\left(\sec\frac{y}{x}\right)=C$$

$$\ell n\left(xy\right)+\ell n\left(\cos\frac{y}{x}\right)=C$$

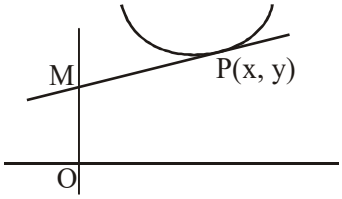
$$xy\cos\frac{y}{x}=C$$

22. Find the curve for which any tangent intersects the y–axis at the point equidistant from the point of tangency and the origin.

उस वक्र का समीकरण ज्ञात कीजिए जिसकी कोई सपर्श रेखा y अक्ष को उस बिन्दु पर काटती है जो मूल बिन्दु तथा स्पर्श बिन्दु से समदूरस्थ है।

Ans. $x^2 + y^2 = cx$

Sol.



$PM = OM$

$Y - y = m(X - x)$

$M(0, y - mx)$

$$\sqrt{x^2 + m^2x^2} = (y - mx)$$

$x^2 + m^2x^2 = y^2 + m^2x^2 - 2mxy$

$2mxy = y^2 - x^2$

$$m = \frac{y^2 - x^2}{2xy}$$

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

$y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + x\frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$= \frac{v^2 - 1 - 2v^2}{2v}$$

$$x\frac{dv}{dx} = -\frac{(1 + v^2)}{2v}$$

$$\int\left(\frac{2v}{1 + v^2}\right)dv = \int -\frac{dx}{x}$$

$\ell n(1 + v^2) = -\ell n\, x + \ell n\, C$

$$\ell n\left(1+\frac{y^2}{x^2}\right)+\ell nx = \ell nC$$

$$\ell n\left(\frac{x^2+y^2}{x}\right) = \ell nC$$

$x^2 + y^2 = cx$

23. $(x - y)\, dy = (x + y + 1)\, dx$

Ans. $\text{arc tan}\,\dfrac{2y+1}{2x+1} = \ell n\, c\sqrt{x^2+y^2+x+y+\dfrac{1}{2}}$

Sol. $(x - y)\, dy = (x + y + 1)\, dx$

$x - y = 0$

$x + y + 1 = 0$

$2x + 1 = 1$

$$x = \frac{-1}{2} \qquad y = \frac{-1}{2}$$

$$x = X - \frac{1}{2} \qquad y = Y - \frac{1}{2}$$

$(X - Y)dY = (X - Y)\, dX$

$$\frac{dY}{dX} = \frac{X+Y}{X-Y}$$

$Y = vX$

$$\frac{dY}{dX} = v + X\frac{dv}{dX}$$

$$X\frac{dv}{dX} = \frac{1+v-v+v^2}{1-v}$$

$$\int \frac{1-v}{1+v^2}dv = \int \frac{dX}{X}$$

$$\int \frac{dv}{1+v^2} - \int \frac{1}{2}\frac{2vdx}{1+v^2} = \int \frac{dX}{X}$$

$$\tan^{-1} v - \frac{1}{2}\ln\left(1+v^2\right) = \ln X + C$$

$$\tan^{-1}\left(\frac{Y}{X}\right)=\ln\sqrt{\frac{X^2+Y^2}{X^2}}X+c$$

$$\tan^{-1}\left(\frac{Y}{X}\right)=\ln\left(c\sqrt{X^2+Y^2}\right)$$

$$\tan^{-1}\left(\frac{y+\frac{1}{2}}{x+\frac{1}{2}}\right)=\ln\left(c\sqrt{\left(x+\frac{1}{2}\right)+\left(y+\frac{1}{2}\right)^2}\right)$$

$$\tan^{-1}\left(\frac{2y+1}{2x+1}\right)=\ln\left(c\sqrt{x^2+y^2+x+y+\frac{1}{2}}\right)$$

24. $\frac{dy}{dx}=\frac{x+y+1}{2x+2y+3}$

Ans. $x+y+\frac{4}{3}=ce^{3(x-2y)}$

Sol. $\frac{dy}{dx}=\frac{x+y+1}{2x+2y+3}$

$x+y+1=t$

$1+\frac{dy}{dx}=\frac{dt}{dx}$

$\frac{dt}{dx}-1=\frac{t}{2t+1}$

$\frac{dt}{dx}=\frac{t}{2t+1}+1=\frac{t+2t+1}{2t+2}$

$\frac{dt}{dx}=\frac{3t+1}{2(t+1)}$

$\frac{3t+3}{3t+1}dt=\frac{3dx}{2}$

$\frac{3t+1+2}{3t+1}dt=\frac{3}{2}dx$

$\int\left(1+\frac{2}{3t+1}\right)dt=\int\frac{3}{2}dx$

$$t+\frac{2}{3}\ln(3t+1)=\frac{3}{2}x+C$$

$$x+y+1+\frac{2}{3}\ln(3x+3y+4)=\frac{3}{2}x+C$$

$$\frac{3x+3y+3+2\ln(3x+3y+4)}{3}=\frac{3x+c}{2}$$

$$6x+6y+6+4\ln(3x+3y+4)=9x+C$$

$$4\ln(3x+3y+4)=3x-6y+C$$

$$4\ln\left[3\left(x+y+\frac{4}{3}\right)\right]=3x-6+C$$

$$4\ln 3+4\ln\left(x+y+\frac{4}{3}\right)=3(x-2y)+C$$

$$\left(x+y+\frac{4}{3}\right)=c(e)^{3(x-2y)}$$

$$x+y+\frac{4}{3}=c.e^{3(x-2y)}$$

25. Show that the curve such that the distance between the origin and the tangent at an arbitrary point is equal to the distance between the origin and the normal at the same point, $\sqrt{x^2+y^2}=ce^{\pm\tan^{-1}\frac{y}{x}}$

वक्र के किसी बिन्दु पर स्पर्श रेखा की मूल बिन्दु से दूरी, उस बिन्दु पर अभिलम्ब एवं मूल बिन्दु के बीच की दूरी के बराबर है। सिद्ध करो कि oØ d kl eha j.kg$ $\sqrt{x^2+y^2}=ce^{\pm\tan^{-1}\frac{y}{x}}$

Ans.

Sol. Equation of tangent $Y-y=\frac{dy}{dx}(X-x)$

$$\text{distance from origin}=\left|\frac{-x\frac{dy}{dx}+y}{\sqrt{1+\left(\frac{dy}{dx}\right)^2}}\right|$$

Equation of normal $\quad Y-y=\frac{-1}{\frac{dy}{dx}}(X-x)$

$$\Rightarrow Y\frac{dy}{dx}-y\frac{dy}{dx}=-X+x$$

$$\text{distance from origin} = \left| \frac{x + y\frac{dy}{dx}}{\sqrt{1+\left(\frac{dy}{dx}\right)^2}} \right|$$

Now $\left|-x\frac{dy}{dx} + y\right| = \left|x + y\frac{dy}{dx}\right|$

either $-x\frac{dy}{dx} + y = x + y\frac{dy}{dx}$

or $\quad x\frac{dy}{dx} - y = x + y\frac{dy}{dx}$

$\Rightarrow \quad (x+y)\frac{dy}{dx} = y - x$

or $\quad (x-y)\frac{dy}{dx} = x + y$

$\Rightarrow \quad \frac{dy}{dx} = \frac{y-x}{y+x}$ or $\quad \frac{dy}{dx} = \frac{x+y}{x-y}$

Put $\quad y = vx$

$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{v-1}{v+1}$ or $\quad v + x\frac{dv}{dx} = \frac{1+v}{1-v}$

$\Rightarrow \quad \grave{o}\frac{v+1}{1+v^2}\,dv = \int\frac{-dx}{x}$ or $\quad \int\frac{v-1}{1+v^2}\,dv = \int\frac{-dx}{x}$

$\Rightarrow \quad \frac{1}{2}\,\ell n\,|1+v^2| + \tan^{-1} v = -\,\ell nx + \ell nc$

or $\frac{1}{2}\,\ell n\,|1+v^2| - \tan^{-1} v = \ell nc - \ell nx$

Hence solution will be

$\frac{1}{2}\,\ell n\,|1+v^2| + \ell nx = \pm\tan^{-1} v + \ell nc$

$x\sqrt{1+v^2} = ke^{\pm\tan^{-1} v}$

$\Rightarrow \sqrt{x^2+y^2} = ce^{\pm\tan^{-1} y/x}$

LINEAR DIFFERENTIAL EQUATION

26. Let $y = y(t)$ be a solution to the differential equation $y' + 2ty = t^2$, then find $\lim\limits_{t\to\infty} \frac{y}{t}$.

यदि $y = y(t)$ अवकल समीकरण $y' + 2ty = t^2$ का एक हल हो तो $\lim\limits_{t\to\infty} \frac{y}{t}$ ज्ञात कीजिए।

Ans. $\frac{1}{2}$

Sol. $y' + 2ty = t^2$

I.F. $= e^{\int 2t} = e^{t^2}$

$$\int e^{t^2} y' + 2te^{t^2} y = \int t^2 e^{t^2}$$

$$ye^{t^2} = \frac{1}{2}\int t(2t)e^{t^2} dt$$

$$\Rightarrow \frac{1}{2}\left[te^{t^2} - \frac{1}{2}\int e^{t^2} dt\right] + C$$

$$y = \frac{t}{2} - \frac{1}{2}e^{-t^2}\int e^{t^2} dt + Ce^{-t^2}$$

$$\lim_{t\to\infty} \frac{y}{t} = \lim_{t\to\infty} \frac{\frac{t}{2} + 2Ce^{-t^2} - e^{-t^2}\int e^{t^2} dt}{2t}$$

$$= \lim_{t\to\infty} \frac{1}{2} + \frac{2C - \int (e)^{t^2} dt}{2t(e)^{t^2}}$$

$$= \frac{1}{2} + \lim_{t\to\infty} \frac{0 - e^{t^2}}{2e^{t^2} + 4t^2 e^{t^2}}$$

$$= \frac{1}{2} + \lim_{t\to\infty} \frac{-1}{2 + 4t^2}$$

$$= \frac{1}{2} + 0$$

$$= \frac{1}{2}$$

27. $\frac{dy}{dx} + \frac{x}{1+x^2} y = \frac{1}{2x(1+x^2)}$

Ans. $y\sqrt{1+x^2} = c + \frac{1}{2} ln\left[\tan\frac{1}{2}\arctan x\right]$ Another form is $y\sqrt{1+x^2} = c + \frac{1}{2} ln \frac{\sqrt{1+x^2}-1}{x}$

Sol. $\frac{dy}{dx} + \frac{x}{1+x^2} y = \frac{1}{2x\left(1+x^2\right)}$

I.F. $= e^{\frac{1}{2}\int\frac{2x}{1+x^2}dx} = e^{\ln\sqrt{1+x^2}} \sqrt{1+x^2}$

$\int \sqrt{1+x^2}\frac{dy}{dx} + \frac{x}{\sqrt{1+x^2}} y = \int \frac{x}{2x^2\sqrt{1+x^2}} dx$

$1+x^2 = t^2$

$2xdx = 2tdt$

$xdx = tdt$

$y\sqrt{1+x^2} = \frac{1}{2}\int \frac{tdt}{\left(t^2-1\right)t} + C$

$y\sqrt{1+x^2} \Rightarrow \frac{1}{2}\int \frac{dt}{(t-1)(t+1)} + C$

$= \frac{1}{2}\times\frac{1}{2}\int \frac{(t+1)-(t-1)}{(t-1)(t+1)} dt + C$

$\Rightarrow \frac{1}{4}\int\left(\frac{dt}{t-1} - \frac{dt}{t+1}\right) + C$

$= \frac{1}{4}\left[\ln\left(\frac{t-1}{t+1}\right)\right] + C$

$y\sqrt{1+x^2} = \frac{1}{4}\ln\left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1}\right) + C$

$= \frac{1}{4}\ln\left(\frac{\left(\sqrt{1+x^2}-1\right)^2}{1+x^2-1}\right) + C$

$$y\sqrt{1+x^2} = \frac{1}{2}\ln\left(\frac{\sqrt{1+x^2}-1}{x}\right)+C$$

28. $(1-x^2)\frac{dy}{dx} + 2xy = x(1-x^2)^{1/2}$

Ans. $y = c(1-x^2) + \sqrt{1-x^2}$

Sol. $(1-x^2)\frac{dy}{dx} + 2xy = x(1-x^2)^{1/2}$

$$\frac{dy}{dx}+\frac{2x}{1-x^2}y = \frac{x}{\sqrt{1-x^2}}$$

$$\text{I.F.} = e^{\int\frac{2x}{1-x^2}dx} \Rightarrow e^{-\ln\left(1+x^2\right)}$$

$$= \frac{1}{1-x^2}$$

$$\int\frac{1}{1-x^2}\frac{dy}{dx}+\frac{2x}{\left(1-x^2\right)^2}y = \int\frac{x}{\left(1-x^2\right)^{3/2}}$$

$1 - x^2 = 1 \Rightarrow -2xdx = dt$

$$\frac{y}{1-x^2} = \frac{-1}{2}\int\frac{dt}{(t)^{3/2}}+C$$

$$\frac{y}{1-x^2} = -\frac{1}{2}\cdot\frac{(t)^{\frac{-3}{2}+1}}{\frac{-3}{2}+1}+C$$

$$= \frac{-1}{2}\frac{(t)^{\frac{-1}{2}}}{\frac{-1}{2}}+C$$

$$\frac{y}{\left(1-x^2\right)} = \frac{1}{\sqrt{t}}+C$$

$$\frac{y}{1-x^2} = \frac{1}{\sqrt{1-x^2}}+C$$

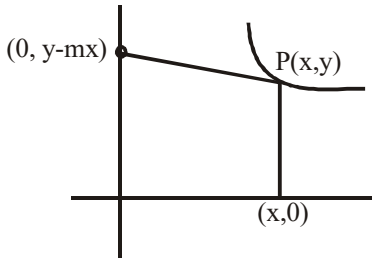
$$y = \sqrt{1-x^2}+C\left(1-x^2\right)$$

29. Find the curve such that the area of the trapezium formed by the co–ordinate axes, ordinate of an arbitrary point and the tangent at this point equals half the square of its abscissa.

यदि वक्र के किसी बिन्दु कोटि, इस बिन्दु पर स्पर्श रेखा तथा निर्देशी अक्षों से बने चर्तुभुज का क्षेत्रफल, स्पर्श बिन्दु के भुज के वर्ग के आधे के बराबर हो तो वक्र का समीकरण बताओं।

Ans. $y = cx^2 \pm x$

Sol.



$(Y - y) = m(X - x)$

$X = 0$

$Y = y - mx$

$$\text{Area} = \frac{1}{2}(y + y - mx) \times x = \frac{x^2}{2}$$

$(2y - mx) = x$

$$m = \frac{2y - x}{x}$$

$$\frac{dy}{dx} = \frac{2y - x}{x}$$

$y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + \frac{xdv}{dx} = \frac{2v - 1}{1}$$

$$\frac{xdv}{dx} = v - 1$$

$$\int \frac{dv}{v - 1} = \int \frac{dv}{x} \Rightarrow \ln|v - 1| = \ln x + c$$

$$\ln\left(\frac{y}{x}-1\right)-\ln x = c$$

$$\ln\left(\frac{y-x}{x^2}\right) = \ln c$$

$$y - x = cx^2$$

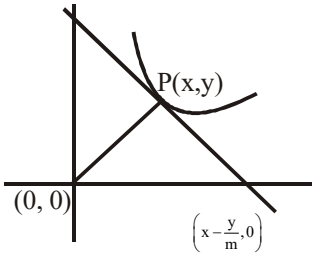
$$y = x + cx^2$$

30. A curve in the first quadrant is such that the area of the triangle formed in the first quadrant by the x-axis, a tangent to the curve at any of its point P and radius vector of the point P is 2 sq. units. If the curve passes through (2, 1), find the equation of the curve.

प्रथम चतुर्थांश में x-अक्ष, वक्र के किसी बिन्दु P पर स्पर्श रेखा तथा बिन्दु P के त्रिज्या सदिश से बनी त्रिभुज का क्षेत्रफल 2 वर्ग इकाई है। यदि वक्र (2, 1) से गुजरता हो तो वक्र का समीकरण ज्ञात कीजिए।

Ans. $xy = 2$

Sol.



$$Y - y = m(X - x)$$

$$\frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ x & y & 1 \\ x-\frac{y}{m} & 0 & 1 \end{vmatrix} = 2$$

$$\left|1(0-y)\left(\left(x-\frac{y}{m}\right)\right)\right| = \pm 4$$

$$y\left(x-\frac{y}{m}\right) = \pm 4$$

$$x-\frac{y}{m} = \frac{4}{y}$$

$$x-\frac{4}{y}=\frac{y}{m}\Rightarrow m=\frac{y^2}{xy-4}$$

$$\frac{dy}{dx}=\frac{y^2}{xy-4}$$

$$\frac{dx}{dy}+\left(-\frac{1}{y}\right)^{x}=\frac{-4}{y^2}$$

$$\text{I.F.}=e^{\int-\frac{1}{y}dy}=e^{-\ln y}=\frac{1}{y}$$

$$\int\frac{1}{y}\frac{dx}{dy}+\left(\frac{-1}{y^2}\right)(x)=\int\frac{-4}{y^3}$$

$$\frac{x}{y}=-4\frac{y^{-3+1}}{-3+1}+C$$

$$\frac{x}{y}=\frac{2}{y^2}+C$$

(2, 1)

$$\frac{1}{2}=\frac{2}{1}+C$$

C = 0

$$\frac{x}{y}=\frac{2}{y^2}$$

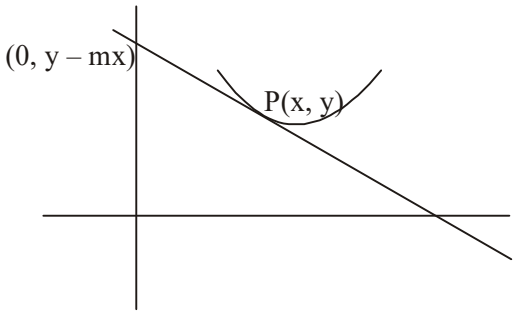
xy = 2

31. Find the curve possessing the property that the intercept , the tangent at any point of a curve cuts off on the y–axis is equal to the square of the abscissa of the point of tangency.

यदि वक्र के किसी बिन्दु पर स्पर्श रेखा का y अक्ष पर अन्तःखण्ड, स्पर्श बिन्दु पर भुज के वर्ग के बराबर हो तो वक्र का समीकरण ज्ञात कीजिए।

Ans. $y = cx - x^2$

Sol. $Y - y = m(X - x)$



$X = 0$

$Y = y - mx$

$|y - mx| = x^2$

$y - mx = x^2$

$mx = y - x^2$

$$\frac{dy}{dx} = \frac{y}{x} - x$$

$$\frac{dy}{dx} + \left(\frac{-1}{x}\right)y = -x$$

$$\text{I.F.} = e^{\int \frac{-1}{x}dx} = e^{-\ln x} = \frac{1}{x}$$

$$\int \frac{1}{x}\frac{dy}{dx} + \left(\frac{-1}{x^2}\right)y = \int -1$$

$$\frac{y}{x} = -x + c$$

$$y = cx - x^2$$

32. $x\,\frac{dy}{dx} - y = 2x^2 \operatorname{cosec} 2x$

Ans. $y = cx + x \ln \tan x$

Sol. $$\frac{xdy}{dx} - y = 2x^2 \operatorname{cosec} 2x$$

$$\frac{dy}{dx} + \left(\frac{-1}{x}\right)y = 2x \operatorname{cosec} 2x$$

$$\text{I.F.} = e^{\int \frac{-1}{x}dx} = e^{-\ln x} = \frac{1}{x}$$

$$\int \frac{1}{x}\frac{dy}{dx} + \left(\frac{-1}{x^2}\right)y = \int 2\,\text{cosec}\,2x$$

$$\frac{y}{x} = \int \frac{2}{2\sin x \cos x}\,dx$$

$$\frac{y}{x} = \int \frac{dx}{\frac{\sin x}{\cos x}\cos^2 x}$$

$$\frac{y}{x} = \int \frac{\sec^2 dx}{\tan x}$$

$\tan x = t$

$\sec^2 x\,dx = dt$

$$\frac{y}{x} = \ln \tan x + c$$

$y = x \ln \tan x + cx$

33. $(1 + y^2)\,dx = (\tan^{-1} y - x)dy$

Ans. $x = ce^{-\arctan y} + \text{arc tan } y - 1$

Sol. $(1 + y^2)dx = (\tan^{-1} y - x)\,dy$

$$\frac{dx}{dy} = \frac{\tan^{-1} y}{1+y^2} - \frac{x}{1+y^2}$$

$$\frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{\tan^{-1} y}{1+y^2}$$

$$\text{I.F.} = e^{\int \frac{dy}{1+y^2}} = e^{\tan^{-1} y}$$

$$\int e^{\tan^{-1} y}\frac{dx}{dy} + \frac{e^{\tan^{-1} y}}{1+y^2}x = \int \frac{e^{\tan^{-1} y}\tan^{-1} y}{1+y^2}$$

$\tan^{-1} y = t$

$$\frac{1}{1+y^2}dy = dt$$

$$xe^{\tan^{-1} y} = \int te^t dt$$

$$= te^t - e^t + c$$

$$= te^t - e^t + c$$

$$xe^{\tan^{-1} y} = \left(\tan^{-1} y - 1\right)e^{\tan^{-1} y} + c$$

$$x = \tan^{-1} y - 1 + ce^{-\tan^{-1} y}$$

34. Find the differentiable function which satisfies the equation $f(x) = -\int_0^x f(t)\tan t\,dt + \int_0^x \tan(t-x)\,dt$ where $x \in (-\pi/2, \pi/2)$

समीकरण $f(x) = -\int_0^x f(t)\tan t\,dt + \int_0^x \tan(t-x)\,dt$ को संतुष्ट करने वाला अवकलनीय फलन ज्ञात कीजिए जहाँ पर $x \in (-\pi/2, \pi/2)$

Ans. $\cos x - 1$

Sol. $$f(x) = -\int_0^x f(t)\tan t\,dt + \int_0^x \tan(t-x)\,dt$$

$$f(x) = -\int_0^x f(t)\tan t dt + \left[\ln\sec\left(t-x\right)\right]_0^x$$

$$= -\int_0^x f(t)\tan t dt - \ln\sec x$$

$$f'(x) = -f(x)\tan x - \frac{1}{\sec x}\sec x\tan x$$

$$\frac{dy}{dx} = -y\tan x - \tan x$$

$$\frac{dy}{dx} = \tan x\left[1+y\right]$$

$$\int \frac{dy}{1+y} = \int -\tan x dx$$

$\ln(1+y) = -\ln(\sec x) + \ln c$

$\ln(1+y) = \ln(c \cos x)$

$1 + y = c \cos x$

$(0, 0)$

$1 + 0 = c(1)$

$c = 1$

$1 + y = \cos x$

$y = \cos x = 1$

35. Find all functions $f(x)$ defined on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with real values and has a primitive F(x) such that

$$f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1+\sin x)^2}. \text{ Find } f(x).$$

$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ पर परिभाषित वास्तविक मान वाले सभी फलनों f (x) को ज्ञात कीजिए जिनका आद्य (पूर्वग) F(x) है तथा

$$f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1+\sin x)^2}$$

Ans. $f(x) = -\frac{2\cos x}{(1+\sin x)^2} - Ce^{-\sin x} \cdot \cos x$

Sol. $\int f(x)dx = F(x) + c$

$f(x) = F'(x)$

let $F(x) = y$

$$\frac{dy}{dx} + \cos x \,.\, y = \frac{\sin 2x}{(1+\sin x)^2}$$

I.F. $= e^{\sin x}$

$$y.e^{\sin x} = 2\int \frac{e^{\sin x}.\sin x \cos x}{(1+\sin x)^2} dx + c$$

$$y\,e^{\sin x} = 2\int \frac{e^t\left(t+1-1\right)}{(1+t)^2}dt \qquad [\text{Put } \sin x = t]$$

$$= 2\int e^t\left\{\frac{1}{t+1}+\frac{-1}{(1+t)^2}\right\}+c$$

$$y\,e^{\sin x} = \frac{2e^t}{t+1}+c \Rightarrow y\,e^{\sin x} = \frac{2e^{\sin x}}{\sin x+1}+c$$

$$\Rightarrow y = \frac{2}{\sin x+1}+c.\,e^{-\sin x}$$

$$f(x) = \frac{dy}{dx} = \frac{-2\cos x}{(\sin x+1)^2} - c.e^{-\sin x}\cos x$$

36. A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of soluble lawn fertilizer runs into the tank at the rate of 1 lit/min, and the mixture is pumped out of the tank at the rate of 3 litres/min. Find the time when the amount of fertilizer in the tank is maximum.

किसी टंकी में 100 लीटर ताजा पानी है। 1 ग्राम प्रतिलीटर लॉन उर्वरक विलेय का विलयन, टंकी में 1 लीटर 1 मिनट की गति से जाता है तथा यह मिश्रण टंकी से 3 लीटर प्रति मिनट की गति से बाहर निलकता है। समय ज्ञात कीजिए जबकि उर्वरक की मात्रा टंकी में अधिकतम है।

Ans. $27\frac{7}{9}$ minutes

Sol. amount of fertilizer at time t

$= t - 3t = -2t$

Volume at $t = 100 - 2t$

Let amount of fertilizer is 'y'

$$\frac{dy}{dt} = 1-\left(\frac{y}{100-2t}\right)^3$$

$$\frac{dy}{dt}+\frac{3y}{100-2t}dt = 1$$

$$\text{I. F.} = e^{\int\frac{3dt}{100-2t}} = e^{-\frac{3}{2}\ln|100-2t|}$$

$\Rightarrow (100-2t)^{-3/2}$

$$y(100-2t)^{-3/2} = \int(100-2t)^{-3/2}dt$$

$$\Rightarrow \frac{(100-2t)^{-1/2}}{-\frac{1}{2}\times -2}+c$$

$$y(100-2t)^{-\frac{3}{2}} = \frac{(100-2t)^{-\frac{1}{2}}}{1}+c$$

$t = 0\ y(0) = 0$

$$c = \frac{-1}{10}$$

$$y = 100-2t-\frac{1}{10}(100-2t)^{\frac{3}{2}}$$

$$\frac{dy}{dt} = -2-\frac{1}{10}\times\frac{3}{2}(100-2t)^{\frac{1}{2}}\times(-2) = 0$$

$$-2-\frac{3}{10}(100-2t)^{\frac{1}{2}} = 0$$

$$\frac{9(100-2t)}{100} = 4$$

$9(100 - 2t) = 400$

$900 - 18t = 400$

$18t = 500$

$$t = \frac{250}{9}$$

$$t = 27\frac{7}{9}\ \text{min}$$

37. A tank with a capacity of 1000 litres originally contains 100 gms of salt dissolved in 400 litres of water. Beginning at time t = 0 and ending at time t = 100 minutes, water containing 1 gm of salt per litre enters the tank at the rate of 4 litres per minute, and the well mixed solution is drained from the tank at a rate of 2 litre/minute. Find the differential equation for the amount of salt *y* in the tank at time t.

1000 लीटर क्षमता के किसी टंकी में प्रारम्भ में 400 लीटर पानी में 100 ग्राम लवण घोला गया। t = 0 से प्रारम्भ तथा t = 100 मिनट पर समाप्त, 1 ग्राम लवण प्रति लीटर वाला पानी, टंकी में 4 लीटर प्रति मीनट की दर से पहुँचता है। इस प्रकार अच्छी तरह से मिश्रित विलयन टंकी से 2 लीटर/मिनट की दर से बाहर जाता है। किसी समय t पर, लवण की मात्रा y के लिये अवकल समीकरण ज्ञात कीजिए।

Ans. $\frac{dy}{dt} = 4 - \frac{y}{200+t}$

Sol. Amount of salt at a time t = 4t – 2t = 24

Volume at time 't' = 400 + 2t

Let amount of salt = y

$$\frac{dy}{dt} = 4 - \frac{y}{400+2t} \times 2$$

$$\frac{dy}{dt} = 4 - \frac{y}{200+t}$$

38. A curve passes through (2, 0) and the slope of tangent at point P (x, y) equals $\frac{(x+1)^2 + y - 3}{(x+1)}$. Find the equation of the curve and area enclosed by the curve and the x-axis in the fourth quadrant.

एक वक्र बिन्दु (2, 0) से गुजरता है तथा बिन्दु P (x, y) पर ढाल $\frac{(x+1)^2 + y - 3}{(x+1)}$ है। वक्र का समीकरण तथा चतुर्थ चतुर्थांश में x-अक्ष एवं वक्र के मध्य का क्षेत्रफल ज्ञात कीजिये।

Ans. $y = x^2 - 2x$, area = 4/3 sq. units

Sol. $$\frac{dy}{dx} = \frac{(x+1)^2 + (y-3)}{(x+1)}$$

x + 1 = X y – 3 = Y

$$\frac{dY}{dX} = \frac{X^2 + Y}{X}$$

$$\frac{dY}{dX} + \left(\frac{-1}{X}\right) Y = X$$

$$\text{I.F.} = e^{\int -\frac{1}{X} dX} = e^{-\ln X} = \frac{1}{X}$$

$$\int \frac{1}{X}\frac{dY}{dX} - \frac{1}{X^2} Y = \int 1$$

$$\frac{Y}{X} = X + c$$

$$\frac{y-3}{x+1} = x + 1 + c$$

(2,0)

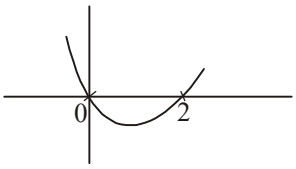
$$-\frac{3}{3} = 2 + 1 + c$$

$c = -4$

$y - 3 = (x + 1)^2 - 4\,(x + 1)$

$y = x^2 + 1 + 2x - 4x - 4 + 3$

$y = x^2 - 2x$



$$A = \int_0^2 \left(x^2 + 2x\right) dx$$

$$\left[\frac{x^3}{3} - x^2\right] = \frac{8}{3} - 4$$

$$\Rightarrow \frac{8-12}{3}$$

$$\Rightarrow \left|\frac{-4}{3}\right|$$

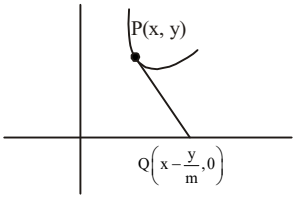
Area = $\frac{4}{3}$ sq. unit

39. If length of tangent at any point on the curve $y = f(x)$ intercepted between the point and the x–axis is of length 1. Find the equation of the curve.

यदि वक्र $y = f(x)$ के किसी बिन्दु पर स्पर्श रेखा बिन्दु तथा x–अक्ष के बीच के अन्तःखण्ड की लम्बाई इकाई है, तो वक्र का समीकरण ज्ञात कीजिये।

Ans. $\sqrt{1-y^2}+\ln\left|\dfrac{1-\sqrt{1-y^2}}{y}\right|=\pm x+c$

Sol.



$Y-y=m(X-x)$

$PQ=1$

$$\left(\frac{x-y}{m}-x\right)^2+(y-0)^2=1^2$$

$$\frac{y^2}{m^2}+y^2=1\Rightarrow\frac{y^2}{m^2}=1-y^2$$

$$m^2=\frac{y^2}{1-y^2}$$

$$\frac{dy}{dx}=\pm\frac{y}{\sqrt{1-y^2}}$$

(+) $$\int\frac{\sqrt{1-y^2}}{y}dy=\int dx$$

$y=\cos\theta$

$dy=-\sin\theta\,d\theta$

$$\int\frac{\sin\theta}{\cos\theta}(-\sin\theta)d\theta=x+c$$

$$-\int\frac{\sin^2\theta}{\cos\theta}d\theta=x+c$$

$$-\int\frac{1-\cos^2\theta}{\cos\theta}d\theta=x+c$$

$$\int(\cos\theta-\sec\theta)d\theta=x+c$$

$$\sin\theta - \ell n\left|\sec\theta - \tan\theta\right| = x + c$$

$$\sqrt{1-y^2} - \ell n\left|\frac{1}{y} - \frac{\sqrt{1-y^2}}{y}\right| = x + c$$

$$\sqrt{1-y^2} - \ell n\left(\left|\frac{1-\sqrt{1-y^2}}{y}\right|\right) = x + c$$

Similarly (–)

$$\sqrt{1-y^2} - \ell n\left|\frac{1-\sqrt{1-y^2}}{y}\right| = -x + c$$

40. (a) Let f be a real valued differentiable function on R (the set of all real numbers) such that f(1) = 1. If the y-intercept of the tangent at any point P(x, y) on the curve y = f(x) is equal to the cube of the abscissa of P, then the value of f(–3) is equal to. **[JEE-2010]**

(b) Let y'(x) + y(x)g'(x) = g(x)g'(x), y(0) = 0, x ∈ R, where f'(x) denotes $\frac{df(x)}{dx}$ and g(x) is a given nonconstant differentiable function on R with g(0) = g(2) = 0. Then the value of y(2) is. **[JEE-2011]**

(a) माना f वास्तविक मान वाला R पर अवकलनीय फलन है जिससे कि f(1) = 1. यदि वक्र y = f(x) के किसी बिन्दु P (x, y) पर स्पर्शी का y-अन्तःखण्ड P के भुज के घन के बराबर हो, तो f(–3) का मान ज्ञात कीजिए। **[JEE-2010]**

(b) माना y'(x) + y(x)g'(x) = g(x)g'(x), y(0) = 0, x ∈ R, जहाँ f'(x), $\frac{df(x)}{dx}$ को दर्शाता है तथा g(x), R पर एक अवकलनीय फलन है जो अचर नहीं है तथा g(0) = g(2) = 0 है, तो y(2) का मान होगा। **[JEE-2011]**

Ans. (a) 9 (b) 0

Sol. Given $y = f(x)$

Tangent at point P(x, y)

$$Y - y = \left(\frac{dy}{dx}\right)_{(x,y)}(X - x)$$

Now y–intercept $\Rightarrow Y = y - x\frac{dy}{dx}$

Given that, $y - x\frac{dy}{dx} = x^3$

$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = -x^2$ is a linear differential equation

with I.F. $= e^{\int -\frac{1}{x}dx} = e^{-\ell nx} = e^{\ell n\left(\frac{1}{x}\right)} = \frac{1}{x}$

Hence, solution is $\frac{y}{x} = \int -x^2 . \frac{1}{x} dx + C$

or $\frac{y}{x} = -\frac{x^2}{2} + C$

Given $f(1) = 1$

Substituting we get, $C = \frac{3}{2}$

so $y = -\frac{x^3}{2} + \frac{3}{2}x$

Now $f(-3) = \frac{27}{2} - \frac{9}{2} = 9$

BERNOULLI'S EQUATION

41. $2\frac{dy}{dx} - y \sec x = y^3 \tan x$

Ans. $\frac{1}{y^2} = -1 + (c + x) \cot\left(\frac{x}{2} + \frac{\pi}{4}\right)$

Sol. $2\frac{dy}{dx} - y \sec x = y^3 \tan x$

$\frac{2}{y^3}\frac{dy}{dx} - \frac{\sec x}{y^2} = \tan x$

$\frac{-1}{y^2} = t$

$\frac{2}{y^3}\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} - t \sec x = \tan x$

I.F. $= e^{\int -\sec x dx} = e^{-\ln(\sec x - \tan x)}$

$$= \frac{1}{\sec x - \tan x} \times \frac{\sec x + \tan x}{\sec x + \tan x}$$

$$= \sec x + \tan x$$

$$\int \frac{dt}{dx}(\sec x + \tan x) - \sec x(\sec x + \tan x)t = \int \tan x(\sec x + \tan x)$$

$$t(\sec x + \tan x) = \int \sec x \tan x + \int(\sec^2 x - 1) + c$$

$$t(\sec x + \tan x) = \sec x + \tan x - x + c$$

$$\frac{-1}{y^2}(\sec x + \tan x) = (\sec x + \tan x) - (x + c)$$

$$\frac{1}{y^2} = -1 + (c + x)\cot\left(\frac{x}{2} + \frac{\pi}{4}\right)$$

42. $y(2xy + e^x)\,dx - e^x\,dy = 0$

Ans. $y^{-1}e^x = c - x^2$

Sol. $y(2xy + e^x)\,dx - e^x\,dy = 0$

$$\frac{y(2xy + e^x)}{e^x} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = y + 2xe^{-x}y^2$$

$$\frac{dy}{dx}\frac{1}{y^2} - \frac{1}{y} = 2xe^{-x}$$

$$-\frac{1}{y} = t$$

$$\frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + t = 2xe^{-x}$$

I.F. = $e^{\int dx} = e^x$

$$\int e^x \frac{dt}{dx} + te^x = \int 2x$$

$$te^x = x^2 + c$$

$$-\frac{e^x}{y} = x^2 + c$$

$y^{-1} e^x = c - x^2$

CHANGE OF VARIABLE BY A SUITABLE SUBSTITUTION

43. $(x - y^2)\, dx + 2xy\, dy = 0$

Ans. $y^2 + x\, ln\, ax = 0$

Sol. $(x - y^2)\, dx + 2xy\, dy = 0$

$xdx - y^2dx + 2xy\, dy = 0$

$$\frac{2xydy - y^2dx}{x^2} = \frac{-xdx}{x^2}$$

$$\int d\left(\frac{y^2}{x}\right) = \int \frac{-1}{x} dx$$

$$\frac{y^2}{x} = -\ln x + \ln c$$

$y^2 = -x \ln x$

$y^2 + x \ln ax = 0$

44. $x \frac{dy}{dx} + y \ln y = xye^x$

Ans. $x\, ln\, y = e^x (x - 1) + c$

Sol. $x \frac{dy}{dx} + y \ln y = xye^x$

$xdy + y \ln ydx = xye^x dx$

$\frac{x}{y} dy + \ln ydx = xe^x dx$

$$\int d(x \ln y) = \int x e^x dx$$

$$x \ln y = e^x (x - 1) + c$$

45. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1 + x)\, e^x \sec y$

Ans. $\sin y = (e^x + c)(1 + x)$

Sol. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1 + x)\, e^x \sec y$

$$\cos y \frac{dy}{dx} - \frac{\sin y}{1+x} = (1+x) e^x$$

$$\int \frac{1}{(1+x)} \cos y \frac{dy}{dx} - \frac{\sin y}{(1+x)^2} = \int e^x$$

$$\frac{\sin y}{1+x} = e^x + c$$

$$\sin y = (1 + x)(e^x + c)$$

46. $\frac{dy}{dx} = \frac{e^y}{x^2} - \frac{1}{x}$

Ans. $cx^2 + 2xe^{-y} = 1$

Sol. $\frac{dy}{dx} = \frac{e^y}{x^2} - \frac{1}{x} \Rightarrow \frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}$

Multiply by e^{-y}

$$e^{-y} \frac{dy}{dx} + \frac{e^{-y}}{x} = \frac{1}{x^2}$$

$e^{-y} = t$

$$e^{-y} \frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

$$\frac{dt}{dx} - \frac{t}{x} = \frac{-1}{x^2}$$

I.F. $= e^{\int \frac{-t}{x} dx} = e^{-\ln x} = \frac{1}{x}$

$$\int \frac{1}{x}\frac{dt}{dx} - \frac{t}{x^2} = \int \frac{-1}{x^3}$$

$$\frac{t}{x} = -\frac{x^{-3+1}}{-3+1} + C$$

$$\frac{e^{-y}}{x} = \frac{1}{2x^2} + C$$

$2xe^{-y} = 1 + 2cx^2$

47. $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$

Ans. $e^y = c \cdot \exp(-e^x) + e^x - 1$

Sol. $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$

$$\frac{dy}{dx} = \frac{e^x}{e^y}\left(e^x - e^y\right)$$

$$e^y \frac{dy}{dx} = e^{2x} - e^{x+y}$$

$$e^y \frac{dy}{dx} + e^x e^y = e^{2x}$$

$e^y = t$

$e^y = t$

$$e^y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + e^x t = e^{2x}$$

I.F. $= e^{\int e^x dx} = e^{e^x}$

$$\int e^{e^x} \frac{dt}{dx} + e^x e^{e^e} t = \int e^{e^x} \cdot (e^x)^2$$

$e^x = p$

$e^x\,dx = dp$

$te^{e^x} = \int pe^p dp$

$te^{e^x} = e^p(p-1) + C$ `

$e^y e^{e^x} = e^{e^x}(e^x - 1) + c \Rightarrow e^y = e^x - 1 + ce^{-e^x}$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

48. Identify the statement(s) which is/are True :

(A) $f(x, y) = e^{y/x} + \tan\frac{y}{x}$ is a homogeneous function of degree zero

(B) $x \cdot \ln\frac{y}{x}\,dx + \frac{y^2}{x}\sin^{-1}\frac{y}{x}\,dy = 0$ is a homogeneous differential equation of degree one

(C) $f(x, y) = x^2 + \sin x \cdot \cos y$ is not homogeneous function

(D) $(x^2 + y^2)\,dx - (xy^2 - y^3)\,dy = 0$ is a homegeneous differential equation

निम्नलिखित में कौन(से) कथन सत्य है :

(A) ि;ग एलद्धत्र मे$^{y/x} + \tan\frac{y}{x}$ एक समघातीय फलन है जिसकी घात शून्य है

(B) ग ण $\ln\frac{y}{x}\,dx + \frac{y^2}{x}\sin^{-1}\frac{y}{x}\,dy = 0$ एक समघातीय अवकल समीकरण है, जिसकी घात एक है

(C) ि;ग एलद्धत्र ग$^2 + \sin x \cdot \cos y$ एक समघातीय फलन नहीं है

(D) ;ग$^2 + y^2)\,dx - (xy^2 - y^3)\,dy = 0$ एक समघातीय अवकल समीकरण है

Ans. ABC

Sol. (A) $f(x, y) = e^{y/x} + \tan\frac{y}{x}$

(B) $x \cdot \ln\frac{y}{x}\,dx + \frac{y^2}{x}\sin^{-1}\frac{y}{x}\,dy = 0$

(C) $f(x, y) = x^2 + \sin x \cdot \cos y$

(D) $(x^2 + y^2)\,dx - (xy^2 - y^3)\,dy = 0$

49. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} . \sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \to 0$ as $x \to \infty$ then the statement which is correct is :

(A) $\lim\limits_{x \to 0} f(x) = 1$ (B) $\int\limits_0^{\pi/2} f(x)\,dx$ is less than $\frac{\pi}{2}$

(C) $\int\limits_0^{\pi/2} f(x)\,dx$ is greater than unity (D) $f(x)$ is an odd function

फलन $y = f(x)$ अवकल समीकरण $\frac{dy}{dx} . \sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ को सन्तुष्ट करता है तथा यदि $y \to 0, x \to \infty$ तो निम्नलिखित esa sdkSl kd Hkku l R; gS

(A) $\lim\limits_{x \to 0} f(x) = 1$ (B) $\int\limits_0^{\pi/2} f(x)\,dx < \frac{\pi}{2}$

(C) $\int\limits_0^{\pi/2} f(x)\,dx$ ईकाई से बड़ा होगा (D) f;gkfdएक विषम फलन है

Ans. ABC

Sol. Hint : $f(x) = \frac{\sin x}{x}$

50. The tangent at any point P on a curve $f(x, y) = 0$ cuts the y-axis at T. If the distance of the point T from P equals the distance of T from the origin then the curve with this property represents a family of circles. Which of the following is/are CORRECT?

(A) Any arbitrary line $y = mx$ cuts every member of this family at the points where the slopes of these members are equal.

(B) $f(x, y) = 0$ is orthogonal to the family of circles $x^2 + y^2 - ky = 0 \ \forall \ k \in R$

(C) If $f(x, y) = 0$ passes through (2, 2) then the intercept made by its director circle on the y-axis is equal to 8.

(D) If $f(x, y) = 0$ passes through (–1, 1) then image of its centre in the line $y = x$, is (1, 0)

वक्र $f(x, y) = 0$ के किसी बिन्दु P पर स्पर्श रेखा y-अक्ष को बिन्दु T पर काटती है। यदि बिन्दु T की P से दूरी तथा मूल बिन्दु से दूरी l eku gS rks oØ] oÙkksa dk ,sl ½fud k, ½ftldk lHkh' kkZkgSA fuEufy f[kr esa sdkSl kd Hkku l R; gS

(A) कोई रेखा $y = mx$ इस कुल के प्रत्येक सदस्य को उस बिन्दु पर काटती है जहाँ प्रत्येक सदस्य का ढाल समान है

(B) f;gkfdलम्ब> 0एवृत्तों $x^2 + y^2 - ky = 0 \ \forall \ k \in R$ के कुल से लम्बकोणीय है

(C) यदि $f(x, y) = 0$, (2, 2) से गुजरता है तब इसके नियामक वृत्त द्वारा y-अक्ष पर काटे खण्ड की लम्बाई 8 है

(D) यदि $f(x, y) = 0$, (–1, 1) से गुजरता है, तो $y = x$ रेखा के सापेक्ष केन्द्र का प्रतिबिम्ब (1, 0) है

Ans. AB

Sol. We have equation of tangent is $(Y - y) = m(X - x)$

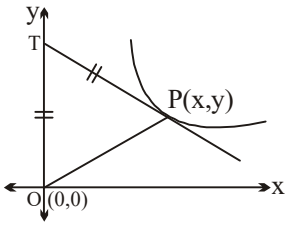
Put $X = 0$, we get $Y = y - mx$.

Now $(OT)^2 = (PT)^2$ (given)

$\Rightarrow (y - mx)^2 = x^2 + m^2x^2$

$\Rightarrow y^2 + m^2x^2 - 2mxy = x^2 + m^2x^2$

$\therefore \dfrac{dy}{dx} = \dfrac{y^2 - x^2}{2xy}$



Put $y^2 = t \Rightarrow 2y\dfrac{dy}{dx} = \dfrac{dt}{dx} \Rightarrow x\dfrac{dt}{dx} = t - x^2 \Rightarrow \dfrac{dt}{dx} - \dfrac{t}{x} = -x$

$\therefore$ I.F. $= e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \dfrac{1}{x}$

The solution is given by $t\left(\dfrac{1}{x}\right) = \int -dx = -x + C \quad \Rightarrow t = -x^2 + Cx$

Hence $y^2 + x^2 = Cx$.

If this is passing through $(2, 2) \Rightarrow C = 4 \Rightarrow x^2 + y^2 = 4x \Rightarrow (x - 2)^2 + y^2 = 4$

It's director circle will be $(x - 2)^2 + y^2 = 8$.

Put $x = 0$, $y^2 = 4 \Rightarrow y = 2$ or -2

Intercept on y-axis is 4.

This represents a family of circles with centre at y-axis and passing through origin.

Verify other alternatives.

51. Let C be the family of curves $f(x, y, c) = 0$ (no member of C is x-axis) such that length of subnormal at any point P(x, y) on the curve C is equal to four times that of the length of subtangent at the same point. Which of the following statement(s) is(are) correct?

(A) Equation of the line with positive y-intercept passing through (4, 2) and perpendicular to the curve C is $x + 2y = 8$.

(B) Orthogonal trajectory of C is family of parallel lines having gradient ± 2

(C) Order and degree of the differential equation of family of curves C are 1 and 2 respectively

(D) Differential equation of family of curves is $2y' \pm x = 0$

माना C वक्रो के कुल $f(x, y, c) = 0$ को दर्शाता है (कोई सदस्य C x-अक्ष नहीं है)। C के किसी सदस्य पर स्थित किसी बिन्दु P(x, y) पर अधोलम्ब की लम्बाई, अधःस्पर्श रेखा की लम्बाई का चार गुना है, निम्न में से कौनसे कथन सत्य है?

(A) ;4ए2द्धसे जाने वाली धनात्मक y-अन्तःखण्ड की एक रेखा जो वक्र C के लम्बवत् है, का समीकरण $x + 2y = 8$ है

(B) ढ के लम्बवतीय पथ एक ± 2 प्रवणता की समान्तर रेखाओं का कुल है

(C) वक्र C के कुल का अवकल समीकरण की कोटि एवं घात क्रमशः 1 एवं 2 है

(D) वक्रो के कुल अवकल समीकरण $2y' \pm x = 0$ हैं

Ans. AC

Sol. Given that $\left| y\frac{dy}{dx} \right| = \left| \frac{4y}{\frac{dy}{dx}} \right|$ at (2, 1) $\Rightarrow \frac{dy}{dx} = \pm 2$ at (2, 1)

$\therefore$ Equation of tangent having positive slope is $y = 2x - 3$.

Now verify alternatives.

52. Let C be a curve such that the tangent at any point P on it meets x-axis and y-axis at A and B respectively. If BP : PA = 3:1 and the curve passes through the point (1, 1), then :

(A) The curve passes through $\left(2, \frac{1}{8}\right)$

(B) Equation of normal to the curve at (1, 1) is $3y - x = 2$

(C) The differential equation for the curve is $3y' + xy = 0$

(D) The differential equation for the curve is $xy' + 3y = 0$

माना C एक वक्र है, जिसके किसी बिन्दु P पर स्पर्शी x एवं y-अक्ष पर क्रमशः A एवं B पर मिलती है। यदि BP : PA = 3:1 तथा वक्र बिन्दु (1, 1) से गुजरता हो,तो :

(A) वक्र बिन्दु $\left(2, \frac{1}{8}\right)$ से गुजरता है

(B) वक्र के बिन्दु (1, 1) पर अभिलम्ब का समीकरण $3y - x = 2$ है

(C) वक्र का अवकल समीकरण $3y' + xy = 0$ है

(D) वक्र का अवकल समीकरण $xy' + 3y = 0$ है

Ans. ABD

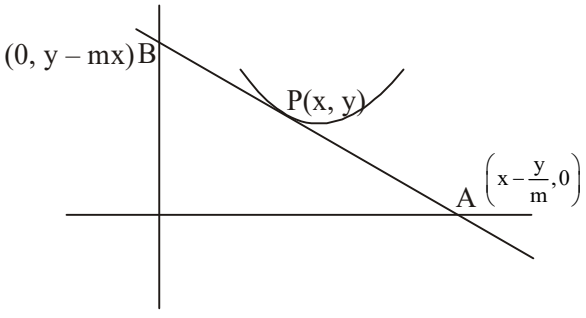
Sol. $Y - y = m(X - x)$

$$\frac{BP}{PA} = \frac{1}{3}$$

$$\frac{\sqrt{x^2 + m^2x^2}}{\sqrt{\left(\frac{y}{m}\right)^2 + y^2}} = \frac{3}{1}$$

$$\frac{x^2 + m^2x^2}{\left(\frac{y}{m}\right)^2 + y^2} = 9$$

$$\Rightarrow x^2 + m^2x^2 = 9\left(\frac{y}{m}\right)^2 + 9y^2$$



$$\left(1+m^2\right)x^2 = 9y^2\left(\frac{1+m^2}{m^2}\right)$$

$$m^2 = \frac{9y^2}{x^2}$$

$$\frac{dy}{dx} = \pm\frac{3y}{x}$$

$$\int\frac{dy}{y} = \int -3\frac{dx}{x} \qquad \int\frac{dy}{y} = \int\frac{3dx}{x}$$

$\ln y = -3\ln x + c$ $\qquad$ $\ln y = 3\ln x + c$

$(1, 1)$ $\qquad$ $(1, 1)$

$c = 0$ $\qquad$ $c = 0$

$$y = \frac{1}{x^3} \qquad \frac{dy}{dx} = x^2$$

$$\frac{dy}{dx} = \frac{-3}{x^2} \qquad (1, 1)$$

$(1, 1)$ $\qquad$ (3)

$x = -3$ $\qquad$ $y - 1 = \frac{-1}{3}x - 1$

$$m_1 = \frac{1}{3} \qquad 3y - 3 = -x - 1$$

$$x + 3y = 4$$

$$y = 1 = \frac{1}{3}(1-1)$$

$3y - 3 = x - 1$

$3y - x = 2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 53 to 55

A curve $y = f(x)$ satisfies the differential equation $(1 + x^2)\frac{dy}{dx} + 2yx = 4x^2$ and passes through the origin.

वक्र $y = f(x)$ अवकल समीकरण $(1 + x^2)\frac{dy}{dx} + 2yx = 4x^2$ को सन्तुष्ट करता है तथा मूल बिन्दु से गुजरता है।

53. The function $y = f(x)$:

(A) is strictly increasing $\forall\ x \in R$ (B) is such that it has a minima but no maxima

(C) is such that it has a maxima but no minima (D) has no inflection point

फलन $y = f(x)$ होगा :

(A) निरन्तर वर्धमान होगा $\forall\ x \in R$. (B) एक निम्निष्ठ परन्तु कोई भी उच्चिष्ठ नहीं

(C) एक उच्चिष्ठ परन्तु कोई निम्निष्ठ नहीं (D) कोई नति परिवर्तन बिन्दु नहीं

Ans. A

Sol. $\frac{dy}{dx} + \left(\frac{2x}{1+x^2}\right)y = \frac{4x^2}{1+x^2}$

I.F. $= e^{\int \frac{2x}{1+x^2}dx} = e^{ln(1+x^2)} = (1+x^2)$

$\therefore \quad y(1 + x^2) = \int 4x^2\,dx = \frac{4x^3}{3} + C$

passing through $(0, 0) \Rightarrow \quad C = 0$

$\therefore \quad y = \frac{4x^3}{3(1+x^2)}$

$$\frac{dy}{dx} = \frac{4}{3}\left[\frac{(1+x^2)3x^2 - x^3 \cdot 2x}{(1+x^2)^2}\right] = \frac{4}{3}\left[\frac{3x^2 + x^4}{(1+x^2)^2}\right] = \frac{4x^2(3+x^2)}{3(1+x^2)^2}$$

hence $\frac{dy}{dx} > 0\ \ \forall\ x \neq 0;$

$\frac{dy}{dx} = 0$ at $x = 0$ and it does not change sign $\Rightarrow$ $x = 0$ is the point of inflection Ans.

$y = f(x)$ is increasing for all $x \in R$

$\xi \to \infty;\ y \to \infty \quad ; \quad \xi \to -\infty;\ y \to -\infty$

Area enclosed by $y = f^{-1}(x)$, x-axis and ordinate at $x = \frac{2}{3}$

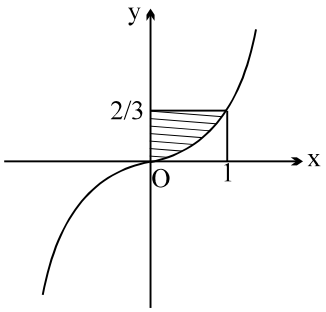
$$A = \frac{2}{3} - \frac{4}{3}\int_0^1 \frac{x^3}{1+x^2}dx$$

put $\quad 1 + x^2 = t \quad \Rightarrow \quad 2x\,dx = dt$

$$A = \frac{2}{3} - \frac{2}{3}\int_1^2 \frac{(t-1)}{t}dt = \frac{2}{3} - \frac{2}{3}\int_1^2 \left(1 - \frac{1}{t}\right)dt$$

$$= \frac{2}{3} - \frac{2}{3}\left[t - \ln t\right]_1^2 = \frac{2}{3} - \frac{2}{3}\left[(2 - \ln 2) - 1\right]$$

$$= \frac{2}{3} - \frac{2}{3}\left[1 - \ln 2\right] = \frac{2}{3}\ln 2$$



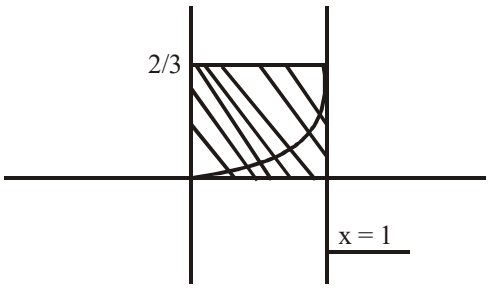
54. The area enclosed by $y = f^{-1}(x)$, the x-axis and the ordinate at $x = 2/3$ is :

$y = f^{-1}(x)$, x-अक्ष तथा कोटि जहाँ $x = 2/3$ है द्वारा घिरा क्षेत्रफल होगा :

(A) $2\ln 2$ (B) $\frac{4}{3}\ln 2$ (C) $\frac{2}{3}\ln 2$ (D) $\frac{1}{3}\ln 2$

Ans. C

Sol.



$y = f^{-1}(x)$

$$y = \frac{4x^3}{3\left(1 + x^2\right)}$$

$$x = \frac{2}{3}$$

area $A = \frac{2}{3} - \frac{4}{3}3\int_0^2 \frac{x^3}{1 + x^1}dx$

$1 + x^2 = t$

$2x\,dx = dt$

$$= \frac{2}{3} - \frac{2}{3}\int_1^2 \frac{t-1}{t}dt$$

$$= \frac{2}{3}\left[1 - \int_1^2 \left(1 - \frac{1}{t}\right) dt\right]$$

$$= \frac{2}{3}\left[1 - (t - \ln t)^2\right] = \frac{2}{3}[1 - 2 + \ln 2 + 1 - 0]$$

$$= \frac{2}{3}\ln 2$$

55. For the function $y = f(x)$ which one of the following does not hold good?

(A) $f(x)$ is a rational function
(B) $f(x)$ has the same domain and same range
(C) $f(x)$ is a transcidental function
(D) $y = f(x)$ is a bijective mapping

फलन $y = f(x)$ के लिए निम्नलिखित में से कौनसा सत्य नहीं है?

(A) ि; गद्धएक परिमेय फलन है
(B) ि; गद्धका प्रान्त तथा परिसर समान है
(C) ि; गद्धअबीजीय फलन है
(D) लत्र ि; गद्धएकैकी आच्छादक है

Ans. C

Sol.

Paragraph for Question Nos. 56 to 58

Let C be a curve $y = f(x)$ passing through $M\left(-\sqrt{3}, 1\right)$ such that the y-intercept of the normal at any point P(x, y) on the curve C is equal to the distance of P from the origin.

माना y = f(x) एक वक्र C है जो $M\left(-\sqrt{3}, 1\right)$ से इस प्रकार गुजरता है कि वक्र C के किसी बिन्दु P(x, y) पर अभिलम्ब का y-अन्तःखण्ड, मूल बिन्दु से P की दूरी के बराबर है।

56. Which one of the following statement is correct?

(A) f(x) has exactly two points of inflection
(B) f(x) has maxima but no minima
(C) The equation $f(x) = -2$ has two distinct real solutions
(D) Inclination of the tangent line drawn to f(x) at (1, 0) is $\tan^{-1}1$

निम्न में से कौनसा कथन सत्य है?

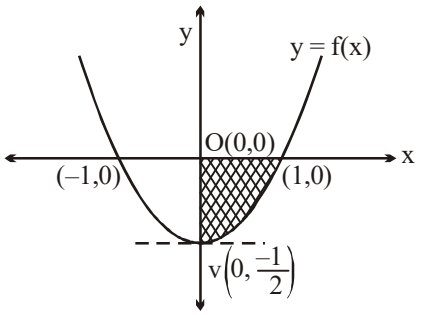
(A) ि; गद्धके केवल दो नतिपरिवर्तन बिन्दु हैं
(B) ि; गद्धका उच्चिष्ठ है, लेकिन निम्निष्ठ नहीं
(C) ि; गद्धत्र 2 के दो भिन्न वास्तविक हल है
(D) ि; गद्धके (1, 0) पर स्पर्शी का x- अक्ष के साथ कोण $\tan^{-1}1$ है

Ans. D

Sol. Equation of normal (N) : $Y - y = \frac{-1}{m}(X - x)$

Put $X = 0 \Rightarrow Y = y + \frac{x}{m}$

hence y-intercept of the normal is $y + \frac{x}{\frac{dy}{dx}}$.



So, $y + \frac{x}{\frac{dy}{dx}} = \sqrt{x^2 + y^2} \Rightarrow \int \frac{ydy + xdx}{\sqrt{x^2 + y^2}} = \int dy$

$\Rightarrow \frac{1}{2}\int \frac{d\left(x^2 + y^2\right)}{\sqrt{x^2 + y^2}} = \int dy \Rightarrow \sqrt{x^2 + y^2} = y + k$

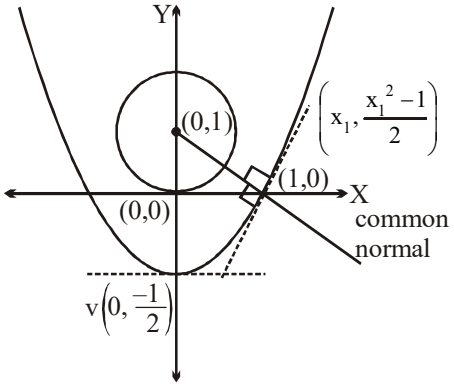
As $M\left(-\sqrt{3}, 1\right)$ satisfy it, so $k = 1 \Rightarrow x^2 + y^2 = y^2 + 2y + 1 \Rightarrow x^2 = 2\left(y + \frac{1}{2}\right)$

(i) From graph $\left.\frac{dy}{dx}\right]_{(1,0)} = 1 \Rightarrow \tan\theta = 1 \Rightarrow \theta = \tan^{-1}(1)$ Ans.

(ii) Clearly, required area

$$= \left|\int_0^1 \frac{x^2 - 1}{2}\,dx\right| = \left|\frac{1}{2}\left(\frac{x^2}{3} - x\right)_0^1\right| = \left|\frac{1}{2}\left(\frac{1}{3} - 1\right)\right| = \frac{1}{3} \text{ (square units).}$$

(iii) Clearly, $\frac{-1}{x_1} = \frac{\left(\frac{x_1^2 - 1}{2}\right) - 1}{x_1 - 0}$



$\Rightarrow x_1^3 - x_1 = 0 \Rightarrow x_1(x_1^2 - 1) = 0$

$\therefore \quad x_1 = -1, 0, 1$

Now, distance of (1, 0) from (0, 1) $= \sqrt{2}$

So, shortest distance between the curve C and the circle $= \sqrt{2} - 1$ (units).

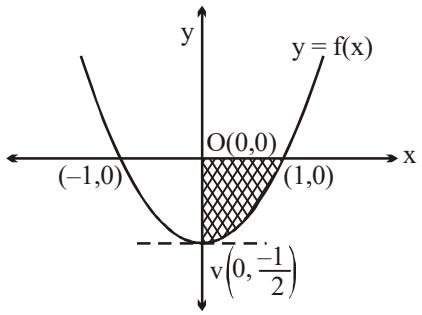
57. The area enclosed by the curve C and the co-ordinate axes is equal to :

वक्र C एवं निर्देशांक्षो द्वारा घिरा क्षेत्रफल होगा :

(A) $\frac{1}{6}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Ans. D

Sol.



$$A = \int_0^1 \left(\frac{x^2}{2} - \frac{1}{2} \right) dx$$

$$= \frac{1}{2}\left[\frac{x^3}{3} - 2 \right]_0^1$$

$$= \frac{1}{2}\left(\frac{1}{3} - 1 \right)$$

$$= \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$$

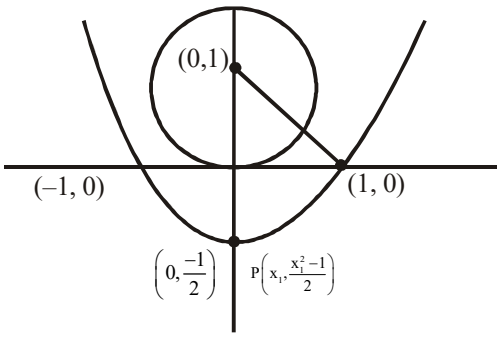
58. The shortest distance between the curve C and the circle $x^2 + y^2 - 2y = 0$ is equal to :

वक्र C एवं वृत्त $x^2 + y^2 - 2y = 0$ के बीच न्यूनतम दूरी होगी :

(A) $\frac{1}{2}$ (B) $\sqrt{2} - 1$ (C) $\sqrt{6} - 1$ (D) $\sqrt{5} - 1$

Ans. B

Sol.



$$\frac{dy}{dx} = x$$

$$m = x_1$$

$$m_N = \frac{-1}{x_1}$$

$$-\frac{1}{x_1} = \frac{\frac{x_1^2-1}{2}-1}{x_1}$$

$$\frac{-1}{x_1} = \frac{x^2-3}{2x_1}$$

$$-2x_1 = x_1^3 - 3x_1$$

$$x_1^3 - x_1 = 0$$

$$x_1\left(x_1^2-1\right) = 0$$

$$x = 0, \pm 1$$

(1, 0), (0, 1)

Shortest distance = $\sqrt{2}-1$

MATCH THE COLUMN TYPE QUESTION(S)

59. Match the following column :

Column-I	**Column-II**
(A) Solution of differential equation $[3x^2y + 2xy - e^x(1+x)]\,dx + (x^3+x^2)dy = 0$ is	(P) $y^2(x^2+1+ce^{x^2}) = 1$
(B) Solution of differential equation $ydx - xdy - 3xy^2\, e^{x^2}dx = 0$ is	(Q) $(x^2+x^3)y - xe^x = c$
(C) Solution of differential equation $\frac{dy}{dx} = xy(x^2y^2-1)$ is	(R) $\frac{x}{y} - \frac{3}{2}e^{x^2} = c$
(D) Solution of differential equation $\frac{dy}{dx}(x^2y^3+xy) = 1$ is	(S) $\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$
	(T) $\frac{2}{x} = 1 - y^2 + ce^{-y/2}$

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) अवकल समीकरण $[3x^2y + 2xy - e^x(1+x)]\,dx + (x^3+x^2)dy = 0$ का हल है	(P) $y^2(x^2+1+ce^{x^2}) = 1$
(B) अवकल समीकरण $ydx - xdy - 3xy^2\, e^{x^2}dx = 0$ का हल है	(Q) $(x^2+x^3)y - xe^x = c$
(C) अवकल समीकरण $\frac{dy}{dx} = xy(x^2y^2-1)$ का हल है	(R) $\frac{x}{y} - \frac{3}{2}e^{x^2} = c$
(D) अवकल समीकरण $\frac{dy}{dx}(x^2y^3+xy) = 1$ का हल है	(S) $\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$
	(T) $\frac{2}{x} = 1 - y^2 + ce^{-y/2}$

Ans. A – Q, B – R, C – P, D – S

Sol. (A) $[3x^2y + 2xy - e^x(1+x)]\,dx + (x^3+x^2)dy = 0$

$y(3x^2+2x) + (x^3+x^2)dy = e^x(1+x)dx$

$$\int d\left[y\left(x^3+x^2\right)\right] = \int (x+1)e^x dx$$

$y(x^3+x^2) = (x+1)e^x - e^x + c$

$y(x^3+x^2) - xe^x = c$

(B) $ydx - xdy - 3xy^2\ e^{x^2}dx = 0$

$ydx - xdy = 3xy^2 3^{x^2}dx$

$$\int \frac{ydx - xdy}{y^2} = \int 3xe^{x^2}dx$$

$x^2 = t$

$2xdx = dt$

$$\frac{x}{y} = \frac{3}{2}\int e^t dt$$

$$\frac{x}{y} = \frac{3}{2}e^{x^2} + C$$

(C) $\frac{dy}{dx} = xy(x^2y^2 - 1)$

$$\frac{dy}{dx} = x^3y^3 - xy$$

$$\frac{dy}{dx} + xy = x^3y^3$$

$$\frac{1}{y^2} = t$$

$$-2y^{-2-1}\frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{-1}{2}\frac{dt}{dx} + tx = x^3$$

I.F. $= e^{\int -2xdx} = e^{-x^2}$

$$\int e^{-x^2}\frac{dt}{dx} - 2xe^{-x^2}t = \int -2x^3e^{-x^2}$$

$-x^2 = P$

$\Rightarrow -2xdx = dp$

$$te^{-x^2} = -\int Pe^p dp$$

$$\frac{e^{-x^2}}{y^2} = -e^P(P-1) + C$$

$$\frac{e^{-x^2}}{y^2} = -e^{-x^2}(-x^2 - 1) + C$$

$$\frac{1}{y^2} = x^2 + 1 + ce^{x^2}$$

(D) $\frac{dy}{dx}(x^2y^3 + xy) = 1$

$$\frac{dx}{dy} = x^2y^3 + xy$$

$$\frac{dx}{dy} - xy = x^2y^3$$

$$\frac{1}{x^2}\frac{dx}{dy} - \frac{y}{x} = y^3$$

$$\frac{-1}{x} = t$$

$$\frac{1}{x^2}\frac{dx}{dy} = \frac{dt}{dy}$$

$$\frac{dt}{dy} + ty = y^3$$

I.F. = $e^{\int y} = e^{\frac{y^2}{2}}$

$$\int e^{\frac{y^2}{2}}\frac{dy}{dy} + tye^{\frac{y^3}{2}} = \int y^3e^{\frac{y^3}{2}}$$

$$te^{\frac{y^3}{2}} = \int 2Pe^P dP + C$$

$$te^{\frac{y^2}{2}} = 2P^P(P-1) + C$$

$$\frac{-e^{\frac{y^2}{2}}}{x} = 2e^{\frac{y^2}{2}}\left(\frac{y^2}{2}-1\right)+C$$

$$\frac{-1}{x} = 2\left(\frac{y^2-2}{2}\right)+ce^{\frac{-y^2}{2}}$$

$$\frac{1}{x} = 2-y^2+ce^{-\frac{y^2}{2}}$$

Exercise J-M

1. Let $y = y(x)$ be the solution of the differential equation $x^3\,dy + (xy - 1)dx = 0$, $x > 0$, $y\left(\frac{1}{2}\right) = 3 - e$. Then $y(1)$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $x^3\,dy + (xy - 1)dx = 0$, $x > 0$, $y\left(\frac{1}{2}\right) = 3 - e$ का हल $y = y(x)$ है। तो $y(1)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $2 - e$ (B) e (C) 1 (D) 3

Ans. C

Sol. $\frac{dy}{dx} = \frac{1-xy}{x^3} = \frac{1}{x^3} - \frac{y}{x^2}$

$\frac{dy}{dx} + \frac{y}{x^2} = \frac{1}{x^3}$

If $= e^{\int \frac{1}{x^2}dx} = e^{-\frac{1}{x}}$

$y \cdot e^{-\frac{1}{x}} = \int e^{-\frac{1}{x}} \cdot \frac{1}{x^3}dx$ (put $-\frac{1}{x} = t$)

$y = \frac{1}{x} + 1 + Ce^{\frac{1}{x}}$

Where C is constant

Put $x = \frac{1}{2}$

$3 - e = 2 + 1 + Ce^2$

$C = -\frac{1}{e}$

$y(1) = 1$

2. Let $y = y(x)$ be the solution of the differential equation $(x^2 - 3y^2)\,dx + 3xy\,dy = 0$, $y(1) = 1$. Then $6\,y^2(e)$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $(x^2 - 3y^2)\,dx + 3xy\,dy = 0$, $y(1) = 1$ का हल $y = y(x)$ है। तब $6\,y^2(e)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $3e^2$ (B) e^2 (C) $\frac{3}{2}e^2$ (D) $2e^2$

Ans. D

Sol. D.E. $\rightarrow \frac{2ydy}{dx} - \frac{2y^2}{x} = \frac{-2x}{3}$

Let $y^2 = t \Rightarrow 2y\frac{dy}{dx} = \frac{dt}{dx}$

D.E. $\rightarrow \frac{dt}{dx} - \frac{2t}{x} = -\frac{2x}{3}$

I.F. $= e^{\int -\frac{2}{x}dx} = e^{-2en|x|} = e^{tn\frac{1}{x^2}} = \frac{1}{x^2}$

Solution $\Rightarrow t \cdot \frac{1}{x^2} = \int \frac{1}{x^2}\left(\frac{-2x}{3}\right)dx + c \Rightarrow \frac{y^2}{x^2} = \frac{-2}{3}\ell n|x| + c$

$\therefore x = 1, y = 1 \Rightarrow 1 = 0 + c \Rightarrow c = 1$

$\Rightarrow \frac{y^2}{x^2} = -\frac{2}{3}\ell n|x| + 1$

$x = e \Rightarrow \frac{y^2(e)}{e^2} = \frac{-2}{3} + 1 \Rightarrow y^2(e) = \frac{e^2}{3} \Rightarrow 6y^2(e) = 2e^2$

3. Let y = y(x) be the solution curve of the differential equation $\frac{dy}{dx} = \frac{y}{x}\left(1 + xy^2(1 + \log_e x)\right)$, x > 0, y(1) =3. Then $\frac{y^2(x)}{9}$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $\frac{dy}{dx} = \frac{y}{x}\left(1 + xy^2(1 + \log_e x)\right)$, x > 0, y(1) =3 का हल y = y(x) है। तो $\frac{y^2(x)}{9}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{x^2}{7 - 3x^2\left(2 + \log_e x^2\right)}$

(B) $\frac{x^2}{2x^3\left(2 + \log_e x^3\right) - 3}$

(C) $\frac{x^2}{3x^3\left(1 + \log_e x^2\right) - 2}$

(D) $\frac{x^2}{5 - 2x^3\left(2 + \log_e x^3\right)}$

Ans. D

Sol. $\frac{dy}{dx} = \frac{y}{x}\cdot\left(1 + xy^2(1 + \log_e x)\right), x > 0$

$\frac{dy}{dx} = \frac{y}{x} + y^3\cdot\left(1 + \log_e x\right)$

$\frac{1}{y^3}\cdot\frac{dy}{dx} = \frac{1}{x\cdot y^2} + \left(1 + \log_e x\right)$

$\frac{1}{y^3}\cdot\frac{dy}{dx} - \frac{1}{x}\cdot\frac{1}{y^2} = \left(1 + \ln x\right)$

Let $\frac{1}{y^2}=t$

$-\frac{2}{y^3}\cdot\frac{dy}{dx}=\frac{dt}{dx}$

$\therefore -\frac{1}{2}\frac{dt}{dx}-\frac{t}{x}=(1+\ln x)$

$\frac{dt}{dx}+\frac{2t}{x}=-2(1+\ln x)$

if $e^{\int\frac{2}{x}dx}=x^2$

$\therefore tx^2=\int -2(1+\ln x)x^2\cdot dx$

$tx^2=-2\left[(1+\ln x)\frac{x^3}{3}-\int\frac{x^2}{3}dx\right]+C$

$\frac{x^2}{y^2}=-2\left[\frac{x^3}{3}(1+\ln x)-\frac{x^3}{9}\right]+C$ ...(1)

$\because y(1)=3$

$\frac{1}{9}=-2\left[\frac{1}{3}(1+0)-\frac{1}{9}\right]+C$

$c=\frac{5}{9}$

put $c=\frac{5}{9}$

$\Rightarrow\frac{y^2(x)}{9}=\frac{x^2}{5-2x^3\left(2+\log_e x^3\right)}$

4. Let $y=y(t)$ be a solution of the differential equation

$\frac{dy}{dt}+\alpha y=\gamma e^{-\beta t}$

where, $\alpha>0,\beta>0$ and $\gamma>0$. Then $\lim\limits_{t\to\infty}y(t)$ **[JEE(Main)Jan.2023]**

(A) is -1 (B) is 0 (C) does not exist (D) is 1

माना अवकल समीकरण

$\frac{dy}{dt}+\alpha y=\gamma e^{-\beta t}$

जहाँ $\alpha > 0, \beta > 0$ तथा $\gamma > 0$ है, का हल $y = y(t)$ है। तो $\lim_{t\to\infty} y(t)$ **[JEE(Main)Jan.2023]**

(A) –1 है (B) 0 है (C) का अस्तित्व नहीं है (D) 1 है

Ans. B

Sol. I.F. $= e^{\int adt} = e^{at}$

Solution $y.e^{at} = \int e^{at} \cdot \gamma \cdot e^{-at} dt + C$

$\Rightarrow ye^{\alpha t} = \gamma \int e^{(\alpha-\beta)t} dt + c$

$\Rightarrow y(t) \cdot e^{\alpha t} = \gamma \frac{e^{(\alpha-\beta)t}}{(\alpha-\beta)} + c$

$\Rightarrow y(t) = \gamma \frac{e^{-\beta t}}{(\alpha-\beta)} + ce^{-\alpha t}$

$\lim_{t\to\infty} y(t) = \lim_{t\to\infty} \left(\frac{\gamma \cdot e^{-\beta t}}{(\alpha-\beta)} - c \cdot e^{-at} \right)$

$= 0 - 0 = 0$

5. Let $y = f(x)$ be the solution of the differential equation $y(x + 1)dx - x^2dy = 0, y(1) = e$. Then $\lim_{x\to 0^+} f(x)$ is equal to: **[JEE(Main)Jan.2023]**

माना $y = f(x)$ अवकल समीकरण $y(x + 1)dx - x^2dy = 0, y(1) = e$ का हल है। तब $\lim_{x\to 0^+} f(x)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) e^2 (B) $\frac{1}{e}$ (C) 0 (D) $\frac{1}{e^2}$

Ans. C

Sol. $\frac{x+1}{x^2} dx = \frac{dy}{y}$

$\ln x - \frac{1}{x} = \ln y + c$

$(1, e)$

$c = -2$

$\ln x - \frac{1}{x} = \ln y - 2$

$y = e^{\ln x - \frac{1}{x} + 2}$

$= \lim_{x\to 0^+} e^{\ln x - \frac{1}{x} + 2}$

$= e^{-\infty}$

$= 0$

6. Let $y = y(x)$ be the solution of the differential equation $x \log_e x \frac{dy}{dx} + y = x^2 \log_e x$, $(x > 1)$. If $y(2) = 2$, then $y(e)$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $x \log_e x \frac{dy}{dx} + y = x^2 \log_e x$, $(x > 1)$ का हल $y = y(x)$ है। यदि $y(2) = 2$ है, तो $y(e)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{1+e^2}{2}$ (B) $\frac{1+e^2}{4}$ (C) $\frac{4+e^2}{4}$ (D) $\frac{2+e^2}{2}$

Ans. C

Sol. $x \log_e x \frac{dy}{dx} + y = x^2 \log_e x, (x > 1)$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x \ln x} = x$

Linear differential equation

I.F. = $e^{\int \frac{1}{x \ln x} dx} = |\ln x|$

$\therefore$ Solution of differential equation

$y|\ln x| = \int x|\ln x| dx$

$\left|\ln x^2\right| \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx$

$\Rightarrow y|\ln x| = |\ln x|\left(\frac{x^2}{2}\right) - \frac{x^2}{4} + c$

For constant

$y(2) = 2 \Rightarrow c = 1$

So, $y(x) = \frac{x^2}{2} - \frac{x^2}{4|\ln x|} + \frac{1}{|\ln x|}$

Hence, $y(e) = \frac{e^2}{2} - \frac{e^2}{4} + 1 = 1 + \frac{e^2}{4}$

7. Let the solution curve $y = y(x)$ of the differential equation

$\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} y = 2x \exp\left\{\frac{x^3 - \tan^{-1} x^3}{\sqrt{(1+x^6)}}\right\}$ pass through the origin. Then $y(1)$ is equal to :

[JEE(Main)Jan.2023]

माना अवकल समीकरण $\frac{dy}{dx}-\frac{3x^5\tan^{-1}(x^3)}{(1+x^6)^{3/2}}y=2x\exp\left\{\frac{x^3-\tan^{-1}x^3}{\sqrt{(1+x^6)}}\right\}$ का हल वक्र $y = y(x)$ मूल बिंदु से होकर जाता है।

r ksy(1) बराबर है : **[JEE(Main)Jan.2023]**

(A) $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$ (B) $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$ (C) $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$ (D) $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$

Ans. D

Sol. $\frac{dy}{dx}-\frac{3x^5\tan^{-1}x^3}{\left(1+x^6\right)^{3/2}}y=2x\exp\left\{\frac{x^3-\tan^{-1}x^3}{\sqrt{1+x^6}}\right\}$

$\text{I.f.}=e^{-\int\frac{3x^5\tan^{-1}x^3}{\left(1+x^6\right)^{3/2}}dx}$

$=e^{-\int\frac{3x^2.x^3\tan^{-1}x^3}{\left(1+x^6\right)^{3/2}}dx}$

Let $\tan^{-1}x^3=t\Rightarrow\frac{3x^2}{1+x^6}dx=dt$

$=e^{-\int\frac{x^3}{\sqrt{1+x^6}}.\frac{3x^2\tan^{-1}x^3}{\left(1+x^6\right)}dx}$

$=e^{-\int\frac{\tan t}{\sec t}.t.dt}$

$=e^{-\int t\sin t.dt}=e^{(t\cos t-\sin t)}$

$=e^{\left(\frac{\tan^{-1}x^3}{\sqrt{1+x^6}}-\frac{x^3}{\sqrt{1+x^6}}\right)}$

Hence solutions of the differential equation is

$y.e^{\frac{\tan^{-1}x^3}{\sqrt{1+x^6}}-\frac{x^3}{\sqrt{1+x^6}}}=\int 2\times e^{\frac{x^3-\tan^{-1}x^3}{\sqrt{1+x^6}}}.e^{\frac{\tan^{-1}x^3-x^3}{\sqrt{1+x^6}}}.dx+C$

$= x^2 + C$

Passes through $(0, 0)\Rightarrow C = 0$

Hence $y(1)=e^{\frac{1}{\sqrt{2}}\left(1-\frac{\pi}{4}\right)}$

8. The solution of the differential equation $\frac{dy}{dx}=-\left(\frac{x^2+3y^2}{3x^2+y^2}\right)$, $y(1) = 0$ is : **[JEE(Main)Jan.2023]**

अवकल समीकरण $\frac{dy}{dx}=-\left(\frac{x^2+3y^2}{3x^2+y^2}\right)$, $y(1)=0$ का हल है : **[JEE(Main)Jan.2023]**

(A) $\log_e|x+y|-\frac{xy}{(x+y)^2}=0$ (B) $\log_e|x+y|+\frac{xy}{(x+y)^2}=0$

(C) $\log_e|x+y|+\frac{2xy}{(x+y)^2}=0$ (D) $\log_e|x+y|-\frac{2xy}{(x+y)^2}=0$

Ans. C

Sol. Let $y=vx$

$$v+x\frac{dv}{dx}=-\left(\frac{1+3v^2}{3+v^2}\right)$$

$$\Rightarrow \frac{v^2+3}{(v+1)^3}dv+\frac{1}{x}dx=0$$

$$\Rightarrow \int\frac{4}{(v+1)^3}dv+\int\frac{1}{v+1}dv-\int\frac{2}{(v+1)^2}dv+\int\frac{1}{x}dx=0$$

$$\Rightarrow \frac{-2}{(v+1)^2}+\ln|v+1|+\frac{2}{v+1}+\ln|x|=c$$

$$\Rightarrow \frac{-2x^2}{(x+y)^2}+\ln\left|\frac{x+y}{x}\right|+\frac{2x}{x+y}+\ln|x|=c$$

$$\Rightarrow \frac{2xy}{(x+y)^2}+\ln|x+y|=c$$

$\because y(1)=0$

$\Rightarrow c=0$

$$\Rightarrow \frac{2xy}{(x+y)^2}+\ln|x+y|=0$$

9. Let a differentiable function f satisfy $f(x)+\int_3^x\frac{f(t)}{t}dt=\sqrt{x+1}, x\geq 3$. Then $12f(8)$ is equal to :

[JEE(Main)Jan.2023]

माना एक अवकलनीय फलन f, $f(x)+\int_3^x\frac{f(t)}{t}dt=\sqrt{x+1}, x\geq 3$ को संतुष्ट करता है। तो $12f(8)$ बराबर है :

[JEE(Main)Jan.2023]

(A) 1 (B) 34 (C) 19 (D) 17

Ans. D

Sol. $f'(x)+\frac{f(x)}{x}=\frac{1}{2\sqrt{x+1}}$

$\frac{dy}{dx}+\frac{y}{x}=\frac{1}{2\sqrt{x+1}}$

I.F. $=e^{\int\frac{1}{x}dx}=x$

$y.x=\int x\frac{1}{2\sqrt{x+1}}dx+c$ put $x+1=t^2$, $dx=2tdt$

$y.x=\frac{1}{2}\int\frac{\left(t^2-1\right)2tdt}{t}+c$

$y.x=\frac{1}{2}\left(\frac{2}{3}t^3-2t\right)+c$

$y.x=\frac{\left(\sqrt{x+1}\right)^3}{3}-\sqrt{x+1}+c$

now $f(3)=2$

$6=\frac{(2)^3}{3}-2+c$

$8-\frac{8}{3}=c$

$\frac{16}{3}=c$

$\therefore f(x)=\frac{1}{x}\left(\frac{\left(\sqrt{x+1}\right)^3}{3}-\left(\sqrt{x+1}\right)+\frac{16}{3}\right)$

$\therefore 12f(8)=\frac{12}{8}\left(\frac{27}{3}-3+\frac{16}{3}\right)$

$=\frac{3}{2}\left(6+\frac{16}{3}\right)=17$

10. Let y = y(x) be the solution of the differential equation $(3y^2 - 5x^2)y\,dx + 2x(x^2 - y^2)\,dy = 0$ such that y(1) = 1. Then $|(y(2))^3 - 12y(2)|$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $(3y^2 - 5x^2)y\,dx + 2x(x^2 - y^2)\,dy = 0$, y(1) = 1 का हल y = y(x) है तो $|(y(2))^3 - 12y(2)|$ बराबर है: **[JEE(Main)Jan.2023]**

(A) $32\sqrt{2}$ (B) $16\sqrt{2}$ (C) 32 (D) 64

Ans. A

Sol. $(3y2 - 5x^2)\, ydx + 2x\,(x^2 - y^2)\, dy = 0$

$$\Rightarrow \frac{dy}{dx} = \frac{y\left(5x^2 - 3y^2\right)}{2x\left(x^2 - y^2\right)}$$

This is homogeneous

Let $y = tx \Rightarrow \frac{dy}{dx} = t + x\frac{dt}{dx}$

$$t + x\frac{dt}{dx} = \frac{(tx)\left(5x^2 - 3t^2x^2\right)}{2x\left(x^2 - t^2x^2\right)} = \frac{t\left(5 - 3t^2\right)}{2\left(1 - t^2\right)}$$

$$x\frac{dt}{dx} = \frac{5t - 3t^3}{2 - 2t^2} - t = \frac{3t - t^3}{2 - 2t^2}$$

$$\int \frac{\left(1 - t^2\right)}{3t - t^3} dt = \int \frac{1}{x} dx$$

Let $3t - t^3 = z \Rightarrow = 3(1 - t^2)dt = dz$

11. If $y = y(x)$ is the solution curve of the differential equation $\frac{dy}{dx} + y\tan x = x\sec x$, $0 \le x \le \frac{\pi}{3}$, $y(0) = 1$, then $y\left(\frac{\pi}{6}\right)$ is equal to : **[JEE(Main)Jan.2023]**

यदि अवकल समीकरण $\frac{dy}{dx} + y\tan x = x\sec x$, $0 \le x \le \frac{\pi}{3}$, $y(0) = 1$ का हल वक्र $y = y(x)$ है, तो $y\left(\frac{\pi}{6}\right)$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi}{12} - \frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$ (B) $\frac{\pi}{12} + \frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$

(C) $\frac{\pi}{12} - \frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$ (D) $\frac{\pi}{12} + \frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$

Ans. C

Sol. Here I.F. = sec x

Then solution of D.E.

$y(\sec x) = x\tan x - \ln(\sec x) + c$

Given $y(0) = 1 \Rightarrow c = 1$

$\therefore\ y(\sec x) = x\tan x - \ln(\sec x) + 1$

At $x = \frac{\pi}{6}$, $y = \frac{\pi}{12} + \frac{\sqrt{3}}{2}\ln\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}$

$$y = \frac{\pi}{12} + \frac{\sqrt{3}}{2}\left(\log\frac{\sqrt{3}}{2} + \log e\right)$$

$$= \frac{\pi}{12} - \frac{\sqrt{3}}{2}\log\left(\frac{2}{e\sqrt{3}}\right)$$

12. Let $\alpha x = \exp\left(x^{\beta}y^{\gamma}\right)$ be the solution of the differential equation $2x^2y\,dy - (1-xy^2)dx = 0, x > 0,\ y(2) = \sqrt{\log_e 2}$. Then $\alpha + \beta - \gamma$ equals : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $2x^2y\,dy - (1-xy^2)dx = 0, x > 0,\ y(2) = \sqrt{\log_e 2}$ का हल $\alpha x = \exp\left(x^{\beta}y^{\gamma}\right)$ है। तो $\alpha + \beta - \gamma$: **[JEE(Main)Jan.2023]**

(A) -1 (B) 1 (C) 0 (D) 3

Ans. B

Sol. $\alpha x = e^{x^{\beta}.y^{\gamma}}$

$$2x^2y\frac{dy}{dx} = 1 - x.y^2 \qquad y^2 = t$$

$$x^2\frac{dt}{dx} = 1 - xt$$

$$\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2} \qquad \text{I.F.} = e^{\ln x} = x$$

$$t(x) = \int \frac{1}{x^2}.x\,dx$$

$y^2 . x = \ln x + C$

$\therefore 2.\ln 2 = \ln 2 + C$

$\therefore C = \ln 2$

Hence, $xy^2 = \ln 2x$

$\therefore 2x = e^{x.y^2}$

Hence $\alpha = 2, \beta = 1, \gamma = 2$

13. Let $f: R \to R$ be differentiable function such that $f'(x) + f(x) = \int_0^2 f(t)\,dt$. If $f(0) = e^{-2}$, then $2f(0) - f(2)$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $f: R \to R$ एक अवकलनीय फलन है, जिसके लिए ताकि $f'(x) + f(x) = \int_0^2 f(t)\,dt$ है। यदि $f(0) = e^{-2}$ है, तो $2f(0) - f(2)$ का मान है। **[JEE(Main)Jan.2023]**

Ans. 1

Sol. $\frac{dy}{dx} + y = k$

$y \cdot e^x = k \cdot e^x + c$

$f(0) = e^{-2}$

$\Rightarrow c = e^{-2} - k$

$\therefore y = k + (e^{-2} - k)\, e^{-x}$

now $k = \int_0^2 \left(k + \left(e^{-2} - k\right)e^{-x}\right)dx$

$\Rightarrow k = e^{-2} - 1$

$\therefore y = (e^{-2} - 1) + e^{-x}$

$f(2) = 2e^{-2} - 1,\ f(0) = e^{-2}$

$2f(0) - f(2) = 1$

14. If the solution curve f(x, y)=0 of the differential equation $\left(1 + \log_e x\right)\frac{dx}{dy} - x \log_e x = e^y, x > 0$, passes through the points (1, 0) and (α, 2) then α^α is equal to : **[JEE(Main)April,2023]**

यदि अवकल समीकरण $\left(1 + \log_e x\right)\frac{dx}{dy} - x \log_e x = e^y, x > 0$ का हल वक्र f(x, y) = 0 बिंदुओं (1, 0) तथा (α, 2) से होकर t kr kgrs α^α बराबर है : **[JEE(Main)April,2023]**

(A) e^{2e^2} (B) e^{e^2} (C) $e^{2e^{\sqrt{2}}}$ (D) $e^{\sqrt{2}e^2}$

Ans. A

Sol. $(1 + \ln x)\frac{dx}{dy} - x \ln x = e^y$

Let $x \ln x = t$

$(1 + \ln x)\frac{dx}{dy} = \frac{dt}{dy}$

$\frac{dt}{dy} - t = e^y$

$\text{If } = e^{\int -dy} = e^{-y}$

$t \cdot e^{-y} = \int e^y e^{-y} dy + c$

$te^{-y} = y + c$

$x \ln x e^{-y} = y + c$

$x \ln x = ye^y + ce^y$

$(1,0)0 = C$

$\Rightarrow x \ln x = ye^y$

$\Rightarrow \alpha \ln \alpha = 2e^2$

$\alpha^\alpha = e^{2e^2}$

15. The slope of tangent at any point (x, y) on a curve $y = y(x)$ is $\frac{x^2+y^2}{2xy}, x > 0$. If $y(2) = 0$, then a value of $y(8)$ is:

[JEE(Main)April,2023]

वक्र $y = y(x)$ के किसी भी बिंदु (x, y) पर स्पर्श रेखा की प्रवणता $\frac{x^2+y^2}{2xy}, x > 0$ है। यदि $y(2) = 0$ है, तो $y(8)$ का एक मान है :

[JEE(Main)April,2023]

(A) $-4\sqrt{2}$ (B) $4\sqrt{3}$ (C) $2\sqrt{3}$ (D) $-2\sqrt{3}$

Ans. B

Sol. $\frac{dy}{dx} = \frac{x^2+y^2}{2xy}$

Let $y = \upsilon x \therefore \frac{dy}{dx} = \upsilon + x\frac{d\upsilon}{dx}$ $\therefore \upsilon + x\frac{d\upsilon}{dx} = \frac{x^2+\upsilon^2x^2}{2x.\upsilon x}$

$x\frac{d\upsilon}{dx} = \frac{1+\upsilon^2}{2\upsilon} - \upsilon = \frac{1-\upsilon^2}{2\upsilon} \Rightarrow \frac{2\upsilon}{1-\upsilon^2}d\upsilon = \frac{dx}{x} \Rightarrow \int\frac{2\upsilon d\upsilon}{\upsilon^2-1} + \int\frac{dx}{x} = 0$

$\Rightarrow \ell n\left(\upsilon^2-1\right) + \ell nx = \ell nc \Rightarrow x\left(\upsilon^2-1\right) = c \Rightarrow x\frac{y^2-x^2}{x^2} = c \Rightarrow y^2 - x^2 = cx$

Now $f(2) = 0 \therefore c = -2$

$\therefore y^2 - x^2 = -2x$ $\therefore x = 8$ then $y^2 - 64 = -16$

$y^2 = 48$

$y = f(8) = 4\sqrt{3}$

16. Let $y = y(x)$ be a solution curve of the differential equation $(1 - x^2y^2)dx = ydx + xdy$. If the line $x = 1$ intersects the curve $y = y(x)$ at $y = 2$ and the line $x = 2$ intersects the curve $y = y(x)$ at $y = \alpha$, then a value of α is :

[JEE(Main)April,2023]

माना अवकल समीकरण $(1 - x^2y^2)dx = ydx + xdy$ का हल वक्र $y = y(x)$ है। यदि रेखा $x = 1$, वक्र $y = y(x)$ को $y = 2$ पर काटती gS rFkk js[kk $x = 2$, वक्र $y = y(x)$ को $y = \alpha$ पर काटती है, तो α का एक मान है : **[JEE(Main)April,2023]**

(A) $\frac{3e^2}{2\left(3e^2+1\right)}$ (B) $\frac{3e^2}{2\left(3e^2-1\right)}$ (C) $\frac{1+3e^2}{2\left(3e^2-1\right)}$ (D) $\frac{1-3e^2}{2\left(3e^2+1\right)}$

Ans. C

Sol. $dx = \frac{d(xy)}{1-(xy)^2} \Rightarrow 2dx = \frac{d(xy)}{1-xy} + \frac{d(xy)}{1+xy}$

$$2x + c = \ell n\left|\frac{1+xy}{1-xy}\right| \Rightarrow \left|\frac{xy+1}{xy-1}\right| = e^c \cdot e^{2x}$$

$\because y(1) = 2$

$\Rightarrow 3 = e^c \cdot e^2$

$\Rightarrow e^c = \frac{3}{e^2}e^{2x}$

So at $x = 2$

$\Rightarrow \left|\frac{2\alpha+1}{2\alpha-1}\right| = 3e^2$

Case : 1 for +ve sign

$\Rightarrow \frac{2\alpha+1}{2\alpha-1} = \frac{3e^2}{1}$

$\Rightarrow \frac{4\alpha}{2} = \frac{3e^2+1}{3e^2-1}$

$\Rightarrow \alpha = \frac{1}{2}\left(\frac{3e^2+1}{3e^2-1}\right)$

Case: 2 for (–ve)sign not required

17. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{5}{x\left(x^5+1\right)}y = \frac{\left(x^5+1\right)^2}{x^7}, x > 0$. If $y(1) = 2$, then $y(2)$ is equal to : **[JEE(Main)April,2023]**

माना अवकल समीकरण $\frac{dy}{dx} + \frac{5}{x\left(x^5+1\right)}y = \frac{\left(x^5+1\right)^2}{x^7}, x > 0$ का हल $y = y(x)$ है। यदि $y(1) = 2$ है, तो $y(2)$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{679}{128}$ (B) $\frac{637}{128}$ (C) $\frac{697}{128}$ (D) $\frac{693}{128}$

Ans. D

Sol. $I.F = e^{\int\frac{5dx}{x\left(x^5+1\right)}} = e^{\frac{5x^{-6}}{\left(x^{-5}+1\right)}}$

$Put, 1 + x^{-5} = t \Rightarrow -5x^{-6}dx = dt$

$\Rightarrow e^{\int\frac{-dt}{t}} = \frac{1}{t} = \frac{x^5}{1+x^5}$

$y \cdot \frac{x^5}{1+x^5} = \int \frac{x^5}{\left(1+x^5\right)} \times \frac{\left(1+x^5\right)^2}{x^7}dx$

$$= \int x^3 dx + \int x^{-2} dx$$

$$y \cdot \frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + c$$

Given than : $x = 1 \Rightarrow y = 2$

$$2 \cdot \frac{1}{2} = \frac{1}{4} - 1 + c$$

$$c = \frac{7}{4}$$

$$y \cdot \frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + \frac{7}{4}$$

Now put, $x = 2$

$$y \cdot \left(\frac{32}{33}\right) = \frac{21}{4}$$

$$y = \frac{693}{128}$$

18. Let $y = y(x)$, $y > 0$, be a solution curve of the differential equation $\left(1+x^2\right)dy = y(x-y)dx$. If $y(0) = 1$ and $y(2\sqrt{2}) = \beta$, then : **[JEE(Main)April,2023]**

माना अवकल समीकरण $\left(1+x^2\right)dy = y(x-y)dx$ का एक हल वक्र $y = y(x)$, $y > 0$ है। यदि $y(0) = 1$ तथा $y(2\sqrt{2}) = \beta$ हैं, r ks **[JEE(Main)April,2023]**

(A) $e^{\beta^{-1}} = e^{-2}(3+2\sqrt{2})$ (B) $e^{3\beta^{-1}} = e(3+2\sqrt{2})$

(C) $e^{3\beta^{-1}} = e(5+\sqrt{2})$ (D) $e^{\beta^{-1}} = e^{-2}(5+\sqrt{2})$

Ans. B

Sol. $y(x-y)dx = \left(1+x^2\right)dy$

$$\frac{y(x-y)}{1+x^2} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{1+x^2}y - \frac{y^2}{1+x^2}$$

$$\frac{dy}{dx} - \frac{x}{1+x^2}y = \frac{-y^2}{1+x^2}$$

$$\frac{-1}{y^2}\frac{dy}{dx} + \frac{x}{1+x^2}\frac{1}{y} = \frac{1}{1+x^2}$$

Let $\frac{1}{y} = t$

$$\therefore \frac{-1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$\therefore \frac{dt}{dx} + \frac{x}{1+x^2}t = \frac{1}{1+x^2} \text{ linear in } t$$

I.F. $= e^{\int \frac{x}{1+x^2}dx} = e^{\frac{1}{2}\ln\left(1+x^2\right)} = \sqrt{1+x^2}$

$\therefore$ solution is

$$t\sqrt{1+x^2} = \int \frac{\sqrt{1+x^2}}{1+x^2}dx + c$$

$$\frac{1}{y}\sqrt{1+x^2} = \int \frac{1}{\sqrt{1+x^2}}dx + c$$

$$\frac{1}{y}\sqrt{1+x^2} = \ln\left(x + \sqrt{x^2+1}\right) + c$$

$$y(0) = 1 \quad \Rightarrow \quad 1 = 0 + c \quad \Rightarrow \quad c = 1$$

So at $x = 2\sqrt{2}$ we have $\frac{3}{\beta} = \ln(2\sqrt{2}+3) + 1 \Rightarrow 3\beta^{-1} = \ln e(2\sqrt{2}+3)$

$$\Rightarrow e^{3\beta^{-1}} = e(3+2\sqrt{2})$$

19. Let $y = y_1(x)$ and $y = y_2(x)$ be the solution curves of the differential equation $\frac{dy}{dx} = y + 7$ with initial conditions $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then the curves $y = y_1(x)$ and $y = y_2(x)$ intersect at :

[JEE(Main)April,2023]

(A) two points (B) infinite number of points

(C) no point (D) one point

माना अवकल समीकरण $\frac{dy}{dx} = y + 7$ के प्रारंभिक प्रतिबंधो $y_1(0) = 0$ तथा $y_2(0) = 1$ के साथ हल वक्र क्रमशः $y = y_1(x)$ तथा $y = y_2(x)$ हैं। तो वक्र $y = y_1(x)$ तथा $y = y_2(x)$: **[JEE(Main)April,2023]**

(A) दो बिन्दुओं पर मिलते हैं (B) अनंत बिन्दुओं पर मिलते हैं

(C) किसी भी बिन्दु पर नहीं मिलते (D) एक बिन्दु पर मिलते हैं

Ans. C

Sol. $\frac{dy}{dx} = y + 7 \Rightarrow \frac{dy}{dx} - y = 7$

I.F. $= e^{-x}$

$$ye^{-x} = \int 7e^{-x}dx$$

$$\Rightarrow ye^{-x} = -7e^{-x} + c$$

$$\Rightarrow y = -7 + ce^{x}$$

$-7+7e^x=-7+8e^x \Rightarrow e^x=0$

No solution.

20. Let x = x(y) be the solution of the differential equation

$2(y+2)\log_e(y+2)dx+\left(x+4-2\log_e(y+2)\right)dy=0, y>-1$ with $x\left(e^4-2\right)=1$. Then $x\left(e^9-2\right)$ is equal to :

[JEE(Main)April,2023]

यदि x = x(y) अवकल समीकरण $2(y+2)\log_e(y+2)dx+\left(x+4-2\log_e(y+2)\right)dy=0, y>-1$ का हल है जहाँ $x\left(e^4-2\right)=1$ हो, तो $x\left(e^9-2\right)$ का मान बराबर होगा :

[JEE(Main)April,2023]

(A) $\frac{10}{3}$ (B) $\frac{4}{9}$ (C) 3 (D) $\frac{32}{9}$

Ans. D

Sol. $2(y+2)\ln(y+2)dx+(x+4-2\ln(y+2))dy=0$

$$2\ln\left(y+2\right)+\left(x+4-2\ln\left(y+2\right)\right)\frac{1}{y+2}\cdot\frac{dy}{dx}=0$$

let $\ln(y+2)=t$

$$\frac{1}{y+2}\cdot\frac{dy}{dx}=\frac{dt}{dx}$$

$$2t+\left(x+4-2t\right)\cdot\frac{dt}{dx}=0$$

$$\left(x+4-2t\right)\frac{dt}{dx}=-2t$$

$$\frac{dx}{dt}=\frac{2t-4-x}{2t}$$

$$\frac{dx}{dt}+\frac{x}{2t}=\frac{2t-4}{2t}$$

$$x\cdot t^{1/2}=\int\frac{2t-4}{2t}\cdot t^{1/2}\cdot dt$$

$$x\cdot t^{1/2}=\int\left(t^{1/2}-\frac{2}{t^{1/2}}\right)\cdot dt$$

$$xt^{\frac{1}{2}}=\frac{t^{\frac{3}{2}}}{\frac{3}{2}}-2\cdot\frac{t^{\frac{1}{2}}}{\frac{1}{2}}+C$$

$$x\cdot t^{\frac{1}{2}}=\frac{2t^{\frac{3}{2}}}{3}-4t^{\frac{1}{2}}+C$$

$$x = \frac{2}{3}\cdot t - 4 + C\cdot t^{-\frac{1}{2}}$$

$$x = \frac{2}{3}\ln(y+2) - 4 + C\cdot(\ln(y+2))^{-\frac{1}{2}}$$

Put $y = e^4 - 2$, $x = 1$

$$1 = \frac{2}{3}\times 4 - 4 + C\times\frac{1}{2}$$

$$\frac{C}{2} = 5 - \frac{8}{3} = \frac{7}{3}$$

$$\Rightarrow C = \frac{14}{3}$$

$$x = \frac{2}{3}\times 9 - 4 + \frac{14}{3}\times\frac{1}{3}$$

$$= 2 + \frac{14}{9}$$

$$= \frac{32}{9}$$

21. Let $y = y(x)$ be a solution of the differential equation $(x\cos x)dy + (xy\sin x + y\cos x - 1)dx = 0, 0 < x < \frac{\pi}{2}$. If $\frac{\pi}{3}y\left(\frac{\pi}{3}\right) = \sqrt{3}$, then $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right)\right|$ is equal to _____. **[JEE(Main)April,2023]**

माना अवकल समीकरण $(x\cos x)dy + (xy\sin x + y\cos x - 1)dx = 0, 0 < x < \frac{\pi}{2}$ का हल $y = y(x)$ है। यदि $\frac{\pi}{3}y\left(\frac{\pi}{3}\right) = \sqrt{3}$ है, तब $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right)\right|$ का मान है _____ **[JEE(Main)April,2023]**

Ans. B

Sol. $(x\cos x)dy + (xy\sin x + y\cos x - 1)dx = 0, 0 < x < \frac{\pi}{2}$

$$\frac{dy}{dx} + \left(\frac{x\sin x + \cos x}{x\cos x}\right)y = \frac{1}{x\cos x}$$

IF = x secx

$$y\cdot x\sec x = \int\frac{x\sec x}{x\cos x}dx = \tan x + c$$

Since $y\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{\pi}$ Hence $c = \sqrt{3}$

Hence $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right)+y'\left(\frac{\pi}{6}\right)\right|=|-2|=2$

22. Let a curve $y=f(x), x\in(0,\infty)$ pass through the points $P\left(1,\frac{3}{2}\right)$ and $Q\left(a,\frac{1}{2}\right)$. If the tangent at any point R(b, f(b)) to the given curve cuts the y-axis at the point S(0, c) such that bc = 3, then $(PQ)^2$ is equal to _____.

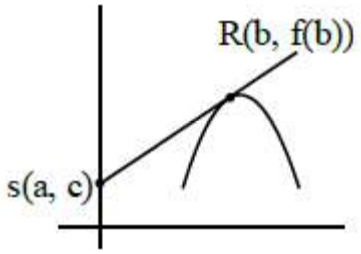
[JEE(Main)April,2023]

माना एक वक्र $y=f(x), x\in(0,\infty)$ बिंदुओं $P\left(1,\frac{3}{2}\right)$ तथा $Q\left(a,\frac{1}{2}\right)$ से होकर जाता है। यदि दिए गए वक्र के किसी भी बिंदु R(b, f(b)) पर स्पर्श रेखा y-अक्ष को बिंदु S(0, c) पर काटती है जबकि bc = 3 तो $(PQ)^2$ बराबर है _____

[JEE(Main)April,2023]

Ans. 5

Sol.



Equation of tangent at R(b, f(2)) is

$y-f(b)=f'(b).(x-b)$

which passes through (0, c)

$\Rightarrow c-f(b)=f'(b)\cdot(-b)$

$\Rightarrow \frac{3}{b}-f(b)=f'(b)\cdot(-b)$

$\Rightarrow bf'(b)-f(b)=-\frac{3}{b}$

$\Rightarrow \frac{bf'(b)-f(b)}{b^2}=-\frac{3}{b^3}$

$\Rightarrow d\left(\frac{f(b)}{b}\right)=-\frac{3}{b^3}\Rightarrow \frac{f(b)}{b}=\frac{3}{2b^2}+\lambda$

Which passes through (1, 3/2)

$\Rightarrow \frac{3}{2}=\frac{3}{2}+\lambda\Rightarrow\lambda=0$

$\Rightarrow f(b)=\frac{3}{2b}$

$f(a)=\frac{1}{2}\Rightarrow\frac{1}{2}=\frac{3}{2b}\Rightarrow b=3$

$\Rightarrow c = 1 \Rightarrow Q(3, 1/2)$

$\Rightarrow PQ^2 = 2^2 + (1)^2 = 5$

23. If the solution curve of the differential equation $\left(y - 2\log_e x\right)dx + \left(x \log_e x^2\right)dy = 0, x > 1$ passes through the points $\left(e, \frac{4}{3}\right)$ and (e^4, α), then α is equal to _____. **[JEE(Main)April,2023]**

यदि अवकल समीकरण $\left(y - 2\log_e x\right)dx + \left(x \log_e x^2\right)dy = 0, x > 1$ का हल वक्र बिंदुओं $\left(e, \frac{4}{3}\right)$ तथा (e^4, α) से होकर t kr kgrk gS] rks α बराबर है _____ **[JEE(Main)April,2023]**

Ans. 3

Sol. $\because \quad (y - 2\ln x)dx + (2x \ln x)dy = 0.$

$$2x \ln x \frac{dy}{dx} + y = 2\ln x$$

$$\frac{dy}{dx} + \frac{y}{2x \ln x} = \frac{1}{x}$$

$$\therefore \quad \text{I.F.} = e^{\int \frac{1}{2x \ln x} dx} = \sqrt{\ln x}$$

$\therefore$ Solution of the equation is :

$$y \cdot \sqrt{\ln x} = \int \frac{\sqrt{\ln x}}{x} dx$$

$$\therefore \quad y \cdot \sqrt{\ln x} = \frac{2}{3}(\ln x)^{\frac{3}{2}} + C \quad \ldots \text{(i)}$$

$\therefore$ eq. (i) passes through point $\left(e, \frac{4}{3}\right)$.

$$\therefore C = \frac{2}{3}$$

$$\therefore y\sqrt{\ln x} = \frac{2}{3}(\ln x)^{\frac{3}{2}} + \frac{2}{3} \quad \ldots \text{(ii)}$$

This equation passes through point (e^4, α)

$\therefore \alpha = 3.$

24. Let the solution curve $x = x(y), 0 < y < \frac{\pi}{2}$, of the differential equation

$\left(\log_e(\cos y)\right)^2 \cos y\, dx - \left(1 + 3x \log_e(\cos y)\right)\sin y dy = 0$ satisfy $x\left(\frac{\pi}{3}\right) = \frac{1}{2\log_e 2}$. If

$x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$, where m and n are coprime, then mn is equal to _____.

माना अवकल समीकरण $\left(\log_e(\cos y)\right)^2 \cos y\,dx - \left(1+3x\log_e(\cos y)\right)\sin y dy = 0$, का हल वक्र x = x(y), $0 < y < \frac{\pi}{2}$; $x\left(\frac{\pi}{3}\right) = \frac{1}{2\log_e 2}$ को संतुष्ट करता है। यदि $x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$ है, जहाँ m तथा n असहभाज्य है, तो mn बराबर है :

[JEE(Main)April,2023]

Ans. 12

Sol. $\cos y \ln^2 \cos y dx = (1+3x\ln\cos y)\sin y dy$

$$\frac{dx}{dy} = \tan y\left(\frac{3x}{\ln\cos y} + \frac{1}{\ln^2\cos y}\right)$$

$$\frac{dx}{dy} - \left(\frac{3\tan y}{\ln\cos y}\right)x = \frac{\tan y}{\ln^2\cos y}$$

$$IF = e^{\int\frac{-3\sin y}{\ln\cos y\cos y}dy}$$

$$\ln\cos y = t$$

$$\frac{1}{\cos y}(-\sin y)dy = dt$$

$$If = e^{\int\frac{3}{t}dt} = e^{3\ln t} = t^3 = \ln^3\cos y$$

$$\text{solution is } x\cdot\ln^3\cos y = \frac{\sin y}{\cos y}\cdot\ln\cos y dy + C$$

$$x\ln^3\cos y = \frac{-\ln^2\cos y}{2} + C$$

$$x\left(\frac{\pi}{3}\right) = \frac{1}{2\ln 2} \text{ so } \frac{1}{2\ln 2}\times\ln^3\left(\frac{1}{2}\right) = -\frac{\ln^2\left(\frac{1}{2}\right)}{2} + C$$

$$C = 0$$

$$y = \frac{\pi}{6} \Rightarrow x\ln^3\frac{\sqrt{3}}{2} = -\frac{1}{2}\ln^2\frac{\sqrt{3}}{2} + 0$$

$$x = -\frac{1}{2\ln\left(\frac{\sqrt{3}}{2}\right)}$$

$$x = \frac{1}{\ln\frac{4}{3}} = \frac{1}{\ln 4 - \ln 3}$$

mn = 12

25. Let the tangent at any point P on a curve passing through the points (1, 1) and $\left(\frac{1}{10},100\right)$, intersect positive x-axis and y-axis at the points A and B respectively. If PA : PB = 1: k and y = y(x) is the solution of the differential equation $e^{\frac{dy}{dx}} = kx + \frac{k}{2}, y(0) = k$, then $4y(1) - 5\log_e 3$ is equal to _____. **[JEE(Main)April,2023]**

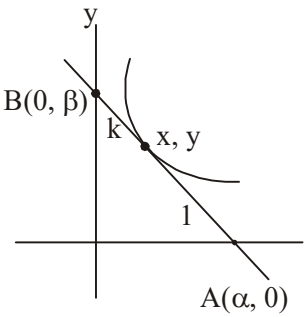
माना बिंदुओं (1, 1) तथा $\left(\frac{1}{10},100\right)$ से होकर जाने वाले एक वक्र के किसी बिंदु P पर स्पर्श रेखा धनात्मक x-अक्ष तथा y-अक्षों को Øe' l?fcaqlA तथा B पर काटती है। यदि PA : PB = 1: k है तथा अवकल समीकरण $e^{\frac{dy}{dx}} = kx + \frac{k}{2}, y(0) = k$ का हल y = y(x) है, तो $4y(1) - 5\log_e 3$ बराबर है : **[JEE(Main)April,2023]**

Ans. Bonus

Sol. $Y - y = \frac{dy}{dx}(X - x)$

$Y = 0$

$X = \frac{-ydx}{dy} + x$



$\frac{k\alpha + 0}{k+1} = 0, \quad \alpha = \frac{k+1}{k}x$

$\frac{k+1}{k}x = -y\frac{dx}{dy} + x$

$x + \frac{x}{k} = -y\frac{dx}{dy} + x$

$x\frac{dy}{dx} + ky = 0 \quad \frac{dy}{dx} + \frac{k}{x}y = 0$

$y.x^k = C$

$C = 1$

$$100\cdot\left(\frac{1}{10}\right)^{k}=1$$

$K=2$

$$\frac{dy}{dx}=\ln(2x+1)$$

$$y=\frac{(2x+1)}{2}(\ln(2x+1)-1)+c$$

$$2=\frac{1}{2}(0-1)+C$$

$$C=2+\frac{1}{2}=\frac{5}{2}$$

$$y(1)=\frac{3}{2}(\ln 3-1)+\frac{5}{2}$$

$$=\frac{3}{2}\ln 3+1$$

$$4y(1)=6\ln 3+4$$

$$4y(1)-5\ln 3=4+\ell\ln 3$$

26. If y = y(x) is the solution of the differential equation $\frac{dy}{dx}+\frac{4x}{\left(x^2-1\right)}y=\frac{x+2}{\left(x^2-1\right)^{\frac{5}{2}}},x>1$ such that $y(2)=\frac{2}{9}\log_e(2+\sqrt{3})$ and $y(\sqrt{2})=\alpha\log_e(\sqrt{\alpha}+\beta)+\beta-\sqrt{\gamma},\alpha,\beta,\gamma\in\mathbb{N}$, then $\alpha\beta\gamma$ is equal to _____.

[JEE(Main)April,2023]

यदि $\frac{dy}{dx}+\frac{4x}{\left(x^2-1\right)}y=\frac{x+2}{\left(x^2-1\right)^{\frac{5}{2}}},x>1$ का हल y = y(x) है तथा $y(2)=\frac{2}{9}\log_e(2+\sqrt{3})$ और $y(\sqrt{2})=\alpha\log_e(\sqrt{\alpha}+\beta)+\beta-\sqrt{\gamma},\alpha,\beta,\gamma\in\mathbb{N}$ हैं, तो $\alpha\beta\gamma$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 6

Sol. $\frac{dy}{dx}+\frac{4x}{\left(x^2-1\right)}y=\frac{x+2}{\left(x^2-1\right)^{\frac{5}{2}}},x>1$ which is linear in y

I.f. $e^{\int\frac{4x}{x^2-1}dx}=e^{2\ln\left(x^2-1\right)}=\left(x^2-1\right)^2$

Solution is $y\cdot\left(x^2-1\right)^2=\int\frac{x+2}{\left(x^2-1\right)^{\frac{5}{2}}}\left(x^2-1\right)^2dx+c$

$$\Rightarrow y\left(x^2-1\right)^2=\int\frac{x+2}{\sqrt{x^2-1}}dx+c$$

$$\Rightarrow y\left(x^2-1\right)^2=\sqrt{x^2-1}+2\ln\left(x+\sqrt{x^2-1}\right)+c$$

$$\therefore y(2)=\frac{2}{9}\ln\left(2+\sqrt{3}\right)$$

$$\therefore \frac{2}{9}\ln\left(2+\sqrt{3}\right)\cdot 9=\sqrt{3}+2\ln\left(2+\sqrt{3}\right)+c$$

$$c=-\sqrt{3}$$

$$\therefore y\left(x^2-1\right)^2=\sqrt{x^2-1}+2\ln\left(x+\sqrt{x^2-1}\right)-\sqrt{3}$$

Put $x=\sqrt{2}$

$$\therefore y\left(\sqrt{2}\right)\cdot 1=1+2\ln\left(\sqrt{2}+1\right)-\sqrt{3}$$

$$y\left(\sqrt{2}\right)=2\ln\left(\sqrt{2}+1\right)+1-\sqrt{3}$$

$$=\alpha\ln\left(\sqrt{\alpha}+\beta\right)+\beta-\sqrt{\gamma}$$

$\alpha = 2, \beta = 1, \gamma = 3$ $\quad\therefore \alpha\cdot\beta\cdot\gamma = 6$

27. Let $x = x(t)$ and $y = y(t)$ be solutions of the differential equations $\frac{dx}{dt}+ax=0$ and $\frac{dy}{dt}+by=0$ respectively, a, b $\in$ R. Given that $x(0) = 2$; $y(0) = 1$ and $3y(1) = 2x(1)$, the value of t, for which $x(t) = y(t)$, is :
[JEE(Main)Jan.2024]

माना अवकल समीकरणों $\frac{dx}{dt}+ax=0$ तथा $\frac{dy}{dt}+by=0$, a, b $\in$ R के हल क्रमशः x = x(t) तथा y = y(t) हैं। यदि $x(0) = 2$; $y(0) = 1$ तथा $3y(1) = 2x(1)$ हैं, तो $x(t) = y(t)$ के लिए t का मान है : **[JEE(Main)Jan.2024]**

(A) $\log_3 4$ (B) $\log_{\frac{2}{3}} 2$ (C) $\log_4 3$ (D) $\log_{\frac{4}{3}} 2$

Ans. D

Sol. $\frac{dx}{dt}+ax=0$

$$\Rightarrow \int\frac{1}{x}dx=-\int a\,dt$$

$\Rightarrow \ln|x| = -at + c$

$\Rightarrow x = \pm e^c \cdot e^{-at} = \lambda e^{-at}$

$\Rightarrow x(0) = 2 \quad \Rightarrow \lambda = 2$

$\Rightarrow x = 2e^{-at}$ _____(1)

Now, $\frac{dy}{dt}+by=0$

$\Rightarrow \int \frac{1}{y} dy = -\int b\, dt \quad \Rightarrow ln(y) = -bt + c$

$\Rightarrow y = \pm e^c \cdot e^{-bt} = \mu \cdot e^{-bt}$

$y(0) = 1$

$\therefore \mu = e^{-bt}$

$3y(1) = 2x(1)$

$3e^{-b} = 2(2e^{-a}) = 4e^{-a}$

$e^{-b+a} = \frac{4}{3}$

$\Rightarrow x(t) = y(t)$

$\Rightarrow e^{-bt} = 2e^{-at}$

$\Rightarrow e^{-bt+at} = 2$

$\Rightarrow e^{(-b+a)t} = 2$

$\Rightarrow \left(\frac{4}{3}\right)^t = 2$

$\Rightarrow t\ell n\frac{4}{3} = \ell n2$

$\Rightarrow t = \log_{\frac{4}{3}} 2$

28. If $y = y(x)$ is the solution curve of the differential equation $\left(x^2-4\right)dy - \left(y^2-3y\right)dx = 0, x > 2, y(4) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals : **[JEE(Main)Jan.2024]**

यदि अवकल समीकरण $\left(x^2-4\right)dy - \left(y^2-3y\right)dx = 0, x > 2, y(4) = \frac{3}{2}$ का हल वक्र $y = y(x)$ है तथा इस वक्र की प्रवणता fd l hHhx के लिए शून्य नहीं है, तो $y(10)$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{3}{1+2\sqrt{2}}$ (B) $\frac{3}{1+(8)^{1/4}}$ (C) $\frac{3}{1-2\sqrt{2}}$ (D) $\frac{3}{1-(8)^{1/4}}$

Ans. B

Sol. $(x^2-4)dy - (y^2-3y)dx = 0$

$\Rightarrow \int \frac{dy}{y^2-3y} = \int \frac{dx}{x^2-4}$

$\Rightarrow \frac{1}{3}\int \frac{y-(y-3)}{y(y-3)}dy = \int \frac{dx}{x^2-4}$

$\Rightarrow \frac{1}{3}(\ell n|y-3| - \ell n|y|) = \frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right| + C$

$$\Rightarrow \frac{1}{3}\ell n\left|\frac{y-3}{y}\right| = \frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right| + C$$

At $x = 4,\ y = \frac{3}{2}$

$$\therefore C = \frac{1}{4}\ell n3$$

$$\therefore \frac{1}{3}\ell n\left|\frac{y-3}{y}\right| = \frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right| + \frac{1}{4}\ell n(3)$$

At $x = 10$

$$\frac{1}{3}\ell n\left|\frac{y-3}{y}\right| = \frac{1}{4}\ell n\left|\frac{2}{3}\right| + \frac{1}{4}\ell n(3)$$

$$\ell n\left|\frac{y-3}{y}\right| = \ell n2^{3/4}, \forall x > 2, \frac{dy}{dx} < 0$$

as $y(4) = \frac{3}{2} \Rightarrow y \in (0,3)$

$$-y + 3 = 8^{1/4} \cdot y$$

$$y = \frac{3}{1+8^{1/4}}$$

29. A function $y = f(x)$ satisfies $f(x)\sin 2x + \sin x - (1+\cos^2 x)f'(x) = 0$ with condition $f(0) = 0$. Then, $f\left(\frac{\pi}{2}\right)$ is equal to : **[JEE(Main)Jan.2024]**

एक फलन $y = f(x)$ के लिए $f(x)\sin 2x + \sin x - (1+\cos^2 x)f'(x) = 0$ तथा $f(0) = 0$ हैं। तो $f\left(\frac{\pi}{2}\right)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 2 (B) 1 (C) 0 (D) – 1

Ans. B

Sol. $$\frac{dy}{dx} - \left(\frac{\sin 2x}{1+\cos^2 x}\right)y = \frac{\sin x}{1+\cos^2 x}$$

$$\text{If} = e^{-\int \frac{\sin 2x}{1+\cos^2 x}dx} = e^{\ln\left(1+\cos^2 x\right)} = 1+\cos^2 x$$

So general solution of DE

$$y\left(1+\cos^2 x\right)=\int \frac{\sin x}{1+\cos^2 x}\cdot\left(1+\cos^2 x\right)dx$$

$$y\left(1+\cos^2 x\right)=-\cos x+c$$

$$y\left(1+\cos^2 x\right)+\cos x=c$$

were $y(0)=0$

$0+1=c$

$$y\left(1+\cos^2 x\right)+\cos x=1$$

$$y=\frac{1-\cos x}{1+\cos^2 x}$$

$$y\left(\frac{\pi}{2}\right)=\frac{1-0}{1+0}=1$$

30. If $\sin\left(\frac{y}{x}\right)=\log_e|x|+\frac{\alpha}{2}$ is the solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx}=y\cos\left(\frac{y}{x}\right)+x$ and $y(1)=\frac{\pi}{3}$, then α^2 is equal to : **[JEE(Main)Jan.2024]**

यदि अवकल समीकरण $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx}=y\cos\left(\frac{y}{x}\right)+x$, $y(1)=\frac{\pi}{3}$, का हल $\sin\left(\frac{y}{x}\right)=\log_e|x|+\frac{\alpha}{2}$ है, तो α^2 बराबर है : **[JEE(Main)Jan.2024]**

(A) 3 (B) 9 (C) 12 (D) 4

Ans. A

Sol. Given differential equation is $x\cos\frac{y}{x}\frac{dy}{dx}-y\cos\frac{y}{x}=x$

$$\Rightarrow \cos\frac{y}{x}\left(x\frac{dy}{dx}-y\right)=x$$

$$\Rightarrow \cos\frac{y}{x}\left(\frac{x\frac{dy}{dx}-y\cdot 1}{x^2}\right)=\frac{1}{x}$$

Let $\frac{y}{x}=t$

$$\cos t\frac{dt}{dx}=\frac{1}{x}$$

$$\cos t\,dt=\frac{1}{x}dx$$

on integeration $\sin t = \ell n|x| + c$

$y(1) = \frac{\pi}{3} \Rightarrow \sin\frac{\pi}{3} = 0 + c$

$\Rightarrow c = \frac{\sqrt{3}}{2}$

$\therefore$ solution is $\sin\frac{y}{x} = \ell n|x| + \frac{\sqrt{3}}{2}$

31. Let y = y(x) be the solution of the differential equation $\sec x dy + \{2(1-x)\tan x + x(2-x)\}dx = 0$ such that $y(0) = 2$. Then $y(2)$ is equal to : **[JEE(Main)Jan.2024]**

माना अवकल समीकरण $\sec x dy + \{2(1-x)\tan x + x(2-x)\}dx = 0$, $y(0) = 2$ का हल $y = y(x)$ है। तो $y(2)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 1 (B) $2\{\sin(2) + 1\}$ (C) 2 (D) $2\{1 - \sin(2)\}$

Ans. C

Sol. $dy = \frac{2(x-1)\tan x + x(x-2)}{\sec x}dx$

$\Rightarrow \int dy = \int 2(x-1)\sin x + (x^2 - 2x)\cos x$

$\Rightarrow y = 2[(x-1)(-\cos x) - (1)(-\sin x)] + [(x^2-2x)(\sin x) - (2x-2)(-\cos x) + (2)(-\sin x)] + c$

$y(0) = 2[1] + [-2] + C$

$y(0) = 2$

$C = 2$

$y(2) = 2(1 + \sin 2) - 2\sin 2 = 2$

32. Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x(\sec x - \sin x \tan x)}$, $x \in \left(0, \frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right) = 2$. Then $y\left(\frac{\pi}{3}\right)$, is : **[JEE(Main)Jan.2024]**

माना अवकल समीकरण $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x(\sec x - \sin x \tan x)}$, $x \in \left(0, \frac{\pi}{2}\right)$ के हल $y = y(x)$ के लिए $y\left(\frac{\pi}{4}\right) = 2$ है। तो $y\left(\frac{\pi}{3}\right)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $\sqrt{3}\left(2 + \log_e\sqrt{3}\right)$ (B) $\frac{\sqrt{3}}{2}\left(2 + \log_e 3\right)$ (C) $\sqrt{3}\left(2 + \log_e 3\right)$ (D) $\sqrt{3}\left(1 + 2\log_e 3\right)$

Ans. A

Sol. $$\frac{dy}{dx}=\frac{\sin x+y\cos x}{\sin x\cdot\cos x\left(\frac{1}{\cos x}-\sin x\frac{\sin x}{\cos x}\right)}$$

$$=\frac{\sin x+y\cos x}{\sin x\left(1-\sin^2 x\right)}$$

$$\frac{dy}{dx}=\sec^2 x+y\cdot 2\left(\text{cosec}2x\right)$$

$$\frac{dy}{dx}-2\text{cosec}\left(2x\right)\cdot y=\sec^2 x$$

$$\frac{dy}{dx}+p\cdot y=Q$$

$$\text{I.F.}=e^{\int p dx}=e^{\int -2\,\text{cosec}(2x)dx}$$

Let $2x=t$

$$2\frac{dx}{dt}=1$$

$$dx=\frac{dt}{2}$$

$$=e^{-\int \text{cosec}(t)dt}$$

$$=e^{-\ell n\left|\tan\frac{t}{2}\right|}$$

$$=e^{-\ell n|\tan x|}=\frac{1}{|\tan x|}$$

$$y(\text{IF})=\int Q(\text{IF})dx+c$$

$$\Rightarrow y\frac{1}{|\tan x|}=\int \sec^2 x\cdot\frac{1}{|\tan x|}+c$$

$$y\cdot\frac{1}{|\tan x|}=\int \frac{dt}{|t|}+c \qquad \text{for } \tan x=t$$

$$y\cdot\frac{1}{|\tan x|}=\ell n|t|+c$$

$$y=|\tan x|\left(\ell n|\tan x|+c\right)$$

Put $x=\frac{\pi}{4},\ y=2$

$2=\ell n\,1+c\Rightarrow c=2$

$y=|\tan x|\,(\ell n\,|\tan x|+2)$

$$y\left(\frac{\pi}{3}\right)=\sqrt{3}\left(\ell n\sqrt{3}+2\right)$$

33. The solution curve of the differential equation $y\frac{dx}{dy}=x\left(\log_e x-\log_e y+1\right)$, $x>0$, $y>0$ passing through the point (e, 1) is : **[JEE(Main)Jan.2024]**

बिन्दु (e, 1) से होकर जाने वाला, अवकल समीकरण $y\frac{dx}{dy}=x\left(\log_e x-\log_e y+1\right)$, $x>0$, $y>0$ का हल वक्र है : **[JEE(Main)Jan.2024]**

(A) $\left|\log_e\frac{y}{x}\right|=y^2$ (B) $\left|\log_e\frac{y}{x}\right|=x$ (C) $2\left|\log_e\frac{x}{y}\right|=y+1$ (D) $\left|\log_e\frac{x}{y}\right|=y$

Ans. D

Sol. $y\frac{dx}{dy}=x\left(\ell nx-\ell ny+1\right)$

$$\frac{dx}{dy}=\frac{x}{y}\left(\ell n\frac{x}{y}+1\right)$$

Let $x=vy\Rightarrow\frac{dx}{dy}=v+y\frac{dv}{dy}$

$$\Rightarrow v+y\frac{dv}{dy}=v\ell nv+v\ \Rightarrow y\frac{dv}{dy}=v\ell nv\Rightarrow\int\frac{dv}{v\ell nv}=\int\frac{dy}{y}$$

$$\Rightarrow\ell n\left(\ell nv\right)=\ell ny+\ell nc\Rightarrow\ell nv=yc\Rightarrow\ell n\left(\frac{x}{y}\right)=yc$$

passes through (e, 1) $\Rightarrow\ell ne=c\Rightarrow c=1\Rightarrow y=\ell n\left(\frac{x}{y}\right)\Rightarrow y=\left|\ell n\left(\frac{x}{y}\right)\right|\ \left(\because x>0,y>0\right)$

34. The temperature T(t) of a body at time t = 0 is 160° F and it decreases continuously as per the differential equation $\frac{dT}{dt}=-k\left(T-80\right)$, where K is is a positive constant. If T(15) = 120°F, then T(45) is equal to : **[JEE(Main)Jan.2024]**

एक वस्तु का तापामान T(t), समय t = 0 पर 160° F है तथा यह अवकल समीकरण $\frac{dT}{dt}=-k\left(T-80\right)$ के अनुसार लगातार कम हो j gkg\$t gkjK धनात्मक अचर है। यदि T(15) = 120°F है, तो T(45) बराबर है : **[JEE(Main)Jan.2024]**

(A) 80° F (B) 85° F (C) 95° F (D) 90° F

Ans. D

Sol. $\int\frac{dT}{T-80}=\int-kdt$

$\ell n\,|T-80| = -kt + c$

$At = 0 \quad T = 160^\circ \quad \Rightarrow c = \ell n\,80$

$\ell n\,|T-80| = -kt + \ell n 80$

$\ell n\left|\frac{T-80}{80}\right| = -kt$

$T = 80 + 80e^{-kt}$

when $t = 15$, and $T = 120$ $\Rightarrow k = \frac{1}{15}\ell n\,2$

when $t = 45$, $T = 80 + 80e^{-\ell n 8} = 90^\circ$

35. Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$, y(0) = 1. Then, $\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ equals: **[JEE(Main)Jan.2024]**

माना अवकलन समीकरण $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$, y(0) = 1 का हल y = y(x) है। तो $\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ बराबर gS **[JEE(Main)Jan.2024]**

(A) $\frac{4}{4+\sqrt{e}}$ (B) $\frac{3}{3-\sqrt{e}}$ (C) $\frac{1}{2-\sqrt{e}}$ (D) $\frac{2}{1+\sqrt{e}}$

Ans. C

Sol. $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$

Put $x + y = t$

$\Rightarrow \frac{dy}{dx} = \frac{dt}{dx} - 1$

$\frac{dt}{dx} - 1 = 2x(t)^3 - xt$

$\Rightarrow \frac{dt}{2t^3 - t} = x\,dx$

$\int \frac{1}{2t^3 - t}\,dt = \int x\,dx$

$\Rightarrow \int \frac{t}{2t^4 - t^2}\,dt = \int x\,dx$

$t^2 = z$

$2t\,dt = dz$

$$\frac{1}{2}\int\frac{dz}{2z^2-z}=\int x dx$$

$$\ell n\left|\frac{z-\frac{1}{2}}{z}\right|=x^2+c$$

$$\ell n\left|\frac{(x+y)^2-\frac{1}{2}}{(x+y)^2}\right|=x^2+c$$

$$y(0)=1\Rightarrow c=\ell n\left(\frac{1}{2}\right)$$

$$\Rightarrow\frac{(x+y)^2-\frac{1}{2}}{(x+y)^2}=e^{x^2}\times\frac{1}{2}$$

$$\Rightarrow\frac{(x+y)^2-\frac{1}{2}}{(x+y)^2}=\sqrt{e}\times\frac{1}{2}$$

$$\Rightarrow(x+y)^2=\frac{1}{2-\sqrt{e}}$$

36. Let α be a non-zero real number. Suppose $f:R\to R$ is a differentiable function such that $f(0)=2$ and $\lim\limits_{x\to-\infty}f(x)=1$. If $f'(x)=\alpha f(x)+3$, for all $x\in R$, then $f(-\log_e 2)$ is equal to : **[JEE(Main)Jan.2024]**

माना α एक शून्येत्तर वास्तविक संख्या है। माना एक अवकलनीय फलन $f:R\to R$ के लिए $f(0)=2$ तथा $\lim\limits_{x\to-\infty}f(x)=1$ हैं। यदि $f'(x)=\alpha f(x)+3$, $x\in R$ है, तो $f(-\log_e 2)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 9 (B) 5 (C) 7 (D) 3

Ans. Bonus

Sol. $f(0)=2,\quad \lim\limits_{x\to-\infty}f(x)=1$

$$f'(x)=\alpha f(x)+3$$

$$\frac{dy}{dx}=\alpha y+3$$

$$\int\frac{dy}{\alpha y+3}=\int dx$$

$$\frac{1}{\alpha}\ell n(\alpha y+3)=x+C$$

$$\alpha y+3=ce^{\alpha x}$$

$$\because \lim_{x\to-\infty} f(x)=1 \qquad \Rightarrow \alpha>0 \text{ but } \alpha=-3$$

So given data contradicts the information

37. If the solution of the differential equation $(2x + 3y - 2)\,dx + (4x + 6y - 7)\,dy = 0$, $y(0) = 3$, is $\alpha x + \beta y + 3\log_e|2x + 3y - \gamma| = 6$, then $\alpha + 2\beta + 3\gamma$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि अवकल समीकरण $(2x + 3y - 2)\,dx + (4x + 6y - 7)\,dy = 0$, $y(0) = 3$ का हल $\alpha x + \beta y + 3\log_e|2x + 3y - \gamma| = 6$ है, तो $\alpha + 2\beta + 3\gamma$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 29

Sol. $(2x + 3y - 2)dx + (4x + 6y - 7)dy = 0$

Let $2x+3y=t$

$$2+3\frac{dy}{dx}=\frac{dt}{dx}$$

$$\frac{1}{3}\left[\frac{dt}{dx}-2\right]=-\left(\frac{t-2}{2t-7}\right)$$

$$\frac{dt}{dx}-2=-\left(\frac{3t-6}{2t-7}\right)$$

$$\frac{dt}{dx}=2-\frac{3t-6}{2t-7}$$

$$\frac{dt}{dx}=\frac{t-8}{2t-7}$$

$$\int\left(\frac{2t-7}{t-8}\right)dt=\int dx$$

$$=\int 2dt+\int\frac{a}{t-8}dt=\int dx$$

$$=2t+9\ln|t-8|=x+c$$

$$=2(2x+3y)+9|2x+3y-8|=x+c$$

38. If the solution curve, of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passing through the point (2, 1) is

$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|$, then $5\beta + \alpha$ is equal to _____.

[JEE(Main)Jan.2024]

यदि अवकलन समीकरण $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ का हल वक्र, जो बिंदु (2, 1) से होकर जाता है,

$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|$ है, तो $5\beta+\alpha$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 11

Sol. $\frac{dy}{dx}=\frac{x+y-2}{x-y}$

$x+y-2=0$ and $x-y=0$ intersect at (1, 1)

$x=X+1,\ y=Y+1$

$\Rightarrow \frac{dY}{dX}=\frac{X+Y}{X-Y}$

Put $Y=tX \rightarrow \frac{dY}{dX}=t+X\frac{dt}{dX}$

$\Rightarrow t+X\frac{dt}{dX}=\frac{1+t}{1-t}$

$X\frac{dt}{dX}=\frac{1+t^2}{1-t}$

$\frac{dX}{X}=\frac{1-t}{1+t^2}dt$

$\ell n|X|=\tan^{-1}t-\frac{1}{2}\ln|1+t^2|$

$\ell n|x-1|=\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ell n\left|1+\left(\frac{y-1}{x-1}\right)^2\right|+C$

Passes through (2, 1) $\Rightarrow C=0$

$\Rightarrow \tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ell n\left(1+\left(\frac{y-1}{x-1}\right)^2\right)=\ell n|x-1|$

$\Rightarrow 5\beta+\alpha=11$

39. If the solution curve $y=y(x)$ of the differential equation $(1+y^2)(1+\log_e x)\,dx+x\,dy=0,\ x>0$ passes through the

point (1, 1) and $y(e)=\frac{\alpha-\tan\left(\frac{3}{2}\right)}{\beta+\tan\left(\frac{3}{2}\right)}$, then $\alpha+2\beta$ is_____. **[JEE(Main)Jan.2024]**

यदि अवकल समीकरण $(1+y^2)(1+\log_e x)\,dx + x\,dy = 0,\ x>0$ का हल वक्र $y = y(x)$, बिन्दु (1, 1) से होकर जाता है तथा

$y(e)=\frac{\alpha-\tan\left(\frac{3}{2}\right)}{\beta+\tan\left(\frac{3}{2}\right)}$ है, तो $\alpha+2\beta$ बराबर है_____. **[JEE(Main)Jan.2024]**

Ans. 3

Sol. $\left(1+y^2\right)(1+\ell n\,x)dx + xdy = 0$

$$\left(\frac{1+\ell n\,x}{x}\right)dx = -\frac{dy}{1+y^2}$$

$$\int\left(\frac{1+\ell n\,x}{x}\right)dx = \int -\frac{dy}{1+y^2}$$

$$\tan^{-1} y = -\frac{(1+\ln x)^2}{2} + C$$

curve passes thraugh $(1, 1) \Rightarrow C = \frac{\pi}{4}+\frac{1}{2}$

$$y = \tan\left(\frac{\pi}{4}+\frac{1}{2}-\frac{(1+\ell n\,x)^2}{2}\right)$$

$$y(e) = \tan\left(\frac{\pi}{4}+\frac{1}{2}-2\right)$$

$$y(e) = \tan\left(\frac{\pi}{4}-\frac{3}{2}\right) = \frac{1-\tan\left(\frac{3}{2}\right)}{1+\tan\left(\frac{3}{2}\right)}$$

Hence $\alpha = \beta = 1$

$\alpha + 2\beta = 3$

40. Let $f(x)=\sqrt{\lim\limits_{r\to x}\left\{\frac{2r^2\left[(f(r))^2-f(x)f(r)\right]}{r^2-x^2}-r^3e^{\frac{f(r)}{r}}\right\}}$ be differentiable in $(-\infty,0)\cup(0,\infty)$ and $f(1)=1$. Then the value of e^a, such that $f(a)=0$, is equal to _____. **[JEE(Main)Jan.2024]**

माना $f(x)=\sqrt{\lim\limits_{r\to x}\left\{\frac{2r^2\left[(f(r))^2-f(x)f(r)\right]}{r^2-x^2}-r^3e^{\frac{f(r)}{r}}\right\}}$ अंतराल $(-\infty,0)\cup(0,\infty)$ में अवकलनीय है तथा $f(1)=1$ है। यदि $f(a)=0$ है, तो e^a का मान है _____. **[JEE(Main)Jan.2024]**

Ans. 2

Sol. $f(x)=\sqrt{\lim\limits_{r\to x}\left\{\frac{2r^2\left[(f(r))^2-f(x)f(r)\right]}{r^2-x^2}-r^3e^{\frac{f(r)}{r}}\right\}}$

$\Rightarrow (f(x))^2=xf(x)f'(x)-x^3e^{\frac{f(x)}{x}}$ by L.H. rule

$\Rightarrow y^2=xy\frac{dy}{dx}-x^3e^{\frac{y}{x}}$

$\Rightarrow \frac{y}{x}=\frac{dy}{dx}-\frac{x^2}{y}e^{\frac{y}{x}}$

Let $y=vx\therefore \frac{dy}{dx}=v+x\frac{dv}{dx}$

$\Rightarrow v=v+x\frac{dv}{dx}-\frac{x^2}{vx}e^v$

$\Rightarrow x\frac{dv}{dx}-\frac{x}{v}e^v=0$

$\Rightarrow ve^{-v}dv-dx=0$

$\Rightarrow (x+c)e^v+v+1=0$ (Integration)

$\Rightarrow (x+c)e^{y/x}+\frac{y}{x}+1=0$

$\therefore y(1)=1\therefore c=-1-\frac{2}{e}$

$\therefore\left(x-1-\frac{2}{e}\right)e^{y/x}+\frac{y}{x}+1=0$

$f(a)=0\Rightarrow x=a,y=0\therefore\left(a-1-\frac{2}{e}\right)+1=0$

$ae - e - 2 + e = 0$

$ae = 2$

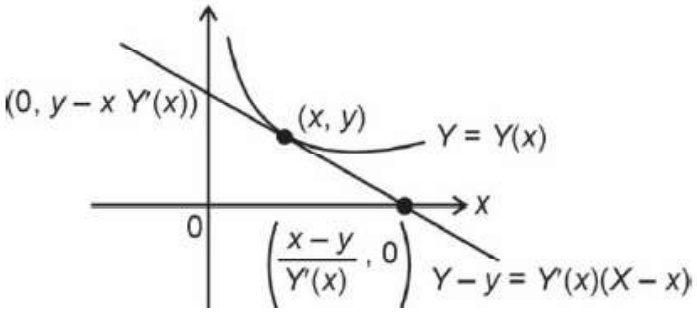
41. Let Y=Y(X) be a curve lying in the first quadrant such that the area enclosed by the line $Y - y = Y'(x)(X - x)$ and the co-ordinate axes, where (x, y) is any point on the curve, is always $\frac{-y^2}{2Y'(x)} + 1$, $Y'(x) \neq 0$. If $Y(1) = 1$, then $12Y(2)$ equals _____. **[JEE(Main)Jan.2024]**

माना प्रथम चतुर्थांश में एक वक्र Y=Y(X) इस प्रकार है कि रेखा $Y - y = Y'(x)(X - x)$ तथा निर्देशांक अक्षों से घिरे क्षेत्र का क्षेत्रफल $\frac{-y^2}{2Y'(x)} + 1$, $Y'(x) \neq 0$ है, (x, y) वक्र पर कोई बिन्दु है। यदि $Y(1) = 1$ है, तो $12Y(2)$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 20

Sol.



$$\Rightarrow \frac{1}{2}\left(y - xY(x)\right)\left(x - \frac{y}{Y(x)}\right) = -\frac{y^2}{2Y(x)} + 1$$

$$\left(y - xY'(x)\right)\left(xY'(x) - y\right) = -y^2 + 2Y'(x)$$

$$yxY'(x) - y^2 - x^2\left(Y'(x)\right) + xyY'(x) = -y^2 + 2Y'(x)$$

$$\left(y - x\frac{dy}{dx}\right)\left(x\frac{dy}{dx} - y\right) = -y^2 + 2\frac{dy}{dx}$$

$$-y^2 - x^2\left(\frac{dy}{dx}\right)^2 + 2xy\left(\frac{dy}{dx}\right) = -y^2 + 2\frac{dy}{dx}$$

$$-x^2\frac{dy}{dx} + 2xy = 2$$

$$\Rightarrow \frac{dy}{dx} + \left(-\frac{2}{x}\right)y = \left(-\frac{2}{x^2}\right)$$

$$\text{I.F.} = \frac{1}{x^2}$$

$\Rightarrow$ Solution is $\frac{y}{x^2}=\frac{-2\cdot x^{-3}}{-3}+c$

$\because y(1)=1$

$\Rightarrow c=\frac{1}{3}$

$\frac{y}{x^2}=\frac{2}{3x^3}+\frac{1}{3}$, when $x=2$, $y=4\left(\frac{1}{12}+\frac{1}{3}\right)=\frac{5}{3}$

$\Rightarrow 12y(2)=20$

42. Let y = y (x) be the solution of the differential equation $\sec^2 x dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0$, $0<x<\frac{\pi}{2}, y\left(\frac{\pi}{4}\right)=0$. If $y\left(\frac{\pi}{6}\right)=\alpha$, then $e^{8\alpha}$ is equal to_____. **[JEE(Main)Jan.2024]**

माना अवकल समीकरण $\sec^2 x dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0$, $0<x<\frac{\pi}{2}$, $y\left(\frac{\pi}{4}\right)=0$ का हल y = y(x) है। यदि $y\left(\frac{\pi}{6}\right)=\alpha$ है, तो $e^{8\alpha}$ बराबर है_____ **[JEE(Main)Jan.2024]**

Ans. 9

Sol. $\sec^2 x dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0$

Put $\tan x=t$

$\sec^2 x dx=dt$

$dt+\left(e^{2y}t^2+t\right)dy=0$

$\frac{1}{t^2}\times\frac{dt}{dy}+\frac{1}{t}=-e^{2y}$

Put $\frac{1}{t}=u$

$-\frac{1}{t^2}\frac{dt}{dy}=\frac{du}{dy}$

$-\frac{du}{dy}+u=-e^{2y}$

$\frac{du}{dy}-u=e^{2y}$

I.f. $=e^{-\int dy}=e^{-y}$

$u\cdot e^{-y}=\int e^{-y}e^{2y}dy$

43. If x= x(t) is the solution of the differential equation $(t+1)dx = (2x + (t+1)^4)dt$, $x(0) = 2$, then $x(1)$ equals _____.
[JEE(Main)Jan.2024]

यदि अवकल समीकरण $(t + 1)dx = (2x + (t + 1)^4)dt$, $x(0) = 2$ का हल $x = x(t)$ है, तो $x(1)$ बराबर है _____
[JEE(Main)Jan.2024]

Ans. 14

Sol. $(t + 1)dx = (2x + (t + 1)^4)dt$

$$\frac{dx}{dt} - \frac{2x}{(t+1)} = (t+1)^3$$

$$IF = e^{-\int \frac{2}{t+1} dt} = \frac{1}{(t+1)^2}$$

$$\frac{x}{(t+1)^2} = \int (t+1)\, dt$$

$$\frac{x}{(t+1)^2} = \frac{t^2}{2} + t + C$$

Now $x(0) = 2$

$\Rightarrow C = 2$

$$\therefore x = \left(\frac{t^2}{2} + t + 2\right)(t+1)^2$$

$$x(1) = \left(\frac{1}{2} + 1 + 2\right)(1+1)^2$$

$$= \frac{7}{2} \times 4 = 14$$

44. If $\frac{dx}{dy} = \frac{1 + x - y^2}{y}$, $x(1) = 1$, then $5x(2)$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि $\frac{dx}{dy} = \frac{1 + x - y^2}{y}$, $x(1) = 1$ है, तो $5x(2)$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 5

Sol. $$\frac{dx}{dy} - \frac{x}{y} = \frac{1 - y^2}{y}$$

Integration factor $= e^{\int -\frac{1}{y} dy} = \frac{1}{y}$

$$x \cdot \frac{1}{y} = \int \frac{1-y^2}{y^2} dy$$

$$\frac{x}{y} = \frac{-1}{y} - y + c$$

$$x = -1 - y^2 + cy$$

$$x(1) = 1$$

$$1 = -1 - 1 + c \Rightarrow c = 3$$

$$x = -1 - y^2 + 3y$$

$$5x(2) = 5(-1 - 4 + 6)$$

$$= 5$$

45. If the solution $y = y(x)$ of the differential equation $(x^4 + 2x^3 + 3x^2 + 2x + 2)dy - (2x^2 + 2x + 3)dx = 0$ satisfies $y(-1) = -\frac{\pi}{4}$, then $y(0)$ is equal to : **[JEE(Main)April,2024]**

यदि अवकल समीकरण $(x^4 + 2x^3 + 3x^2 + 2x + 2)dy - (2x^2 + 2x + 3)dx = 0$ का हल $y = y(x)$, $y(-1) = -\frac{\pi}{4}$ को संतुष्ट d j r kgSतो $y(0)$ बराबर है : **[JEE(Main)April,2024]**

(A) 0 (B) $-\frac{\pi}{12}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Ans. D

Sol. $$\int dy = \int \frac{\left(2x^2 + 2x + 3\right)}{x^4 + 2x^3 + 2x + 2} dx$$

$$y = \int \frac{\left(2x^2 + 2x + 3\right)}{\left(x^2 + 1\right)\left(x^2 + 2x + 2\right)} dx$$

$$y = \int \frac{dx}{x^2 + 2x + 2} + \int \frac{dx}{x^2 + 1}$$

$$y = \tan^{-1}(x + 1) + \tan^{-1}x + C$$

$$y(-1) = -\frac{\pi}{4}$$

$$-\frac{\pi}{4} = 0 - \frac{\pi}{4} + C \qquad \Rightarrow C = 0$$

$$\Rightarrow y = \tan^{-1}(x + 1) + \tan^{-1}x$$

$$y(0) = \tan^{-1} 1 = \frac{\pi}{4}$$

46. Let $y = y(x)$ be the solution of the differential equation $(x^2+4)^2dy + (2x^3y + 8xy - 2)dx = 0$. If $y(0) = 0$, then $y(2)$ is equal to : **[JEE(Main)April,2024]**

माना $y = y(x)$ अवकल समीकरण $(x^2 + 4)^2dy + (2x^3y + 8xy - 2)dx = 0$ का हल है। यदि $y(0) = 0$ है, तो $y(2)$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{\pi}{32}$ (B) $\frac{\pi}{16}$ (C) 2π (D) $\frac{\pi}{8}$

Ans. A

Sol. $\frac{dy}{dx} + \left(\frac{2x}{x^2+4}\right)y = \frac{2}{\left(x^2+4\right)^2}$

I.F. $= e^{\int \frac{2x}{x^2+4}dx} = e^{\ln\left(x^2+4\right)} = x^2+4$

solution of differential equation $y\left(x^2+4\right) = \int \frac{2}{x^2+4}dx \Rightarrow y\left(x^2+4\right) = \tan^{-1}\left(\frac{x}{2}\right) + C$

But $y(0) = 0 \therefore c = 0$

$y\left(x^2+4\right) = \tan^{-1}\left(\frac{x}{2}\right)$

at $x = 2 \rightarrow y = \frac{\pi/4}{8} = \frac{\pi}{32}$

47. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + 2y = \sin(2x)$, $y(0) = \frac{3}{4}$, then $y\left(\frac{\pi}{8}\right)$ is equal to : **[JEE(Main)April,2024]**

यदि अवकल समीकरण $\frac{dy}{dx} + 2y = \sin(2x)$, $y(0) = \frac{3}{4}$ का हल $y = y(x)$ है, तो $y\left(\frac{\pi}{8}\right)$ बरारबर है : **[JEE(Main)April,2024]**

(A) $e^{-\pi/4}$ (B) $e^{-\pi/8}$ (C) $e^{\pi/8}$ (D) $e^{\pi/4}$

Ans. A

Sol. $\frac{dy}{dx} + 2y = \sin 2x$

IF $= e^{\int 2dx} = e^{2x}$

$y.e^{2x} = \int e^{2x} \sin 2x\, dx + c$

$$= \frac{e^{2x}}{8}(2\sin 2x - 2\cos 2x) + c$$

$$y(0) = \frac{3}{4}$$

$$\frac{3}{4} = \frac{1}{8}(-2) + c \Rightarrow c = 1$$

Put $x = \frac{\pi}{8}$

$$y = \frac{1}{8} \times 2\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) + e^{-\pi/4}$$

$$y = e^{-\pi/4}$$

48. The differential equation of the family of circles passing through the origin and having center at the line $y = x$ is :
[JEE(Main)April,2024]

मूल बिन्दु से होकर जाने वाले तथा रेखा $y = x$ पर केन्द्र के वृत्त कुल का अवकल समीकरण है : **[JEE(Main)April,2024]**

(A) $(x^2 - y^2 + 2xy)dx = (x^2 - y^2 - 2xy)dy$ (B) $(x^2 + y^2 + 2xy)dx = (x^2 - y^2 + 2xy)dy$

(C) $(x^2 + y^2 - 2xy)dx = (x^2 + y^2 + 2xy)dy$ (D) $(x^2 + y^2 + 2xy)dx = (x^2 + y^2 - 2xy)dy$

Ans. A

Sol. $C \equiv x^2 + y^2 + gx + gy = 0$ .....(1)

$2x + 2yy' + g + gy' = 0$

$$g = -\left(\frac{2x + 2yy'}{1 + y'}\right)$$

put in (1)

$$x^2 + y^2 - \left(\frac{2x + 2yy'}{1 + y'}\right)(x + y) = 0$$

$$(x^2 - y^2 - 2xy)y' = x^2 - y^2 + 2xy$$

49. Let $y = y(x)$ be the solution of the differential equation $(2x\log_e x)\frac{dy}{dx} + 2y = \frac{3}{x}\log_e x, x > 0$ and $y(e^{-1}) = 0$.
Then, $y(e)$ is equal to : **[JEE(Main)April,2024]**

माना $y = y(x)$ अवकल समीकरण $(2x\log_e x)\frac{dy}{dx} + 2y = \frac{3}{x}\log_e x, x > 0$ तथा $y(e^{-1}) = 0$ का हल है, तो $y(e)$ बराबर है :
[JEE(Main)April,2024]

(A) $-\frac{3}{2e}$ (B) $-\frac{3}{e}$ (C) $-\frac{2}{e}$ (D) $-\frac{2}{3e}$

Ans. B

Sol. $\left(1+x^2\right)\cdot\frac{dy}{dx}+y=e^{\tan^{-1}x}$

Let $\tan^{-1}x=t$

$$\frac{1}{1+x^2}\cdot dx=dt$$

$$\frac{dy}{dt}+y=e^t$$

$$If=e^{\int 1\cdot dt}=e^t$$

$$y\cdot e^t=\frac{e^{2t}}{2}+C$$

$$y\cdot e^{\tan^{-1}x}=\frac{e^{2\tan^{-1}x}}{2}+C$$

$$x=1 \quad y=0 \quad C=-\frac{e^{\pi/2}}{2}$$

$$y\cdot\tan^{-1}x=\frac{e^{2\tan^{-1}x}}{2}-\frac{e^{\pi/2}}{2}$$

$$x=0\Rightarrow y=\frac{1}{2}\left(1-e^{\pi/2}\right)$$

50. Let $y=y(x)$ be the solution of the differential equation $\left(1+x^2\right)\frac{dy}{dx}+y=e^{\tan^{-1}x}$, $y(1)=0$. Then $y(0)$ is :

[JEE(Main)April,2024]

माना $y=y(x)$ अवकल समीकरण $\left(1+x^2\right)\frac{dy}{dx}+y=e^{\tan^{-1}x}$, $y(1)=0$ का हल है, तो $y(0)$ है :

[JEE(Main)April,2024]

(A) $\frac{1}{2}\left(1-e^{\pi/2}\right)$ (B) $\frac{1}{4}\left(e^{\pi/2}-1\right)$ (C) $\frac{1}{4}\left(1-e^{\pi/2}\right)$ (D) $\frac{1}{2}\left(e^{\pi/2}-1\right)$

Ans. A

Sol. $\left(1+x^2\right)\cdot\frac{dy}{dx}+y=e^{\tan^{-1}x}$

Let $\tan^{-1}x=t$

$$\frac{1}{1+x^2}\cdot dx = dt$$

$$\frac{dy}{dt} + y = e^t$$

$$\text{If} = e^{\int 1\cdot dt} = e^t$$

$$y\cdot e^t = \frac{e^{2t}}{2} + C$$

$$y\cdot e^{\tan^{-1}x} = \frac{e^{2\tan^{-1}x}}{2} + C$$

$$x = 1 \quad y = 0 \quad C = -\frac{e^{\pi/2}}{2}$$

$$y\cdot \tan^{-1}x = \frac{e^{2\tan^{-1}x}}{2} - \frac{e^{\pi/2}}{2}$$

$$x = 0 \quad \Rightarrow y = \frac{1}{2}\left(1 - e^{\pi/2}\right)$$

51. Suppose the solution of the differential equation $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ represents a circle passing through origin. Then the radius of this circle is : **[JEE(Main)April,2024]**

मान लीजिए कि अवकल समीकरण $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ का हल मूल बिन्दु से गुजरने वाले वृत्त को दर्शाता है, तो इस वृत्त d hf=kT; kgS **[JEE(Main)April,2024]**

(A) $\frac{\sqrt{17}}{2}$ (B) 2 (C) $\sqrt{17}$ (D) $\frac{1}{2}$

Ans. A

Sol. $$\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - y(2\alpha+\beta) + 4\alpha}$$

$$\beta x dy - (2\alpha+\beta)ydy + 4\alpha dy = (2+\alpha)xdx - \beta y dx + 2dx$$

$$\beta(xdy + ydx) - (2\alpha+\beta)ydy + 4\alpha dy = (2+\alpha)xdx + 2dx$$

$$\beta xy - \frac{(2\alpha+\beta)y^2}{2} + 4\alpha y = \frac{(2+\alpha)x^2}{2}$$

$\Rightarrow \beta = 0$ for this to be circle

$$(2+\alpha)\frac{x^2}{2} + \alpha y^2 + 2x - 4\alpha y = 0$$

coeff. of $x^2 = y^2$ $\Rightarrow 2 + a = 2a$

$\Rightarrow \alpha = 2$

i.e. $2x^2 + 2y^2 + 2x - 8y = 0$

$x^2 + y^2 + x - 4y = 0$

$$rd = \sqrt{\frac{1}{4}+4} = \frac{\sqrt{17}}{2}$$

52. Let $y = y(x)$ be the solution of the differential equation $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$, $y(0) = 1$. Then $y\left(\frac{\pi}{4}\right)$ is equal to : **[JEE(Main)April,2024]**

माना $y = y(x)$ अवकल समीकरण $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$ का हल है एवं $y(0) = 1$ है तब $y\left(\frac{\pi}{4}\right)$ बराबर होगा: **[JEE(Main)April,2024]**

(A) $\frac{2}{e}$ (B) $\frac{2}{e^2}$ (C) $\frac{1}{e^2}$ (D) $\frac{1}{e}$

Ans. D

Sol. $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$

$$\frac{dy}{1+y^2} = -\frac{e^{\tan x}dx}{\cos^2 x\left(1+e^{2\tan x}\right)}$$

Let $e^{\tan x} = t$

then $e^{\tan x} \cdot \sec^2 x\,dx = dt$

$$\Rightarrow \tan^{-1} y = -\frac{dt}{\left(1+t^2\right)}$$

$\Rightarrow \tan^{-1}y = -\tan^{-1}(t) + C$

$\Rightarrow \tan^{-1}y + \tan^{-1}(e^{\tan x}) = C$

$y(0) = 1$

$\Rightarrow \tan^{-1}(1) + \tan^{-1}(1) = C$

$$C = \frac{\pi}{2}$$

$$\rightarrow \tan^{-1} y + \tan^{-1}\left(e^{\tan x}\right) = \frac{\pi}{2}$$

$$\rightarrow y\left(\frac{\pi}{4}\right) = ?$$

$$\Rightarrow \tan^{-1}(y) + \tan^{-1}(e) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}\left(\frac{y+e}{1-ye}\right) = \frac{\pi}{2}$$

$$\frac{y+e}{1-ye} = \tan\left(\frac{\pi}{2}\right)$$

$$y = \frac{1}{e}$$

53. Let f(x) be a positive function such that the area bounded by y = f(x), y = 0 from x = 0 to x = a > 0 is $e^{-a} + 4a^2 + a - 1$. Then the diffrential equation, whose general solution is $y = c_1 f(x) + c_2$, where c_1 and c_2 are arbitrary constants, is : **[JEE(Main)April,2024]**

माना f(x) एक धनात्मक फलन इस प्रकार है कि y = f(x), y = 0 द्वारा x = 0 से x = a > 0 में परिबद्ध क्षेत्रफल $e^{-a} + 4a^2 + a - 1$ है। तब अवकल समीकरण जिसका सामान्य $y = c_1 f(x) + c_2$ है, जहाँ c_1 तथा c_2 स्वैच्छिक अचर है, होगा : **[JEE(Main)April,2024]**

(A) $\left(8e^x - 1\right)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

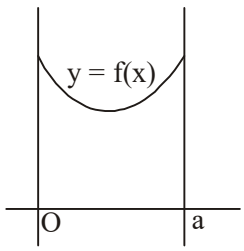
(B) $\left(8e^x + 1\right)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

(C) $\left(8e^x + 1\right)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

(D) $\left(8e^x - 1\right)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Ans. C

Sol. $\int_0^a f(x)\,dx = e^{-a} + 4a^2 + a - 1$



$f(a) = -e^{-a} + 8a + 1$

$$y = C_1\left(1 + 8x - \frac{1}{e^x}\right) + C_2$$

$$\frac{dy}{dx} = C_1\left(8 + \frac{1}{e^x}\right) \Rightarrow \frac{\left(\frac{dy}{dx}\right)}{\left(8 + \frac{1}{e^x}\right)} = C_1$$

$$\frac{d^2y}{dx^2}\left(8 + \frac{1}{e^x}\right) - \left(\frac{-1}{e^x}\right)\frac{dy}{dx} = 0$$

54. Let y= y(x) be the solution curve of the differential equation $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$, $y(1) = 0$. Then $y\left(\sqrt{3}\right)$ is equal to : **[JEE(Main)April,2024]**

माना अवकल समीकरण $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$, $y(1) = 0$ का हल वक्र $y = y(x)$ है। तो $y\left(\sqrt{3}\right)$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{12}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Ans. C

Sol. $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$

$\Rightarrow \sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$

Let $z = \tan y$

$\frac{dz}{dx} = \sec^2 y \frac{dy}{dx}$

$\Rightarrow \frac{dz}{dx} + 2xz = x^3$

I.F. $= e^{x^2}$

$\Rightarrow z.e^{x^2} = \int e^{x^2} \cdot x^3 dx + c$

$\Rightarrow \tan y \cdot e^{x^2} = \frac{1}{2}\left(x^2 e^{x^2} - e^{x^2}\right) + c$

$\Rightarrow \tan(0) \cdot e = \frac{1}{2}(1 \cdot e - e) + c$

$\Rightarrow c = 0$

$\Rightarrow \tan y = \frac{x^2 - 1}{2}$

$$f(x)=\tan^{-1}\left(\frac{x^2-1}{2}\right)\Rightarrow f\left(\sqrt{3}\right)=\frac{\pi}{4}$$

55. The solution curve, of the differential equation $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$, passing through the point (0, 1) is a conic, whose vertex lies on the line : **[JEE(Main)April,2024]**

अवकल समीकरण $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$ का हल वक्र, जो (0, 1) से होकर जाता है, एक शांकव है। इस शांकव का शीर्ष निम्न में से किस j§[kkij gS% **[JEE(Main)April,2024]**

(A) $2x+3y=6$ (B) $2x+3y=9$ (C) $2x+3y=-6$ (D) $2x+3y=-9$

Ans. B

Sol. $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$

$2ydy+3dx=5dy$

$y^2+3x=5y+c$, (0, 1)

$1+0=5+c$

$c=-4$

$y^2-5y=-3x-4$

$y^2-5y+\frac{25}{4}-\frac{25}{4}=3x-4$

$\left(y-\frac{5}{2}\right)^2=-3x+\frac{9}{4}$

$\left(y-\frac{5}{2}\right)^2=-3\left(x-\frac{3}{4}\right)$

$\left(\frac{3}{4},\frac{5}{2}\right)$ satisfies by $2x+3y=9$

56. The solution of the differential equation $(x^2+y^2)dx-5xy\,dy=0$, $y(1)=0$, is : **[JEE(Main)April,2024]**

अवकल समीकरण $(x^2+y^2)dx-5xy\,dy=0$, $y(1)=0$ का हल है : **[JEE(Main)April,2024]**

(A) $|x^2-4y^2|^5=x^2$ (B) $|x^2-2y^2|^5=x^2$ (C) $|x^2-4y^2|^6=x$ (D) $|x^2-2y^2|^6=x$

Ans. A

Sol. $(x^2+y^2)dx=5xydy$

$$\frac{dy}{dx}=\frac{x^2+y^2}{5xy}$$

Let $y=vx$

$$\frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$\frac{1+v^2}{5v}=v+x\frac{dv}{dx}$$

$$x\frac{dv}{dx}=\frac{1+v^2}{5v}-v$$

$$x\frac{dv}{dx}=\frac{1-4v^2}{5v}$$

$$\frac{5v}{1-4v^2}dv=\frac{dx}{x}$$

$$\int\frac{dx}{x}=-\frac{5}{8}\int\frac{-8v}{1-4v^2}dv$$

$$\ell nx=-\frac{5}{8}\ell n\left|1-4v^2\right|+\ell nc$$

$$\ell nx+\frac{5}{8}\ell n\left|1-\frac{4y^2}{x^2}\right|=\ell nc \qquad \because\ y(1)=0$$

So $l$nc = 0

$$8\ell nx+5\ell n\left|\frac{x^2-4y^2}{x^2}\right|=0$$

$8lnx + 5ln\,|x^2 - 4y^2| - 10lnx = 0$

$ln\,|x^2 - 4y^2|^5 - lnx^2 = 0$

$|x^2 - 4y^2|^5 = x^2$

57. Let $\int_0^x\sqrt{1-\left(y'(t)\right)^2}dt=\int_0^x y(t)dt$, $0\le x\le 3$, $y\ge 0$, $y(0)=0$. Then at $x=2$, $y''+y+1$ is equal to :

[JEE(Main)April,2024]

माना $\int_0^x\sqrt{1-\left(y'(t)\right)^2}dt=\int_0^x y(t)dt$, $0\le x\le 3$, $y\ge 0$, $y(0)=0$ हैं। तो $x=2$ पर $y''+y+1$ बराबर है :

[JEE(Main)April,2024]

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{2}$

Ans. A

Sol. $\sqrt{1-\left(y'(x)\right)^2} = y(x)$

$$1-\left(\frac{dy}{dx}\right)^2 = y^2$$

$$\left(\frac{dy}{dx}\right)^2 = 1-y^2$$

$$\frac{dy}{\sqrt{1-y^2}} = dx \text{ OR } \frac{dy}{\sqrt{1-y^2}} = -dx$$

$\Rightarrow \sin^{-1}y = x + c, \sin^{-1}y = -x + c$

$x = 0, y = 0 \quad \Rightarrow c = 0$

$\sin^{-1}y = x$, as $y \geq 0$

$\sin x = y$

$$\Rightarrow \frac{dy}{dx} = \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\sin x$$

$\Rightarrow -\sin x + \sin x + 1 = 1$

58. Let the solution $y = y(x)$ of the differential equation $\frac{dy}{dx} - y = 1 + 4\sin x$ satisfy $y(\pi) = 1$. Then $y\left(\frac{\pi}{2}\right) + 10$ is equal to _____. **[JEE(Main)April,2024]**

माना अवकल समीकरण $\frac{dy}{dx} - y = 1 + 4\sin x$ के हल $y = y(x)$ के लिए $y(\pi) = 1$ है। तो $y\left(\frac{\pi}{2}\right) + 10$ बराबर है _____

[JEE(Main)April,2024]

Ans. 7

Sol. $\frac{dy}{dx} - y = 1 + 4\sin x$ linear differential equation

I.F. $= e^{\int -dx} = e^{-x}$

So solution of linear differential equation is

$$ye^{-x} = \int e^{-x}(1 + 4\sin x)dx$$

$$ye^{-x} = -e^{-x} + 4\int e^{-x}\sin x dx$$

$$ye^{-x} = -e^{-x} + 4\frac{e^{-x}}{2}\left[-\sin x - \cos x\right] + c$$

$$y = -1 - 2\left(\sin x + \cos x\right) + ce^{x}$$

Now $y(\pi) = 1$

$\Rightarrow 1 = -1 - 2(0 - 1) + ce^{\pi}$

Hence $y = -1 - 2(\sin x + \cos x)$

$$\Rightarrow y\left(\frac{\pi}{2}\right) + 10 = -1 - 2(1 + 0) + 10$$

$= 7$

59. Let $y = y(x)$ be the solution of the differential equation $(x + y + 2)^2\, dx = dy$; $y(0) = -2$. Let the maximum and minimum values of the function $y = y(x)$ in $\left[0, \frac{\pi}{3}\right]$ be α and β, respectively. If $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$, $\gamma, \delta \in Z$, then $\gamma + \delta$ equals _____. **[JEE(Main)April,2024]**

माना $y = y(x)$ अवकल समीकरण $(x + y + 2)^2\, dx = dy$; $y(0) = -2$ का एक हल है। माना $\left[0, \frac{\pi}{3}\right]$ में फलन $y = y(x)$ के उच्चतम r Hkkfuur e eku Øe'k% α तथा β हैं। यदि $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$, $\gamma, \delta \in Z$ तो $\gamma + \delta$ बराबर है _____

[JEE(Main)April,2024]

Ans. 31

Sol. $\frac{dy}{dx} = (x + y + 2)^2$

Put $x + y + 2 = t$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - 1 = t^2$$

$\tan^{-1} t = x + c$

$\tan^{-1}(x + y + z) = x + c$

$\Rightarrow (x = 0, y = -2)$

$C = 0$

$y = \tan x - x - 2$

$$\frac{dy}{dx} = \tan^2 x \geq 0$$

$\forall x \in \left[0, \frac{\pi}{3}\right]$

$\Rightarrow y_{min} = y(0) = -2 = \beta$

And $y_{max} = y\left(\frac{\pi}{3}\right) = \sqrt{3} - 2 - \frac{\pi}{3} = \alpha$

Now, $(3\alpha + \pi)^2 + \beta^2$

$\left(3\sqrt{3} - 6 - \pi + \pi\right)^2 + 4$

$27 + 36 - 36\sqrt{3} + 4$

$67 - 36\sqrt{3}$

So, $\gamma = 67$

$\delta = -36$

$\Rightarrow \gamma + \delta = 31$

60. Let f be a differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim\limits_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $2 f(2) + 3 f(3)$ is equal to _____. **[JEE(Main)April,2024]**

माना अंतराल $(0, \infty)$ में एक अवकलनीय फलन f है, जिसके लिए $f(1) = 1$ है तथा प्रत्येक $x > 0$ के लिए $\lim\limits_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ है। तो $2 f(2) + 3 f(3)$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 24

Sol. $f(1) = 1$

$\lim\limits_{x \to \infty} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$

$\lim\limits_{x \to \infty} \frac{2 + f(x) - x^2 f'(t)}{1} = 1$

$= 2xy - x^2 \frac{dy}{dx} = 1$

$= \frac{dy}{dx} - \frac{2y}{x} = \frac{-1}{x^2}$

I.f. $e^{-\int \frac{2}{x} dx} = e^{-2\ell nx} = \frac{1}{x^2}$

$y \cdot \frac{1}{x^2} = \int \frac{-1}{x^4} dx$

$$\frac{y}{x^2}=\frac{-x^{-4+1}}{-4+1}+C$$

$$\frac{y}{x^2}=\frac{1}{3x^3}+C$$

$\because f(1)=1$

$$\frac{1}{1}=\frac{1}{3}+C \Rightarrow C=1-\frac{1}{3}=\frac{2}{3}$$

$$y=\frac{1}{3x}+\frac{2}{3}x^2$$

Now $2f(2)+3f(3)=2\times\left(\frac{1}{6}+\frac{8}{3}\right)+3\times\left(\frac{1}{9}+6\right)$

$$=2\times\left(\frac{1+16}{6}\right)+3\times\left(\frac{54+1}{9/3}\right)=\frac{17}{3}+\frac{55}{3}=\frac{72}{3}=24$$ Ans.

61. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{\left(1+x^2\right)}}$; $y(0)=0$. Then the area enclosed by the curve $f(x)=y(x)e^{\frac{1}{\left(1+x^2\right)}}$ and the line $y-x=4$ is_____. **[JEE(Main)April,2024]**

माना अवकल समीकरण $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{\left(1+x^2\right)}}$; $y(0)=0$ का हल $y = y(x)$ है। तो वक्र $f(x)=y(x)e^{\frac{1}{\left(1+x^2\right)}}$ तथा रेखा $y - x = 4$ से घिरे क्षेत्र का क्षेत्रफल है_____ **[JEE(Main)April,2024]**

Ans. 18

Sol. $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{1+x^2}}$; $y(0)=0$

I.F. of linear differential equation,

$$\text{I.F.}=e^{\int\frac{2x}{\left(1+x^2\right)^2}dx}=e^{\left(\frac{-1}{1+x^2}\right)}$$

$$\Rightarrow y\left(e^{\left(\frac{-1}{1+x^2}\right)}\right)=\int x\cdot e^{\frac{1}{1+x^2}}\cdot e^{\left(\frac{-1}{1+x^2}\right)}dx$$

$$=\frac{x^2}{2}+c$$

$\Rightarrow y(0) = 0 \Rightarrow 0(e^{-1}) = c \Rightarrow c = 0$

$$\Rightarrow y = \frac{e^{\frac{1}{1+x^2}} \cdot x^2}{2}$$

Area between curve $ye^{\left(\frac{-1}{1+x^2}\right)} = \frac{x^2}{2}$ and $y - x = 4$

$\Rightarrow 2(x + 4) = x^2 \Rightarrow x^2 - 2x - 8 = 0$

$\Rightarrow (x - 4)(x + 2) = 0$

$$\int_{-2}^{4}\left[(x+4)-\frac{x^2}{2}\right]dx = \frac{x^2}{2} + 4x - \frac{x^3}{6}\Bigg|_{-2}^{4}$$

$$= \left(8+16-\frac{64}{6}\right) - \left(2-8+\frac{8}{6}\right)$$

$= 30 - 12 = 18$

62. Let a conic C pass through the point (4, –2) and P(x, y), $x \geq 3$, be any point on C. Let the slope of the line touching the conic C only at a single point P be half the slope of the line joining the points P and (3, –5). If the focal distance of the point (7, 1) on C is d, then 12d equals _____. **[JEE(Main)April,2024]**

माना कोई शंकु C बिन्दु (4, –2) से होकर जाता है तथा C पर कोई बिन्दु P(x, y), $x \geq 3$ है। माना शंकु C को मात्र एक बिन्दु P पर Li ' kZd j usokyh hj s[kkd hi ø. k rFkk fcUnq P तथा (3, –5) को मिलाने वाली रेखा की प्रवणता की आधी है। यदि C पर स्थित बिन्दु (7, 1) की नाभीय दूरी d है, तो 12d बराबर है _____ **[JEE(Main)April,2024]**

Ans. 75

Sol. P(x, y) and $x \geq 3$

Slope of line at P(x, y) will be $\frac{dy}{dx} = \frac{1}{2}\left(\frac{y+5}{x-3}\right)$

$$\Rightarrow 2\frac{dy}{(y+5)} = \frac{1}{(x-3)}dx$$

$\Rightarrow 2\ell n(y+5) = \ell n(x-3) + C$

Passes through (4, –2)

$\Rightarrow 2\ell n(3) = \ell n(1) + C$

$\Rightarrow C = 2\ell n(3)$

$\Rightarrow 2\ell n(y+5) = \ell n(x-3) + 2\ell n(3)$

$$\Rightarrow 2\left(\ln\left(\frac{y+5}{3}\right)\right)=\ell n\left(x-3\right)$$

$$\Rightarrow\left(\frac{y+5}{3}\right)^2=\left(x-3\right)$$

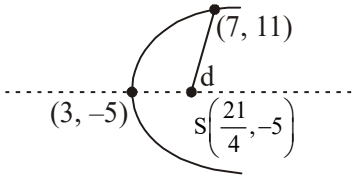
$$\Rightarrow\left(y+5\right)^2=9\left(x-3\right)$$

$$\downarrow$$

Parabola

$4a = 9$

$$a=\frac{9}{4}$$



$$d=\sqrt{\left(\frac{7}{4}\right)^2+6^2}$$

$$d=\frac{\sqrt{625}}{4}$$

$$d=\frac{25}{4}$$

$12d = 75$

63. If the solution y(x) of the given differential equation $(e^y + 1)\cos x dx + e^y \sin x\, dy = 0$ passes through the point $\left(\frac{\pi}{2}, 0\right)$, then the value of $e^{y\left(\frac{\pi}{6}\right)}$ is equal to _____. **[JEE(Main)April,2024]**

यदि अवकल समीकरण $(e^y + 1)\cos x dx + e^y \sin x\, dy = 0$ का हल y(x) बिन्दु $\left(\frac{\pi}{2}, 0\right)$ से गुजरता है, तो $e^{y\left(\frac{\pi}{6}\right)}$ का मान बराबर gS_____ **[JEE(Main)April,2024]**

Ans. 3

Sol. $(e^y + 1)\cos x\, dx + e^y \sin x dy = 0$

$$\cot x dx+\frac{e^y}{e^y+1}dy=0$$

$$\int\cot x dx+\int\frac{e^y}{e^y+1}dy=0$$

$\ln|\sin x| + \ln|e^y + 1| = \ln c$

$|\sin x\,(e^y + 1)| = c$

Put & $x = \frac{\pi}{2}$ $y = 0$

$c = 2$

$\sin x\,(e^y + 1) = 2$

$e^y + 1 = 2\,\text{cosec}\,x$

$e^y = 2\,\text{cosec}\,x - 1$

$e^{y\left(\frac{\pi}{6}\right)} = 2\,\text{cosec}\,\frac{\pi}{6} - 1$

$= 2 \times 2 - 3 = 3$

64. Let $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in N$ be the solution of the differential equation $xdy - ydx + xy(xdy + ydx) = 0$, $y(1) = 2$. Then $\alpha + \beta$ is equal to _____. **[JEE(Main)April,2024]**

माना अवकल समीकरण $xdy - ydx + xy(xdy + ydx) = 0$, $y(1) = 2$ का हल $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in N$ है। तो $\alpha + \beta$ बराबर gS_____. **[JEE(Main)April,2024]**

Ans. 4

Sol. $\alpha|x| = |y|\,e^{yx-\beta}$, $\alpha, \beta \in N$

$xdy - ydx + xy(xdy + ydx) = 0$

$\frac{dy}{y} - \frac{dx}{x} + (xdy + ydx) = 0$

$\ell n|y| - \ell n|x| + xy = c$

$y(1) = 2$

$\ell n|2| - 0 + 2 = c$

$c = 2 + \ell n2$

$\ell n|y| - \ell n|x| + xy = 2 + \ell n2$

$\ell n|x| = \ell n\left|\frac{y}{2}\right| - 2 + xy$

$|x| = \left|\frac{y}{2}\right|e^{xy-2}$

$2|x| = |y|e^{xy-2}$

$\alpha = 2$ $\beta = 2$ $\alpha + \beta = 4$

65. For a differentiable function $f: R \to R$, suppose $f'(x) = 3f(x) + \alpha$, where $\alpha \in R$, $f(0) = 1$ and $\lim_{x\to-\infty} f(x) = 7$.

Then $9f\left(-\log_e 3\right)$ is equal to _____. **[JEE(Main)April,2024]**

एक अवकलनीय फलन $f : R \to R$ के लिए, माना $f'(x) = 3f(x) + \alpha$, जहाँ $\alpha \in R$, $f(0) = 1$ तथा $\lim\limits_{x\to-\infty} f(x) = 7$ हैं। तो $9f\left(-\log_e 3\right)$ बराबर है _____. **[JEE(Main)April,2024]**

Ans. 61

Sol. $f'(x) = 3f(x) + \alpha$

$$\Rightarrow \frac{dy}{3y+\alpha} = dx$$

$$\Rightarrow \frac{1}{3}\ell n\left(3y+\alpha\right) = x + C$$

$$y(0) = 1 \Rightarrow C = \frac{1}{3}\ell n\left(3+\alpha\right)$$

$$\frac{1}{3}\ell n\left(\frac{3y+\alpha}{3+\alpha}\right) = x$$

$$\Rightarrow y = \frac{1}{3}\left(\left(3+\alpha\right)e^{3x} - \alpha\right) = f(x)$$

$$\lim_{x\to-\infty} f\left(x\right) = 7 \Rightarrow \alpha = -21$$

$$\Rightarrow f\left(x\right) = 7 - 6e^{3x}$$

$$9f\left(-\ell n3\right) = 61$$

Exercise J - A

1. If y (x) satisfies the differential equation y' – y tan x = 2x sec x and y(0) = 0, then : **[JEE-2012]**

   यदि y (x) अवकल समीकरण y' – y tan x = 2x sec x तथा y(0) = 0, को सन्तुष्ट करता है तो : **[JEE-2012]**

   (A) $y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{8\sqrt{2}}$ (B) $y'\left(\frac{\pi}{4}\right)=\frac{\pi^2}{18}$ (C) $y\left(\frac{\pi}{3}\right)=\frac{\pi^2}{9}$ (D) $y'\left(\frac{\pi}{3}\right)=\frac{4\pi}{3}+\frac{2\pi^2}{3\sqrt{3}}$

Ans. AD

Sol. $\frac{dy}{dx} - y\tan x = 2x\sec x$

I.F. $= e^{\int -\tan x\,dx} = \cos x$

$\therefore$ Equation reduces to

$y.\cos x = \int 2x.\sec x.\cos x dx$

$\Rightarrow y\cos x = x^2 + C$

$\because y(0) = 0 \Rightarrow 0 = 0 + C$

$\therefore y\cos x = x^2$

$\Rightarrow y(x) = x^2\sec x$

$\therefore y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{16}\sqrt{2}=\frac{\pi^2}{8\sqrt{2}}$ ($\therefore$ (A) is correct)

$y\left(\frac{\pi}{3}\right)=\frac{\pi^2}{9}.2=\frac{2\pi^2}{9}$ ($\therefore$ (C) is wrong)

Also $y'(x) = 2x\sec x + x^2\sec x\tan x$

$\Rightarrow y'\left(\frac{\pi}{4}\right)=\frac{\pi}{2}.\sqrt{2}+\frac{\pi^2\sqrt{2}}{16}$ ($\therefore$ (B) is wrong)

and $y'\left(\frac{\pi}{3}\right)=2.\frac{\pi}{3}.2+\frac{\pi^2}{9}.2.\sqrt{3}$

$=\frac{4\pi}{3}+\frac{2\pi^2}{3\sqrt{3}}$ ($\therefore$ (D) is correct)

2. A curve passes through the point $\left(1,\frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x}+\sec\left(\frac{y}{x}\right)$, $x>0$.

   Then the equation of the curve is : **[JEE(Adv.)2013]**

एक वक्र बिन्दु $\left(1,\frac{\pi}{6}\right)$ से गुजरता है। माना कि प्रत्येक बिन्दु (x, y) पर वक्र की प्रवणता $\frac{y}{x}$ + sec $\left(\frac{y}{x}\right)$, x > 0 है, तब वक्र का

l ehdj .kgS **[JEE(Adv.)2013]**

(A) $\sin\left(\frac{y}{x}\right)=\log x+\frac{1}{2}$ (B) $\cos ec\left(\frac{y}{x}\right)=\log x+2$ (C) $\sec\left(\frac{2y}{x}\right)=\log x+2$ (D) $\cos\left(\frac{2y}{x}\right)=\log x+\frac{1}{2}$

Ans. A

Sol. $\frac{dy}{dx}=\frac{y}{x}+\sec\frac{y}{x}$

Let $y=vx$

$\frac{dy}{dx}=v+x\frac{dv}{dx}$

$v+\frac{xdv}{dx}=v+\sec v$

$\cos v\,dv=\frac{dx}{x}$

$\sin v=\ell nx+c$

$\sin\left(\frac{y}{x}\right)=\ell nx+c$

$\because$ passing through $\left(1,\frac{\pi}{6}\right)$

$\Rightarrow \sin\frac{\pi}{6}=c\Rightarrow c=\frac{1}{2}$

$\therefore\ \sin\frac{y}{x}=\ell nx+\frac{1}{2}$

3. The function y = f(x) is the solution of the differential equation $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{x^4+2x}{\sqrt{1-x^2}}$ in (–1, 1) satisfying f(0) = 0.

Then : $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx$ is : **[JEE(Adv.)2014]**

यदि फलन y = f(x) अवकल समीकरण $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{x^4+2x}{\sqrt{1-x^2}}$ का अन्तराल (–1, 1) में हल है जहाँ f(0) = 0 है तो $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx$ होगा:

[JEE(Adv.)2014]

(A) $\frac{\pi}{3}-\frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3}-\frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6}-\frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6}-\frac{\sqrt{3}}{2}$

Ans. B

Sol. $\text{I.F.} = e^{\int \frac{x}{x^2-1}dx} = e^{\frac{1}{2}\int \frac{2x}{x^2-1}dx} = e^{\frac{1}{2}\ln|x^2-1|} = e^{\frac{1}{2}\ln(1-x^2)} = \sqrt{1-x^2}$

$$\therefore y\sqrt{1-x^2} = \int \frac{x^4+2x}{\sqrt{1-x^2}} \times \sqrt{1^2-x^2}dx + c$$

$$y\sqrt{1^2-x^2} = \frac{x^5}{5}+x^2+c$$

$x = 0, y = 0 \Rightarrow c = 0$

$$y = \frac{\frac{x^5}{5}+x^2}{\sqrt{1-x^2}}$$

$$\therefore I = \int_0^{\frac{\sqrt{3}}{2}} \left(\frac{\frac{x^5}{5}+x^2}{\sqrt{1-x^2}} + \frac{\frac{-x^5}{5}+x^2}{\sqrt{1-x^2}} \right) dx = 2\int_0^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}}dx$$

$x = \sin\theta$

$dx = \cos\theta d\theta$

$$= 2\int_0^{\frac{\pi}{3}} \frac{\sin^2\theta\cos\theta}{\cos\theta}d\theta$$

$$= \int_0^{\frac{\pi}{3}} (1-\cos 2\theta)d\theta = \left(\theta - \frac{1}{2}\sin 2\theta\right)\Big|_0^{\frac{\pi}{3}} = \frac{\pi}{3}-\frac{1}{2}\times\frac{\sqrt{3}}{2} = \frac{\pi}{3}-\frac{\sqrt{3}}{4}$$

4. Let y (x) be a solution of the differential equation $(1+e^x)y' + ye^x = 1$. If y (0) = 2, then which of the following statements is (are) true? **[JEE(Adv.)2015]**

(A) y (–4) = 0 (B) y (–2) = 0

(C) y(x) has a critical point in the interval (–1, 0) (D) y(x) has no critical point in the interval (–1, 0)

माना कि y (x) अवकल समीकरण $(1+e^x)y' + ye^x = 1$ का हल है। यदि y (0) = 2 तब निम्नलिखित कथनों में से कौन सा सही है ? **[JEE(Adv.)2015]**

(A) y (–4) = 0 (B) y (–2) = 0

(C) ल गद्धका एक क्रांतिक बिन्दु अंतराल (–1, 0) में है (D) ल गद्धका कोई भी क्रांतिक बिन्दु अंतराल (–1, 0) में नहीं है

Ans. AC

Sol. $\left(1+e^x\right)\frac{dy}{dx}+ye^x=1$

$$\frac{dy}{dx}+\frac{e^x}{1+e^x}y=\frac{1}{1+e^x}$$

$$\text{I.F.}=e^{\int\frac{e^x}{1+e^x}dx}=e^{\ln\left(1+e^x\right)}=1+e^x$$

complete solution

$$y.\left(1+e^x\right)=\int 1dx$$

$(1 + e^x)y = x + c$

$x = 0, y = 2 \quad \Rightarrow \quad c = 4$

$(1 + e^x)y = x + 4$

$$y=\frac{x+4}{e^x+1}$$

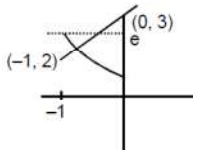
$x = -4, y = 0$

$$x=-2,\ y=\frac{2}{x^{-2}+1}$$

$$\frac{dy}{dx}=\frac{\left(e^x+1\right).1-\left(x+4\right)e^x}{\left(e^x+1\right)^2}$$

$$\frac{e^x\left(-x-3\right)+1}{\left(e^x+1\right)^2}$$

$\frac{dy}{dx}=0 \quad \Rightarrow \quad x+3=e^{-x}$



$$e^x=\frac{1}{x+3}$$

5. Consider the family of all circles whose centers lie on the straight line y = x. If this family of circles is represented by the differential equation Py'' + Qy' + 1 = 0, where P, Q are functions of x, y and y' (here $y'=\frac{dy}{dx}$, $y''=\frac{d^2y}{dx^2}$), then which of the following statements is (are) true? **[JEE(Adv.)2015]**

उन सभी वृत–कुल को विचार कीजिए जिनके केन्द्र सरल रेखा $y = x$ पर स्थित है। यदि इस वृत–कुल के सभी वृत, अवकल समीकरण $Py'' + Qy' + 1 = 0$ से निरूपित होते है, जहाँ P, Q इस प्रकार है कि वे x, y और y' के फलन है (यहाँ $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$), तब निम्नलिखित कथनों में से कौन कौन से सही है ? **[JEE(Adv.)2015]**

(A) $P = y + x$ (B) $P = y - x$

(C) $P + Q = 1 - x + y + y' + (y')^2$ (D) $P - Q = x + y - y' - (y')^2$

Ans. BC

Sol. $(x-\alpha)^2 + (y-\alpha)^2 = r^2$

$x^2 + y^2 - 2\alpha x - 2\alpha y + 2\alpha^2 - r^2 = 0$

$2x + 2yy' - 2\alpha - 2\alpha y' = 0$ ...(i)

$\Rightarrow \alpha = \frac{x + yy'}{1 + y'}$ ...(ii)

again diff. w.r.t.

$2 + 2(y')2 + 2yy'' - 2\alpha y'' = 0$

$\Rightarrow 1 + (y')^2 + yy' - \left(\frac{x + yy'}{1 + y'}\right) y'' = 0$

$\Rightarrow 1 + y' + (y')^2 + (y')^3 + yy'' + yy'y' - xy'' - yy'y'' = 0$

$\Rightarrow (y - x)y'' + (1 + y + (y')^2)y' + 1 = 0$

$\Rightarrow P = y - x,\ Q = 1 + y' + (y')^2$

Ans. (B, C)

Note : P and Q will not be unique function

$Py'' + Qy' + Ry' - Ry' + 1 = 0$

$\Rightarrow \frac{Py'}{1 - Ry''} + \frac{Qy'}{1 - Ry'} + 1 = 0$ Hence new P & Q can be obtained

So it can be a controversial problem.

6. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4)\frac{dy}{dx} - y^2 = 0,\ x > 0$, passes through the point (1, 3). Then the solution curve : **[JEE(Adv.)2016]**

(A) intersects $y = x + 2$ exactly at one point (B) intersects $y = x + 2$ exactly at two points

(C) intersects $y = (x + 2)^2$ (D) does NOT intersect $y = (x + 3)^2$

माना कि अवकल समीकरण $(x^2 + xy + 4x + 2y + 4)\frac{dy}{dx} - y^2 = 0,\ x > 0$, का एक हल वक्र बिंदु (1, 3) से गुजरता है। तब वह हल oØ : **[JEE(Adv.)2016]**

(A) लꣳ ग़ 2 को ठीक एक बिंदु पर प्रतिच्छेद करता है (B) लꣳ ग़ 2 को ठीक दो बिन्दुओं पर प्रतिच्छेदित करता है

(C) लꣳ ; ग़ 2द्धै को प्रतिच्छेदित करता है (D) लꣳ ; ग़ 3द्धै को प्रतिच्छेदित नहीं करता है

Ans. AD

Sol. $\left[(x+2)^2 + y(x+2)\right]\frac{dy}{dx} - y^2 = 0$

Put $x + 2 = t$

$\frac{dt}{dy} = \frac{t^2 + ty}{y^2}$ which is homogeneous differental equation

By solving it

$\ln y = \frac{-y}{x+2} + C$, $C = 1 + \ln 3$ since $y(1) = 3$

$\Rightarrow \ln\left(\frac{y}{3}\right) + \frac{y}{x+2} = 1$

Now : Verify the alternatives.

7. Let $f : (0,\infty) \to R$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0, \infty)$ and $f(1) \neq 1$.. Then :

[JEE(Adv.)2016]

(A) $\lim\limits_{x\to 0+} f'\left(\frac{1}{x}\right) = 1$ (B) $\lim\limits_{x\to 0+} x f\left(\frac{1}{x}\right) = 2$ (C) $\lim\limits_{x\to 0+} x^2 f'(x) = 0$ (D) $|f(x)| \leq 2$ for all $x \in (0, 2)$

माना कि $f : (0,\infty) \to R$ एक अवलकनीय फलन ऐसा है कि सभी $x \in (0, \infty)$ के लिए $f'(x) = 2 - \frac{f(x)}{x}$, और $f(1) \neq 1$ है। तब :

[JEE(Adv.)2016]

(A) $\lim\limits_{x\to 0+} f'\left(\frac{1}{x}\right) = 1$ (B) $\lim\limits_{x\to 0+} x f\left(\frac{1}{x}\right) = 2$ (C) $\lim\limits_{x\to 0+} x^2 f'(x) = 0$ (D) सभी $x \in (0, 2)$ के लिए $|f(x)| \leq 2$

Ans. A

Sol. $f'(x) = 2 - \frac{f(x)}{x}$ or $\frac{dy}{dx} + \frac{y}{x} = 2$

I.F. $= e^{\int \frac{1}{x} dx} = x$

$\Rightarrow y \cdot x = x^2 + c \Rightarrow f(x) = x + \frac{c}{x}$, $c \neq 0$ as $f(1) \neq 1$

(A) $\lim\limits_{x\to 0^+} f'\left(\frac{1}{x}\right) = \lim\limits_{x\to 0^+}\left(1-cx^2\right) = 1$

(B) $\lim\limits_{x\to 0^+}\left(xf\left(\frac{1}{x}\right)\right) = \lim\limits_{x\to 0^+}\left(1+cx^2\right) = 1$

(C) $\lim\limits_{x\to 0^+}\left(x^2 f'(x)\right) = \lim\limits_{x\to 0^+}\left(x^2\left(1-\frac{c}{x^2}\right)\right) = -c \pm 0$

(D) $\lim\limits_{x\to 0^+} f(x) = +\infty$ or $-\infty$, so for $c \pm 0$ f(x) is unbounded function for $x \in (0, 2)$

8. If y = y(x) satisfies the differential equation $8\sqrt{x}\left(\sqrt{9+\sqrt{x}}\right)dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx$, x > 0 and $y(0) = \sqrt{7}$, then $y(256) =$ **[JEE(Adv.)2017]**

यदि y = y(x) अवकलनीय समीकरण $8\sqrt{x}\left(\sqrt{9+\sqrt{x}}\right)dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx$, x > 0 को सन्तुष्ट करता है तथा $y(0) = \sqrt{7}$ है, तब $y(256) =$ **[JEE(Adv.)2017]**

(A) 80 (B) 9 (C) 16 (D) 3

Ans. D

Sol. $\frac{dy}{dx} = \frac{\left(\sqrt{4+\sqrt{9+x}}\right)^{-1}}{8\sqrt{x}\sqrt{9+\sqrt{x}}}$

$dy = \frac{1}{\sqrt{4+\sqrt{9+\sqrt{x}}}} \cdot \frac{1}{\sqrt{9+\sqrt{x}}} \cdot \frac{1}{8\sqrt{x}} dx$

Let $4+\sqrt{9+\sqrt{x}} = t \Rightarrow \frac{1}{2\sqrt{9+\sqrt{x}}} \times \frac{1}{2\sqrt{x}} dx = dt$

$\int dy = \int \frac{1}{\sqrt{t}} \cdot \frac{1}{2} dt$

$y = \sqrt{t} + c$

$y = \sqrt{4+\sqrt{9+\sqrt{x}}} + c$

at $x = 0,\ y = \sqrt{7}$

$\Rightarrow \sqrt{7} = \sqrt{7} + c \Rightarrow c = 0$

at $x = 256 \Rightarrow y = \sqrt{4 + \sqrt{9 + \sqrt{256}}} = 3$

9. If $f : R \to R$ is a differentiable function such that $f'(x) > 2f(x)$ for all $x \in R$, and $f(0) = 1$, then : **[JEE(Adv.)2017]**

(A) $f(x)$ is increasing in $(0, \infty)$ (B) $f'(x) < e^{2x}$ in $(0, \infty)$

(C) $f(x) > e^{2x}$ in $(0, \infty)$ (D) $f(x)$ is decreasing in $(0, \infty)$

यदि $f : R \to R$ इस प्रकार का अवकलनीय फलन है, कि सभी $x \in R$ के लिये $f'(x) > 2f(x)$ एवं $f(0) = 1$ है, तब : **[JEE(Adv.)2017]**

(A) $(0, \infty)$ में $f(x)$ वर्धमान है (B) $(0, \infty)$ में $f'(x) < e^{2x}$

(C) $(0, \infty)$ में $f(x) > e^{2x}$ (D) $(0, \infty)$ में $f(x)$ ह्रासमान है

Ans. AC

Sol. $f'(x) - 2f(x) > 0$

$\Rightarrow \frac{d}{dx}\left(f(x).e^{-2x}\right) > 0 \quad \Rightarrow \quad g(x) = f(x).e^{-2x}$ is an increasing function

for $x > 0$, $g(x) > g(0)$

$\Rightarrow f(x).e^{-2x} > 1 \quad \Rightarrow \quad f(x) > e^{2x}$

Now $f'(x) > 2f(x) > 2.e^{2x}$

10. Let $f : R \to R$ and $g : R \to R$ be two non-constant differentiable functions. If $f'(x) = \left(e^{(f(x)-g(x))}\right) g'(x)$ for all $x \in R$ and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$ (C) $g(1) > 1 - \log_e 2$ (D) $g(1) < 1 - \log_e 2$

माना कि $f : R \to R$ और $g : R \to R$ दो चर और अवकलनीय फलन हैं। यदि $f'(x) = \left(e^{(f(x)-g(x))}\right) g'(x)$ सभी $x \in R$ के लिये और $f(1) = g(2) = 1$, तब निम्नलिखित में से कौनसा (से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$ (C) $g(1) > 1 - \log_e 2$ (D) $g(1) < 1 - \log_e 2$

Ans. BC

Sol. $f'(x) = e^{f(x)-g(x)} g'(x) : f(1) = g(2) = 1$

$e^{-f(x)} = e^{-g(x)} + C$

$e^{-f(x)}.f'(x) = e^{-g(x)}.g'(x)$

$\int d\left(e^{-f(x)}\right) = \int d\left(e^{-g(x)}\right)$

$e^{-f(x)} = e^{-g(x)} + c$

$x = 1 \quad \frac{1}{e} = e^{-g(1)} + c$

$x = 2 \quad e^{-f(2)} = \frac{1}{e} + c$

$\therefore \quad g(1) > 1 - \ln 2$

$e^{-f(2)} = 2e^{-1} - e^{-g(1)}$

$e^{-f(2)} = 2e^{-1} - e^{-g(1)}$

$f(2) > 1 - \ln 2$

$e^{-1} - e^{-f(2)} = e^{-g(1)} - e^{-1} \Rightarrow \quad e^{-g(1)} + e^{-f(2)} = 2e^{-1}$

$e^{-g(1)} < 2e^{-1}$

$-g(1) < \ln 2 - 1$

11. Let $f: (0, \pi) \to R$ be a twice differentiable function such that $\lim\limits_{t \to x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x$ for all $x \in (0, \pi)$. If $f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}$, then which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$ (B) $f(x) < \frac{x^4}{6} - x^2$ for all $x \in (0, \pi)$

(C) There exists $\alpha \in (0, \pi)$ such that $f'(\alpha) = 0$ (D) $f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$

माना कि $f: (0, \pi) \to R$ एक ऐसा द्विअवकलनीय फलन है कि $\lim\limits_{t \to x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x$ सभी $x \in (0, \pi)$ के लिये। यदि $f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}$, तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$ (B) $f(x) < \frac{x^4}{6} - x^2$ सभी $x \in (0, \pi)$ के लिये

(C) एक ऐसे $\alpha \in (0, \pi)$ का अस्तित्व है जिसके लिये $f'(\alpha) = 0$ (D) $f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$

Ans. BCD

Sol. $\lim\limits_{t \to x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x$

$$\frac{f(x)\cos x - f'(x)\sin x}{\sin^2 x} = 1$$

$$-d\left(\frac{f(x)}{\sin x}\right) = 1$$

$$\frac{f(x)}{\sin x} = -x + c \qquad \because f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12} \Rightarrow c = 0 \Rightarrow f(x) = -x\sin x$$

(D) $f(x) + f''(x) = -2\cos x$

$$f\left(\frac{\pi}{2}\right) + f''\left(\frac{\pi}{2}\right) = 0$$

(A) $f\left(\frac{\pi}{4}\right) = \left(-\frac{\pi}{4\sqrt{2}}\right)$

(C) f(x) is continuous and differentiable and $f(0) = f(x) = 0$

Using LMVT $f'(c) = 0$ for some $x \in (0, \pi)$

(B) $g(x) = -x\sin x + x^2 - \frac{x^4}{6}$

$$g'(x) = f'(x) + 2x - \frac{2x^3}{3}$$

$g''(x) = f''(x) + 2x - 2x^2$

$g'''(x) = 3\sin x + x\cos x - 4x = 3(\sin x - x) + x(\cos x - 1)$

$\Rightarrow g'''(x) < 0 \quad \Rightarrow g''(x)$ is decreasing

for $x > 0 \qquad g''(x) < g''(0) \Rightarrow g''(x) < 0$

hence g'(x) is decreasing

for $x > 0 \qquad g'(x) < g'(0) \quad \Rightarrow g'(x) < 0$

hence $g(x) < 0$

for $x > 0 \qquad g(x) < g(0) \quad \Rightarrow g(x) < 0$

Hence $f(x) < \frac{x^4}{6} - x^2 \forall x \in (0, \pi)$

12. Let $f : R \to R$ be a differentiable function with f(0) = 0. If y = f(x) satisfies the differential equation $\frac{dy}{dx} = (2 + 5y)(5y - 2)$, then the value of $\lim_{x \to -\infty} f(x)$ is _____. **[JEE(Adv.)2018]**

माना कि $f : R \to R$ एक ऐसा अवकलनीय फलन है जिसके लिये $f(0) = 0$. यदि $y = f(x)$, अवकल समीकरण $\frac{dy}{dx} = (2 + 5y)(5y - 2)$ को सन्तुष्ट करता है, तब $\lim_{x \to -\infty} f(x)$ का मान है। **[JEE(Adv.)2018]**

Ans. 0.4

Sol. $\frac{dy}{dx} = (5y+2)(5y-2)$

$$\frac{1}{25}\int \frac{dy}{\left(y+\frac{2}{5}\right)\left(y-\frac{2}{5}\right)} = \int dx$$

$$\frac{1}{25}\cdot\frac{5}{4}\ln\left|\frac{y-\frac{2}{5}}{y+\frac{2}{5}}\right| = x+c$$

$$\frac{1}{20}\ln\left|\frac{5y-2}{5y+2}\right| = x+c$$

at $x = 0, y = 0 \Rightarrow c = 0$

Hence $\frac{2-5y}{2+5y} = e^{20x}$

$\frac{2-5y}{2+5y} = e^{20x}$, $\lim\limits_{x\to-\infty} e^{20x} = 0$ $\Rightarrow \lim\limits_{x\to-\infty} y = \frac{2}{5} = 0.4$

13. Let $f: R \to R$ be a differentiable function with $f(0) = 1$ and satisfying the equation $f(x+y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in R$. Then, the value of $\log_e(f(4))$ is _____. **[JEE(Adv.)2018]**

माना कि $f: R \to R$ एक ऐसा अवकलनीय फलन है जिसके लिये $f(0) = 1$, और जो सभी $x, y \in R$ के लिये समीकरण $f(x+y) = f(x)f'(y) + f'(x)f(y)$ को सन्तुष्ट करता है। तब $\log_e(f(4))$ का मान है। **[JEE(Adv.)2018]**

Ans. 2

Sol. $f(x+y) = f(x) \cdot f'(y) + f'(x) \cdot f(y)$

substituting $x = y = 0$, we get

$f(0) = 2f'(0)$ $\Rightarrow f'(0) = \frac{1}{2}$

Now substituting $y = 0$

$f(x) = f(x) \cdot f'(0) + f'(x) \cdot f(0)$

$\Rightarrow f'(x) = \frac{f(x)}{2}$

$\Rightarrow f'(x) = \lambda e^{\frac{x}{2}}$ $\Rightarrow f(x) = e^{\frac{x}{2}}$ (as $f(0) = 1$)

Now $\ln\left(f(x)\right)=\frac{x}{2} \quad \Rightarrow \ln(f(4))=2$

14. Let Γ denote a curve $y=y(x)$ which is in the first quadrant and let the point (1,0) on it.Let the tangent to Γ at a point P intersect the y-axis at Y_P. If PY_P has length 1 for each point P on Γ, then which of the following options is/ are correct : **[JEE(Adv.)2019]**

माना कि Γ एक वक्र $y=y(x)$ है जो प्रथम चतुर्थांश (first quadrant) में है और माना कि बिन्दु (1,0) उस पर स्थित है। माना कि Γ कें fc Uh qP पर खिचीं गयी स्पर्श रेखा (tangent) y-अक्ष को Y_p पर प्रतिच्छेद (intersect) करती है, यदि Γ के प्रत्येक बिन्दु P के लिए PY_P की लम्बाई 1 है। तब निम्न में से कौन सा (से) कथन सही है : **[JEE(Adv.)2019]**

(A) $xy'+\sqrt{1-x^2}=0$

(B) $y=-\log_e\left(\frac{1+\sqrt{1-x^2}}{x}\right)+\sqrt{1-x^2}$

(C) $y=\log_e\left(\frac{1+\sqrt{1-x^2}}{x}\right)-\sqrt{1-x^2}$

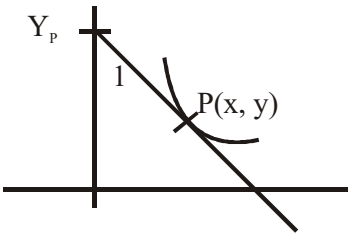
(D) $xy'-\sqrt{1-x^2}=0$

Ans. AC

Sol. Equation of tangent at P

$Y-y=m(X-x)$

$Y_p(0, y-mx)$



$PY_p=1$

$\sqrt{x^2+m^2x^2}=1$

$x^2+m^2x^2=1$

$m^2=\frac{1-x^2}{x^2}$

$m=\pm\sqrt{\frac{1-x^2}{x^2}}$

$\frac{dy}{dx}=\frac{\sqrt{1-x^2}}{x}$

$\int dy=\int\frac{\sqrt{1-x^2}}{x}dx$

Put $x=\sin\theta$

$y=\int\frac{\cos^2\theta}{\sin\theta}d\theta$

$\frac{dy}{dx}=-\frac{\sqrt{1-x^2}}{x}$

$\int dy=-\int\frac{\sqrt{1-x^2}}{x}dx$

Put $x=\sin\theta$

$y=\ell n\left(\frac{1+\sqrt{1-x^2}}{x}\right)-\sqrt{1-x^2}+c$

$\Rightarrow c=0$

$$= \int \frac{1-\sin^2\theta}{\sin\theta} d\theta$$

$$= \int (\cos ec\theta - \sin\theta) d\theta$$

$$= -\ell n(\cos ec\theta + \cot\theta) + \cos\theta + c$$

$$= -\ell n\left(\frac{1+\sqrt{1-x^2}}{x}\right) + \sqrt{1-x^2} + c$$

$\Rightarrow c = 0$

(Rejected because curve is lying in IV Quadrant)

15. Let b be a nonzero real number. Suppose $f: R \to R$ is a differentiable function such that $f(0) = 1$. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$ for all $x \in R$, then which of the following statements is/are TRUE?

[JEE(Adv.)2020]

(A) If $b > 0$, then f is an increasing function
(B) If $b < 0$, then f is a decreasing function
(C) $f(x) f(-x) = 1$ for all $x \in R$
(D) $f(x) - f(-x) = 0$ for all $x \in R$

माना कि b एक शून्येतर वास्तविक संख्या है। मान लिजिये कि $f: R \to R$ एक ऐसा अवकलनीय फलन है जिसके लिये $f(0) = 1$ है। यदि f का अवकलज f' समीकरण $f'(x) = \frac{f(x)}{b^2 + x^2}$ को सभी $x \in R$ के लिये संतुष्ट करता है, तब निम्न में से कौनसा (से) कथन सही g$(हैं)?

[JEE(Adv.)2020]

(A) यदि $b > 0$ है, तब f एक वर्धमान फलन है
(B) यदि $b < 0$ है, तब f एक ह्यसमान फलन है
(C) सभी $x \in R$ के लिए $f(x) f(-x) = 1$ है
(D) सभी $x \in R$ के लिए $f(x) - f(-x) = 0$ है

Ans. AC

Sol. f is non zero

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{1}{x^2 + b^2} dx$$

$$ln(f(x)) = \frac{1}{b}\tan^{-1}\frac{x}{b} + c$$

$$0 = 0 + c \quad \{\because f(0) = 1\}$$

$$f(x) = e^{\frac{1}{b}\tan^{-1}\frac{x}{b}}$$

$$f(-x) = e^{\frac{-1}{b}\tan^{-1}\frac{x}{b}}$$

(A) f(x) is positive function for all $x \in R$

$$f'(x) = \frac{f(x)}{x^2 + b^2}$$

$f'(x)$ is also positive function

so f(x) is increasing function

(d) $f(x) - f(-x) = e^{\frac{1}{b}\tan^{-1}\frac{x}{b}} - e^{\frac{-1}{b}\tan^{-1}\frac{x}{b}} \neq 0$

for all $x \in R$

(c) $f(x)\, f(-x) = 1$

$e^0 = 1$

$1 = 1$

16. For any real numbers α and β let $y_{\alpha,\beta}$, $x \in \mathbb{R}$, be the solution of the differential equation $\frac{dy}{dx} + \alpha y = xe^{\beta x}$, $y(1) = 1$. Let $S = \{y_{\alpha,\beta}(x) : \alpha, \beta \in \mathbb{R}\}$. Then which of the following functions belong(s) to the set S?

[JEE(Adv.)2021]

किन्हीं वास्तविक संख्याओं α व β के लिए, माना $y_{\alpha,\beta}$, $x \in \mathbb{R}$ अवकल समीकरण $\frac{dy}{dx} + \alpha y = xe^{\beta x}$, $y(1) = 1$ का हल है। माना $S = \{y_{\alpha,\beta}(x) : \alpha, \beta \in \mathbb{R}\}R$, तब निम्न में से कौनसा/कौनसे फलन समुच्चय S से सम्बन्धित है/है? **[JEE(Adv.)2021]**

(A) $f(x) = \frac{x^2}{2}e^{-x} + \left(e - \frac{1}{2}\right)e^{-x}$

(B) $f(x) = -\frac{x^2}{2}e^{-x} + \left(e + \frac{1}{2}\right)e^{-x}$

(C) $f(x) = \frac{e^x}{2}\left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right)e^{-x}$

(D) $f(x) = \frac{e^x}{2}\left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right)e^{-x}$

Ans. AC

Sol. $\frac{dy}{dx} + \alpha y = xe^{\beta x}$

I.F. $= e^{\int \alpha dx} = e^{\alpha x}$

$$y.e^{\alpha x} = \int e^{\alpha x}.\left(xe^{\beta x}\right)dx + C$$

$$y.e^{\alpha x} = \int \underset{I}{x}.\underset{II}{e^{(\alpha+\beta)x}}\,dx + C$$

$$y.e^{\alpha x} = \frac{x.e^{(\alpha+\beta)x}}{(\alpha+\beta)} - \int \frac{e^{(\alpha+\beta)x}}{\alpha+\beta}dx + C$$

$$y.e^{\alpha x} = e^{(\alpha+\beta)x}\left[\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2}\right] + C$$

$y(1) = 1$

$$\Rightarrow e^{\alpha} = e^{\alpha+\beta}\left[\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2}\right] + C$$

$$C = \left(1 - e^{\beta}\left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2}\right)\right)e^{\alpha}$$

$$y.e^{\alpha x} = e^{(\alpha+\beta)x}\left[\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2}\right] + e^{\alpha}\left(1 - e^{\beta}\left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2}\right)\right)$$

$$y = e^{\beta x}\left(\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2}\right) + \left(e^{\alpha} - e^{\alpha+\beta}\left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2}\right)\right)e^{-\alpha x}$$

for $\alpha = \beta = 1$

$$y = \frac{e^x}{2}\left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right)e^{-x}$$

for $\alpha = 1, \beta = -1$

$$\frac{dy}{dx} + y = x.e^{-x}$$

IF $e^{\int dx} = e^x$

$$y.e^x = \int x dx = \frac{x^2}{2} + C$$

$y(1) = 1$

$$|\vec{OB} \times \vec{OC}| = \frac{9}{2}$$

$$y.e^x = \frac{x^2}{2} + e - \frac{1}{2}$$

$$y = \frac{x^2}{2}.e^{-x} + \left(e - \frac{1}{2}\right)e^{-x}$$

As (A, C)

17. If y(x) is the solution of the differential equation $xdy - (y^2 - 4y)dx = 0$ for $x > 0$, $y(1) = 2$, and the slope of the curve y = y(x) is never zero, then the value of $10\,y(\sqrt{2})$ is _____. **[JEE(Adv.)2022]**

यदि $X > 0$ के लिए, y(x) अवकल समीकरण $xdy - (y^2 - 4y)dx = 0$ for $x > 0$, $y(1) = 2$, का हल है एवं वक्र y = y(x) का ढाल कभी भी शून्य नहीं होता है, तब $10\,y(\sqrt{2})$ का मान _____ है। **[JEE(Adv.)2022]**

Ans. 8

Sol. $xdy - \left(y^2 - 4y\right)dx = 0, x > 0$

$$\int \frac{dy}{y^2 - 4y} = \int \frac{dx}{x}$$

$$\int \left(\frac{1}{y-4} - \frac{1}{y}\right)dy = 4\int \frac{dx}{x}$$

$$\log_e |y-4| - \log_e |y| = 4\log_e x + \log_e c$$

$$\frac{|y-4|}{|y|} = cx^4 \xrightarrow{(1,2)} c = 1$$

$$|y-4| = |y|\,x^4$$

C–1 and C–2

$y - 4 = yx^4$ $\qquad$ $y - 4 = -yx^4$

$y = \frac{4}{1-x^4}$ $\qquad$ $y = \frac{4}{1+x^4}$

y(1)=ND (rejected) $\qquad$ $y(1) = 2$

$$y(\sqrt{2}) = \frac{4}{5} \Rightarrow 10y(\sqrt{2}) = 8$$

18. For $x \in R$, let the function y(x) be the solution of the differential equation $\frac{dy}{dx} + 12y = \cos\left(\frac{\pi}{12}x\right), y(0) = 0$. Then, which of the following statements is/are TRUE? **[JEE(Adv.)2022]**

(A) y(x) is an increasing function

(B) y(x) is decreasing function

(C) There exists a real number β such that the line $y = \beta$ intersects the curve y = y(x) at infinitely many points

(D) y(x) is a periodic function

माना कि $x \in R$ के लिए, फलन y(x) अवकल समीकरण $\frac{dy}{dx} + 12y = \cos\left(\frac{\pi}{12}x\right), y(0) = 0$ का हल है। तब निम्न कथनों में से कौनसा सत्य हैं? **[JEE(Adv.)2022]**

(A) ल गद्धएक वर्धमान फलन है

(B) ल गद्धएक ह्यसमान फलन है

(C) एक ऐसी वास्तविक संख्या β है जिसके लिए रेखा $y = \beta$, वक्र $y = y(x)$ को अनंत बिंदुओं पर प्रतिच्छेदित करती है

(D) ल गद्धएक आवर्ती फलन है

Ans. C

Sol. $\frac{dy}{dx} + 12y = \cos\left(\frac{\pi x}{12}\right)$

$$\text{I.F.} = e^{12x} \Rightarrow y.e^{12x} = \int e^{12x} \cdot \cos\left(\frac{\pi x}{12}\right) dx + C$$

$$\Rightarrow y \cdot e^{12x} = \frac{e^{12x}}{12^2 + \left(\frac{\pi}{12}\right)^2}\left[12\cos\frac{\pi x}{12} + \frac{\pi}{12}\sin\left(\frac{\pi x}{12}\right)\right.$$

$$\because y(0) = 0 \Rightarrow C = -\frac{12}{12^2 + \left(\frac{\pi}{12}\right)^2}$$

$$\text{So } y = \frac{1}{\lambda}[\underbrace{12\cos\left(\frac{\pi x}{12}\right) + \frac{\pi}{12}\sin\left(\frac{\pi x}{12}\right)}_{f_1(x)} - 12e^{-12x}]$$

$$\frac{dy}{dx} = \frac{1}{\lambda}[\underbrace{-\pi\sin\left(\frac{\pi x}{12}\right) + \frac{\pi^2}{12^2}\cos\left(\frac{\pi x}{12}\right)}_{f_2(x)} + 12e^{-12x}]$$

When x is large then $12e^{-12x}$ tends to zero.

But $f_2(x)$ varies in $\left[-\sqrt{\pi^2 + \left(\frac{\pi}{12}\right)^4}, \sqrt{\pi^2 + \left(\frac{\pi}{12}\right)^4}\right]$

Hence $\frac{dy}{dx}$ is changing its sign.

So y(x) is non monotonic for all real number.

Also when x is very large then again $-12e^{-12x}$ is almost zero but $f_1(x)$ is periodic, so there exist some β for which $y = \beta$ intersect $y = y(x)$ at infinitely many points.

19. For $x \in R$, let y(x) be a solution of the differential equation $\left(x^2 - 5\right)\frac{dy}{dx} - 2xy = -2x\left(x^2 - 5\right)^2$ such that $y(2) = 7$. Then the maximum value of the function y(x) is_____. **[JEE(Adv.)2023]**

माना कि y(x), $x \in R$ निम्नलिखित अवकल समीकरण (differential equation) $\left(x^2-5\right)\frac{dy}{dx}-2xy=-2x\left(x^2-5\right)^2$ का एक , s kgy gSft 1 d sfy , y(2) = 7 है। तब फलन y(x) का महत्तम मान (maximum value) है _____ **[JEE(Adv.)2023]**

Ans. 16

Sol. $\frac{dy}{dx}-\frac{2x}{\left(x^2-5\right)}y=-2x\left(x^2-5\right)$

$\text{If} = e^{-\int\frac{2x}{x^2-5}dx} = e^{-\ell n\left|x^2-5\right|} = \frac{1}{\left|x^2-5\right|}$

Solution of DE

$$y.\frac{1}{\left|x^2-5\right|}=\int\frac{-2x\left(x^2-5\right)}{\left|x^2-5\right|}dx+C$$

$$\Rightarrow y.\frac{1}{\left|x^2-5\right|}=-x^2\left(\frac{x^2-5}{\left|x^2-5\right|}\right)+C$$

$y(2) = 7$

$\Rightarrow 7 = 4 + C \quad \Rightarrow \boxed{C = 3}$

So $y = -x^2(x^2 - 5) + 3\,|x^2 - 5|$

if $-\sqrt{5} < x < \sqrt{5}$	If $\|x\| > \sqrt{5}$
$y = (5 - x^2)(x^2 + 3)$	$y = (x^2 - 5)(-x^2 + 3)$
$= -x^4 + 2x^2 + 15$	$= -x^4 + 8x^2 - 15$
$= -[x^4 - 2x^2 - 15]$	$= -[x^4 - 8x^2 + 15]$
$= -[(x^2 - 1)^2 - 16]$	$= -[(x^2 - 4)^2 - 1]$
$= 16 - (x^2 - 1)^2$	$= 1 - (x^2 - 4)^2$

So max at $x = \pm 1$

and $\boxed{y_{max} = 16}$ $\boxed{\text{Ans} = 16}$

20. Let f(x) be a continuously differentiable function on the interval $(0, \infty)$ such that $f(1) = 2$ and $\lim\limits_{t\to x}\frac{t^{10}f(x)-x^{10}f(t)}{t^9-x^9}=1$ for each $x > 0$. Then, for all $x > 0$, f(x) is equal to : **[JEE(Adv.)2024]**

माना कि f(x), अंतराल (interval) $(0, \infty)$ में सांतत्य रूप से अवकलनीय (continuously differentiable) एक इस प्रकार का फलन (function) है कि $f(1) = 2$, तथा प्रत्येक $x > 0$ के लिए $\lim\limits_{t\to x} \frac{t^{10}f(x) - x^{10}f(t)}{t^9 - x^9} = 1$ है। तब सभी $x > 0$ के लिए $f(x)$ बराबर है :

[JEE(Adv.)2024]

(A) $\frac{31}{11x} - \frac{9}{11}x^{10}$ (B) $\frac{9}{11x} + \frac{13}{11}x^{10}$ (C) $\frac{-9}{11x} + \frac{31}{11}x^{10}$ (D) $\frac{13}{11x} + \frac{9}{11}x^{10}$

Ans. B

Sol. $\lim\limits_{t\to x} \frac{t^{10}f(x) - x^{10}f(t)}{t^9 - x^9} = 1 \qquad (x > 0)$

Apply L-H rule

$\lim\limits_{t\to x} \frac{10t^9 f(x) - x^{10}f'(t)}{9t^8} = 1$

$\Rightarrow 10x\, f(x) - x^2 f'(x) = 9$

$x^2 \frac{dy}{dx} - 10xy = -9$

$\frac{dy}{dx} - \frac{10}{x} = -\frac{9}{x^2}$

$I.F. = e^{\int -\frac{10}{x}dx} = \frac{1}{x^{10}}$

$y \times \frac{1}{x^{10}} = \frac{-9}{-11(x^{11})} + C$

$\frac{y}{x^{10}} = \frac{9}{11x^{11}} + C$

$f(1) = 2$ put $x = 1, y = 2$

$2 = \frac{9}{11} + C$

$C = \frac{13}{11}$

$\frac{y}{x^{10}} = \frac{9}{11x^{11}} + \frac{13}{11}$

$y = \frac{9}{11x} + \frac{13}{11}x^{10}$

21. For all $x > 0$, let $y_1(x)$, $y_2(x)$ and $y_3(x)$ be the functions satisfying

$$\frac{dy_1}{dx} - (\sin x)^2 y_1 = 0,\ y_1(1) = 5,$$

$$\frac{dy_2}{dx} - (\cos x)^2 y_2 = 0, y_2(1) = \frac{1}{3},$$

$$\frac{dy_3}{dx} - \left(\frac{2 - x^3}{x^3}\right) y_3 = 0, y_3(1) = \frac{3}{5e},$$

respectively. Then

$$\lim_{x \to 0} \frac{y_1(x) y_2(x) y_3(x) + 2x}{e^{3x} \sin x}$$

is equal to _____. **[JEE(Adv.)2025]**

सभी $x > 0$ के लिए, मान लीजिए कि $y_1(x)$, $y_2(x)$ और $y_3(x)$ इस प्रकार के फलन हैं जो क्रमशः

$$\frac{dy_1}{dx} - (\sin x)^2 y_1 = 0,\ y_1(1) = 5,$$

$$\frac{dy_2}{dx} - (\cos x)^2 y_2 = 0, y_2(1) = \frac{1}{3},$$

$$\frac{dy_3}{dx} - \left(\frac{2 - x^3}{x^3}\right) y_3 = 0, y_3(1) = \frac{3}{5e},$$

को संतुष्ट करते हैं। तब

$$\lim_{x \to 0} \frac{y_1(x) y_2(x) y_3(x) + 2x}{e^{3x} \sin x}$$

बराबर _____ है। **[JEE(Adv.)2025]**

Ans. 2

Sol. $\frac{dy_1}{dx} - (\sin x)^2 y_1 = 0$

$P = -\sin^2 x$; $Q = 0$

I.F. $= e^{-\int \sin^2 x dx}$

$= e^{-\int \frac{1-\cos 2x}{2} dx}$

$= e^{-\frac{x}{2}+\frac{\sin 2x}{4}}$

$y_1 . = e^{-\frac{x}{2}+\frac{\sin 2x}{4}} = c_1$

$y_1 = c_1 e^{\frac{x}{2}-\frac{\sin 2x}{4}}$

$5 = c_1 e^{\frac{1}{2}-\frac{\sin 2}{4}}$

$c_1 = 5e^{\frac{\sin 2}{4}-\frac{1}{2}}$

$y_1 = 5e^{\frac{\sin 2-2}{4}} . e^{\frac{x}{2}-\frac{\sin 2x}{4}}$ .....(A)

$\frac{dy_2}{dx} - \cos^2 x \cdot y_2 = 0$

$P = -\cos^2 x \quad Q = 0$

I.F. $= e^{-\int \cos^2 x dx}$

$= e^{-\int \frac{1+\cos 2x}{2} dx}$

$= e^{-\frac{x}{2}-\frac{\sin 2x}{4}}$

$= e^{-\frac{2x-\sin 2x}{4}}$

$y_2 e^{-\frac{2x-\sin 2x}{4}} = c_2$

$\frac{1}{3} e^{\frac{-2-\sin 2}{4}} = c_2$

$\therefore y_2 = \frac{1}{3} e^{-\frac{2+\sin 2}{4}} \cdot e^{\frac{2x+\sin 2x}{4}}$ .....(B)

$\frac{dy_3}{dx} - \left(\frac{2-x^3}{x^3} \right) y_3 = 0$

$P = \frac{x^3-2}{x^3} \quad Q = 0$

$= 1 - \frac{2}{x^3}$

$I.F = e^{\int\left(1-\frac{2}{x^3}\right)dx}$

$= e^{x+\frac{2.x^{-2}}{+2}}$

$= e^{x+\frac{1}{x^2}}$

$y \cdot e^{x+\frac{1}{x^2}} = c_3$

$\frac{3}{5e}e^2 = c_3$

$\therefore y_3 = \frac{3}{5}e^{\left(x+\frac{1}{x^2}\right)}$

$$\lim_{x\to 0}\frac{5e^{\frac{\sin 2-2}{4}} \cdot e^{\frac{x}{2}-\frac{\sin 2x}{4}} \cdot \frac{1}{3}e^{-\frac{2-\sin 2}{4}} \cdot e^{\frac{2x+\sin 2x}{4}} \cdot \frac{3}{5}e \cdot e^{-x-\frac{1}{x^2}} + 2x}{\sin x}$$

$$\lim_{x\to 0}\frac{e^{-1+1} \cdot e^{x-x-\frac{1}{x^2}} + 2x}{\sin x}$$

$$\lim_{x\to 0}\frac{e^{-\frac{1}{x^2}} + 2x}{\sin x}$$

$\Rightarrow 2$ Ans.

22. Let y(x) be the solution of the differential equation $x^2\frac{dy}{dx} + xy = x^2 + y^2, \quad x > \frac{1}{e}$, satisfying $y(1) = 0$.

Then the value of $2\frac{(y(e))^2}{y\left(e^2\right)}$ is _____. **[JEE(Adv.)2025]**

मान लीजिए y(x), अवकल समीकरण (differential equation) $x^2\frac{dy}{dx} + xy = x^2 + y^2, \quad x > \frac{1}{e}$, का वह हल (solution) है जो

$y(1) = 0$ को संतुष्ट करता है। तब $2\frac{(y(e))^2}{y\left(e^2\right)}$ का मान _____ है। **[JEE(Adv.)2025]**

Ans. 0.75

Sol. Put $y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

So, $x^2\left(v + \frac{xdv}{dx}\right) + x^2v = x^2\left(1 + v^2\right)$

$$v + \frac{xdv}{dx} + v = 1 + v^2$$

$$\frac{xdv}{dx} = 1 + v^2 - 2v$$

$$\int \frac{dv}{(v-1)^2} = \int \frac{dx}{x}$$

$$-\frac{1}{v-1} = \ell n|x| + C$$

$$\frac{x}{x-y} = \ell n|x| + C$$

$y(1) = 0$

$\Rightarrow C = 1$

Now, $\frac{x}{x-y} = \ell nx + 1$

$$\frac{x}{x-y} = \ell nex$$

$$y(e) = \frac{e}{2} \text{ \& } y\left(e^2\right) = \frac{2e^2}{3}$$

$$\therefore \frac{2\left(y(e)\right)^2}{y\left(e^2\right)} = \frac{\frac{2e^2}{4}}{\frac{2e^2}{3}} = \frac{3}{4} = 0.75$$

23. Let $y : (-\infty, \infty) \to (0, \infty)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{e^{5x}y^3 + y^3}{e^x + e^x y^4}$, satisfying $y(0) = \frac{1}{\sqrt{2}}$. Then the value of $y(\log_e 2)$ is : **[JEE(Adv.)2026]**

मान लीजिए कि $y : (-\infty, \infty) \to (0, \infty)$, अवकल समीकरण $\frac{dy}{dx} = \frac{e^{5x}y^3 + y^3}{e^x + e^x y^4}$ का वह हल है जो $y(0) = \frac{1}{\sqrt{2}}$ को संतुष्ट करता g§r c $y(\log_e 2)$ का मान होगा : **[JEE(Adv.)2026]**

(A) $\sqrt{\frac{5+\sqrt{35}}{2}}$ (B) $\sqrt{\frac{7+\sqrt{53}}{2}}$ (C) $\frac{7+\sqrt{53}}{2}$ (D) $\frac{5+\sqrt{35}}{2}$

Ans. B

Sol. $\frac{dy}{dx} = \frac{\left(e^{5x}+1\right)y^3}{e^x\left(1+y^4\right)}$

$$\int \frac{1+y^4}{y^3}dy = \int \frac{e^{5x}+1}{e^x}dx$$

$$\Rightarrow \frac{-1}{2y^2} + \frac{y^2}{2} = \frac{e^{4x}}{4} - e^{-x} + C$$

$$y(0) = \frac{1}{\sqrt{2}} \Rightarrow -1 + \frac{1}{4} = \frac{1}{4} - 1 + C \Rightarrow C = 0$$

$$-\frac{1}{2y^2} + \frac{y^2}{2} = \frac{e^{4x}}{4} - \frac{1}{e^x}$$

$$x = \log_e^2 \Rightarrow \frac{-1}{2y^2} + \frac{y^2}{2} = \frac{e^{4\log_e 2}}{4} - \frac{1}{e^{\log_e 2}}$$

$$-\frac{1}{2y^2} + \frac{y^2}{2} = 4 - \frac{1}{2} = \frac{7}{2}$$

$-1 + y^4 = 7y^2$

$y^x - 7y^2 - 1 = 0$

$$y^2 = \frac{7 \pm \sqrt{49+4}}{2}$$

$$y^2 = \frac{7+\sqrt{53}}{2}$$

$$y = \sqrt{\frac{7+\sqrt{53}}{2}}$$

24. Let $y = f(x)$ be the real valued function defined on the interval $(0, \infty)$, satisfying $y(1) = 0$ and the differential equation

$$x\frac{dy}{dx} = y - x^3$$

Then which of the following statements is (are) **TRUE** : **[JEE(Adv.)2026]**

(A) The function f has a local minimum at $x = \frac{1}{\sqrt{3}}$

(B) The function f has a local maximum at $x = \frac{1}{\sqrt{3}}$

(C) The function f is increasing in the interval $(1, 2)$

(D) If $g(x) = 4x^3 - 5x^2 + \frac{3}{2}x$ for $x > 0$, then the number of elements in the set $\{x \in (0,\infty) : f(x) = g(x)\}$ is 2

मान लीजिए कि $y = f(x)$, अंतराल $(0, \infty)$ पर परिभाषित वह वास्तविक मान फलन है, जो $y(1) = 0$ और अवकल समीकरण

$$x\frac{dy}{dx} = y - x^3$$ को संतुष्ट करता है।

r c fuEufy f[kr d Fkuksa esa sdlkSu k(से) सत्य है (हैं) : **[JEE(Adv.)2026]**

(A) फलन f का, $x = \frac{1}{\sqrt{3}}$ पर एक स्थानीय निम्नतम है

(B) फलन f का, $x = \frac{1}{\sqrt{3}}$ पर एक स्थानीय उच्चतम है

(C) फलन f, अंतराल $(1, 2)$ में वर्धमान है

(D) यदि $x > 0$ के लिए $g(x) = 4x^3 - 5x^2 + \frac{3}{2}x$ है, तब समुच्चय $\{x \in (0,\infty) : f(x) = g(x)\}$ में अवयवों की संख्या 2 है।

Ans. BD

Sol. $\frac{xdy}{dx} = y - x^3$

$$\frac{dy}{dx} - \frac{1}{x}y = -x^2$$

$$\text{I.F.} = e^{\int -\frac{1}{x}dx} \Rightarrow e^{-\ell n(x)} \Rightarrow \frac{1}{x}$$

Solution $$y \times \frac{1}{x} = \int \frac{1}{x}\left(-x^2\right)dx + C$$

$$\frac{y}{x} = -\frac{x^2}{2} + C$$

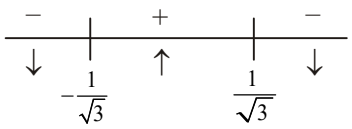
$$y = -\frac{x^3}{2} + Cx$$

$$y(1) = 0 \Rightarrow c = \frac{1}{2}$$

$$y = -\frac{x^3}{2} + \frac{x}{2}$$

$$\frac{dy}{dx} = -\frac{3x^2}{2} + \frac{1}{2}$$

$$\Rightarrow -\frac{3}{2}\left(x - \frac{1}{\sqrt{3}}\right)\left(x + \frac{1}{\sqrt{3}}\right)$$



Local maxima at $x = \frac{1}{\sqrt{3}}$

Local minima at $x = -\frac{1}{\sqrt{3}}$

For f(x) = g(x)

$$-\frac{x^3}{2} + \frac{x}{2} = 4x^3 - 5x^2 + \frac{3}{2}x$$

$9x^3 - 10x^2 + 2x = 0$

$x(9x^2 - 10x + 2) = 0$

$$x = 0,\ x = \frac{5 \pm \sqrt{7}}{2}$$

For x > 0, only 2 values of x.

CHALLENGER EXERCISES
Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. A function $y = f(x)$ satisfies the differential equation $f(x) \cdot \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$ with initial condition $y(0) = 0$. The value of $f\left(\frac{\pi}{6}\right)$ is equal to :

फलन $y = f(x)$, अवकल समीकरण $f(x) \cdot \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$ को सन्तुष्ट करता है। जबकि प्रारम्भिक अवस्था esa $y(0) = 0$. तो $f\left(\frac{\pi}{6}\right)$ का मान होगा :

(A) 1/5 (B) 3/5 (C) 4/5 (D) 2/5

Ans. D

Sol. $y \sin 2x - \cos x + (1 + \sin^2 x)\frac{dy}{dx} = 0$ where $y = f(x)$

$$\frac{dy}{dx} + \left(\frac{\sin 2x}{1+\sin^2 x}\right)y = \frac{\cos x}{1+\sin^2 x}$$

I.F. $= e^{\int \frac{\sin 2x}{1+\sin^2 x} dx} = e^{\int \frac{dt}{t}} = e^{ln(1+\sin^2 x)} = 1 + \sin^2 x$ (by putting $1 + \sin^2 x = t$)

$y(1 + \sin^2 x) = \int \cos x \, dx$

$y(1 + \sin^2 x) = \sin x + C$; $(y(0) = 0) \Rightarrow C = 0$

hence, $y = \frac{\sin x}{1+\sin^2 x}$

$y\left(\frac{\pi}{6}\right) = \frac{2}{5}$

2. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y, where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole. If t is measured in minutes and $k = \frac{1}{15}$ then the time to drain the tank if the water is 4 meter deep to start with is :

उर्ध्वाधर बेलनाकार टंकी से, उसके आधार पर स्थित छेद (hole) से, पानी बाहर निकल रहा है। यह ज्ञात है कि टंकी में पानी का तल गिरने की दर किसी भी समय पर पानी के गहराई y के वर्ग मूल के समानुपाती है। जहाँ समानुपाती स्थिरांक $k > 0$ गुरूत्वीय त्वरण का तथा छेद (hole) के आकार पर निर्भर करता है। यदि $k = \frac{1}{15}$ तथा t मिनट में हो तो टंकी के खाली होने में लगा समय कितना होगा जहाँ शुरू में VkSj he sa ikuh dk xgjkbZ 4 मीटर का है :

(A) 30 min (B) 45 min (C) 60 min (D) 80 min

Ans. C

Sol. $\frac{dy}{dt} = -k\sqrt{y}$; when $t = 0$; $y = 4$

$$\int_4^0 \frac{dy}{\sqrt{y}} = -k\int_0^t dt$$

$$2\sqrt{y}\Big|_4^0 = -kt = -\frac{t}{15}$$

$$0 - 4 = -\frac{t}{15} \Rightarrow \quad t = 60 \text{ minutes} \Rightarrow (C)$$

3. The equation of the trajectories which is orthogonal to the family of curves $\cos y = ae^{-x}$ is :

(A) $\sin y = c\,e^{x}$ (B) $\cos y = c\,e^{x}$ (C) $\sin y = c\,e^{-x}$ (D) None of these

वक्र $\cos y = ae^{-x}$ के लम्बवतीय पथ का समीकरण है :

(A) $\sin y = c\,e^{x}$ (B) $\cos y = c\,e^{x}$ (C) $\sin y = c\,e^{-x}$ (D) कोई नहीं

Ans. C

Sol. $+\sin y \frac{dy}{dx} = +ae^{-x}$

$$\sin y\frac{dy}{dx} = \cos y \qquad \Rightarrow \qquad \frac{dy}{dx} = \cot y$$

equation to the orthogonal trajectory will be obtained by solving the differential equation

$$-\frac{dx}{dy} = \cot y$$

$c - x = \ln(\sin y) \Rightarrow \sin y = c\,e^{-x}$

4. A function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} - y = \cos x - \sin x$, with initial condition that y is bounded when $x \to \infty$. The area enclosed by $y = f(x)$, $y = \cos x$ and the y-axis in the 1st quadrant :

फलन $y = f(x)$, अवकल समीकरण $\frac{dy}{dx} - y = \cos x - \sin x$, को सन्तुष्ट करता है, तथा दिया है कि $x \to \infty$ पर परिबद्ध है, तो प्रथम

pr dqk=ke a $y = f(x)$, $y = \cos x$ तथा y-अक्ष द्वारा घिरा क्षेत्रफल होगा :

(A) $\sqrt{2}-1$ (B) $\sqrt{2}$ (C) 1 (D) $\frac{1}{\sqrt{2}}$

Ans. A

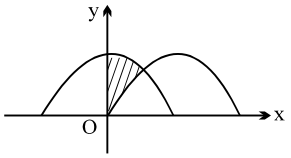
Sol. I.F. $= e^{-x}$

$\therefore \quad ye^{-x} = \int e^{-x}(\cos x - \sin x)\,dx \qquad$ put $\quad -x = t$

$= -\int e^{t}(\cos t + \sin t)\,dt$

$= -e^{t}\sin t + c$

$y\,e^{-x} = e^{-x}\sin x + c$



since y is bounded when $x \to \infty \;\Rightarrow\; c = 0$

$\therefore \quad y = \sin x$

Area $= \int_0^{\pi/4}(\cos x - \sin x)\,dx = \sqrt{2}-1 \;\Rightarrow$ (A)

5. A function $f(x)$ satisfying $\int_0^1 f(tx)\,dt = n f(x)$, where $x > 0$, is :

यदि फलन $f(x)$ इस प्रकार है कि $\int_0^1 f(tx)\,dt = n f(x)$ जहाँ $x > 0$ है, तो :

(A) $f(x) = c \cdot x^{\frac{1-n}{n}}$ (B) $f(x) = c \cdot x^{\frac{n}{n-1}}$ (C) $f(x) = c \cdot x^{\frac{1}{n}}$ (D) $f(x) = c \cdot x^{(1-n)}$

Ans. A

Sol. $\int_0^1 f(t\,x)\,dt = n \cdot f(x)$

put $t\,x = y \;\Rightarrow dt = \frac{1}{x}\,dy$

$\therefore \quad \frac{1}{x}\int_0^x f(y)\,dy = n\,f(x)$

$\therefore \quad \int_0^x f(y)\,dy = x \cdot n \cdot f(x)$

Differentiating

$f(x) = n\,[\,f(x) + x\,f'(x)\,]$

$f(x)\,(1-n) = n\,x\,f'(x)$

$$\therefore \quad \frac{f'(x)}{f(x)} = \frac{1-n}{n\,x}$$

Integrating $\quad \ln f(x) = \left(\frac{1-n}{n}\right) \ln c\,x = \ln (cx)^{\frac{1-n}{n}}$

$$\therefore \quad f(x) = c.x^{\frac{1-n}{n}}$$

6. A curve $y = f(x)$ passing through the point $\left(1, \frac{1}{\sqrt{e}}\right)$ satisfies the differential equation $\frac{dy}{dx} + x\,e^{-\frac{x^2}{2}} = 0$. Then which of the following does not hold good?

(A) $f(x)$ is differentiable at $x = 0$

(B) $f(x)$ is symmetric w.r.t. the origin

(C) $f(x)$ is increasing for $x < 0$ and decreasing for $x > 0$

(D) $f(x)$ has two inflection points

वक्र $y = f(x)$ बिन्दु $\left(1, \frac{1}{\sqrt{e}}\right)$ से गुजरता है, तथा अवकल समीकरण $\frac{dy}{dx} + x\,e^{-\frac{x^2}{2}} = 0$ को सन्तुष्ट करता है, तब निम्नलिखित में से d kSl kd Hkh lR; ugÈ gS%

(A) ग ¾ 0 पर फलन $f(x)$ अवकलनीय है

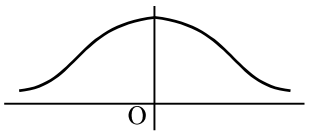
(B) फलन $f(x)$ मूलबिन्दु के सापेक्ष सममित है

(C) फलन $f(x)$, $x < 0$ के लिए वर्धमान तथा $x > 0$ के लिए ह्रासमान है

(D) फलन $f(x)$ के दो नति परिवर्तन बिन्दु हैं

Ans. B

Sol. $y = f(x) = e^{-\frac{x^2}{2}}$. The graph is as shown



7. A curve passing through (2, 3) and satisfying the differential equation $\int_0^x t\,y(t)\,dt = x^2 y(x),\ (x > 0)$ is :

एक वक्र बिन्दु (2, 3) से गुजरता है तथा अवकल समीकरण $\int_0^x t\,y(t)\,dt = x^2 y(x),\ (x > 0),$ को संतुष्ट करता है :

(A) $x^2 + y^2 = 13$ (B) $y^2 = \frac{9}{2}x$ (C) $\frac{x^2}{8} + \frac{y^2}{18} = 1$ (D) $xy = 6$

Ans. D

Sol. differentiate $x\,y(x) = x^2y'(x) + 2xy(x)$

or $x\,y(x) + x^2\,y'(x) = 0$

$$x\frac{dy}{dx} + y = 0$$

$$\ln y + \ln x = \ln c$$

$xy = c \Rightarrow$ (D)

8. A function $y = f(x)$ satisfies $(x+1).f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{(x+1)}$, $\forall x > -1$ If $f(0) = 5$, then $f(x)$ is :

यदि फलन $y = f(x)$ सम्बन्ध $(x+1).f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{(x+1)}$, $\forall x > -1$ को सन्तुष्ट करता है तथा $f(0) = 5$, तो $f(x)$ होगा :

(A) $\left(\frac{3x+5}{x+1}\right).e^{x^2}$ (B) $\left(\frac{6x+5}{x+1}\right).e^{x^2}$ (C) $\left(\frac{6x+5}{(x+1)^2}\right).e^{x^2}$ (D) $\left(\frac{5-6x}{x+1}\right).e^{x^2}$

Ans. B

Sol. $f'(x) - \frac{2x(x+1)}{x+1}f(x) = \frac{e^{x^2}}{(x+1)^2}$

I. F. $= e^{\int -2x\,dx} = e^{-x^2}$

$\therefore$ $f(x).e^{-x^2} = \int \frac{dx}{(x+1)^2} \Rightarrow$ $f(x).e^{-x^2} = -\frac{1}{x+1} + C$

at $x = 0$, $f(0) = 5 \Rightarrow C = 6$

$\therefore$ $f(x) = \left(\frac{6x+5}{x+1}\right).e^{x^2}$

9. The solution of primitive integral equation $(x^2+y^2)dy = xy\,dx$, is $y = y(x)$. If $y(1) = 1$ and $y(x_0) = e$, then x_0 is :

अवकल समीकरण $(x^2+y^2)dy = xy\,dx$ का हल $y = y(x)$, यदि $y(1) = 1$ एवं $y(x_0) = e$, को संतुष्ट करता हो, तो x_0 का मान है :

(A) $\sqrt{2(e^2-1)}$ (B) $\sqrt{2(e^2+1)}$ (C) $\sqrt{3}\,e$ (D) $\sqrt{\frac{e^2+1}{2}}$

Ans. C

Sol. $\frac{dy}{dx}=\frac{xy}{x^2+y^2}$ This is homogeneous so put $y=vx$

$$\frac{dy}{dx}=v+\frac{dv}{dx}x$$

$$v+\frac{dv}{dx}x=\frac{xvx}{x^2+(vx)^2}=\frac{v}{1+v^2}$$

or $$x\frac{dv}{dx}=\frac{v}{1+v^2}-\frac{v}{1}=\frac{-v^3}{1+v^2}$$

$$\int-\frac{(1+v^2)dv}{v^3}=\int\frac{dx}{x} \text{ or } -\int\frac{dv}{v^3}-\int\frac{dv}{v}=\int\frac{dx}{x}$$

or $$\frac{1}{2v^2}-\log v=\log x+c$$

put $v=\frac{y}{x}\Rightarrow\frac{x^2}{2y^2}=\log\frac{y}{x}+\log x+c=\log y+c$ given $y(1)=1$

$$\frac{1}{2}=c \qquad \Rightarrow\frac{x^2}{2y^2}=\log y+\frac{1}{2}$$

Now put $x=x_0,\ y=e$

$$\frac{x_0^2}{2e^2}=1+\frac{1}{2}=\frac{3}{2} \text{ or } x_0^2=\frac{6e^2}{2}=3e^2$$

$$\Rightarrow \quad x_0=\sqrt{3}\,.e$$

10. For the primitive integral equation $ydx+y^2dy=xdy$; $x\in R$, $y>0$, $y=y(x)$, $y(1)=1$, then $y(-3)$ is :

आद्य समाकल समीकरण $ydx+y^2dy=xdy$; $x\in R$, $y>0$, $y=y(x)$, $y(1)=1$ के लिये $y(-3)$ होगा :

(A) 3 (B) 2 (C) 1 (D) 5

Ans. A

Sol. $\frac{xdy-ydx}{y^2}=dy$ or $\frac{ydx-xdy}{y^2}=-dy$

or $$\int d\left(\frac{x}{y}\right)=-\int dy$$

$\frac{x}{y}=-y+c$ Given $y(1)=1\Rightarrow 1=-1+c\Rightarrow c=2$

$\frac{x}{y} = -y + 2$ Now y(–3), $\frac{-3}{y} = -y + 2$

or $y^2 - 2y - 3 = 0$

$y^2 - 3y + y - 3 = 0 = (y - 3)(y + 1) = 0$

$y = 3$ or $y = -1$ But $y > 0$

$\therefore \quad y = 3$

11. The solution of the differential equation $2x^2y\frac{dy}{dx} = \tan(x^2y^2) - 2xy^2$ given $y(1) = \sqrt{\frac{\pi}{2}}$ is :

अवकल समीकरण $2x^2y\frac{dy}{dx} = \tan(x^2y^2) - 2xy^2$ का हल होगा, दिया गया है $y(1) = \sqrt{\frac{\pi}{2}}$:

(A) $\sin(x^2y^2) - 1 = 0$

(B) $\cos\left(\frac{\pi}{2} + x^2y^2\right) + x = 0$

(C) $\sin(x^2y^2) = e^{x-1}$

(D) $\sin(x^2y^2) = e^{2(x-1)}$

Ans. C

Sol. $2x^2y\frac{dy}{dx} = \tan(x^2y^2) - 2xy^2$

$2x^2y\frac{dy}{dx} + 2xy^2 = \tan\left(x^2y^2\right)$

$x^2y^2 = t$

$2xy^2 + 2x^2y\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} = \tan t$

$\int \cot t\,dt = \int dx$

$\ln|\sin t| = x + c$

$\ln\left(\sin\left(x^2y^2\right)\right) = x + c$

$\left(1, \sqrt{\frac{\pi}{2}}\right)$

$$\ln\left[\sin\frac{\pi}{2}\right]=1+c$$

$0 = 1 + c \Rightarrow c = -1$

$\sin(x^2y^2) = (e)^{x-1}$

12. The general solution of the differential equation $\frac{dy}{dx}+x(x+y)=x(x+y)^3-1$ is : (where C is arbitary constant.)

अवकल समीकरण $\frac{dy}{dx}+x(x+y)=x(x+y)^3-1$ का व्यापक हल होगा : (जहाँ C एक स्वेच्छ अचर है।)

(A) $\ell n\left|\frac{(x+y+1)(x+y-1)}{(x+y)^4}\right|=x^2+C$

(B) $\ell n\left|\frac{(x+y+1)(x+y-1)}{(x+y)^2}\right|=x^2+C$

(C) $2\ell n\left|\frac{(x+y+1)(x+y-1)}{(x+y)^2}\right|=x^2+C$

(D) $\ell n\left|\frac{(x+y+1)(x+y-1)}{(x+y)^2}\right|=x+C$

Ans. B

Sol. $\frac{dy}{dx}+x(x+y)=x(x+y)^3-1$

$1+\frac{dy}{dx}+x(x+y)=x(x+y)^3$

$x + y = t$

$1+\frac{dy}{dx}=\frac{dt}{dx}$

$\frac{dt}{dx}+xt=xt^3$

$\frac{dt}{dx}=x\left(t^3-t\right)$

$\frac{dt}{t\left(t^2-1\right)}=xdx$

$\frac{dt}{t(t-1)(t+1)}=\frac{A}{t}+\frac{B}{t-1}+\frac{C}{t+1}$

$1 = A(t - 1)(t + 1) + B(t)(t + 1) + C(t)(t - 1)$

$t = 0$ $t = 1$ $t = -1$

$A = -1$ $B = \frac{1}{2}$ $C = \frac{1}{2}$

$$\int\left(-\frac{dt}{t}+\frac{1}{2}\frac{dt}{t-1}+\frac{1}{2}\frac{dt}{t+1}\right)=\int x dx$$

$$\int\frac{-2dt}{t}+\int\frac{dt}{t-1}+\int\frac{dt}{t+1}=\int 2x dx$$

$$-2n|t|+\ln|t-1|+\ln(t+1)=x^2+c$$

$$\ell n\left(\frac{(t-1)(t+1)}{t^2}\right)=x^2+c$$

$$\ell n\left[\frac{(x+y-1)(x+y+1)}{(x+y)^2}\right]=x^2+c$$

13. The solution of the equation $\frac{dy}{dx}+\frac{1}{x}\tan y=\frac{1}{x^2}\tan y\sin y$ is :

समीकरण $\frac{dy}{dx}+\frac{1}{x}\tan y=\frac{1}{x^2}\tan y\sin y$ का हल होगा :

(A) $2y = \sin y(1 - 2cx^2)$ (B) $2x = \cot y(1 + 2cx^2)$

(C) $2x = \sin y(1 + 2cx^2)$ (D) $2x\sin y = 1 - 2cx^2$

Ans. C

Sol. $\frac{dy}{dx}+\frac{1}{x}\tan y=\frac{1}{x^2}\tan y\sin y$

$$\cot y\,\mathrm{cosec}\,y\frac{dy}{dx}+\frac{1}{x}\mathrm{cosec}\,y=\frac{1}{x^2}$$

$\mathrm{cosec}\,y = t$

$$\mathrm{cosec}\,y\cot y\frac{dy}{dx}=\frac{-dt}{dx}$$

$$\frac{-dt}{dx}+\frac{t}{x}=\frac{1}{x^2}$$

$$\frac{dt}{dx}+\left(\frac{-1}{x}\right)t=\frac{-1}{x^2}$$

$$\text{I.F.} = e^{\int -\frac{1}{x}dx} = e^{-\ell nx} = \frac{1}{x}$$

$$\int \frac{1}{x}\frac{dt}{dx} - \frac{1}{x^2}t = \int -\frac{1}{x^3}$$

$$\frac{t}{x} = -\frac{x^{-3+1}}{-3+1} + C$$

$$\frac{\cos ecy}{x} = \frac{1}{2x^2} + C$$

$$2x\cos ecy = 1 + 2Cx^2$$

$$2x = \left(1 + 2Cx^2\right)\sin y$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

14. The equation of the curve such that the distance between the origin and the tangent line at an arbitrary point is equal to the distance between the origin and the normal at the same point is : (where c is an arbitrary constant)

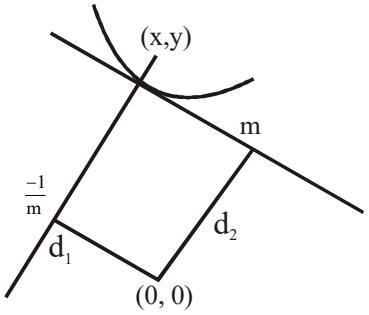
उस वक्र का समीकरण, जिस पर खीचीं गई स्पर्श रेखा पर स्थित किसी बिन्दु की मूल बिन्दु से दूरी, उसी बिन्दु पर खींचे अभिलम्ब पर मूल fcUnq ls yEc dh yEckbZ eku gS (जहाँ c प्राचल है)

(A) $x^2 + y^2 = c \cdot e^{\text{arc tan }(y/x)}$ (B) $\sqrt{x^2 + y^2} = c \cdot e^{-\text{arc tan }(y/x)}$

(C) $\sqrt{x^2 + y^2} = c \cdot e^{\text{arc tan }(y/x)}$ (D) $x^2 + y^2 = c \cdot e^{-\text{arc tan }(y/x)}$

Ans. BC

Sol.



$$Y - y = -\frac{1}{m}(X - x)$$

$X + mY - my - x = 0$ $d_1 = d_2$

$(0, 0)$

$$d_1 = \left|\frac{x + my}{\sqrt{m^2 + 1}}\right|$$ $Y - y = m(X - x)$

$mX - Y - mx + y = 0$

$(0, 0)$

$$d_2 = \left|\frac{y - mx}{\sqrt{m^2 + 1}}\right|$$

$$\frac{|x + my|}{\sqrt{m^2 + 1}} = \frac{|y - mx|}{\sqrt{m^2 + 1}}$$

$x + my = \pm (y - mx)$

$x + my = y - mx$ $x + my = mx - y$

$m(y + x) = y - x$ $m(x - y) = x + y$

$$\frac{dy}{dx} = \frac{y - x}{x + y}$$ $$\frac{dy}{dx} = \frac{x + y}{x - y}$$

$y = vx$

$$v + \frac{x dv}{dx} = \frac{v - 1}{v + 1}$$ $$v + \frac{x dv}{dx} = \frac{v + 1}{1 - v}$$

$$\frac{x dv}{dx} = \frac{v - 1 - v^2 - v}{v + 1}$$ $$\frac{x dv}{dx} = \frac{v + 1}{1 - v} - v = \frac{v + 1 - v + v^2}{1 - v} = \frac{v^2 + 1}{1 - v}$$

$$\frac{x dv}{dx} = -\frac{\left(v^2 + 1\right)}{v + 1}$$ $$\int \frac{1 - v}{v^2 + 1} dv = \int \frac{dx}{x}$$

$$\frac{v + 1}{v^2 + 1} dv = -\frac{dx}{x}$$ $$\int \frac{dv}{v^2 + 1} - \frac{1}{2}\int \frac{2v}{v^2 + 1} dv = \ln x + C$$

$$\int \frac{v dv}{v^2 + 1} + \int \frac{dv}{v^2 + 1} = -\ln x + c$$ $$\tan^{-1} v - \ln\sqrt{v^2 + 1} = \ln x + C$$

$$\frac{1}{2}\ell n\left(v^2 + 1\right) + \tan^{-1} v = -\ell n x + C$$ $$\tan^{-1}\left(\frac{y}{x}\right) + c = \ln\sqrt{y^2 + x^2}$$

$$\sqrt{x^2+y^2} = ce^{\tan^{-1}\left(\frac{y}{x}\right)}$$

$$\ell n\sqrt{\frac{y^2+x^2}{x^2}} + \ln x = -\tan^{-1}\frac{y}{x} + c$$

$$\ell n\left(\sqrt{x^2+y^2}\right) + \ln C = -\tan^{-1}\left(\frac{y}{x}\right)$$

$$\sqrt{x^2+y^2} = ce^{-\tan^{-1}\left(\frac{y}{x}\right)}$$

15. Let $\frac{dy}{dx} + y = f(x)$ where y is a continuous function of x with $y(0) = 1$ and $f(x) = \begin{cases} e^{-x} & \text{if } 0 \le x \le 2 \\ e^{-2} & \text{if } x > 2 \end{cases}$. Which of the following hold(s) good :

यदि $\frac{dy}{dx} + y = f(x)$ जहाँ y, x में सतत् फलन है, $y(0) = 1$ तथा $f(x) = \begin{cases} e^{-x} & \text{यदि } 0 \le x \le 2 \\ e^{-2} & \text{यदि } x > 2 \end{cases}$ तब निम्नलिखित में कौनसे dHkuh; gS

(A) $y(1) = 2e^{-1}$ (B) $y'(1) = -e^{-1}$ (C) $y(3) = -2e^{-3}$ (D) $y'(3) = -2e^{-3}$

Ans. ABD

Sol. $\frac{dy}{dx} + y = f(x)$

I.F. $= e^x$

$ye^x = \int e^x f(x)\,dx + C$

now if $0 \le x \le 2$ then $ye^x = \int e^x e^{-x}\,dx + C \Rightarrow ye^x = x + C$

$x = 0,\ y(0) = 1, \quad C = 1$

$\therefore \quad ye^x = x + 1 \quad ....(1)$

$y = \frac{x+1}{e^x}; \quad y(1) = \frac{2}{e}$ Ans. $\Rightarrow$ (A) is correct

$y' = \frac{e^x - (x+1)e^x}{e^{2x}}$

$y'(1) = \frac{e-2e}{e^2} = \frac{-e}{e^2} = -\frac{1}{e}$ Ans. $\Rightarrow$ (B) is correct

if $x > 2$

$ye^x = \int e^{x-2}\,dx$

$ye^x = e^{x-2} + C$

$y = e^{-2} + Ce^{-x}$

as y is continuous

$\therefore \quad \lim\limits_{x\to 2} \dfrac{x+1}{e^x} = \lim\limits_{x\to 2} \left(e^{-2} + Ce^{-x}\right)$

$3e^{-2} = e^{-2} + Ce^{-2} \quad \Rightarrow \quad C = 2$

$\therefore$ for $x > 2$

$y = e^{-2} + 2e^{-x}$ hence $y(3) = 2e^{-3} + e^{-2} = e^{-2}(2e^{-1} + 1)$

$y' = -2e^{-x}$

$y'(3) = -2e^{-3}$ Ans. $\Rightarrow$ (D) is correct

16. If y(x) satisfies the differential equation $\dfrac{dy}{dx} = \sin 2x + 3y\cot x$ and $y\left(\dfrac{\pi}{2}\right) = 2$ then which of the following statement(s) is/are correct?

(A) $y\left(\dfrac{\pi}{6}\right) = 0$ (B) $y'\left(\dfrac{\pi}{3}\right) = \dfrac{9-3\sqrt{2}}{2}$

(C) y(x) increases in the interval $\left(0, \dfrac{\pi}{2}\right)$ (D) $\int\limits_{-\pi/2}^{\pi/2} y(x)dx = x$

यदि y(x) अवकल समीकरण $\dfrac{dy}{dx} = \sin 2x + 3y\cot x$ तथा $y\left(\dfrac{\pi}{2}\right) = 2$ को संतुष्ट करता है, तो निम्न में से कौनसा/कौनसे कथन

l R, gkx8

(A) $y\left(\dfrac{\pi}{6}\right) = 0$ (B) $y'\left(\dfrac{\pi}{3}\right) = \dfrac{9-3\sqrt{2}}{2}$

(C) y(x) अन्तराल $\left(0, \dfrac{\pi}{2}\right)$ में वर्धमान होगा (D) $\int\limits_{-\pi/2}^{\pi/2} y(x)dx = x$

Ans. A

Sol. $\dfrac{dy}{dx} = \sin 2x + 3y\cot x$

$\dfrac{dy}{dx} + y(-3\cot x) = \sin 2x$

I.E. $= e^{\int -3\cot x dx} = e^{-3\ln\sin x}$

$= \frac{1}{\sin^3 x}$

$\int \frac{1}{\sin^3 x}\frac{dy}{dx} + y\left(\frac{-3\cot x}{\sin^3 x}\right) = \int \frac{2\sin x \cos x}{\sin^2 x}$

$\frac{y}{\sin^3 x} = 2\int \frac{dx}{t^2}; \qquad t = \sin x$

$\frac{y}{\sin^3 x} = \frac{-2}{\sin x} + C \qquad \left(\frac{\pi}{2}, 2\right)$

$C = +4 \qquad y = -2\sin^2 x + 4\sin^3 x$

$f\left(\frac{\pi}{6}\right) = -2\times\frac{1}{4} + 4\times\frac{1}{8}$

$= -\frac{1}{2} + \frac{1}{2} = 0$

$f'(x) = -4\sin x\cos x + 12\sin^2\cos x = 4\sin x\cos x(3\sin x - 1)$

$\Rightarrow -4 + \frac{\sqrt{3}}{2}\times\frac{1}{2} + 12\times\frac{3}{4}\times\frac{1}{2}$

$\Rightarrow -\sqrt{3} + \frac{9}{2}$

$\Rightarrow \frac{9-2\sqrt{3}}{2}$

17. Solve : $\tan y \frac{dy}{dx} + \tan x = \cos y \cdot \cos^3 x$:

$\tan y \frac{dy}{dx} + \tan x = \cos y \cdot \cos^3 x$ को हल होगा :

(A) $\sec x \sec y = \frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) + C$ (B) $((x + \sin x\cos x) + 2C)\cos x\cos y = 2$

(C) $\cos x\cos y = \frac{1}{2}(x + \sin x\cos x) + C$ (D) $\sec x\sec y = \frac{1}{2}(x + \cos x) + C$

Ans. AB

Sol. $\tan y \frac{dy}{dx} + \tan x = \cos y \cdot \cos^3 x$

$$\frac{\sin y}{\cos y}\frac{dy}{dx} + \frac{\sin x}{\cos x} = \cos y \cos^3 x$$

$$\cos x \sin y \frac{dy}{dx} + \sin x \cos y = \cos^2 y \cos^4 x$$

$\div \cos^2 x \cos^2 y$

$$\sec x \tan y \sec y \frac{dy}{dx} + \sec y \tan x \sec x = \cos^2 x = \frac{1+\cos 2x}{2}$$

$$\sec x \sec y = \int \frac{1+\cos 2x}{2} dx$$

$$\sec x \sec y = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

$$\frac{1}{\cos x \cos y} = \frac{x}{2} + \frac{\sin x \cos x}{2} + C$$

$2 = (x + \sin x \cos x + 2c) \cos x \cos y$

18. The solution(s) of the differential equation $9(2-y)^2 \left(\frac{dy}{dx}\right)^2 = 4(3-y)$ is /are :

अवकल समीकरण $9(2-y)^2 \left(\frac{dy}{dx}\right)^2 = 4(3-y)$ का हल होगा :

(A) $(3-x)^2 = y^2 (y + c)$ (B) $(x + c)^2 = y^2 (3 - y)$

(C) $y = 0$ (D) $y = 3$

Ans. BD

Sol. $9(2-y)^2 \left(\frac{dy}{dx}\right)^2 = 4(3-y)$

$$3(2-y)\frac{dy}{dx} = \pm 2\left(\sqrt{3-y}\right)$$

$$+ \Rightarrow 3(2-y)\frac{dy}{dx} = 2\sqrt{3-y}$$

$$\frac{1+2-y-1}{\sqrt{3-y}}dy = \frac{2}{3}dx$$

$$\frac{(3-y)-1}{\sqrt{3-y}}dy = \frac{2}{3}dx$$

$$\int\left(\sqrt{3-y} - \frac{1}{\sqrt{3-y}}\right)dy = \int\frac{2}{3}dx$$

$$\frac{(3-y)^{3/2}}{-\frac{3}{2}} - \frac{\sqrt{3-y}}{-\frac{1}{2}} = \frac{2x}{3} + C$$

$$\Rightarrow 2\sqrt{3-y} - \frac{2}{3}(3-y)^{3/2} = \frac{2x}{3} + C$$

$$\Rightarrow 3\sqrt{3-y} - (3-y)^{3/2} = x + c$$

$$\left(\sqrt{3-y}\right)(3-3+y) = x + c$$

$$y\sqrt{3-y} = x + c$$

$$y^2(3-y) = (x+c)^2$$

$$- \Rightarrow 3(2-y)\frac{dy}{dx} = -2\sqrt{3-y}$$

$$\frac{1+2-y-1}{\sqrt{3-y}}dy = -\frac{2}{3}dx$$

$$\int\left(\sqrt{3-y} - \frac{1}{\sqrt{3-y}}\right)dy = \int -\frac{2}{3}dx$$

$$\frac{(3-y)^{3/2}}{-\frac{3}{2}} - \frac{\sqrt{3-y}}{-\frac{1}{2}} = -\frac{2x}{3} + C$$

$$\frac{2(3-y)^{3/2}}{3} - 2\sqrt{3-y} = \frac{2x}{3} + C$$

$(3-y)^{3/2} - 3\sqrt{3-y} = x + c$

$\left(\sqrt{3-y}\right)[3-y-3]x + c$

$-y\sqrt{3-y} = x + c$

19. Let $y = f(x)$ be a real valued function satisfying $x\frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, then :

(A) $f(x)$ is minimum at $x = 1$ (B) $f(x)$ is maximum at $x = 1$

(C) $f(3) = 5$ (D) $f(2) = 3$

माना $y = f(x)$ एक वास्तविक मान फलन जो $x\frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, को सन्तुष्ट करता है तो :

(A) ग ञ 1 पर f(x) न्यूनतम है (B) ग ञ 1 पर f(x) अधिकतम है

(C) $f(3) = 5$ (D) $f(2) = 3$

Ans. AC

Sol. $x\frac{dy}{dx} = x^2 + y - 2$

$\frac{dy}{dx} = x + \frac{y}{x} - \frac{2}{x}$

$\frac{dy}{dx} + \left(\frac{-1}{x}\right)y = x - \frac{2}{x}$

I. F. $= e^{\int -\frac{1}{x}dx} = e^{-\ell nx} = \frac{1}{x}$

$\int \frac{1}{x}\frac{dy}{dx} + \left(-\frac{1}{x^2}\right)y = \int 1 - \frac{2}{x^2}$

$\frac{y}{x} = x + \frac{2}{x} + C$

(1,1)

$1 = 1 + 2 + C$

$C = -2$

$y = x^2 + 2 - 2x$

$y = x^2 - 2x + 2$

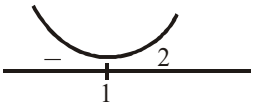
$x = 3$ $x = 2$

$y = 9 - 6 + 2$ $\qquad$ $y = 4 - 4 + 2$

$= 5$ $\qquad$ $y = 2$

$y' = 2x - 2$

$= 2(x - 1)$



20. Solution of differential equation $x\cos x\left(\frac{dy}{dx}\right) + y(x\sin x + \cos x) = 1$ is : (where c is arbitrary constant)

(A) $xy = \sin x + c\cos x$ $\qquad$ (B) $xy\sec x = \tan x + c$

(C) $xy + \sin x + c\cos x = 0$ $\qquad$ (D) None of these

अवकल समीकरण $x\cos x\left(\frac{dy}{dx}\right) + y(x\sin x + \cos x) = 1$ का हल होगा : (जहाँ c एक स्वच्छ अचर है)

(A) $xy = \sin x + c\cos x$ $\qquad$ (B) $xy\sec x = \tan x + c$

(C) $xy + \sin x + c\cos x = 0$ $\qquad$ (D) इनमें से कोई नहीं

Ans. AB

Sol. $x\cos x\left(\frac{dy}{dx}\right) + y(x\sin x + \cos x) = 1$

$$\frac{dy}{dx} + y\left(\frac{2\sin x + \cos x}{x\cos x}\right) = \frac{1}{x\cos x}$$

I. $f = e^{\int \frac{x\sin x + \cos x}{x\cos x}dx}$

$= e^{\int \tan x + \frac{1}{x}} = e^{\ell n\sec x + \ell nx}$

$= x\sec x$

$$\int x\sec x\frac{dy}{dx} + (x\sec x)y\left(\frac{2\sin x + \cos x}{2\cos x}\right) = \int \frac{x\sec x}{x\cos x}dx$$

$yx\sec x = \tan x + c$

$xy = \sin x + c\cos x$

21. If $y = y(x)$ is solution of differential equation $(1 - x^2)\frac{dx}{dy} + xy = x$ such that $y(1) = 1$ and $g(x) = (y - 1)^2$, then :

(A) g(x) is an even function

(B) $\lim_{x\to 1}\frac{g(x)}{(x-1)^2}=2$

(C) g(x) has only two local minima

(D) g(x) = 1 has four solution

यदि y = y(x) अवकल समीकरण $(1-x^2)\frac{dx}{dy}+xy=x$ का हल इस प्रकार है कि y(1) = 1 तथा $g(x)=(y-1)^2$ हो, तब :

(A) ह; गद्धसम फलन है

(B) $\lim_{x\to 1}\frac{g(x)}{(x-1)^2}=2$

(C) ह; गद्धके केवल दो स्थानीय निम्निष्ठ है

(D) ह; गद्ध> 1 के चार हल है

Ans. ABCD

Sol. $(1-x^2)\frac{dx}{dy}+xy=x$

$$\left(1-x^2\right)\frac{dx}{dy}=x\left(1-y\right)$$

$$\left(\frac{1-x^2}{x}\right)dx=\left(1-y\right)dy$$

$$y-\frac{y^2}{2}=\ln x-\frac{x^2}{2}+C$$

(1, 1)

$$1-\frac{1}{2}=0-\frac{1}{2}+C$$

c = 1

$$y-\frac{y^2}{2}=\ell nx-\frac{x^2}{2}+1$$

$2y - y^2 = 2\ln x - x^2 + 2$

$y^2 - 2y = x^2 - \ln x^2 - 2$

$(y - 1)^2 = x^2 - \ln x^2 - 1$

$g(x) = x^2 - \ln x^2 - 1$

Even

$$\lim_{x\to 1}\frac{x^2-\ln x^2-1}{(x-1)^2}=\frac{2x-\frac{1}{x^2}\times 2x}{2(x-1)}$$

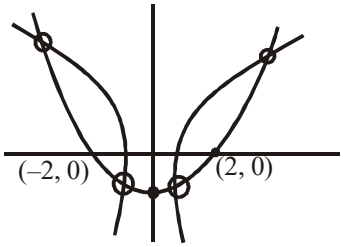
$$\Rightarrow \frac{2x\left(1-\frac{1}{x^2}\right)}{2(x-1)}$$

$$\Rightarrow \frac{x(x^2-1)}{(x-1)x^2}=\frac{(x-1)(x+1)}{(x-1)x}$$

$$\lim_{x\to 1}\frac{x+1}{x}=2$$

$x^2-\ln x^2-1=1$ $\qquad$ $g'(x)=2x-\frac{2x}{x^2}$

$x^2=2(\ln x+2)$ $\qquad$ $=2\left(\frac{x^2-1}{x}\right)=\frac{(x-1)(x+1)}{x}$



$x^2=2(\ln x+2)$

$$y=\frac{x^2}{2}-2=\ln|x|$$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. $(x^3 + y^2 + 2)\,dx + 2y\,dy = 0$

Ans. $y^2 = 3x^2 - 6x - x^3 + ce^{-x} + 4$

Sol. $(x^3 + y^2 + 2)dx + 2ydy = 0$ _____(1)

Let $y^2 + 2 = v$

$\Rightarrow 2ydy = dv$

Substituting values in equation (1)

$\Rightarrow (x^3 + v)dx + dv = 0$

$$\Rightarrow \frac{dv}{dx} + v = -x^3$$

$$\text{I.F.} = e^{\int 1dx} = e^x$$

$$\Rightarrow v \cdot e^x = \int -x^3 e^x dx + C$$

$$\Rightarrow v \cdot e^x = -\left(x^3 - 3x^2 + 6x - 6\right)e^x + C$$

$$\Rightarrow v = -x^3 + 3x^2 - 6x + 6 + ce^{-x}$$

$$y^2 = -x^3 + 3x^2 - 6x + 4 + ce^{-x}$$

2. $$\left(\frac{dy}{dx}\right)^2 - (x+y)\frac{dy}{dx} + xy = 0$$

Ans. $y = ce^x$; $y = c + \frac{x^2}{2}$

Sol. $$\left(\frac{dy}{dx} - y\right)\left(\frac{dy}{dx} - x\right) = 0$$

$$\frac{dy}{dx} = y \quad \text{or} \quad \frac{dy}{dx} = x$$

$$\Rightarrow \ell ny = x + c \quad \text{or} \quad y = \frac{x^2}{2} + c$$

$$\Rightarrow y = ke^x$$

3. $\frac{dy}{dx} = \frac{y^2 - x}{2y\,(x+1)}$

Ans. $y^2 = -1 + (x+1)\ \ell n\ \frac{c}{x+1}$

Sol. $\frac{dy}{dx} = \frac{y^2 - x}{2y(x+1)}$

$\Rightarrow 2y\frac{dy}{dx} - \frac{y^2}{x+1} = \frac{-x}{x+1}$

putting $y^2 = t$

$\Rightarrow 2y\frac{dy}{dx} = \frac{dt}{dx}$

$\frac{dt}{dx} - \frac{t}{x+1} = \frac{-x}{1+x}$

I.F. $= e^{-\int \frac{1}{x+1}dx} = e^{\ell n \frac{1}{x+1}} = \frac{1}{x+1}$

$\frac{t}{x+1} = \int \frac{1}{(1+x)}\cdot\left(\frac{-x}{1+x}\right)dx$

$= \int\left(\frac{1}{(1+x)^2} - \frac{1}{(1+x)}\right)dx$

$\frac{t}{1+x} = \frac{-1}{(1+x)} - \ell n(1+x) + \ell nC$

$t = -1 + (1+x)\,\ell n\left(\frac{C}{1+x}\right)$

$y^2 = -1 + (1+x)\,\ell n\left(\frac{C}{1+x}\right)$

4. $(1 - xy + x^2y^2)\,dx = x^2\,dy$

Ans. $y = \frac{1}{x}\tan\left(\ell n\,|cx|\right)$

Sol. $xy = t \Rightarrow x\frac{dy}{dx} + y = \frac{dt}{dx}$

$\Rightarrow x\frac{dy}{dx} = \frac{dt}{dx} - \frac{t}{x}$

$\Rightarrow (1 - t + t^2) = x\left(\frac{dt}{dx} - \frac{t}{x}\right)$

$\Rightarrow \frac{(1 - t + t^2)}{x} + \frac{t}{x} = \frac{dt}{dx} \Rightarrow \frac{1 + t^2}{x} = \frac{dt}{dx}$

$\Rightarrow \frac{dt}{1 + t^2} = \frac{dx}{x} \quad \Rightarrow \quad \tan^{-1}t = \ell n|x| + \ell nc$

$\Rightarrow \tan^{-1}(xy) = \ell n|cx|$

$xy = \tan \ell n|xc|$

5. $(x^2 + y)\frac{dy}{dx} = 6x$

Ans. $y = 3\ ln(x^2 + y + 3) + C$

Sol. $\left(x^2 + y\right)\frac{dy}{dx} = 6x$

$\Rightarrow 6x\frac{dx}{dy} = \left(x^2 + y\right)$

Let $x^2 = t \Rightarrow 2x\frac{dx}{dy} = \frac{dt}{dy}$

$\Rightarrow 3\frac{dt}{dy} = \left(t + y\right)$

$\Rightarrow \frac{dt}{dy} - \frac{t}{3} = \frac{y}{3}$

$\text{I.F.} = e^{\int -\frac{1}{3}dy} = e^{-\frac{y}{3}}$

$t \cdot e^{-\frac{y}{3}} = \int e^{-\frac{y}{3}}\frac{y}{3}dy = -\left(y + 3\right)e^{-\frac{y}{3}} + C$

$\Rightarrow x^2 = -\left(y + 3\right) + Ce^{\frac{y}{3}}$

6. $\frac{dy}{dx} - y\,ln2 = 2^{\sin x}.(\cos x - 1)\,ln2$, y being bounded when $x \to +\infty$.

$\frac{dy}{dx} - y\,ln2 = 2^{\sin x}.(\cos x - 1)\,ln2$, $x \to +\infty$ के लिए y परिबद्ध है

Ans. $y = 2^{\sin x}$

Sol. $\frac{dy}{dx} - y\log_e 2 = 2^{\sin x}(\cos x - 1)\log_e 2$

This is linear differential equation.

$I.F. = e^{-\log_e 2\int dx} = e^{-x\log_e 2} = 2^{-x}$

Solution is

$y2^{-x} = \int 2^{-x}2^{\sin x}(\cos x - 1)\log_e 2dx$

put $\sin x - x = t \Rightarrow (\cos x - 1)dx = dt$

$\therefore y2^{-x} = \log_e 2\int 2^t dt$

$\therefore y2^{-x} = 2^t + c$

$\therefore y = 2^{x+t} + c2^x$

$\therefore y = 2^{\sin x} + c2^x$

7. Find the curve which passes through the point (2, 0) such that the segment of the tangent between the point of tangency and the y–axis has a constant length equal to 2.

बिन्दु वक्र (2, 0) से गुजरने वाला वक्र ज्ञात कीजिए जिसके किसी स्पर्शी का स्पर्श बिन्दु एवं y–अक्ष के मध्य अन्तःखण्ड की लम्बाई 2 हों।

Ans. $y = \pm\left[\sqrt{4 - x^2} + 2\ell n\frac{2 - \sqrt{4 - x^2}}{x}\right]$

Sol. Let the curve be $y = f(x)$ and slope of tangent at any point $P(x, y) = \frac{dy}{dx}$

So, the equation of the tangent :

$Y - y = \frac{dy}{dx}(X - x)$

Let tangent meet y-axis at point N and y-axis, put $X = 0$.

$Y = y - x\frac{dy}{dx}$

So, the coordinates of the point $N = \left(0, y - x\frac{dy}{dx}\right)$

Given that PN = 2.

$$\Rightarrow \sqrt{(0-x)^2 + \left(y - x\frac{dy}{dx} - y\right)^2} = 2$$

$$\Rightarrow x^2 + x^2\left(\frac{dy}{dx}\right)^2 = 4$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{4-x^2}{x^2}$$

$$\Rightarrow \frac{dy}{dx} = \pm\sqrt{\frac{4-x^2}{x^2}}$$

$$\Rightarrow dy = \pm\left(\sqrt{\frac{4-x^2}{x^2}}\right)dx$$

On integrating both sides and let $x = 2\sin\theta$ _____(1)

$$\Rightarrow dx = 2\cos\theta d\theta$$

$$\Rightarrow \int dy = \pm 2\int \frac{\cos^2\theta}{\sin\theta} d\theta$$

$$\Rightarrow \int dy = \pm 2\int \left(\frac{1-\sin^2\theta}{\sin\theta}\right) d\theta$$

$$\Rightarrow \int dy = \pm 2\int (\operatorname{cosec}\theta - \sin\theta)\, d\theta$$

$$\Rightarrow y = \pm 2\left[\ln(\operatorname{cosec}\theta - \cos\theta) - (-\cos\theta)\right] + c$$ _____(2)

Since from equation (1)

$$\sin\theta = \frac{x}{2}$$

$$\Rightarrow \cos\theta = \sqrt{1 - \frac{x^2}{4}} = \frac{1}{2}\sqrt{4-x^2}$$

$$\cos ec\theta = \frac{2}{x}$$

$$\cot\theta = \frac{\sqrt{4-x^2}}{x}$$

Putting all these values in equation (2), we get

$$y = \pm 2\left[\ln\left(\frac{2}{x} - \frac{\sqrt{4-x^2}}{x}\right) + \frac{\sqrt{4-x^2}}{2}\right] + c$$

Since, the curve passes through the point (2,0).

$\Rightarrow c = 0$

Therefore, $y = \pm\left[\sqrt{4-x^2} + 2\ln\frac{2-\sqrt{4-x^2}}{x}\right]$

8. $x\,dy + y\,dx + \frac{x\,dy - y\,dx}{x^2+y^2} = 0$

Ans. $xy + \tan^{-1}\frac{y}{x} = c$

Sol. $xdy + ydx + \frac{xdy - ydx}{x^2+y^2} = 0$

$$\Rightarrow d(xy) + d\tan^{-1}\left(\frac{y}{x}\right) = 0$$

$$\Rightarrow xy + \tan^{-1}\frac{y}{x} = c$$

9. $\frac{y\,dx - x\,dy}{(x-y)^2} = \frac{dx}{2\sqrt{1-x^2}}$, given that y=2 when x = 1.

Ans. $\frac{\sin^{-1}x}{2} - \frac{y}{x-y} = \frac{\pi}{4} + 2$

Sol. $y - \frac{xdy}{dx} = \frac{(x-y)^2}{2\sqrt{1-x^2}}$

$$\Rightarrow (y-x)+x-\frac{xdy}{dx}=\frac{(x-y)^2}{2\sqrt{1-x^2}}$$

$$\Rightarrow \frac{x}{(x-y)^2}\left(1-\frac{dy}{dx}\right)-\frac{1}{(x-y)}=-\frac{1}{2\sqrt{1-x^2}}$$

$$\Rightarrow \frac{1}{(x-y)^2}\left(1-\frac{dy}{dx}\right)-\frac{1}{x(x-y)}=-\frac{1}{2x\sqrt{1-x^2}}$$

Let $-\frac{1}{x-y}=t \Rightarrow \frac{1}{(x-y)^2}\left(1-\frac{dy}{dx}\right)=\frac{dt}{dx}$

$$\Rightarrow \frac{dt}{dx}+\frac{t}{x}=-\frac{1}{2x\sqrt{1-x^2}}$$

$$\text{I.F.}=e^{\int\frac{1}{x}dx}=x$$

$$t\cdot x=-\int\frac{x}{2x\sqrt{1-x^2}}dx$$

$$\Rightarrow t\cdot x=-\frac{1}{2}\int\frac{1}{\sqrt{1-x^2}}dx$$

$$\Rightarrow t\cdot x=-\frac{1}{2}\sin^{-1}x+c$$

$$\Rightarrow -\frac{x}{x-y}=-\frac{1}{2}\sin^{-1}x+c$$

or

$$\Rightarrow -\frac{y}{x-y}=-\frac{1}{2}\sin^{-1}x+c^1$$

Now, (x, y) = (1,2)

$$\Rightarrow \frac{-2}{-1}=-\frac{1}{2}\sin^{-1}1+c^1 \Rightarrow c^1=2+\frac{\pi}{4}$$

$$\frac{\sin^{-1}x}{2}-\frac{y}{x-y}=\frac{\pi}{4}+2$$

10. Find the equation of the curve passing through the orgin if the middle point of the segment of its normal from any point of the curve to the x-axis lies on the parabola $2y^2 = x$.

मूल बिन्दु से गुजरने वाले उस वक्र का समीकरण ज्ञात कीजिए। यदि वक्र के किसी बिन्दु पर खीचें गये लम्ब के बिन्दु एवं x-अक्ष के मध्य भाग का मध्य बिन्दु परवलय $2y^2 = x$ पर पड़ता है।

Ans. $y^2 = 2x + 1 - e^{2x}$

Sol. Equation of normal at any point P(x, y), is

$$\frac{dy}{dx}(Y-y)+(X-x)=0$$

This meets the x-axis at $A\left(x+y\frac{dy}{dx},0\right)$

Midpoint of AP is $\left(x+\frac{1}{2}y\frac{dy}{dx},\frac{y}{2}\right)$ which lies on the parabola $2y^2 = x$.

Therefore, $2\times\frac{y^2}{4}=x+\frac{1}{2}y\frac{dy}{dx}$ or $y^2=2x+y\frac{dy}{dx}$

Put $y^2 = t$ $\Rightarrow 2y\frac{dy}{dx}=\frac{dt}{dx}$

we get $\frac{dt}{dx}-2t=-4x$ (linear)

$I.F.e^{-2\int(dx)}=e^{-2x}$

Therefore, solution is given by

$$te^{-2x}=-4\int xe^{-2x}dx+c$$

$$=-4\left[-\frac{1}{2}xe^{-2x}-\int-\frac{1}{2}e^{-2x}1dx\right]+c$$

$$y^2e^{-2x}=2xe^{-2x}+e^{-2x}+c$$

Since, the curve passes through (0,0), c = –1. Therefore,

$y^2 = 2x + 1 - e^{2x}$ is the equation of the required curve.

11. Find the continuous function which satisfies the relation, $\int_0^x t\, f(x-t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$, for all real number x.

वह सतत् फलन ज्ञात कीजिए जो संबंध $\int_0^x t\, f(x-t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$, को x के सभी वास्तविक मानों के लिये संतुष्ट करता है।

Ans. $f(x) = e^x - \cos x$

Sol. Using $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

$$\Rightarrow \int_0^x tf(x-t)\,dt = \int_0^x (x-t) f(t)\,dt$$

$$\text{Now, } \int_0^x (x-t) f(t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$$

$$\Rightarrow x\int_0^x f(t)\,dt - \int_0^x tf(t)\,dt = \int_0^x f(t)\,dt + \sin x + \cos x - x - 1$$

Differentiating both sides

$$\Rightarrow \int_0^x f(t)\,dt + xf(x) - xf(x) = f(x) + \cos x - \sin x - 1$$

$$\Rightarrow \int_0^x f(t)\,dt = f(x) + \cos x - \sin x - 1 \qquad ...(1)$$

Again differentiating both sides

$$\Rightarrow f(x) = f'(x) - \cos x - \sin x \qquad (y = f(x))$$

$$\Rightarrow \frac{dy}{dx} - y = \sin x + \cos x$$

$$IF = e^{\int -1dx} = e^{-x}$$

$$\Rightarrow ye^{-x} = \int e^{-x}(\sin x + \cos x)\,dx \qquad ...(2)$$

$\int e^{-x}(\sin x + \cos x)dx \rightarrow$ Put $x = -t$ & $dx = -dt$

$\Rightarrow -\int e^{t}(\cos t - \sin t)dt = -e^{t}\cos t \quad \left[\int e^{x}(f(x)) + f'(x)dx = e^{x}f(x)\right]$

Put in equation (2)

$ye^{-x} = -e^{-x}\cos x + c$

$y = ce^{x} - \cos x$

Put $x = 0$ in equation (1), $f(0) = 0$

$y = ce^{x} - \cos x$

Put $(x, y) = (0,0) \Rightarrow 0 = c - 1 \Rightarrow c = 1$

$y = e^{x} - \cos x$

12. Given two curves $y = f(x)$ passing through the point (0, 1) & $y = \int_{-\infty}^{x} f(t)dt$ passing through the point (0, 1/2). The tangents drawn to both curves at the points with equal abscissas intersect on the x–axis. Find the curve f(x).

वक्र $y = f(x)$ बिन्दु (0, 1) से तथा वक्र $y = \int_{-\infty}^{x} f(t)dt$, (0, 1/2) से गुजरते है। दोनों वक्रो पर समान भुज वाले बिन्दुओं पर खीचीं स्पर्श रेखाएं x अक्ष पर मिलती है। वक्र f(x) ज्ञात कीजिए।

Ans. $f(x) = e^{2x}$

Sol. Equation of tangent to the curve $y = f(x)$ is

$Y - y = f'(x)(X - x)$

Equation of tangents to the curve $g(x) = y_1 = \int_{-\infty}^{x} f(t)dt$ is $Y - y = f(x)(X - x)\left(\frac{dy_1}{dx} = g'(x) = f(x)\right)$

Since the tangent with equal abscissa intersect on the x-axis

$\Rightarrow x - \frac{y}{f'(x)} = x - \frac{y_1}{f(x)}$

$\Rightarrow \frac{f(x)}{f'(x)} = \frac{y_1}{f(x)}$

$\Rightarrow \frac{g'(x)}{g(x)} = \frac{g''(x)}{g'(x)}$

$\Rightarrow \ell n\, g(x) = \ln cg'(x)$

$\Rightarrow g(x) = cg'(x)$

$$\Rightarrow \frac{g'(x)}{g(x)} = c$$

$\Rightarrow g(x) = ke^{cx}$

$\Rightarrow f(x) = g'(x) = kce^{cx}$

The curve $y = f(x)$ passes through $(0,1) \Rightarrow kc = 1$

$g(x) = ke^{cx}$ passing through $\left(0, \frac{1}{2}\right)$

$$\Rightarrow \frac{1}{2} = k \Rightarrow kc = 1 \Rightarrow c = 2$$

Hence, $f(x) = e^{2x}$

13. Find the integral curve of the differential equation, $x(1 - x\, l n y).\frac{dy}{dx} + y = 0$ which passes through $\left(1, \frac{1}{e}\right)$.

अवकल समीकरण $x(1 - x\, l n y).\frac{dy}{dx} + y = 0$ का समाकल वक्र ज्ञात कीजिए जो $\left(1, \frac{1}{e}\right)$ से गुजरता है।

Ans. $x(ey + l n\, y + 1) = 1$

Sol. $$x\left(1 - x\ell ny\right)\frac{dy}{dx} + y = 0$$

$$\Rightarrow y\frac{dx}{dy} + x = x^2 \ell ny \qquad \text{_____(1)}$$

Let $\ell n\, y = z$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow y\frac{dx}{dy} = \frac{dx}{dz}$$

$$\Rightarrow \frac{dx}{dz} + x = x^2 z$$

$$\Rightarrow \frac{1}{x^2}\frac{dx}{dz}+\frac{1}{x}=z$$

which is a linear differential equation.

Let $\frac{1}{x}=v$

$$\Rightarrow -\frac{1}{x^2}\frac{dx}{dz}=\frac{dv}{dz}$$

$$\Rightarrow -\frac{dv}{dz}+v=z$$

$$\Rightarrow \frac{dv}{dz}-v=-z$$

which is a linear differential equation.

$\therefore IF = e^{-\int dz} = e^{-z}$

Hence the solution is

$$v.e^{-z}=\int -ze^{-z}dz+c$$

$$\Rightarrow v\cdot e^{-z}=-\left(-z-1\right)e^{-z}+c$$

$$\Rightarrow v=\left(z+1\right)+ce^{z}$$

$$\Rightarrow \frac{1}{x}=\left(\ell ny+1\right)+ce^{\ell ny}$$

$$\Rightarrow \frac{1}{x}=\left(\ell ny+1\right)+cy$$

which is passing through $\left(e,\frac{1}{e}\right)$.

So, c = e

Hence, the required integral curves is $\frac{1}{x}=\left(\ell ny+1\right)+ey$

14. Find all the curves possessing the following property; the segment of the tangent between the point of tangency & the x-axis is bisected at the point of intersection with the y-axis.

वह सभी वक्र ज्ञात कीजिए जिसके लिये स्पर्श बिन्दु एवं x अक्ष के बीच स्पर्श रेखा का अन्तःखण्ड, y-अक्ष पर समद्विभाजित होता है।

Ans. $y^2 = cx$

Sol. Equation of tangent at P(x, y) is

$$Y - y = \frac{dy}{dx}(X - x)$$

It meets x-axis at Q(x, 0)

$$-y = \frac{dy}{dx}(X - x)$$

$$x - y\frac{dx}{dy} = X$$

$$Q \equiv \left(x - y\frac{dx}{dy}, 0\right)$$

PQ bisected at point of intersection at y-axis let R (0,y)

$$\Rightarrow \frac{x + x - y\frac{dx}{dy}}{2} = 0$$

$$2x = y\frac{dx}{dy}$$

$$\int \frac{dy}{y} = \int \frac{dx}{2x}$$

$$\log y = \frac{1}{2}\log x + \log c$$

$y^2 = cx$

15. A curve passing through (1 , 0) such that the ratio of the square of the intercept cut by any tangent off the y-axis to the subnormal is equal to the ratio of the product of the co-ordinates of the point of tangency to the product of square of the slope of the tangent and the subtangent at the same point. Determine all such possible curves.

एक वक्र जो (1 , 0) से गुजरता है, के स्पर्श रेखा द्वारा y-अ़क्ष पर अन्तःखण्ड की लम्बाई का वर्ग एवं अर्धलम्ब का अनुपात, स्पर्श बिन्दु के निर्देशांको के गुणनफल एवं उसी बिन्दु पर स्पर्श रेखा के ढाल का वर्ग तथा अधःस्पर्शी के गुणनफल के अनुपात के बराबर है। ऐसे सभी वक्रो को ज्ञात कीजिए।

Ans. $x = e^{2\sqrt{y/x}}; e^{-2\sqrt{y/x}}$

Sol. Tangent at P(x, y)

$$\Rightarrow Y - y = \frac{dy}{dx}(X - x) \text{ for y intercept } x = 0.$$

$$\Rightarrow Y = y - x\frac{dy}{dx}$$

Given $\frac{Y^2}{my} = \frac{xy}{m^2\frac{y}{m}}$

$$\Rightarrow \left(y - x\frac{dy}{dx}\right)^2 = xy$$

$$\Rightarrow \left(\frac{y}{x} - \frac{dy}{dx}\right)^2 = \frac{y}{x}$$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$\Rightarrow \left(v - v - x\frac{dv}{dx}\right)^2 = v$$

$$\Rightarrow \left|x\frac{dv}{dx}\right| = \sqrt{v} \Rightarrow \int \frac{dv}{\pm\sqrt{v}} = \int \frac{dx}{x}$$

$$\Rightarrow \pm 2\sqrt{v} = \ell nx + \ell nc$$

$$\Rightarrow xc = e^{\pm 2\sqrt{\frac{y}{x}}} \text{ passing from } (1,0)$$

$$\Rightarrow c = 1 \Rightarrow x = e^{\pm 2\sqrt{\frac{y}{x}}}$$

16. A & B are two separate reservoirs of water. Capacity of reservoir A is double the capacity of reservoir B. Both the reservoirs are filled completely with water, their inlets are closed and then the water is released simultaneously from both the reservoirs. The rate of flow of water out of each reservoir at any instant of time is proportional to the quantity of water in the reservoir at that time. One hour after the water is released, the quantity of water in reservoir A is 1.5 times the quantity of water in reservoir B. After how many hours do both the reservoirs have the same quantity of water ?

A एवं B दो भिन्न पानी की टंकिया है। A की क्षमता B की क्षमता से दोगुनी है। दोनों टंकियों को पूरी तरह पानी से भरकर उनके नलों को बंद करके दोनों टंकियों से एक साथ पानी छोड़ा जाता है। किसी टंकी में पानी के बहाव की दर, उस समय टंकी में पानी की मात्रा के समानुपातिक है। पानी खोलने के 1 घण्टे पश्चात् A टंकी में पानी B की 1.5 गुना होता है। कितने घंटे पश्चात् दोनों टंकियों में समान मात्रा में पानी होगा ?

Ans. $T = \log_{4/3} 2$ hrs from the start

Sol. Let at time t hours, the volume of water in A and B be u and v, respectively. From the question,

$$\frac{du}{dt} = -k_1 u \qquad \text{_____(1)}$$

and $\frac{dv}{dt} = -k_2 v$ _____(2)

At $t = 0$, $u = 2v$ and $v = V$ (from the question). Solving Equation (1),

$$\frac{du}{u} = -k_1 dt$$

Integrating both sides, we get

$$\log u = -k_1 t + c_1$$

$$u = e^{-k_1 t + c_1} \qquad \text{_____(3)}$$

When $t = 0$, $u = 2v$. Then

$$2V = e^{c_1}$$

Hence, equation (3) gives

$$u = 2Ve^{-k_1 t} \qquad \text{_____(4)}$$

Similarly, from equation (2), we get

$$v = Ve^{-k_2 t} \qquad \text{_____(5)}$$

From the question, if $v = v_0$ when $t = 1$, then $u = \frac{3}{2} v_0$

Therefore, equation (4) gives

$$\frac{3}{2} v_0 = 2Ve^{-k},$$

and equation (5) gives $v_0 = Ve^{-k_1}$

Dividing these, we get

$$\frac{3}{2}=\frac{2e^{-k_1}}{e^{-k_2}}=2e^{k_2-k_1}$$

Thus,

$$e^{k_2-k_1}=\frac{3}{4} \qquad \text{_____(6)}$$

Let after T hours, the volume of water in A and B be equal. Therefore, equations (4) and (5) gives

$$2Ve^{-k_1T}=Ve^{-k_2T}$$

$$\Rightarrow 2e^{-k_1T}=e^{-k_2T}\Rightarrow 2e^{(k_2-k_1)T}=1$$

Using equation (6), we get

$$2\cdot\left(\frac{3}{4}\right)^{T}=1$$

$$\Rightarrow\left(\frac{4}{3}\right)^{T}=2$$

Therefore,

$$T\log\frac{4}{3}=\log 2\Rightarrow T=\frac{\log 2}{\log\frac{4}{3}}=\log_{\frac{4}{3}}2$$

Hence, after $\log_{\frac{4}{3}}2$ hours, the reservoirs will have the same quantity of water.

17. A tank consists of 50 litres of fresh water. Two litres of brine each litre containing 5 gms of dissolved salt are run into tank per minute; the mixture is kept uniform by stirring, and runs out at the rate of one litre per minute. If 'm' grams of salt are present in the tank after t minute, express 'm' in terms of t and find the amount of salt present after 10 minutes.

किसी टंकी में 50 लीटर ताजा पानी है। एक लीटर में 5 gms घुला लवण वाला 2 लीटर ब्राइन, प्रति मिनट टंकी में जाता है। मिश्रण को हिलाकर एक समान बनाया जाता है, तथा 1 लीटर प्रति मिनट की दर से बाहर निकलता है। यदि t समय पश्चात् टंकी में 'm' ग्राम लवण उपस्थित हो तो 'm' को t के पदों में व्यक्त करो तथा 10 मिनट पश्चात् उपस्थित लवण की मात्रा बताओं।

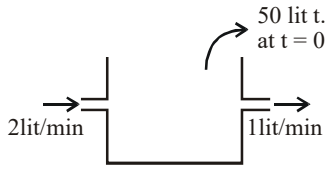
Ans. $y=5t\left(1+\frac{50}{50+t}\right)\text{gms}\,;\,91\frac{2}{3}\text{ gms}$

Sol. Input rate = 10 gm/min

After time t volume of tank = 50 + (2 – 1)t

Concentration of salt in time $t = \frac{m}{50+t}$ gm/lit

Output rate $= \frac{m}{50+t}.1$ gm/min.



$$\frac{dm}{dt} = \frac{-m}{50+t} + 10$$

$$m(50+t) = \int 10(50+t)\,dt$$

$\Rightarrow \quad m(50+t) = 500\,t + 5t^2 + c$

$\Rightarrow \quad (\because \text{ at } t = 0,\ m = 0 \Rightarrow\ c = 0)$

$\Rightarrow \quad m(50+t) = 5\,(100\,t + t^2)$

$$\Rightarrow \quad m = 5t\left(\frac{100+t}{50+t}\right) \text{ gm}$$

18. Find the curve for which the portion of y-axis cut-off between the origin and the tangent varies as cube of the absissa of the point of contact.

किसी वक्र पर स्पर्शी एवं मूल बिन्दु के बीच y-अक्ष का कटा अन्तःखण्ड, स्पर्श बिन्दु के भुज के घन के समानुपातिक है। वक्र का समीकरण बताओं।

Ans. $2y + Kx^3 = cx$

Sol. Equation of tangent at (x, y) is

$$Y - y = \frac{dy}{dx}(X - x), \text{ for y-axis, } X = 0$$

$$\Rightarrow Y = y - x\frac{dy}{dx}$$

Given $\quad y - x\frac{dy}{dx} \propto x^3 \Rightarrow y - x\frac{dy}{dx} = kx^3$

$$\therefore \frac{dy}{dx} - \frac{y}{x} = kx^2 \qquad \left\{IF = \frac{1}{x}\right\}$$

$$\Rightarrow y = \frac{-kx^3}{2} + cx$$

19. Find the equation of the curve which passes through the origin and the tangent to which at every point (x, y) has slope equal to $\frac{x^4 + 2xy - 1}{1 + x^2}$.

मूल बिन्दु से गुजरने वाले उस वक्र का समीकरण ज्ञात कीजिए, जिसके किसी बिन्दु (x, y) पर स्पर्शी की प्रवणता $\frac{x^4+2xy-1}{1+x^2}$ हो।

Ans. $y=(x-2\tan^{-1}x)(1+x^2)$

Sol. It is given that

$$\frac{dy}{dx}=\frac{x^4+2xy-1}{1+x^2}$$

$$\Rightarrow \frac{dy}{dx}-\frac{2x}{1+x^2}y=\frac{x^4-1}{1+x^2}$$

which is a linear differential equation.

$$\therefore IF=e^{-\int\frac{2x}{1+x^2}dx}=e^{-\log\left(x^2+1\right)}=\frac{1}{\left(x^2+1\right)}$$

Hence, the solution is

$$y\cdot\frac{1}{\left(x^2+1\right)}=\int\frac{x^4-1}{\left(x^2+1\right)^2}dx+c$$

$$\Rightarrow \frac{y}{\left(x^2+1\right)}=\int\frac{x^2-1}{\left(x^2+1\right)}dx+c$$

$$=\int\left(1-\frac{2}{x^2+1}\right)dx+c$$

$=(x-2\tan^{-1}x)+c$

20. A right circular cone with radius R and height H contains a liquid which evaporates at a rate proportional to its surface area in contact with air (proportionality constant = k > 0). Find the time after which the cone is empty.

৮ ऊँचाई और R त्रिज्या का एक प्रतिलोम शंकु पैंदे पर रखा हुआ है। इसे एक वाष्पशील द्रव से पूरा भरा जाता है। यदि वाष्पन की दर हवा के सम्पर्क में रहने वाले द्रव के पृष्ठीय क्षेत्रफल के समानुपाती है (समानुपाती नियतांक = k > 0 है), तो वह समय ज्ञात कीजिए जिसमें पूरा द्रव वाष्पित हो जायेगा।

Ans. T = H/k

Sol. Given, liquid evaporates at a rate proportional to its surface area.

$$\Rightarrow \frac{dV}{dt}\propto -S \qquad \text{_____(1)}$$

We know, volume of cone $= \frac{1}{3}\pi r^2 h$

and surface area $= \pi r^2$

or $V = \frac{1}{3}\pi r^2 h$ and $S = \pi r^2$ _____(2)

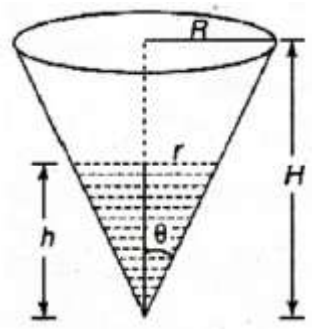
where $\tan\theta = \frac{R}{H}$ and $\frac{r}{h} = \tan\theta$ _____(3)

From equations (ii) and (iii), we get

$V = \frac{1}{3}\pi r^3 \cot\theta$ and $S = \pi r^2$ _____(4)

On substituting equation (4) in equation (1), we get

$$\frac{\pi}{3}\cot\theta \cdot 3r^2 \frac{dr}{dt} = -k\pi r^2$$



$$\Rightarrow \cot\theta \int_R^0 dr = -k\int_0^T dt$$

$$\Rightarrow \cot\theta(0 - R) = -k(T - 0)$$

$$\Rightarrow R\cot\theta = kT$$

$\Rightarrow H = kT$ [From equation (3)]

$$\Rightarrow T = \frac{H}{k}$$

21. If $P(1) = 0$ and $\frac{dP(x)}{dx} > P(x)$ for all $x \geq 1$ then prove that $P(x) > 0$ for all $x > 1$.

यदि $P(1) = 0$ तथा $\frac{dP(x)}{dx} > P(x)\ \ \forall\ x \geq 1$ हो, तो सिद्ध कीजिये कि $P(x) > 0\ \ \forall\ x > 1$.

Ans.

Sol. Given, $P(1) = 0$ and

$$\frac{dP(x)}{dx} - P(x) > 0, \forall x \geq 1 \qquad ...(1)$$

On multiplying by e^{-x} ($e^{-x} > 0$)

$$\Rightarrow e^{-x} \cdot \frac{d}{dx} P(x) + P(x) \cdot \frac{d}{dx} e^{-x} > 0$$

$$\Rightarrow \frac{d}{dx}\left(P(x) \cdot e^{-x}\right) > 0$$

$\Rightarrow P(x) \cdot e^{-x}$ is an increasing function

$$\Rightarrow P(x) \cdot e^{-x} > P(1) \cdot e^{-1}, \forall x \geq 1$$

$$\Rightarrow P(x) > 0, \forall x > 1 \ [P(1) = 0 \text{ and } e^{-x} > 0]$$

22. Let f(x) be a differentiable function in $[-1, \infty)$ and $f(0) = 1$ such that $\lim\limits_{t \to x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$. Find the value of $\lim\limits_{x \to 1} \frac{\ln(f(x)) - \ln 2}{x - 1}$.

माना f(x) अन्तराल $[-1, \infty)$ में अवकलनीय फलन है तथा $f(0) = 1$ इस प्रकार है कि $\lim\limits_{t \to x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$ है, तो $\lim\limits_{x \to 1} \frac{\ln(f(x)) - \ln 2}{x - 1}$ का मान ज्ञात कीजिए।

Ans. 1

Sol. $h = t - (x + 1)$

$$= \lim_{h \to 0} \frac{\left(h + (x+1)\right)^2 f(x+1) - (x+1)^2 f\left(h + (x+1)\right)}{f\left(h + (x+1)\right) + f(x+1)}$$

$$= \lim_{h \to 0} \frac{\left(h^2 + 2h(x+1)\right) f(x+1) - \left[f\left(h + (x+1)\right) - f(x+1)\right](x+1)^2}{f\left(h + (x+1)\right) - f(x+1)}$$

$$= \lim_{h\to 0} \frac{\left(h+2(x+1)\right)f(x+1) - \frac{\left(f\left(h+(x+1)\right)-f(x+1)\right)(x+1)^2}{h}}{\frac{\left(f\left(h+(x+1)\right)-f(x+1)\right)}{h}}$$

$$= \frac{2(x+1)f(x+1) - f'(x+1)(x+1)^2}{f'(x+1)}$$

Now, $\frac{2(x+1)f(x+1) - f'(x+1)(x+1)^2}{f'(x+1)} = 1$

$$\Rightarrow f'(x+1)\left((x+1)^2+1\right) - 2(x+1)f(x+1) = 0$$

Replace $x + 1$ by x

$$\Rightarrow f'(x)\left(x^2+1\right) - 2xf(x) = 0$$

$$\Rightarrow \frac{dy}{dx} - \frac{2xy}{x^2+1} = 0 \Rightarrow \frac{dy}{y} - \frac{2x}{x^2+1}dx = 0$$

$\Rightarrow \ell n\, y - \ell n\,(x^2 + 1) = \ell n\, c$

$\Rightarrow y = c(x^2 + 1)$ at $x = 0, y = 1$

$\Rightarrow c = 1 \quad \Rightarrow y = (x^2 + 1)$

Now, $\lim_{x\to 1} \frac{\ell n f(x) - \ell n 2}{x-1}$

$$= \lim_{x\to 1} \frac{\ell n \frac{x^2+1}{2}}{x-1} = \lim_{x\to 1} \frac{\frac{x^2+1}{2} - 1}{x-1}$$

$$= \frac{1}{2}\lim_{x\to 1} \frac{x^2-1}{x-1} = \frac{1}{2}\lim_{x\to 1}(x+1) = 1$$

23. Let $y = (a \sin x + (b + c) \cos x)\, e^{x+d}$, where a, b, c and d are parameters represent a family of curves, then differential equation for the given family of curves is given by $y'' - \alpha y' + \beta y = 0$, then $\alpha + \beta =$

माना $y = (a \sin x + (b + c) \cos x)\, e^{x+d}$, जहाँ a, b, c तथा d प्राचल है, एक वक्र निकाय को निरूपित करता है, तो दिये गये वक्र fud k; d kv od y 1 ehdj .ky'' – $\alpha y' + \beta y = 0$ है, तो $\alpha + \beta$ =

Ans. 4

Sol. $y = (a \sin x + (b + c) \cos x)\, e^{x+d}$

$y = (A \sin x + B \cos x)\, e^{x}$ ...(1)

$y' = e^{x}(A \sin x + B \cos x) + e^{x} (A \cos x - B \sin x)$

$y' = y + e^{x} (A \cos x - B \sin x)$ ...(2)

$y'' = y' + e^{x} (A \cos x - B \sin x) + e^{x}(-A \sin x - B \cos x)$

$= y' + y' - y - y$

$\Rightarrow y'' - 2y' + 2y = 0, \quad \alpha = 2, \beta = 2$

$\Rightarrow \alpha + \beta = 4$

24. let $y = f(x)$ satisfies the differential equation $xy(1 + y)\, dx = dy$. If $f(0) = 1$ and $f(2) = \dfrac{e^2}{k - e^2}$, then find the value of k.

माना $y = f(x)$ अवकल समीकरण $xy(1 + y)\, dx = dy$ को संतुष्ट करता है। यदि $f(0) = 1$ तथा $f(2) = \dfrac{e^2}{k - e^2}$ है, तो k का मान होगा।

Ans. 2

Sol. $xy(1 + y)dx = dy$

$$\Rightarrow x\,dx = \frac{1}{y(1+y)}\,dy$$

$$\Rightarrow x\,dx = \frac{1+y-y}{y(1+y)}\,dy$$

$$\Rightarrow x\,dx = \frac{1}{y} - \frac{1}{1+y}\,dy$$

$$\Rightarrow \int x\,dx = \int \frac{1}{y} - \frac{1}{1+y}\,dy$$

$$\frac{x^2}{2} = \log y - \log(1+y) + \log C$$

$$f(0) = 1 \Rightarrow \frac{0^2}{2} = \log 1 - \log 2 + \log C \Rightarrow C = 2$$

Now, $x = 2 \Rightarrow \frac{4}{2} = \ell nf(2) - \ell n(1+f(2)) + \ell n2$

$\Rightarrow e^2 = \frac{2f(2)}{1+f(2)} \Rightarrow e^2 = \frac{-2}{1+f(2)}$

$\Rightarrow 1+f(2) = \frac{2}{2-e^2}$

$\Rightarrow f(2) = \frac{e^2}{2-e^2}, k = 2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 25 to 27

Given a function 'g' which has a derivative g′(x) for every real x and which satisfy g′(0) = 2 and $g(x + y) = e^y g(x) + e^x g(y)$ for all x and y.

दिया गया फलन 'g' का सभी वास्तविक x के लिए अवकलन g′(x) है तथा सभी x तथा y के लिए g′(0) = 2 तथा $g(x + y) = e^y g(x) + e^x g(y)$ को संतुष्ट करता है।

25. The function g(x) is :

फलन g(x) है :

(A) $x(2 + xe^x)$ (B) $x(e^x + 1)$ (C) $2xe^x$ (D) $x + \ln(x + 1)$

Ans. C

Sol. Put $x = 0; y = 0 \Rightarrow g(0) = 0$

and $g'(0) = \lim_{h\to 0} \frac{g(h)}{h} = 2$

Now, $g'(x) = \lim_{h\to 0} \frac{g(x+h) - g(x)}{h}$

$= \lim_{h\to 0} \frac{e^h g(x) + e^x g(h) - g(x)}{h}$

$= h(x) \lim_{h\to 0} \left(\frac{e^h - 1}{h}\right) + e^x \lim_{h\to 0} \frac{g(h)}{h}$

$\Rightarrow g'(x) = g(x) + 2e^x$

Let $g(x) = y$ then $\frac{dy}{dx} = y + 2e^x$

$\Rightarrow e^{-x}\frac{dy}{dx} - ye^{-x} = 2$

$\Rightarrow \frac{d}{dx}\left(ye^{-x}\right) = 2$

$\Rightarrow ye^{-x} = 2x + c$

Given if $x = 0, y = 0 \Rightarrow c = 0$

Then $ye^{-x} = 2x \Rightarrow y = 2xe^x$

26. The range of function g(x) is :

फलन g(x) का परिसर है :

(A) R (B) $\left[-\frac{2}{e}, \infty\right)$ (C) $\left[-\frac{1}{e}, \infty\right)$ (D) $[0, \infty)$

Ans. B

Sol. Now $\frac{dy}{dx} = 2\left[e^x + xe^x\right] = 0 \Rightarrow x = -1$

$\Rightarrow$ minima at $x = -1$

$\Rightarrow$ range is $\left[-\frac{2}{e}, \infty\right)$

27. The value of $\lim\limits_{x\to-\infty} g(x)$ is :

(A) 0 (B) 1 (C) 2 (D) Does not exist

$\lim\limits_{x\to-\infty} g(x)$ का मान होगा :

(A) 0 (B) 1 (C) 2 (D) विद्यमान नहीं है

Ans. A

Sol. $g(x) = 2xe^x$

$$\lim_{x\to-\infty} g(x) = 0$$

Paragraph for Question nos. 28 to 30

Let $f: R^+ \to R$ be a differentiable function satisfying $f(x) = e + (1-x)\ln\left(\frac{x}{e}\right) + \int_1^x f(t)\,dt$ for all $x \in R^+$.

माना $f: R^+ \to R$ एक अवकलनीय फलन $f(x) = e + (1-x)\ln\left(\frac{x}{e}\right) + \int_1^x f(t)\,dt$ सभी $x \in R^+$ के लिए संतुष्ट करता है।

28. The value of definite integral $\int_0^1 f(x)\,dx$ is equal to :

निश्चित समाकल $\int_0^1 f(x)\,dx$ का मान होगा :

(A) e (B) e – 2 (C) e + 1 (D) $\frac{e+1}{2}$

Ans. B

Sol. $f'(x) = 0 + (1-x)\left(\frac{1}{x}\right) + \ell n\left(\frac{x}{e}\right)(-1) + f(x)$ (By Newton's Leibnitz rule)

$$\Rightarrow f'(x) = \frac{1}{x} - 1 + (\ell nx - 1)(-1) + f(x)$$

$$\Rightarrow \frac{dy}{dx} - y = \frac{1}{x} - \ell nx$$

$$\Rightarrow ye^{-x} = \int\left(\frac{1}{x} - \ell nx\right)e^{-x}dx$$ (solving of linear differential equation)

$$\Rightarrow ye^{-x} = e^{-x}(\log x) + c$$

Since, $f(1) = e \Rightarrow ee^{-1} = 0 + c \Rightarrow c = 1$

$y = \log x + e^x$

$$\int_0^1 (\ell nx + e^x)dx = \left[x(\ell nx - 1) + e^x\right]_0^1$$

$= -1 + e - 0 - 1 = e - 2$

29. The x-intercept of normal drawn to the curve $y = f(x)$ at point P where $y = f(x)$ crosses the line $x = 1$, is equal to :

$y = f(x)$ के बिन्दु P जहाँ $y = f(x)$ रेखा $x = 1$ को काटता है, पर अभिलम्ब का x अंतःखण्ड होगा :

(A) $e^2 + e - 1$ (B) $\frac{e^2+e+1}{e+1}$ (C) $\frac{e^2-e+1}{e+1}$ (D) $e^2 + e + 1$

Ans. D

Sol. $f(1) = e$

$$\frac{dy}{dx} = \frac{1}{x} + e^x$$

$$\left.\frac{dy}{dx}\right|_{(1,e)} = 1 + e = m_T$$

$$m_n = \frac{-1}{1+e}$$

equation of normal $\quad y - e = -\frac{1}{1+e}(x-1)$

for x intercept $y = 0$

$\Rightarrow x = e^2 + e + 1$

30. Which one of the following statement is correct for function f(x)?

(A) f has maxima but no minima (B) f has minima but no maxima

(C) f has neither maxima nor minima (D) f has both maxima and minima

फलन f(x) के लिए निम्न में से कौनसे कथन सत्य है?

(A) f का उच्चिष्ठ है लेकिन निम्निष्ठ नहीं (B) f का निम्निष्ठ है लेकिन उच्चिष्ठ नहीं

(C) f का न तो उच्चिष्ठ है ना ही निम्निष्ठ है (D) f के दोनों उच्चिष्ट तथा निम्निष्ठ है

Ans. C

Sol. $f'(x) = \frac{1}{x} + e^x$

for $x > 0 \Rightarrow f'(x) > 0$

f is neither maxima nor minima.

MATCH THE COLUMN TYPE QUESTION(S)

31. Match the following column :

Column-I (Differential equation)	**Column-II (Integral curves)**
(A) $y - x\frac{dy}{dx} = y^2 + \frac{dy}{dx}$	(P) $y = A_1x^2 + A_2x + A_3$
(B) $(2x - 10y^3)\frac{dy}{dx} + y = 0$	(Q) $x^2y^2 + 1 = cy$
(C) $\left(\frac{dy}{dx}\right)\left(\frac{d^3y}{dx^3}\right) - 3\left(\frac{d^2y}{dx^2}\right)^2 = 0$	(R) $(x + 1)(1 - y) = cy$
(D) $(x^2y^2 - 1)dy + 2xy^3\,dx = 0$	(S) $x = A_1y^2 + A_2y + A_3$
	(T) $xy^2 = 2y^5 + c$

निम्न कॉलम को सुमेलित कीजिए :

Column-I (अवकल समीकरण**)**	**Column-II (**समाकल वक्र**)**
(A) $y - x\frac{dy}{dx} = y^2 + \frac{dy}{dx}$	(P) $y = A_1x^2 + A_2x + A_3$
(B) $(2x - 10y^3)\frac{dy}{dx} + y = 0$	(Q) $x^2y^2 + 1 = cy$
(C) $\left(\frac{dy}{dx}\right)\left(\frac{d^3y}{dx^3}\right) - 3\left(\frac{d^2y}{dx^2}\right)^2 = 0$	(R) $(x + 1)(1 - y) = cy$
(D) $(x^2y^2 - 1)dy + 2xy^3\,dx = 0$	(S) $x = A_1y^2 + A_2y + A_3$
	(T) $xy^2 = 2y^5 + c$

Ans. A – R, B – T, C – S, D – Q

Sol. (A) $y - x\frac{dy}{dx} = y^2 + \frac{dy}{dx}$

$$\Rightarrow \frac{dy}{dx} = \frac{y(1-y)}{1+x}$$

$$\Rightarrow \int \frac{dy}{y(1-y)} = \int \frac{dx}{1+x}$$

$$\Rightarrow \int\left(\frac{1}{y} - \frac{1}{(y-1)}\right)dy = \int \frac{dx}{1+x}$$

$\Rightarrow \ell ny - \ell n(y-1) = \ell n(1+x) - \ell nC$

$\Rightarrow yc = (1+x)(1-y)$

(B) $\left(2x - 10y^3\right)\frac{dy}{dx} + y = 0$

$\Rightarrow \frac{dx}{dy} = \frac{2x - 10y^3}{-y} \Rightarrow \frac{dx}{dy} + \frac{2x}{y} = 10y^2$

I.F. $= e^{\int \frac{2}{y} dy} = y^2$

$x.y^2 = \int y^2 \left(10y^2\right) dy$

$\Rightarrow xy^2 = 2y^5 + c$

(C) $\frac{dy}{dx}.\frac{d^3y}{dx^3} - 3\left(\frac{d^2y}{dx^2}\right)^2 = 0$

Let $\frac{dy}{dx} = v$ ...(1)

$\Rightarrow v \cdot v'' - 3(v')^2 = 0$

$\Rightarrow v.\frac{d^2v}{dx^2} - 3\left(\frac{dv}{dx}\right)^2 = 0$ ...(2) (v is variable of "x")

Now let $u(v) = \frac{dv}{dx}$ ...(3) [u is variable of v]

$u'(v)\frac{dv}{dx} = \frac{d^2v}{dx^2}$

$\Rightarrow u'(v)u(v) = \frac{d^2v}{dx^2}$ $\left[u'(v) = \frac{du}{dv}\right]$

put in equation (2)

$\Rightarrow v \cdot u' \cdot u - 3u^2 = 0$ or $u(v) = 0$

$\Rightarrow v\frac{du}{dv} - 3u = 0$

$$\Rightarrow \frac{du}{u} - 3\frac{dv}{v} = 0$$

$\Rightarrow u = cv^3$ or $u(v) = 0$

from equation (3)

$\frac{dv}{dx} = c_1 v^3$ or $\frac{dv}{dx} = 0$

$\Rightarrow \int \frac{dv}{v^3} = \int c_1 dx$ or $v = c_1$

$$\Rightarrow -\frac{1}{2v^2} = c_1 x - c_2$$

$\Rightarrow v = \pm \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{c_2 - c_1 x}}$ or $v = c_1$

Now $v = c_1$ or $v = \pm \frac{1}{\sqrt{2}} \frac{1}{\sqrt{c_2 - c_1 x}}$

using equation (1)

$\frac{dy}{dx} = c_1$ or $\frac{dy}{dx} = \pm \frac{1}{\sqrt{2}} \frac{1}{\sqrt{c_2 - c_1 x}}$

$y = c_1 x + c_2$ or $y = \pm \frac{\sqrt{2}\sqrt{c_2 - c_1 x}}{c_1} + c_3$

$y = c_1 x + c_2$ or $(y - c_3)^2 = \frac{2(c_2 - c_1 x)}{c_1^2}$

$\Rightarrow y = c_1 x + c_2$ or $c_1^2 y^2 - 2c_1^2 c_3 y + c_3^2 c_1^2 = 2c_2 - c_1 x$

$$x = \frac{-c_1 y^2 + 2c_1 c_3 y - c_3^2 c_1 + \frac{c_2}{c_1}}{2}$$

$$x = Ay^2 + By + C$$

M-2

$$y'y'''-3(y'')^2=0$$

$$\Rightarrow \frac{y'y'''-(y'')^2}{(y'')^2}=2 \qquad [\text{If } y''\neq 0]$$

$$\Rightarrow \frac{d}{dx}\left(\frac{-y'}{y''}\right)=2$$

$$\Rightarrow \int d\left(\frac{y'}{y''}\right)=\int -2dx$$

$$\Rightarrow \frac{y'}{y''}=-2x+c_1 \qquad \text{Let } y'=v$$

$$\text{then } \frac{v}{v'}=-2x+c_1$$

$$\Rightarrow \int \frac{dv}{v}=\int \frac{dx}{c_1-2x}$$

$$\Rightarrow \ell nv=-\frac{1}{2}\ell n(c_1-2x)+\ell nc_2$$

$$\Rightarrow v=\frac{c_2}{\sqrt{c_1-2x}} \Rightarrow \frac{dy}{dx}=\frac{c_2}{\sqrt{c_1-2x}}$$

$$\Rightarrow \int dy=c_2\int \frac{dx}{\sqrt{c_1-2x}}$$

$$\Rightarrow y=-c_2\sqrt{2}\sqrt{c_2-2x}+c_3$$

$$\Rightarrow (y-c_3)^2=2c_2^2(c_2-2x)$$

$$\Rightarrow y^2-2c_3y+c_3^2=2c_2^3-4c_2^2x$$

$$\Rightarrow 4c_2^2x=-y^2+2c_3y+2c_2^3-c_3^2$$

$$\Rightarrow k_1x=-y^2+k_2y+k_3$$

$$\Rightarrow x=Ay^2+By+C$$

or $y'' = 0 \Rightarrow y = mx + c$

(D) $\left(x^2y^2 - 1\right)dy + 2xy^3dx = 0$

$\Rightarrow x^2y^2dy + 2xy^3dx = dy$

$\Rightarrow x^2dy + 2xydx = \frac{dy}{y^2}$

$\Rightarrow \int d\left(x^2y\right) = \int \frac{dy}{y^2} + C$

$\Rightarrow x^2y = \frac{y^{-1}}{-1} + C$

$\Rightarrow x^2y^2 = -1 + cy$

$\Rightarrow 1 + x^2y^2 = cy$

32. Match the statements/expressions In Column I with the open intervals In Column II : **[JEE-2009]**

Column–I	**Column –II**
(A) Interval contained in the domain of definition of non-zero solutions of the differential equation $(x - 3)^2 y' + y = 0$	(P) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(B) Interval containing the value of the integral $\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)dx$	(Q) $\left(0, \frac{\pi}{2}\right)$
	(R) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$
(C) Interval in which at least one of the points of local maximum of $\cos^2 x + \sin x$ lies	(S) $\left(0, \frac{\pi}{8}\right)$
(D) Interval in which $\tan^{-1}(\sin x + \cos x)$ is increasing	(T) $(-\pi, \pi)$

कॉलम–I में दिए गये प्रकथन/व्यंजको का कॉलम–II में दिए गए विवृत अंतरालों से सुमेल करें : **[JEE-2009]**

Column–I	**Column –II**
(A) अवकल समीकरण $(x - 3)^2 y' + y = 0$ के अशून्य हलों के परिभाषा प्रान्त में समाहित अन्तराल	(P) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(B) अन्तराल जिसमें समाकल $\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)dx$ (Q) $\left(0, \frac{\pi}{2}\right)$

का मान अन्तर्विष्ट है

(C) अन्तराल जिसमें कम से कम एक ऐसा बिन्दु जिस पर (R) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$

$\cos^2 x + \sin x$ स्थानीय अधिकतम हो। (S) $\left(0, \frac{\pi}{8}\right)$

(D) अन्तराल जिसमें $\tan^{-1}(\sin x + \cos x)$ वर्धमान है (T) $(-\pi, \pi)$

Ans. (A)P,Q,S (B) P,T (C) P,Q,R,T (D)S

Sol. (A) $\frac{dy}{dx} = -\frac{y}{(x-3)^2} \Rightarrow \ln y = \frac{1}{x-3} + c$

$\Rightarrow y = e^{\frac{1}{x-3}+c}$, $x \neq 3$.

(B) $I = \int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)\,dx$

Applying $x \to 6-x$

$I = \int_1^5 (5-x)(4-x)(3-x)(2-x)(1-x)\,dx = -I$

$\Rightarrow I = 0$.

(C) $f(x) = \cos^2 x + \sin x$

$f'(x) = -2\cos x \sin x + \cos x$

$\Rightarrow \cos x(-2\sin x + 1) = 0$

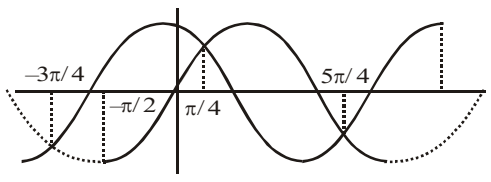
$\cos x = 0$ or $\sin x = \frac{1}{2}$

sign of $f'(x)$ changes from –ve to +ve while $f(x)$ passes through $x = \frac{\pi}{6}, \frac{5\pi}{6}$.

(D) $f(x) = \tan^{-1}(\sin x + \cos x)$

$f(x) = \frac{\cos x - \sin x}{1 + (\sin x + \cos)^2} > 0$

$x \in (-3\pi/4, \pi/4)$



33. Match the statements/expressions given in Column I with the values given in Column II : **[JEE-2009]**

Column–I	**Column –II**
(A) The number of solutions of the equation $xe^{\sin x} - \cos x = 0$ in the interval $\left(0, \frac{\pi}{2}\right)$	(P) 1
(B) Value(s) of k for which the planes $kx + 4y + z = 0$, $4x + ky + 2z = 0$ and $2x + 2y + z = 0$ intersect in a straight line	(Q) 2
(C) Value(s) of k for which $\|x - 1\| + \|x - 2\| + \|x + 1\| + \|x + 2\| = 4k$ has integer solution(s)	(R) 3
(D) If $y' = y + 1$ and $y(0) = 1$, then value(s) of $y(\ln 2)$	(S) 4
	(T) 5

कॉलम–I में दिए गये वक्तव्यों / व्यंजको का कॉलम–II में दिये गये मानों से सुमेल कीजिए : **[JEE-2009]**

Column–I	**Column –II**
(A) अन्तराल $\left(0, \frac{\pi}{2}\right)$ में समीकरण $xe^{\sin x} - \cos x = 0$ के हलों की संख्या	(P) 1
(B) k के मान (मानों) जिनके लिये समतल $kx + 4y + z = 0$, $4x + ky + 2z = 0$ तथा $2x + 2y + z = 0$ एक सरल रेखा में प्रतिच्छेद करते है (हैं)	(Q) 2
(C) k के मान (मानों) जिनके लिये $\|x - 1\| + \|x - 2\| + \|x + 1\| + \|x + 2\| = 4k$ का (के) हल पूर्णांक है (हैं)	(R) 3
(D) यदि $y' = y + 1$ तथा $y(0) = 1$, तब $y(\ln 2)$ का मान (मानों)	(S) 4
	(T) 5

Ans. (A) P; (B) Q,S ; (C) Q,R,S,T ; (D)R

Sol. (A) $x.e^{\sin x} - \cos x = 0$

$\Rightarrow e^{\sin x} = \frac{\cos x}{x}$

Let $f(x) = \frac{\cos x}{x}, g(x) = e^{\sin x}$

According the graph

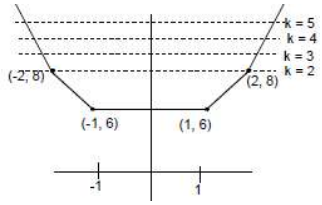
Number of solution is 1.

(B) $\begin{vmatrix} k & 4 & 1 \\ 4 & k & 2 \\ 2 & 2 & 1 \end{vmatrix} = 0$

$\Rightarrow k = 2, 4$

(C) $|x + 1| + |x - 2| + |x + 1| + |x + 2| = 4k$

graph will be



So, solution is k = 2,3,4,5

(D) $\frac{dy}{dx} = y + 1$

$$\int \frac{dy}{y+1} = \int dx$$

$\ln (y + 1) = x + C$

$(x + c)$

$y + 1 = e^{(x+c)}$

$y = c^{x+c} - 1$

$x = 0, y = 1$

$1 = e^{c} - 1$

$e^{c} = 2$

$C = \ln 2$

$y = e^{x + \ln 2} - 1$

for $x = \ln 2$

$e^{2\ln 2} - 1$

$y = 3.$